

AIM Foundations

**Curvature-Driven Modelling, Classification, Runtime Evaluation, Realization,
and Optimization of Structured Railway Alignments**

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8. Juni 2026

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Preface

This manuscript is intended as a long-term scientific foundation for alignment-based information modelling.

It is not merely software documentation, but a structured knowledge base covering:

- intrinsic railway geometry,
- curvature-driven reconstruction,
- runtime evaluation,
- realization processes,
- vehicle-space interpretation,
- optimization,
- and Earth-attached railway geometry.

The AIM framework treats railway alignments not merely as static curves, but as coupled operator-based state systems connecting geometry, construction, runtime interpretation, and operational behaviour.

Executive Summary

Alignment-Based Information Modelling (AIM)

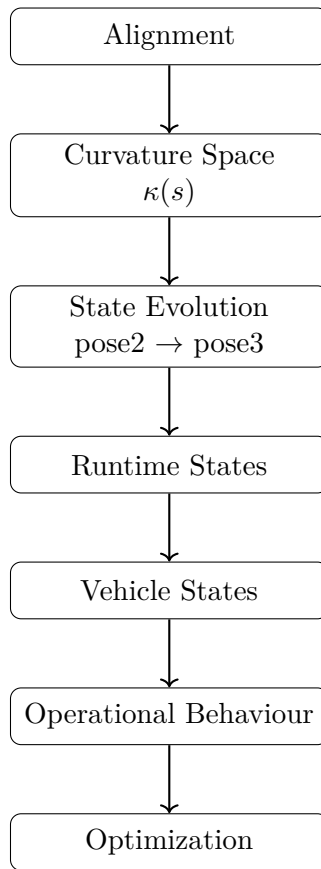


Abbildung 1.: Overview of the AIM framework.

Railway alignments are traditionally described as geometric curves in space. This perspective is insufficient for modern engineering tasks, which require the integration of design, measurement, construction, and optimization.

Alignment is not a curve. It is a curvature-driven system whose geometry emerges from a structured process.

This work introduces *Alignment-Based Information Modelling (AIM)* as a unified framework for this perspective.

—

Core Idea

AIM represents alignments intrinsically by curvature:

$$\kappa(s).$$

This formulation:

- unifies geometry and dynamics,
 - reduces model complexity,
 - enables efficient computation and optimization.
-

From Representation to Process

AIM replaces static modelling by an operator pipeline:

$$\mathcal{P} = \mathcal{W}^{-1} \circ \mathcal{R} \circ \mathcal{M} \circ \mathcal{W} \circ \mathcal{C}.$$

This pipeline connects:

- coordinate systems,
- world geometry,
- intrinsic track coordinates,
- modelling,
- realization,
- and optimization.

Alignment modelling becomes a process.

Hybrid Modelling

AIM combines structure and flexibility:

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s).$$

This hybrid representation:

- preserves engineering intent,

- integrates measurement data,
 - avoids overfitting.
-

Realization as a First-Class Concept

The designed alignment is not identical to the constructed track.

A realization operator

$$\mathcal{R}$$

maps the theoretical model to physical geometry, introducing smoothing and constraints.

Design must therefore be performed in realized space.

Optimization

Alignment design is formulated as a structured nonlinear least-squares problem.

Due to:

- locality in arc-length,
- sparse Jacobians,
- intrinsic block structure,

the problem is efficiently solvable using SQP methods.

World-to-Track Mapping

A central component is the transformation:

$$x \leftrightarrow (s, d).$$

This mapping:

- links measurement data to the model,
 - introduces nonlinear complexity,
 - motivates hybrid and LOD-based strategies.
-

Key Principles

- curvature-centric modelling,
 - separation of model and realization,
 - hybrid parametric–discrete representation,
 - operator-based architecture,
 - optimization in realized space,
 - exploitation of sparsity and structure.
-

Outcome

AIM provides a coherent framework for:

- reconstruction from imperfect data,
 - physically meaningful optimization,
 - integration of design, measurement, and construction,
 - scalable computational implementation.
-

Conclusion

Alignment modelling is not a purely geometric task.

It is the integration of:

- geometry,
- physics,
- data,
- and computation.

AIM formalizes this integration as a structured operator framework, establishing a new foundation for railway alignment engineering.

Core Thesis

Alignment is not a curve.

It is a curvature-driven, realization-aware generative system.

Remark 0.1. Classical railway geometry represents track layouts as curves in Euclidean space. While sufficient for visualization, this perspective is inadequate for design, construction, and dynamic analysis.

Remark 0.2. Within the AIM framework, the fundamental object is not the curve $\gamma(s)$, but the curvature law $\kappa(s)$, complemented by the cant function $\phi(s)$. All geometric quantities are derived by integration.

Remark 0.3. This shifts the modelling paradigm from descriptive geometry to generative structure: alignments are defined by the laws that produce curves, not by the curves themselves.

Remark 0.4. The overall AIM interpretation is summarized schematically in fig. 1.

Key Principles

Curvature as Primary Variable The alignment is defined by curvature $\kappa(s)$. Position $\gamma(s)$ is a derived quantity. This inversion simplifies modelling and exposes the intrinsic structure of railway geometry.

Realization as Operator The physically constructed track differs from the theoretical alignment. This transformation is captured by a realization operator $\mathcal{R}$, which includes discretization, interpolation, and mechanical response. Design must therefore be formulated in realized space.

Hybrid Representation Real-world data and engineering design require both structure and flexibility. A hybrid model combines parametric representations with local corrections, enabling robust reconstruction from imperfect data.

Analytical Structure All relevant quantities arise from arc-length based relations. As a consequence, derivatives with respect to parameters are available in closed form. This eliminates the need for numerical differentiation.

Efficient Optimization Alignment optimization reduces to a structured nonlinear least-squares problem with sparse and banded Jacobians. This enables efficient solution via Gauss–Newton or SQP methods.

Implication

Remark 0.5. The AIM framework unifies geometry, dynamics, realization, and data processing within a single coherent model.

It replaces curve-based thinking by a system-based perspective in which alignment, realization, and optimization are intrinsically linked.

Remark 0.6. This provides a foundation for alignment-based information modelling, capable of bridging theoretical design and practical railway engineering.

AIM Roadmap

Remark 0.7. The present thesis develops Alignment-based Information Modelling (AIM) through a sequence of increasingly richer modelling concepts.

Remark 0.8. The purpose of this roadmap is to provide a global overview before the individual concepts are introduced in detail.

Reading Guide

The development of the thesis follows four major stages.

1. **Geometry.**

Railway alignments are reinterpreted as curvature-driven objects. The central geometric state is introduced as *pose2*, which emerges from curvature integration rather than from explicit coordinate construction.

2. **State.**

The geometric description is extended to spatial and operational contexts through *pose3*, vehicle-space concepts, and runtime state representations.

3. **Reality and Runtime.**

Real railway systems are never ideal. Measurement uncertainty, realization effects, construction tolerances, and operational influences require runtime-aware modelling concepts.

4. **Optimization.**

Once geometry, state, runtime, and realization are unified, railway engineering can be reformulated as an optimization problem in state space rather than purely as a geometric construction problem.

Summary 0.9. The thesis begins with railway geometry, extends geometry into state, extends state into runtime and realization, and finally arrives at optimization-oriented Alignment-based Information Modelling (AIM).

Within the broader perspective proposed here, AIM may be interpreted as one specialization of a more general Process Information Modelling (PIM) framework.

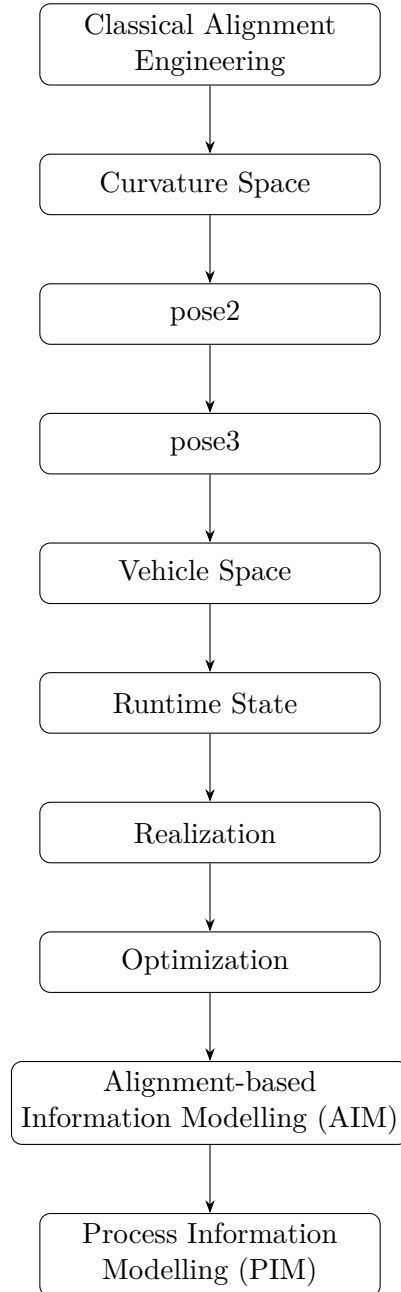


Abbildung 2.: Conceptual roadmap of the AIM framework. The thesis proceeds from intrinsic railway geometry toward runtime, realization, and optimization concepts, culminating in Alignment-based Information Modelling (AIM) as part of a broader Process Information Modelling (PIM) perspective.

Abstract

Railway alignments are traditionally represented as geometric curves in space. This perspective is insufficient for modern engineering tasks, which require the integration of design, measurement, construction, and optimization.

This work introduces *Alignment-Based Information Modelling (AIM)*, a unified framework that reformulates alignment modelling as a curvature-driven and realization-aware system. Instead of treating geometry as a primary object, AIM represents alignments intrinsically through curvature and cant, from which all geometric and dynamic quantities are derived.

A central contribution is the introduction of a realization operator, which models the transformation from theoretical alignment to physically constructed track. This leads to the formulation of alignment design in realized space rather than in purely geometric terms.

The framework combines parametric structure with discrete corrections in a hybrid representation, enabling robust reconstruction from imperfect data. Alignment optimization is formulated as a structured nonlinear least-squares problem with sparse analytical Jacobians, allowing efficient solution via SQP methods.

AIM provides a coherent foundation that unifies geometry, dynamics, data, and computation, establishing a new perspective on railway alignment engineering.

Introduction

1. A Shift in Perspective

Remark 0.1. Railway alignment is commonly described as a geometric curve $\gamma(s)$ embedded in space.

Remark 0.2. This view is sufficient for visualization, but it fails to capture the quantities that actually govern railway design: curvature, cant, dynamics, and realization.

Remark 0.3. In practice, railway tracks are not designed, constructed, or maintained as curves, but through operations acting on curvature and discrete control points.

Summary 0.4. Alignment is not a curve. It is a curvature-driven system whose geometry emerges from its generative structure.

2. Limitations of Current Practice

Remark 0.5. Despite a long tradition of railway engineering, alignment modelling remains fragmented across several domains:

- geometric construction of curves,
- rule-based engineering design,
- data exchange formats such as LandXML and IFC,
- numerical evaluation and simulation.

Remark 0.6. These aspects are typically handled in separate tools and workflows, leading to discontinuities between modelling, analysis, and design.

Remark 0.7. In particular:

- geometry is treated independently of dynamics,
- transition curves are selected from predefined catalogues,
- data formats focus on representation rather than computation,
- optimization is not part of the core modelling paradigm.

Remark 0.8. As a consequence, alignment design lacks a unified internal model that connects data, geometry, dynamics, and realization.

3. Problem Statement

Remark 0.9. The absence of a unified modelling framework leads to fundamental limitations:

- inconsistent interpretation of alignment data,
- separation of geometry and dynamics,
- limited support for optimization-based design,
- high effort for integrating heterogeneous data sources.

Remark 0.10. These issues become increasingly critical as infrastructure systems grow in complexity and demand higher levels of automation, precision, and traceability.

4. AIM: A Unified Framework

Remark 0.11. This work introduces **Alignment-Based Information Modelling (AIM)** as a framework addressing these limitations.

Remark 0.12. The central idea is to represent alignment as a structured, state-based object defined by curvature and cant:

$$(\kappa(s), \phi(s)),$$

from which all geometric and dynamic quantities are derived.

Remark 0.13. Within this framework:

- curvature is the primary modelling variable,
- geometry is obtained by integration,
- transition curves are interpreted as control functions,
- alignment becomes a generative system rather than a static object.

5. Core Claim

Proposition 0.14. *Railway alignment modelling can be reformulated as a curvature-driven, realization-aware, and optimization-ready system.*

Remark 0.15. This reformulation enables a unified treatment of:

- geometry,
- dynamics,
- construction and realization,
- data integration,
- and optimization.

6. Consequences

Remark 0.16. Adopting this perspective leads to several key consequences:

- alignment design becomes an optimization problem,
- realization is treated as an operator acting on the model,
- data reconstruction is formulated as an inverse problem,
- hybrid models bridge parametric structure and measured data,
- analytical derivatives arise naturally from the formulation.

7. Scope of the Work

Remark 0.17. The present manuscript develops the AIM framework across the following dimensions:

- mathematical foundations of curvature-based modelling,
- structured representation of alignment elements,
- classification and interpretation of transition curves,
- modelling of realization and construction processes,
- formulation of optimization problems and solution methods,
- integration of real-world data and coordinate systems,
- application to railway engineering use cases.

8. Contribution

Remark 0.18. The main contributions of this work are:

- a unified state-based alignment model,
- the reinterpretation of transition curves as curvature control functions,
- the introduction of a realization-aware modelling perspective,
- the formulation of alignment design as a structured optimization problem,
- the development of a pipeline connecting data, model, and computation.

Remark 0.19. Together, these contributions establish a coherent framework linking theoretical modelling and practical engineering.

9. Structure of the Manuscript

Remark 0.20. The manuscript is organized into several parts, progressing from conceptual foundations to practical applications:

- motivation and conceptual shift,
- mathematical and state-based foundations,
- modelling and representation,
- realization and system perspective,
- optimization methods,
- engineering constraints and standards,
- applications and outlook.

10. Outlook

Remark 0.21. The AIM framework is intended as a foundation for future development.

Remark 0.22. It opens the path toward:

- optimization-driven design systems,
- deeper integration with BIM,
- network-level alignment modelling,
- interactive and real-time engineering tools.

Remark 0.23. In this sense, the work contributes to a broader transition from static geometry toward computational infrastructure modelling.

Glossary

Core Concepts

Alignment A structured representation of railway track geometry defined primarily through curvature $\kappa(s)$, together with associated state variables such as position, orientation, profile, and cant. An alignment is not merely a curve, but a system linking geometry, dynamics, realization, and operational behaviour.

Arc-Length Domain The intrinsic one-dimensional domain s shared by all synchronized longitudinal alignment bands. Within AIM, arc-length acts as the primary reference structure for geometry, cant, profile, dynamics, and optimization.

Curvature $\kappa(s)$ The fundamental geometric quantity describing the rate of change of direction along the track. Within AIM, curvature is the primary design variable.

In planar formulations this coincides with the standard curvature. On curved surfaces, the physically relevant quantity becomes geodesic curvature.

Cant The rail height difference between both rails, represented as a longitudinal band along the alignment.

The corresponding roll angle

$$\phi(s) = \arctan\left(\frac{u(s)}{g}\right)$$

is derived using the applicable gauge g .

Pose2 A minimal planar alignment state consisting of:

$$\gamma(s), \quad \theta(s).$$

Pose2 forms the reconstruction kernel of planar alignment geometry.

Pose3 A runtime-derived spatial railway state combining:

- planar geometry,
- profile evaluation,
- cant evaluation,
- and spatial frame construction.

State A geometric or physical description of the railway system at a given arc-length position s .

Sparse Alignment Structure

cGeom The central curvature-driven geometric band of the sparse alignment model. It contains:

- fixed elements,
- transition elements,
- pose2-based reconstruction anchors.

Cant Band A longitudinal band storing cant information correlated with fixed alignment elements.

The cant band:

- has no independent stationing system,
- stores rail height differences,
- follows the same normalized curvature family as the associated cGeom element,
- may contain local station offsets.

Profile Band A longitudinal band representing vertical geometry independently from planar curvature geometry.

Fixed Element An alignment element with constant curvature.

Typical special cases are:

- straight segments,
- circular arcs.

Transition Element An alignment element with non-constant curvature evolution.

Transition elements describe controlled changes between curvature states.

Sparse Representation A compact, reconstruction-oriented representation of an alignment using semantically meaningful elements rather than dense sampled geometry.

The sparse representation is reconstruction-oriented rather than sample-oriented.

Runtime Pose3 A pose3 state evaluated dynamically from cGeom, cant, profile, and runtime operators rather than stored explicitly as persistent truth.

Modeling and Representation

Parametric Model A low-dimensional representation of curvature or cant defined by a small set of parameters $\mathbf{T}$, such as transition types or segment lengths.

Sample-Based Model A discrete representation of curvature or geometry defined at a set of points $\{s_i\}$, typically used for measurement data or numerical optimization.

Hybrid Model A combination of parametric structure and discrete corrections:

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s).$$

It allows structured modelling with local flexibility.

Normalized Curvature Law A curvature law defined on the normalized interval

$$u \in [0, 1],$$

separating shape from physical scale.

Operators and Runtime Evaluation

Sampling Operator $\mathcal{S}$ Maps a continuous function to discrete values:

$$\mathcal{S}(\kappa) = \{\kappa(s_i)\}.$$

Interpolation Operator $\mathcal{I}$ Reconstructs a continuous curve or function from discrete data.

Mechanical Operator $\mathcal{M}$ Represents the physical response of the track structure, including smoothing and deformation effects.

Realization Operator $\mathcal{R}$ Maps a theoretical alignment to its physically realized form:

$$\mathcal{R} = \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}.$$

Adjustment Operator $\mathcal{T}$ A local operator representing one construction or maintenance step, for example tamping:

$$\gamma_{k+1} = \mathcal{T}(\gamma_k).$$

Runtime Evaluation Dynamic computation of geometric or physical states from sparse bands and mathematical operators rather than from explicitly stored geometry.

Spaces and Transformations

Track Space A coordinate system aligned with the track, typically parameterized by arc-length s , where geometry and dynamics are defined.

World Space A global coordinate system, often CRS-based, used for georeferencing and data integration.

World-to-Track Transformation The mapping from world coordinates to intrinsic alignment coordinates, typically requiring projection onto the alignment and evaluation of s , d , κ , and cant.

Kilometre Line A longitudinal reference structure used for stationing, independent of the physical geometric alignment.

Optimization

Residual A component of the objective function representing deviation from a desired property, such as smoothness, dynamic equilibrium, or data fit.

Jacobian The matrix of derivatives of residuals with respect to parameters.

Within AIM, Jacobians are preferably derived analytically from arc-length-based relations.

Gauss–Newton Method An optimization method using the approximation

$$H \approx J^T J$$

for solving nonlinear least-squares problems.

Sequential Quadratic Programming (SQP) An iterative optimization framework solving a sequence of quadratic subproblems based on linearized constraints.

Realization and Construction

Realized Alignment The physically constructed track geometry, which differs from the theoretical alignment due to discretization, construction, and mechanical effects.

Control Points Discrete positions $\{s_i\}$ at which geometry is specified or adjusted during construction or maintenance.

Realization Error The deviation between target and realized curvature or geometry.

Inverse Realization Problem The problem of finding a target alignment such that

$$\mathcal{R}(\kappa_{\text{target}}) \approx \kappa_{\text{desired}}.$$

Advanced Geometry

Geodesic Curvature κ_g The intrinsic curvature of a curve on a surface, measuring the change of direction within the surface.

For railway alignment on the Earth, this is the physically relevant curvature.

Normal Curvature κ_n The component of curvature induced by the curvature of the underlying surface, for example the Earth ellipsoid.

Ellipsoidal Alignment An alignment defined directly on a reference ellipsoid rather than in a projected coordinate system.

Surface Frame A local coordinate frame on the ellipsoid consisting of tangent, lateral, and surface-normal directions.

Cant Rotation Rotation of the track cross-section around the tangent direction, defining superelevation.

Effective Lateral Acceleration The physically relevant lateral acceleration acting on a vehicle:

$$a_{\text{eff}} = v^2 \kappa_g - \mathbf{g} \cdot \mathbf{b}_\phi.$$

Pipeline

Pipeline The sequence of transformations from raw data to optimized alignment:

$$\text{data} \rightarrow \text{model} \rightarrow \text{state} \rightarrow \text{realization} \rightarrow \text{optimization}.$$

Failure Modes Typical breakdown points within the pipeline, such as inconsistent data, invalid coordinate systems, unstable projection, or failed optimization.

Design Principles Guiding principles ensuring robustness, interpretability, and consistency across the pipeline.

Teil I.

Why Alignments Matter

Why Alignments Matter

Remark 0.24. This part introduces the conceptual motivation of the AIM framework. It explains why railway alignments cannot be treated as mere geometric curves, but must be understood as structured engineering objects.

Remark 0.25. The central claim is that alignment modelling must integrate geometry, dynamics, realization, and data interpretation within a single coherent framework.

Remark 0.26. The following chapters therefore develop the problem setting, the need for a new perspective, and the guiding vision behind alignment-based information modelling.

1. Motivation

1.1. The Hidden Complexity of Railway Alignments

Remark 1.1. At first glance, a railway alignment appears to be a simple geometric object: a curve connecting two points in space.

Remark 1.2. However, this view is misleading. An alignment is not merely a curve, but the result of a complex interaction between geometry, physics, engineering constraints, and operational requirements.

Remark 1.3. Even small changes in curvature or cant can have significant effects on:

- passenger comfort,
- vehicle stability,
- track wear,
- and operational safety.

1.2. Fragmentation of Current Practice

Remark 1.4. In current engineering practice, alignment design is typically divided into separate domains:

- geometric design (plan and profile),
- cant design,
- dynamic evaluation,
- and regulatory verification.

Remark 1.5. These aspects are often treated sequentially rather than jointly. As a consequence:

- design iterations are required,
- optimal solutions are rarely achieved,
- and knowledge remains fragmented across tools and disciplines.

1.3. Limitations of Existing Models

Remark 1.6. Traditional alignment models are primarily geometric. They describe the track as a sequence of elements such as:

- straight lines,
- circular arcs,
- and transition curves.

Remark 1.7. While this representation is sufficient for construction, it is insufficient for:

- capturing dynamic behaviour,
- integrating measurement data,
- or supporting automated optimization.

Remark 1.8. In particular, curvature is often treated as a derived quantity, although it is the fundamental driver of railway dynamics.

1.4. Need for a Unified Representation

Remark 1.9. A modern alignment model must integrate:

- geometry,
- curvature evolution,
- cant,
- and dynamic effects.

Remark 1.10. Such a model should not treat these aspects independently, but as components of a single coherent system.

Remark 1.11. This requires a shift from element-based descriptions to function-based representations.

1.5. Alignment as a State Trajectory

Remark 1.12. Instead of viewing an alignment as a static geometric object, it can be interpreted as a trajectory of a system evolving along arc length.

Remark 1.13. In this interpretation:

- curvature acts as a control variable,
- cant represents an additional state component,

1. Motivation

- and the resulting motion determines physical behaviour.

Remark 1.14. This perspective naturally connects alignment design with modern concepts from:

- differential geometry,
- numerical analysis,
- and optimal control theory.

1.6. Towards Alignment-Based Information Modelling

Remark 1.15. The observations above motivate the development of a new modelling paradigm: *alignment-based information modelling* (AIM).

Remark 1.16. The central idea of AIM is to treat alignment not as a collection of geometric primitives, but as a structured, functional object defined by curvature and its derived quantities.

Remark 1.17. Within this framework:

- transitions are primary modelling elements,
- dynamics are embedded in the representation,
- and optimization becomes a natural design tool.

1.7. Connection to Practical Engineering

Remark 1.18. Despite its mathematical formulation, the AIM approach is not purely theoretical.

Remark 1.19. It directly addresses practical challenges such as:

- handling imperfect measurement data,
- integrating multiple coordinate systems,
- modelling complex track configurations,
- and supporting modern BIM workflows.

1.8. Objective of this Work

Remark 1.20. The objective of this manuscript is to provide a coherent foundation for alignment-based information modelling.

Remark 1.21. This includes:

1. Motivation

- a mathematical description of alignment structures,
- a unified state-based representation,
- methods for numerical reconstruction,
- integration of engineering constraints,
- and approaches to optimization.

Summary 1.22. Railway alignments are not simple geometric curves, but complex systems governed by curvature, cant, and dynamic effects.

A unified, curvature-driven modelling approach is required to capture this complexity and to enable modern, optimization-based design methods.

This work develops the foundations of such an approach.

2. Problem Statement

2.1. From Motivation to Formalization

Remark 2.1. The previous chapter has argued that railway alignment design is intrinsically a coupled problem involving geometry, dynamics, and engineering constraints.

Remark 2.2. We now formulate this insight as a precise problem statement.

2.2. Core Deficiency of Existing Approaches

Remark 2.3. Current alignment models are predominantly:

- element-based,
- geometry-centered,
- and weakly connected to dynamic behaviour.

Remark 2.4. As a consequence, they exhibit the following structural deficiencies:

- curvature is not treated as a primary design variable,
- cant is handled in a separate design process,
- dynamic quantities are evaluated only after geometric design,
- optimization is indirect and iterative rather than intrinsic.

2.3. Formal Gap

Remark 2.5. Let

$$\gamma : [0, L] \rightarrow \mathbb{R}^2$$

denote a planar alignment.

Remark 2.6. In classical formulations, the curve γ is the primary object, while curvature is derived as

$$\kappa(s) = \theta'(s).$$

Remark 2.7. However, from a dynamic perspective, the situation is reversed:

- curvature determines acceleration,

2. Problem Statement

- curvature variation determines jerk,
- and therefore curvature governs physical behaviour.

Remark 2.8. This reveals a fundamental mismatch between:

- geometric modelling (curve-first),
- and physical interpretation (curvature-first).

2.4. Missing Coupling of State Variables

Remark 2.9. Let

$$\kappa(s), \quad \phi(s)$$

denote curvature and cant.

Remark 2.10. In practice, these variables are strongly coupled through

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 2.11. Despite this coupling:

- κ is typically designed geometrically,
- ϕ is designed in a separate process,
- and their interaction is only checked afterwards.

Remark 2.12. This separation prevents a unified optimization of the system.

2.5. Data and Reality Gap

Remark 2.13. Modern railway projects rely heavily on real-world data, including:

- measurement polylines,
- legacy alignment descriptions,
- and heterogeneous coordinate systems.

Remark 2.14. Existing models struggle to:

- integrate noisy and incomplete data,
- reconcile different coordinate reference systems,
- and maintain a consistent representation across scales.

Remark 2.15. This leads to a disconnect between:

- theoretical alignment models,
- and real-world engineering data.

2.6. Lack of a Unified Optimization Framework

Remark 2.16. Alignment design problems naturally take the form:

$$\min J(\kappa, \phi)$$

subject to geometric and engineering constraints.

Remark 2.17. However, in current practice:

- the objective function is implicit or fragmented,
- constraints are applied in separate stages,
- and optimization is not formulated as a single problem.

Remark 2.18. This prevents the use of systematic numerical optimization methods.

2.7. Target Formulation

Remark 2.19. We aim to formulate alignment design as a unified problem:

$$\text{find } (\kappa, \phi)$$

such that:

- geometric consistency is satisfied,
- engineering constraints are fulfilled,
- and a global objective functional is minimized.

Remark 2.20. This requires:

- a curvature-based representation,
- a coupled state model,
- and a consistent treatment of data and constraints.

2.8. Scope of the Problem

Remark 2.21. The problem addressed in this work includes:

- modelling of alignments via curvature functions,
- integration of cant as a state variable,
- reconstruction of geometry from curvature,

2. Problem Statement

- incorporation of engineering rules as constraints,
- and formulation of optimization problems.

Remark 2.22. The problem explicitly excludes:

- detailed multibody vehicle dynamics,
- full infrastructure modelling,
- and construction-specific considerations.

2.9. Research Questions

Remark 2.23. The following central questions arise:

- How can alignment be represented in a curvature-driven manner?
- How can curvature and cant be modelled as a coupled state?
- How can real-world data be integrated into such a model?
- How can engineering rules be expressed as mathematical constraints?
- How can alignment design be formulated as an optimization problem?

2.10. Working Hypothesis

Remark 2.24. We hypothesize that:

- a curvature-based state representation,
- combined with a unified optimization framework,

provides a more consistent and powerful foundation for railway alignment design.

Remark 2.25. In particular, we assume that:

- transition curves emerge naturally from optimization,
- geometry and dynamics can be unified,
- and real-world data can be systematically integrated.

2.11. Alignment

2.11.1. Motivation

Remark 2.26. In classical geometry, a railway track is often represented as a curve $\gamma(s)$ in space.

Remark 2.27. This representation is insufficient for engineering purposes, as it does not capture the intrinsic quantities governing construction, dynamics, and realization.

Remark 2.28. In particular, curvature and cant are not derived properties, but primary design variables.

2.11.2. Definition

Definition 2.29 (Alignment). An **alignment** is a structured object defined over arc-length $s \in [0, L]$, consisting of the following components:

- a curvature function

$$\kappa : [0, L] \rightarrow \mathbb{R},$$

- a cant function

$$\phi : [0, L] \rightarrow \mathbb{R},$$

- an initial pose

$$(\gamma_0, \theta_0),$$

together with the induced state variables

$$\theta(s) = \theta_0 + \int_0^s \kappa(\sigma) \, d\sigma,$$

$$\gamma(s) = \gamma_0 + \int_0^s (\cos \theta(\sigma), \sin \theta(\sigma)) \, d\sigma.$$

2.11.3. Interpretation

Remark 2.30. An alignment is not defined by its position $\gamma(s)$, but by its curvature $\kappa(s)$.

Remark 2.31. The curve $\gamma(s)$ is a derived quantity obtained by integration.

Remark 2.32. Thus, curvature is the primary design variable, while position is a consequence.

2.11.4. State Structure

Definition 2.33 (Alignment state). The **state** of an alignment at position s is given by

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 2.34. This state couples geometry, orientation, and dynamics in a unified representation.

2. Problem Statement

2.11.5. Structural Nature

Proposition 2.35. *An alignment is a low-dimensional structured object, even though its realization may involve high-dimensional data.*

Remark 2.36. This structure enables:

- efficient storage,
- robust optimization,
- meaningful interpretation.

2.11.6. Relation to Realization

Remark 2.37. The alignment represents the intended geometry.

Remark 2.38. The physically constructed track is given by the realization operator:

$$\gamma_{\text{real}} = \mathcal{R}(\kappa, \phi).$$

Remark 2.39. Thus, the alignment exists in design space, while the track exists in physical space.

2.11.7. Relation to Data

Remark 2.40. Measured data typically provide sampled positions or polylines.

Remark 2.41. An alignment must be reconstructed from such data via inverse problems.

2.11.8. Key Insight

Proposition 2.42. *An alignment is not a curve, but a curvature-driven generative model of a curve.*

2.11.9. Summary

Summary 2.43. The alignment is the central object of railway geometry.

It is defined by curvature and cant, from which all geometric and dynamic quantities are derived.

This perspective shifts the focus from describing curves to defining generative structures, forming the foundation of alignment-based information modelling.

2.12. Summary

Summary 2.44. Current alignment design methods suffer from fragmentation between geometry, dynamics, and data.

2. Problem Statement

This work addresses the problem of developing a unified, curvature-driven framework that integrates these aspects into a single coherent model.

The goal is to enable consistent modelling, analysis, and optimization of railway alignments.

3. Vision

Why Railway is Different

Designing a railway alignment is fundamentally different from designing a road.

A road vehicle can steer.

A train cannot.

A road alignment may tolerate small geometric imperfections. A railway alignment cannot.

For a train:

- *curvature directly determines lateral acceleration,*
- *curvature variation determines jerk,*
- *and both directly affect safety, comfort, and wear.*

As a consequence:

- *geometry is not just shape,*
- *curvature is not just a derived quantity,*
- *and transitions are not optional smoothing elements.*

They are the system.

3.1. From Problem to Paradigm

Remark 3.1. The previous chapters have identified a fundamental limitation of current alignment modelling approaches: the fragmentation of geometry, dynamics, and data.

Remark 3.2. This motivates not only incremental improvements, but a shift in perspective.

3.2. Core Idea

Remark 3.3. The central idea of this work is to treat railway alignment as a *state trajectory* governed by curvature and its associated quantities.

3. Vision

Remark 3.4. Instead of describing alignment as a sequence of geometric elements, we propose to describe it as a function-based object:

$$(\kappa(s), \phi(s)).$$

Remark 3.5. In this view:

- curvature defines geometry,
- cant defines dynamic balancing,
- and their interaction determines physical behaviour.

3.3. Alignment as a Controlled System

Remark 3.6. The alignment is interpreted as a controlled system evolving along arc length:

$$\begin{aligned}\gamma'(s) &= (\cos \theta(s), \sin \theta(s)), \\ \theta'(s) &= \kappa(s), \\ \phi'(s) &= \omega(s).\end{aligned}$$

Remark 3.7. Here:

- $\kappa(s)$ acts as a geometric control,
- $\omega(s)$ governs cant evolution,
- and the resulting state determines both geometry and dynamics.

Remark 3.8. This formulation establishes a direct link between alignment design and optimal control theory.

3.4. Transitions as Primary Objects

Remark 3.9. In classical models, transitions are treated as auxiliary elements connecting geometric primitives.

Remark 3.10. In the proposed framework, transitions become the primary modelling objects, as they define the evolution of curvature.

Remark 3.11. This shifts the focus:

- from static geometry
- to controlled change of state.

3.5. Integration of Reality

Remark 3.12. A central requirement of the proposed approach is compatibility with real-world data.

Remark 3.13. This includes:

- measurement data with uncertainty,
- heterogeneous coordinate systems (CRS),
- legacy alignment descriptions,
- and complex track configurations.

Remark 3.14. The model must therefore support:

- robust reconstruction from imperfect data,
- explicit handling of coordinate transformations,
- and consistent representation across scales.

3.6. Unified Optimization Framework

Remark 3.15. Within this paradigm, alignment design is formulated as an optimization problem:

$$\min J(\kappa, \phi)$$

subject to:

- state dynamics,
- engineering constraints,
- and boundary conditions.

Remark 3.16. This allows:

- direct incorporation of comfort criteria,
- integration of dynamic effects,
- and systematic exploration of design alternatives.

3.7. Towards Alignment-Based Information Modelling

Remark 3.17. The proposed framework forms the basis of *alignment-based information modelling* (AIM).

Remark 3.18. AIM is characterized by:

- curvature-driven representation,
- state-based modelling,
- integration of geometry and dynamics,
- and compatibility with data-driven workflows.

Remark 3.19. In this sense, alignment is not a static object, but a structured, computable entity.

3.8. Implications

Remark 3.20. Adopting this perspective has several implications:

- design becomes a problem of state control,
- transitions emerge from functional requirements,
- optimization replaces iterative manual adjustment,
- and modelling becomes directly compatible with computation.

3.9. Long-Term Perspective

Remark 3.21. The proposed framework is intended as a foundation for future developments, including:

- integration with BIM systems,
- extension to network-level modelling,
- incorporation of vehicle dynamics,
- and real-time optimization.

Remark 3.22. It is not a finished solution, but a structured starting point.

3.10. Summary

Summary 3.23. This work proposes a paradigm shift from element-based alignment modelling to a curvature-driven, state-based representation.

By treating alignment as a controlled system and integrating geometry, dynamics, and data, it establishes a unified foundation for modelling and optimization.

This perspective forms the basis of alignment-based information modelling (AIM).

Teil II.

Mathematical Foundations

Mathematical Foundations

Remark 3.24. The preceding chapters introduced the motivation for AIM and argued that railway engineering cannot be fully understood through traditional object-oriented descriptions alone.

Remark 3.25. However, motivation alone is insufficient.

Before alignments can be represented, analysed, realized, or optimized, a rigorous mathematical foundation must be established.

What do we already know?

Remark 3.26. Railway systems possess an intrinsic geometric structure.

Vehicle motion, cant, stationing, operational behaviour, and engineering constraints all depend on this structure.

Remark 3.27. The previous part identified the need for a modelling framework centred on railway alignments rather than solely on engineering assets.

What is still missing?

Remark 3.28. A modelling framework requires a precise mathematical language.

At this point, concepts such as alignment, geometry, state, and reference systems have been discussed only on an intuitive level.

Remark 3.29. AIM therefore requires a formal foundation capable of describing railway geometry independently of implementation details, software systems, or engineering conventions.

What comes next?

Remark 3.30. This part develops the mathematical basis of AIM.

Particular emphasis is placed on curvature-driven geometry, arc-length-based descriptions, state representations, and the operators that connect geometric, physical, and operational interpretations.

Remark 3.31. These concepts form the foundation upon which all later discussions of runtime behaviour, realization, optimization, and information modelling are built.

Relation to the AIM Roadmap

Remark 3.32. Within the AIM roadmap (see Figure 2), the present part corresponds primarily to the transition from classical alignment engineering toward curvature space, intrinsic geometry, and the first state descriptions.

Remark 3.33. The objective is not yet to describe complete railway systems, but to establish the mathematical building blocks from which such systems can later be constructed.

4. Information Modelling

4.1. Motivation

Remark 4.1. Information models describe, organize, exchange, and interpret engineering knowledge.

Remark 4.2. Different engineering domains emphasize different aspects of reality. Buildings, infrastructure systems, operational systems, and networks do not all share the same modelling centre.

Remark 4.3. The present work introduces Alignment-based Information Modelling (AIM) as a modelling paradigm centred on railway alignments and their associated geometric, spatial, and operational information.

4.2. BIM, AIM, and PIM

Definition 4.4 (Process Information Modelling). In this thesis, the term Process Information Modelling, abbreviated as PIM, is used as an umbrella concept for engineering information modelling approaches that organize information not only around physical assets, but also around processes, state evolution, lifecycle behaviour, and operational context.

Remark 4.5. PIM is not introduced here as an established standard. It is used as a conceptual framing device for relating different information modelling paradigms.

Definition 4.6 (Building Information Modelling). Building Information Modelling, abbreviated as BIM, denotes an asset-oriented information modelling paradigm in which physical objects, their properties, relations, and lifecycle information form the primary organizing structure.

Definition 4.7 (Alignment-based Information Modelling). Alignment-based Information Modelling, abbreviated as AIM, denotes an alignment-oriented information modelling paradigm in which railway alignment, reference systems, admissible states, operational behaviour, and realization processes form the primary organizing structure.

Remark 4.8. Within this thesis, BIM and AIM are interpreted as complementary specializations within a broader process-oriented modelling perspective.

4. Information Modelling

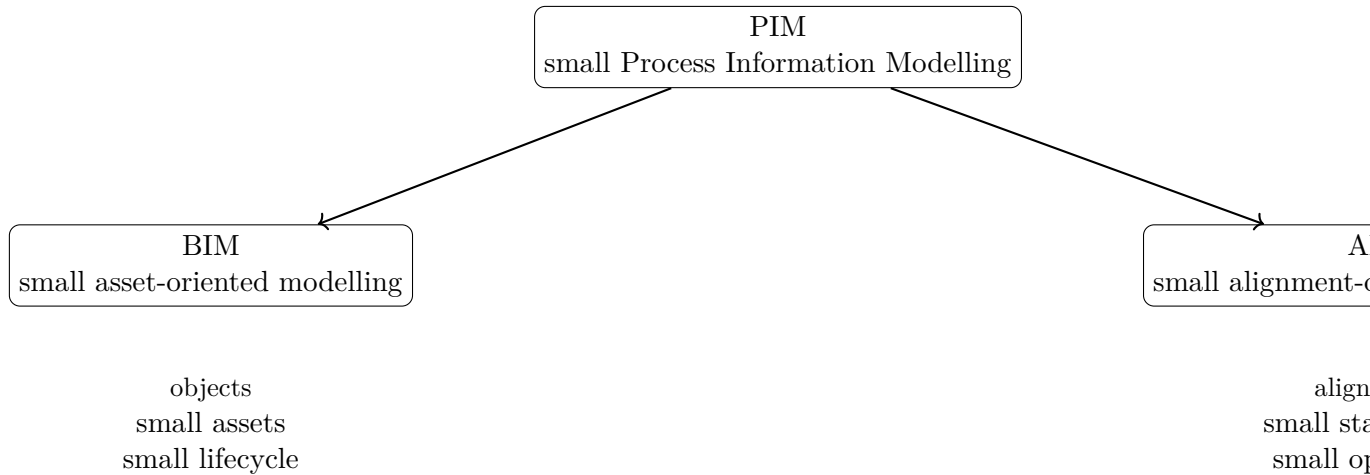


Abbildung 4.1.: BIM and AIM as complementary specializations within the broader PIM perspective used in this thesis.

4.3. Conceptual Position

Remark 4.9. The distinction is not intended as a replacement of BIM by AIM. Rather, AIM complements BIM where infrastructure systems cannot be understood sufficiently through asset description alone.

Remark 4.10. Railway infrastructure is not merely a collection of physical objects. It is an environment enabling and constraining motion, operation, and state evolution.

4.4. Why AIM?

Remark 4.11. Engineering information models are never neutral.

Every modelling paradigm implicitly chooses a central organizing principle around which information is structured.

Remark 4.12. In many traditional information models, physical assets constitute this organizing principle.

Objects, components, assemblies, and their associated properties form the primary modelling entities.

Remark 4.13. This perspective has proven highly successful for buildings and many other engineering disciplines.

However, railway systems exhibit characteristics that cannot be fully explained through assets alone.

Remark 4.14. Railway operation is fundamentally governed by geometry.

Vehicle motion, cant, speed, comfort, safety, operational performance, and infrastructure utilization all depend on the geometric structure of the alignment.

Remark 4.15. The alignment is therefore not merely another infrastructure object.

4. Information Modelling

Instead, it acts as the geometric framework within which railway operation becomes possible.

Definition 4.16 (Primary modelling entity). Within AIM, the alignment is adopted as the primary modelling entity.

Other engineering information is interpreted relative to the alignment, its reference systems, and its induced operational state space.

Remark 4.17. This choice does not imply that assets become unimportant.

Bridges, tunnels, switches, signals, stations, and other infrastructure elements remain essential engineering objects.

However, within AIM they are interpreted as entities associated with, located along, constrained by, or interacting with an alignment.

Remark 4.18. The resulting viewpoint shifts the modelling focus from static objects toward geometry, motion, admissible states, and operational processes.

4.5. Connection to AIM

Summary 4.19. BIM primarily organizes information around assets.

AIM primarily organizes information around alignments.

PIM is used in this thesis as a broader conceptual umbrella under which asset-oriented, lifecycle-oriented, alignment-oriented, and process-oriented modelling perspectives can coexist.

AIM therefore appears not as an isolated alternative to BIM, but as a natural member of a larger family of engineering information modelling approaches.

5. Axiomatic Foundation

5.1. Primitives

Definition 5.1. $s \in \mathbb{R}$ is the arc-length parameter.

Definition 5.2. $\gamma : I \rightarrow \mathbb{R}^2$ is a curve.

Definition 5.3. $\theta : I \rightarrow \mathbb{R}$ is the direction angle.

Definition 5.4. $\kappa : I \rightarrow \mathbb{R}$ is the curvature.

5.2. Axioms

Axiom 5.5.

$$\|\gamma'(s)\| = 1$$

Axiom 5.6.

$$\theta'(s) = \kappa(s)$$

Axiom 5.7.

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s))$$

5.3. Consequence

Proposition 5.8. *Curvature uniquely defines the curve up to initial conditions.*

6. Notation and Terminology

Motivation

Remark 6.1. Railway alignment engineering combines concepts from geometry, surveying, dynamics, construction, realization, runtime evaluation, and operational railway systems.

Remark 6.2. Many inconsistencies in classical railway systems arise not from incorrect mathematics, but from ambiguous terminology and mixed conceptual layers.

Remark 6.3. The AIM framework therefore introduces a canonical notation and terminology system.

Remark 6.4. The purpose of this chapter is:

- to define canonical symbols,
- to stabilize terminology,
- to reduce ambiguity,
- and to separate fundamentally different conceptual layers.

6.1. General Principles

Principle 6.5 (Intrinsic-first principle). Whenever possible, railway geometry is formulated intrinsically using arc-length-based quantities.

Principle 6.6 (Runtime distinction principle). A distinction is made between:

- stored representations,
- runtime-evaluated states,
- and physically realized states.

Principle 6.7 (Target vs realization principle). Target geometry and realized geometry are treated as distinct objects.

Principle 6.8 (Canonical layer principle). AIM separates:

- intrinsic geometry,
- metric realization,

- runtime interpretation,
- physical realization,
- operational behaviour,
- and CRS-dependent embedding

into distinct conceptual layers.

6.2. Canonical Spaces

Definition 6.9 (State space notation).

$$\mathcal{X}$$

denotes the canonical railway state space.

Definition 6.10 (Runtime quantity space notation).

$$\mathcal{Y}$$

denotes the space of runtime-evaluated quantities.

Definition 6.11 (Track-space domain notation).

$$\Omega_{\text{track}}$$

denotes the intrinsic railway coordinate domain.

Definition 6.12 (World-space domain notation).

$$\Omega_{\text{world}}$$

denotes the CRS-dependent spatial world domain.

Definition 6.13 (Curvature space notation).

$$\mathcal{K}$$

denotes the space of admissible curvature evolutions and curvature laws.

6.3. Canonical Coordinates

6.3.1. Arc-Length

Definition 6.14 (Arc-length).

$$s$$

denotes intrinsic arc-length along the alignment.

Remark 6.15. Arc-length is the primary intrinsic coordinate within AIM.

6.3.2. Track Coordinates

Definition 6.16 (Track coordinates). Track-space coordinates are:

$$(s, d, h),$$

where:

- s : longitudinal coordinate,
- d : lateral offset,
- h : vertical offset.

Remark 6.17. Track coordinates belong to Ω_{track} .

Remark 6.18. Additional operational coordinates may be introduced by runtime systems.

6.3.3. World Coordinates

Definition 6.19 (World coordinates). World coordinates are denoted by:

$$x \in \Omega_{\text{world}}.$$

Remark 6.20. World coordinates are CRS-dependent.

Remark 6.21. When required, world-space quantities may explicitly use the notation:

$$x_{\text{w}}.$$

6.4. Geometry Symbols

6.4.1. Alignment Curve

Definition 6.22 (Alignment curve).

$$\gamma(s)$$

denotes the intrinsic alignment geometry.

Remark 6.23. The alignment curve represents the geometric backbone of the railway alignment.

Remark 6.24. Depending on context, $\gamma(s)$ may denote planar geometry, spatial embedding, target geometry, or realized geometry. Such distinctions should be marked explicitly when relevant.

6.4.2. Orientation

Definition 6.25 (Orientation angle).

$$\theta(s)$$

denotes horizontal alignment orientation.

6.4.3. Curvature

Definition 6.26 (Curvature).

$$\kappa(s)$$

denotes curvature as function of arc-length.

Remark 6.27. Curvature is the primary intrinsic geometric quantity within AIM.

6.4.4. Cant

Definition 6.28 (Cant angle).

$$\phi(s)$$

denotes cant rotation.

Remark 6.29. Within AIM, $\phi(s)$ is interpreted geometrically as frame rotation. Engineering source data may instead describe cant as rail height difference; the conversion is a convention-dependent runtime or import operation.

Remark 6.30. Depending on context, $\phi(s)$ may also be interpreted as roll angle of the local railway frame.

6.4.5. Cant Evolution

Definition 6.31 (Cant-rate).

$$\omega(s) = \phi'(s)$$

denotes cant-rate evolution.

6.5. Frame Symbols

6.5.1. Tangent Direction

Definition 6.32 (Tangent vector).

$$\mathbf{t}(s)$$

denotes the local tangent direction.

6.5.2. Lateral Direction

Definition 6.33 (Lateral vector).

$$\mathbf{n}(s)$$

denotes the local lateral direction.

6.5.3. Binormal Direction

Definition 6.34 (Binormal vector).

$$\mathbf{b}(s)$$

denotes the local binormal direction.

6.5.4. Local Frame

Definition 6.35 (Local frame).

$$F(s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s))$$

denotes the local pose3 frame.

6.6. Pose and State Terminology

6.6.1. Pose

Definition 6.36 (Pose). A pose denotes a local intrinsic railway state attached to arc-length position s .

6.6.2. pose2

Definition 6.37 (pose2). The pose2 state is:

$$\text{pose2}(s) = (\gamma(s), \theta(s)).$$

Remark 6.38. pose2 represents intrinsic planar railway geometry.

6.6.3. pose3

Definition 6.39 (pose3). The pose3 state extends pose2 by including spatial orientation and cant.

A minimal intrinsic representation is:

$$\text{pose3}(s) = (\gamma(s), \theta(s), \phi(s)).$$

Remark 6.40. More elaborate runtime pose3 interpretations may additionally include:

- local frames,
- profile evaluation,
- metric realization,
- or operational vehicle-space interpretation.

6.6.4. Controls

Definition 6.41 (Control quantities). The canonical control quantities are:

$$\kappa(s), \quad \omega(s).$$

Remark 6.42. Curvature controls directional evolution, whereas cant-rate controls cant evolution.

6.6.5. State

Definition 6.43 (State). A state denotes a structured railway condition together with its geometric, operational, or physical interpretation.

Remark 6.44. Within AIM, states are not merely stored parameter containers.

6.6.6. Runtime State

Definition 6.45 (Runtime state). A runtime state is a dynamically evaluated operational state generated through runtime operator evaluation.

Remark 6.46. Runtime refers to operational evaluation of alignment-related quantities, independent of implementation details.

6.7. Target, Realized, and Measured Notation

6.7.1. Target Quantities

Definition 6.47 (Target notation). Target quantities use the index:

$$(\cdot)_{\text{target}}.$$

6.7.2. Realized Quantities

Definition 6.48 (Realized notation). Realized quantities use the index:

$$(\cdot)_{\text{real}}.$$

6.7.3. Measured Quantities

Definition 6.49 (Measured notation). Measured quantities use the index:

$$(\cdot)_{\text{meas}}.$$

6.8. Operator Notation

6.8.1. Projection and Transformation Operators

Definition 6.50 (Track-to-world operator notation).

$$\mathcal{P}_{\text{tw}}$$

denotes the track-to-world operator.

Definition 6.51 (World-to-track operator notation).

$$\mathcal{P}_{\text{wt}}$$

denotes the world-to-track projection operator.

Definition 6.52 (Coordinate transformation operator notation).

$$\mathcal{C}$$

denotes coordinate transformation between world-space representations.

6.8.2. Runtime Operators

Definition 6.53 (Evaluation operator notation).

$$\mathcal{E}$$

denotes runtime operator evaluation.

Definition 6.54 (Frame operator notation).

$$\mathcal{F}$$

denotes frame evaluation.

6.8.3. Realization Operators

Definition 6.55 (Sampling operator notation).

$$\mathcal{S}$$

6. Notation and Terminology

denotes sampling from continuous to discrete representation.

Definition 6.56 (Interpolation operator notation).

$$\mathcal{I}$$

denotes reconstruction from discrete data.

Definition 6.57 (Mechanical operator notation).

$$\mathcal{M}$$

denotes mechanical or structural response of the railway system.

Definition 6.58 (Adjustment operator notation).

$$\mathcal{A}$$

denotes a local construction or maintenance adjustment operator.

Definition 6.59 (Realization operator notation).

$$\mathcal{R}$$

denotes the realization operator.

6.9. CRS and Earth Geometry

6.9.1. Geodesic Curvature

Definition 6.60 (Geodesic curvature).

$$\kappa_g$$

denotes geodesic curvature on the Earth surface.

6.9.2. Normal Curvature

Definition 6.61 (Normal curvature).

$$\kappa_n$$

denotes curvature induced by Earth surface geometry.

6.9.3. Earth Surface Normal

Definition 6.62 (Earth surface normal).

$$\mathbf{n}_E$$

denotes the ellipsoid surface normal.

6.10. Vehicle-Space Terminology

6.10.1. Vehicle Space

Definition 6.63 (Vehicle space). Vehicle space denotes the physically occupied operational space of the railway vehicle.

6.10.2. Operational Geometry

Definition 6.64 (Operational geometry). Operational geometry denotes geometry relevant to:

- passenger comfort,
- platform interaction,
- clearance,
- vehicle behaviour,
- and dynamic vehicle-space interaction.

6.11. Canonical Terminology

Definition 6.65 (Preferred terminology). The following terminology is preferred:

alignment	intrinsic railway geometry system
curve	mathematical geometric object
geometry	geometric description
trajectory	motion-related path
pose	local railway state
frame	local spatial interpretation
state	structured railway condition
runtime state	dynamically evaluated operational state
runtime service	operator-based evaluation process
realization	physical construction or maintenance process
realization operator	mapping from target state to realized state
metric realization	metric-dependent spatial interpretation
world space	CRS-based spatial coordinates
track space	intrinsic alignment coordinates
track	physically realized railway structure

6.12. Avoided Ambiguities

Remark 6.66. The following ambiguities should be avoided:

- alignment $\neq$ merely curve,
- realization $\neq$ visualization,
- runtime $\neq$ software process only,
- pose $\neq$ frame,
- geometry $\neq$ realization,
- target geometry $\neq$ realized geometry,
- metric realization $\neq$ physical realization,
- coordinate transformation $\neq$ world-to-track projection,
- projection $\neq$ CRS transformation,
- world coordinates $\neq$ track coordinates.

6.13. Connection to AIM

Summary 6.67. The AIM notation system separates:

- intrinsic geometry,
- metric realization,
- realized geometry,
- runtime states,
- operational vehicle-space interpretation,
- and CRS-dependent world geometry.

This notation framework forms the canonical language of AIM.

7. Core Equations and Canonical Relations

Motivation

Remark 7.1. The AIM framework is based on a compact set of canonical equations that connect geometry, dynamics, realization, runtime interpretation, projection, and optimization.

Remark 7.2. These equations do not represent isolated formulas, but a coupled state description for railway alignment modelling.

Remark 7.3. The purpose of this chapter is not to derive all relations in detail, but to define the canonical equations repeatedly used throughout AIM.

Remark 7.4. The equations collected here serve as:

- canonical reference relations,
- shared notation anchors,
- and structural building blocks of the AIM framework.

7.1. Canonical State Representation

7.1.1. Intrinsic State

Definition 7.5 (Canonical intrinsic state). The canonical intrinsic pose3 state is:

$$x(s) = (\gamma(s), \theta(s), \phi(s)). \quad (7.1)$$

Remark 7.6. The state combines:

- intrinsic geometry,
- orientation,
- and cant evolution.

Remark 7.7. This intrinsic state represents the minimal canonical geometric railway state. More elaborate runtime-evaluated spatial interpretations may extend this representation through frame evaluation, profile integration, metric interpretation, and realization-aware runtime services.

7.1.2. Control Variables

Definition 7.8 (Control variables). The canonical control variables are:

$$u(s) = (\kappa(s), \omega(s)), \quad (7.2)$$

where:

$$\omega(s) = \phi'(s).$$

Remark 7.9. Curvature and cant-rate act as the primary controls of railway state evolution.

7.2. Intrinsic Geometry

7.2.1. Arc-Length State Evolution

Definition 7.10 (State evolution relation). The intrinsic geometric state evolves according to:

$$\gamma'(s) = \mathbf{t}(s). \quad (7.3)$$

Remark 7.11. This relation defines the arc-length parameterization of the alignment.

7.2.2. Curvature Relation

Definition 7.12 (Curvature relation). Directional evolution is governed by:

$$\theta'(s) = \kappa(s). \quad (7.4)$$

Remark 7.13. Curvature is the differential quantity governing directional evolution along the alignment.

7.2.3. Planar Tangent Representation

Definition 7.14 (Planar tangent relation). In the planar case:

$$\mathbf{t}(s) = (\cos \theta(s), \sin \theta(s)). \quad (7.5)$$

Remark 7.15. This is the planar specialization of eq. (7.3).

7.2.4. Canonical Pose2 System

Definition 7.16 (Pose2 system). The canonical planar pose2 system is:

$$\begin{aligned} \gamma'(s) &= \mathbf{t}(s), \\ \theta'(s) &= \kappa(s). \end{aligned} \quad (7.6)$$

Remark 7.17. Together with eq. (7.5), this defines the classical curvature-based planar reconstruction system.

7.3. Cant Evolution

7.3.1. Cant State Equation

Definition 7.18 (Cant evolution). Cant evolution is governed by:

$$\phi'(s) = \omega(s), \quad (7.7)$$

where $\omega(s)$ denotes the cant-rate control.

Remark 7.19. Cant is treated as a state quantity rather than as a secondary annotation.

7.3.2. Canonical Pose3 System

Definition 7.20 (Pose3 system). The canonical pose3 system is:

$$\begin{aligned} \gamma'(s) &= \mathbf{t}(s), \\ \theta'(s) &= \kappa(s), \\ \phi'(s) &= \omega(s). \end{aligned} \quad (7.8)$$

Remark 7.21. The controls of the system are given by eq. (7.2).

Remark 7.22. This formulation interprets railway alignment as a controlled dynamical state system.

7.3.3. Generalized State Form

Definition 7.23 (Generalized state equation). The pose evolution may be written abstractly as:

$$x'(s) = f(x(s), u(s)). \quad (7.9)$$

Remark 7.24. Within AIM, the canonical controls are:

$$u(s) = (\kappa(s), \omega(s)).$$

7.4. Dynamic Relations

7.4.1. Effective Lateral Acceleration

Definition 7.25 (Effective lateral acceleration). The effective lateral acceleration is:

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi. \quad (7.10)$$

Remark 7.26. This relation expresses the coupling between curvature and cant.

Remark 7.27. The relation represents the classical engineering approximation for moderate cant angles and local engineering geometry. More advanced formulations may additionally incorporate:

7. Core Equations and Canonical Relations

- spatial frame dynamics,
- center-of-gravity displacement,
- vehicle suspension,
- and ellipsoid-aware geometry.

7.4.2. Equilibrium Condition

Definition 7.28 (Equilibrium condition). A perfectly balanced alignment satisfies:

$$v^2 \kappa = g \sin \phi. \quad (7.11)$$

Remark 7.29. This relation defines ideal cant balance.

7.4.3. Jerk Relation

Definition 7.30 (Jerk relation). The jerk is:

$$j = v^3 \kappa' - g v \cos \phi \phi'. \quad (7.12)$$

Remark 7.31. The jerk depends jointly on curvature evolution and cant evolution.

7.5. Representation and Projection Relations

7.5.1. Track-to-World Mapping

Definition 7.32 (Track-to-world mapping). The forward transformation is:

$$x_w(s, d) = \gamma(s) + d \mathbf{n}(s). \quad (7.13)$$

Remark 7.33. The vector $\mathbf{n}(s)$ denotes the local lateral direction.

7.5.2. World-to-Track Projection

Definition 7.34 (World-to-track projection). The inverse transformation is:

$$(s^*, d^*) = \arg \min_{s, d} \|x_w - (\gamma(s) + d \mathbf{n}(s))\|. \quad (7.14)$$

Remark 7.35. This is a nonlinear runtime projection problem onto the alignment.

7.5.3. Coordinate Transformation

Definition 7.36 (Coordinate transformation). Coordinate transformation is represented abstractly by:

$$x_2 = \mathcal{C}(x_1). \quad (7.15)$$

Remark 7.37. Coordinate transformation must not be confused with intrinsic railway projection.

7.6. Realization Relations

7.6.1. Sampling Relation

Definition 7.38 (Sampling relation). Continuous geometry may be sampled as:

$$\mathcal{S}(f) = \{f(s_i)\}. \quad (7.16)$$

7.6.2. Interpolation Relation

Definition 7.39 (Interpolation relation). Continuous reconstruction from sampled data is represented by:

$$\mathcal{I}(\{f(s_i)\}) = \tilde{f}(s). \quad (7.17)$$

7.6.3. Realization Operator

Definition 7.40 (Realization operator). The realization operator maps target states to realized states:

$$\mathcal{R} : \text{pose3}_{\text{target}} \rightarrow \text{pose3}_{\text{real}}. \quad (7.18)$$

Remark 7.41. Realization includes:

- construction,
- machine behaviour,
- discretization,
- interpolation,
- and structural response.

7.6.4. Realization Decomposition

Definition 7.42 (Realization decomposition). The realization process may be interpreted as:

$$\mathcal{R} = \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}. \quad (7.19)$$

Remark 7.43. This decomposition separates:

- discretization,
- reconstruction,
- and physical realization effects.

7. Core Equations and Canonical Relations

Remark 7.44. Within AIM, this decomposition forms one of the canonical bridges between:

- analytical geometry,
- construction processes,
- runtime reconstruction,
- and physically realized railway behaviour.

7.6.5. Curvature Realization Relation

Definition 7.45 (Curvature realization). At curvature level:

$$\mathcal{R}(\kappa_{\text{target}}) = \kappa_{\text{real}}. \quad (7.20)$$

7.6.6. Inverse Realization Problem

Definition 7.46 (Inverse realization). The inverse realization problem is:

$$\mathcal{R}(\kappa_{\text{target}}) \approx \kappa_{\text{desired}}. \quad (7.21)$$

Remark 7.47. Alignment design therefore becomes an inverse problem under realization constraints.

7.7. Ellipsoid Geometry

7.7.1. Curvature Decomposition

Definition 7.48 (Curvature decomposition). The total curvature satisfies:

$$\kappa^2 = \kappa_g^2 + \kappa_n^2. \quad (7.22)$$

Remark 7.49. The geodesic curvature κ_g is the physically relevant curvature for railway dynamics on the Earth surface.

7.7.2. Intrinsic Surface Curvature

Definition 7.50 (Geodesic curvature). The intrinsic surface curvature is:

$$\kappa_g = \|\nabla_s \mathbf{t}\|. \quad (7.23)$$

Remark 7.51. This replaces the purely planar curvature interpretation on curved reference surfaces.

7.8. Optimization Relations

7.8.1. General Optimization Problem

Definition 7.52 (Optimization problem). A general realization-aware optimization problem is:

$$\min_u J(\mathcal{R}(x(u))). \quad (7.24)$$

Remark 7.53. The objective is evaluated on realized states rather than purely analytical states.

7.8.2. Least-Squares Structure

Definition 7.54 (Least-squares structure). Many AIM optimization problems take the form:

$$\min_x \frac{1}{2} \|r(x)\|^2. \quad (7.25)$$

7.8.3. Gauss–Newton Approximation

Definition 7.55 (Gauss–Newton approximation). The Gauss–Newton approximation is:

$$H \approx J^T J. \quad (7.26)$$

Remark 7.56. This structure appears naturally in:

- curvature fitting,
- realization matching,
- trajectory optimization,
- and runtime adjustment problems.

7.9. Canonical AIM Interpretation

Proposition 7.57. *Within AIM, railway alignment is interpreted as a spatial state trajectory governed by curvature and cant evolution.*

Proposition 7.58. *Curvature, cant, realization, runtime interpretation, projection, metric interpretation, and optimization must therefore be treated as coupled components of one unified system.*

7.10. Connection to AIM

Summary 7.59. The equations collected in this chapter form the mathematical backbone of AIM.

They define:

7. Core Equations and Canonical Relations

- intrinsic state evolution,
- cant and dynamic relations,
- runtime projection,
- realization mappings,
- ellipsoid-aware geometry,
- and optimization structures.

Together, these equations establish a unified mathematical framework connecting geometry, realization, runtime interpretation, metric embedding, and optimization.

8. State Model

Motivation

Remark 8.1. Railway alignment modelling involves interacting descriptions of geometry, orientation, cant, realization, runtime interpretation, and operational behaviour.

Remark 8.2. Classical alignment theory often treats these aspects separately:

- geometry as curves,
- cant as annotation,
- realization as construction detail,
- and optimization as external post-processing.

Remark 8.3. Within AIM, these components are interpreted as parts of a unified state-based system.

Remark 8.4. The purpose of the state model is:

- to define canonical railway states,
- to distinguish intrinsic and extrinsic representations,
- to separate target and realized states,
- and to establish a consistent basis for runtime services, projection, realization, and optimization.

8.1. State Concept

8.1.1. State Definition

Definition 8.5 (State). A state is the minimal set of variables required to describe the geometric and physical condition of a railway alignment at arc-length position s .

Remark 8.6. Within AIM, states are interpreted intrinsically along the alignment rather than globally in world coordinates.

Remark 8.7. The alignment is therefore interpreted as a spatial state trajectory parameterized by arc-length.

8.1.2. Canonical State Variable

Definition 8.8 (Canonical state). The canonical intrinsic railway state is:

$$x(s) = (\gamma(s), \theta(s), \phi(s)). \quad (8.1)$$

Remark 8.9. The state combines:

- intrinsic geometry,
- orientation,
- and cant evolution.

8.2. Pose2 State

8.2.1. Definition

Definition 8.10 (Pose2 state). The planar pose state is:

$$\text{pose2}(s) = (\gamma(s), \theta(s)). \quad (8.2)$$

Remark 8.11. The pose2 state describes:

- planar position,
- and planar orientation.

8.2.2. Pose2 Evolution

Remark 8.12. The pose2 evolution is governed by:

$$\theta'(s) = \kappa(s), \quad (8.3)$$

together with:

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s)). \quad (8.4)$$

Remark 8.13. Curvature therefore acts as the primary geometric control variable.

8.3. Pose3 State

8.3.1. Definition

Definition 8.14 (Pose3 state). The extended railway pose state is:

$$\text{pose3}(s) = (\gamma(s), \theta(s), \phi(s)). \quad (8.5)$$

Remark 8.15. The pose3 state extends the planar state by including cant.

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Remark 8.16. Cant is interpreted as a rotation around the local tangent direction.

8.3.2. Controls

Definition 8.17 (Pose3 controls). The canonical controls are:

$$u(s) = (\kappa(s), \omega(s)), \quad (8.6)$$

where:

$$\omega(s) = \phi'(s).$$

Remark 8.18. Curvature controls directional evolution, whereas cant-rate controls cross-level evolution.

8.3.3. Canonical Pose3 System

Remark 8.19. The canonical pose3 evolution system is:

$$\begin{aligned} \gamma'(s) &= (\cos \theta(s), \sin \theta(s)), \\ \theta'(s) &= \kappa(s), \\ \phi'(s) &= \omega(s). \end{aligned} \quad (8.7)$$

Remark 8.20. This formulation interprets railway alignment as a controlled dynamical state system.

8.4. Intrinsic and Extrinsic States

8.4.1. Intrinsic State

Definition 8.21 (Intrinsic state). An intrinsic state is defined relative to the alignment itself and uses arc-length s as primary parameter.

Remark 8.22. Examples include:

- curvature,
- cant,
- tangent direction,
- local railway frames.

8.4.2. Extrinsic State

Definition 8.23 (Extrinsic state). An extrinsic state is defined relative to a global coordinate reference system.

Remark 8.24. Examples include:

- WGS84 coordinates,
- projected CRS coordinates,
- BIM placement coordinates,
- sensor measurements.

8.4.3. Intrinsic–Extrinsic Coupling

Remark 8.25. The world–track transformation couples intrinsic and extrinsic states.

Remark 8.26. World-to-track projection therefore acts as a state conversion process between global and alignment-relative representations.

8.5. Target and Realized States

8.5.1. Target State

Definition 8.27 (Target state). The target state describes the analytically intended alignment:

$$\text{pose3}_{\text{target}}. \quad (8.8)$$

8.5.2. Realized State

Definition 8.28 (Realized state). The realized state describes the physically constructed alignment:

$$\text{pose3}_{\text{real}}. \quad (8.9)$$

8.5.3. Realization Relation

Remark 8.29. The realization operator connects both states:

$$\mathcal{R} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}. \quad (8.10)$$

Remark 8.30. The realized state includes:

- construction effects,
- machine behaviour,
- discretization,
- maintenance history,
- and measurement uncertainty.

8.6. Runtime States

8.6.1. Runtime Interpretation

Remark 8.31. Within AIM, states are not static stored objects but runtime-evaluated entities.

Remark 8.32. Runtime systems continuously evaluate:

- world-to-track projections,
- local frames,
- realized geometry,
- dynamic quantities,
- and optimization residuals.

8.6.2. Derived Runtime Quantities

Definition 8.33 (Derived runtime quantities). Derived runtime quantities include:

$$a_{\text{eff}}, \quad j, \quad \kappa_g, \quad d, \quad h. \quad (8.11)$$

Remark 8.34. These are not independent state variables, but quantities derived from the underlying pose and frame structure.

8.7. Sparse and Hybrid States

8.7.1. Sparse Representation

Definition 8.35 (Sparse state). A sparse state is a compact representation using only essential control parameters.

Remark 8.36. Examples include:

- fixed curvature segments,
- transition segments,
- cant transitions,
- anchor poses.

8.7.2. Hybrid Representation

Definition 8.37 (Hybrid state). A hybrid state combines:

- parametric structure,
- and local sampled corrections.

Remark 8.38. Hybrid states combine efficient modelling with local adaptation.

8.8. Optimization States

8.8.1. Optimization Variables

Remark 8.39. Optimization does not necessarily operate on full geometry.

Remark 8.40. Typical optimization variables include:

- curvature parameters,
- transition lengths,
- cant parameters,
- control point positions,
- and realization corrections.

8.8.2. Realization-Aware Optimization

Remark 8.41. Within AIM, optimization is evaluated on realized states:

$$J(\mathcal{R}(\text{pose3})). \tag{8.12}$$

Remark 8.42. Optimization therefore operates indirectly through the realization operator.

8.9. State Hierarchy

Remark 8.43. The AIM framework distinguishes:

- intrinsic states,
- extrinsic states,
- target states,
- realized states,
- runtime states,
- and optimization states.

Remark 8.44. These are not independent models, but interconnected layers of one unified railway state system.

8.10. Canonical Interpretation

Proposition 8.45. *Within AIM, railway alignment is interpreted as a controlled spatial state trajectory evolving along arc-length.*

Proposition 8.46. *Curvature and cant evolution form the primary control structure of this trajectory.*

Proposition 8.47. *Realization, projection, and optimization operate on states rather than on isolated geometric entities.*

8.11. Connection to AIM

Summary 8.48. The AIM state model provides a unified framework linking:

- geometry,
- cant,
- realization,
- runtime projection,
- dynamics,
- and optimization.

It establishes railway alignment as a coupled state-based system rather than merely a collection of curves and coordinates.

9. Operators

Motivation

Remark 9.1. Railway alignment modelling is not merely based on geometric objects, but on transformations acting on states.

Remark 9.2. Within AIM, geometry, realization, projection, runtime evaluation, metric interpretation, and optimization are formulated operator-theoretically.

Remark 9.3. Operators connect:

- intrinsic geometry,
- metric realization,
- world-space embedding,
- sampled representations,
- runtime evaluation,
- physical realization,
- and optimization states.

Remark 9.4. This chapter introduces the canonical operators and operator spaces of the AIM framework.

9.1. General Operator Interpretation

Definition 9.5 (Operator). An operator is a mapping acting on:

- states,
- functions,
- representations,
- or geometric structures.

Remark 9.6. Operators may:

- transform representations,

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- evaluate quantities,
- modify states,
- reconstruct geometry,
- or approximate physical processes.

Principle 9.7 (State-operator principle). Within AIM, operators act on states rather than on isolated geometric objects.

Proposition 9.8. *Within AIM, geometry is not interpreted as static stored data, but as states transformed and interpreted through coupled operator systems.*

9.2. Operator Spaces

Remark 9.9. The canonical spaces used by the operator framework are introduced in chapter 6.

Remark 9.10. The most important spaces are:

$$\mathcal{X}, \quad \mathcal{Y}, \quad \Omega_{\text{track}}, \quad \Omega_{\text{world}}, \quad \mathcal{K}.$$

Remark 9.11. Here:

- $\mathcal{X}$ denotes railway state space,
- $\mathcal{Y}$ denotes runtime quantity space,
- Ω_{track} denotes intrinsic track space,
- Ω_{world} denotes CRS-dependent world space,
- and $\mathcal{K}$ denotes curvature space.

9.3. Operator Categories

Remark 9.12. The AIM framework distinguishes:

- geometric operators,
- representation operators,
- metric operators,
- runtime operators,
- realization operators,
- and optimization operators.

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Remark 9.13. These operator families are conceptually distinct, but operationally coupled.

Principle 9.14 (Operator separation principle). Runtime evaluation, metric realization, coordinate transformation, projection, and physical realization should not be collapsed into one implicit operation.

9.4. Geometric Operators

9.4.1. Curvature Operator

Definition 9.15 (Curvature operator). The directional evolution relation is:

$$\theta'(s) = \kappa(s).$$

Remark 9.16. Curvature acts as the differential generator of directional evolution.

9.4.2. Reconstruction Operator

Definition 9.17 (Reconstruction operator). The intrinsic reconstruction relation is:

$$\gamma'(s) = \mathbf{t}(s).$$

Remark 9.18. Together with curvature evolution, this defines intrinsic alignment reconstruction.

9.4.3. Cant Evolution Operator

Definition 9.19 (Cant evolution operator). Cant evolution is governed by:

$$\phi'(s) = \omega(s).$$

Remark 9.20. Cant-rate acts as the control quantity governing cross-level evolution.

Remark 9.21. The canonical geometric relations are collected in chapter 7.

9.5. Representation Operators

9.5.1. Track-to-World Transformation

Definition 9.22 (Track-to-world operator).

$$\mathcal{P}_{\text{tw}} : \Omega_{\text{track}} \rightarrow \Omega_{\text{world}}.$$

Remark 9.23. The operator maps intrinsic railway coordinates into world-space coordinates.

9.5.2. World-to-Track Projection

Definition 9.24 (World-to-track operator).

$$\mathcal{P}_{\text{wt}} : \Omega_{\text{world}} \rightarrow \Omega_{\text{track}}.$$

Remark 9.25. Inverse projection is generally nonlinear, metric-dependent, and potentially ambiguous.

Principle 9.26 (Projection principle). World-to-track transformation is an inverse runtime projection problem, not a static coordinate conversion.

9.5.3. Coordinate Transformation Operator

Definition 9.27 (Coordinate transformation operator).

$$\mathcal{C} : \Omega_{\text{world}} \rightarrow \Omega_{\text{world}}.$$

Remark 9.28. Coordinate transformations map between CRS-dependent spatial representations.

Remark 9.29. Coordinate transformation does not alter intrinsic railway semantics.

Remark 9.30. Coordinate transformation must not be confused with intrinsic world-to-track projection.

9.6. Metric Operators

Remark 9.31. Within AIM, intrinsic alignment semantics are separated from metric realization.

Principle 9.32 (Metric separation principle). Alignment semantics should not directly depend on a specific metric backend.

Definition 9.33 (Metric realization operator). The metric realization operator is written as:

$$\mathcal{G} : \mathcal{X} \rightarrow \mathcal{X}_{\text{metric}}.$$

Remark 9.34. The operator $\mathcal{G}$ expresses how intrinsic railway states are interpreted within a metric geometry model.

Remark 9.35. Metric interpretation may depend on:

- Euclidean engineering geometry,
- projected CRS geometry,
- ellipsoid geometry,
- or generalized differential-geometric manifolds.

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Remark 9.36. Metric operators affect:

- distance interpretation,
- curvature interpretation,
- projection behaviour,
- frame construction,
- and stationing-related computations.

Principle 9.37 (Metric backend principle). The alignment principle remains stable while the metric realization may change.

9.7. Runtime Operators

9.7.1. Evaluation Operator

Definition 9.38 (Evaluation operator).

$$\mathcal{E} : \mathcal{X} \times \Omega_{\text{track}} \rightarrow \mathcal{Y}.$$

Remark 9.39. Evaluation operators produce:

- runtime geometry,
- frame states,
- dynamic quantities,
- operational quantities,
- and projection results.

Remark 9.40. Runtime operators interpret states. They do not by themselves physically realize or modify the railway infrastructure.

9.7.2. Frame Operator

Definition 9.41 (Frame operator).

$$\mathcal{F} : \mathcal{X}_{\text{metric}} \times \Omega_{\text{track}} \rightarrow \text{Frames}.$$

Remark 9.42. The frame operator evaluates:

$$\mathcal{F}(x, s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s)).$$

Remark 9.43. Frames provide local operational spatial interpretation of alignment states.

Remark 9.44. Frame construction may depend on the selected metric realization. For example, a projected engineering frame and an ellipsoid-attached frame need not coincide.

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9.7.3. Runtime Quantities

Definition 9.45 (Runtime quantities). Typical runtime-derived quantities include:

$$a_{\text{eff}}, \quad j, \quad \kappa_g, \quad \phi, \quad F(s).$$

Remark 9.46. These quantities are evaluated dynamically from underlying states, metric interpretation, and runtime operators.

9.8. Realization Operators

9.8.1. Sampling Operator

Definition 9.47 (Sampling operator).

$$\mathcal{S} : f(s) \mapsto \{f(s_i)\}.$$

Remark 9.48. Sampling converts continuous representations into discrete representations.

9.8.2. Interpolation Operator

Definition 9.49 (Interpolation operator).

$$\mathcal{I} : \{f(s_i)\} \mapsto \tilde{f}(s).$$

Remark 9.50. Interpolation reconstructs continuous approximations from sampled data.

9.8.3. Mechanical Operator

Definition 9.51 (Mechanical operator).

$$\mathcal{M}.$$

Remark 9.52. The mechanical operator represents:

- structural response,
- ballast behaviour,
- track elasticity,
- and mechanical smoothing.

9.8.4. Adjustment Operator

Definition 9.53 (Adjustment operator).

$$\mathcal{A}.$$

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Remark 9.54. The adjustment operator represents localized construction or maintenance actions.

9.8.5. Realization Operator

Definition 9.55 (Realization operator).

$$\mathcal{R} : \mathcal{X}_{\text{target}} \rightarrow \mathcal{X}_{\text{real}}.$$

Principle 9.56 (Realization principle). Realization is interpreted as transformation of alignment states rather than as direct geometric reconstruction.

Remark 9.57. Realization operators describe physical or construction-related transformation. They must be distinguished from runtime operators, which evaluate or interpret states.

9.8.6. Realization Decomposition

Definition 9.58 (Realization decomposition). The realization process may be interpreted as:

$$\mathcal{R} = \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}.$$

Remark 9.59. This decomposition separates:

- discretization,
- reconstruction,
- and physical realization effects.

9.8.7. Iterative Realization

Definition 9.60 (Iterative realization).

$$x_{k+1} = \mathcal{A}(x_k).$$

Remark 9.61. Realization may therefore be interpreted as an iterative operator process.

9.9. Optimization Operators

9.9.1. Residual Operator

Definition 9.62 (Residual operator).

$$r(x).$$

Remark 9.63. Residuals quantify deviations from desired operational, geometric, realization, or metric properties.

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9.9.2. Jacobian Operator

Definition 9.64 (Jacobian operator).

$$J(x) = \frac{\partial r}{\partial x}.$$

9.9.3. Gauss–Newton Approximation

Definition 9.65 (Gauss–Newton approximation).

$$H \approx J^T J.$$

Remark 9.66. This approximation appears naturally in least-squares optimization.

9.9.4. Realization-Aware Optimization

Definition 9.67 (Realization-aware optimization).

$$\min_u J(\mathcal{R}(x(u))).$$

Remark 9.68. Optimization therefore acts indirectly through realization.

9.10. Operator Composition

9.10.1. Runtime Composition

Definition 9.69 (Runtime operator composition). A simplified runtime evaluation chain may be written as:

$$\mathcal{E} \sim \mathcal{P} \circ \mathcal{F} \circ \mathcal{G}.$$

Remark 9.70. This expresses that runtime evaluation may depend on:

- metric realization,
- frame evaluation,
- and projection services.

9.10.2. Realization Composition

Definition 9.71 (Realization operator composition). The physical realization chain is represented by:

$$\mathcal{R} = \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}.$$

Remark 9.72. This separates physical realization from runtime interpretation.

9.10.3. Full Operational Composition

Definition 9.73 (Full operational operator chain). A schematic full AIM evaluation chain may be written as:

$$\mathcal{E}_{\text{full}} \sim \mathcal{P} \circ \mathcal{F} \circ \mathcal{G} \circ \mathcal{R}.$$

Remark 9.74. This expression is not meant as a rigid software pipeline. It describes the conceptual dependency between realization, metric interpretation, frame evaluation, projection, and runtime quantities.

9.11. Runtime versus Realization

Principle 9.75 (Runtime-realization distinction). Runtime operators evaluate states, whereas realization operators transform target states into realized states.

Remark 9.76. Runtime evaluation may operate on target, realized, or measured states. Realization describes how target states become realized states.

Remark 9.77. Confusing runtime evaluation with physical realization leads to ambiguous models and duplicated geometry.

9.12. Canonical Interpretation

Proposition 9.78. *Within AIM, railway alignment modelling is fundamentally operator-based.*

Proposition 9.79. *Operators transform and evaluate states across:*

- *intrinsic geometry,*
- *metric realization,*
- *world-space embedding,*
- *runtime interpretation,*
- *physical realization,*
- *and optimization.*

Proposition 9.80. *Realization, metric interpretation, projection, runtime evaluation, and optimization are interpreted as coupled operator layers acting on railway states.*

9.13. Connection to AIM

Summary 9.81. The AIM operator framework unifies:

- *intrinsic geometry,*

9. Operators

- metric interpretation,
- world–track projection,
- runtime evaluation,
- physical realization,
- and optimization.

Operators form the dynamic transformation architecture connecting all parts of the AIM framework.

10. System Layers

Motivation

Remark 10.1. Railway alignment modelling combines multiple fundamentally different domains.

Remark 10.2. Classical approaches often merge:

- intrinsic geometry,
- coordinate reference systems,
- metric interpretation,
- construction realization,
- runtime evaluation,
- dynamics,
- and operational vehicle interpretation

into a single implicit model.

Remark 10.3. Within AIM, these concerns are separated into interacting conceptual and operational system layers.

Remark 10.4. This separation improves:

- conceptual clarity,
- numerical robustness,
- scalability,
- interoperability,
- runtime flexibility,
- and realization-aware modelling.

10.1. Overview

Definition 10.5 (System layer). A system layer is a coherent subsystem operating on its own:

- state representation,
- operators,
- constraints,
- interpretation rules,
- and runtime assumptions.

Remark 10.6. The AIM framework separates railway alignment modelling into:

- geometry layer,
- CRS layer,
- metric realization layer,
- physical realization layer,
- runtime layer,
- vehicle layer,
- and optimization layer.

Remark 10.7. These layers are conceptually distinct, but operationally coupled.

10.2. Geometry Layer

10.2.1. Role

Remark 10.8. The geometry layer defines the intrinsic semantic structure of the alignment.

Remark 10.9. Its central quantities are:

- arc-length s ,
- curvature $\kappa(s)$,
- cant evolution,
- transition structure,
- and pose evolution laws.

10.2.2. Core Relations

Remark 10.10. The geometry layer is governed by the canonical intrinsic relations defined in chapter 7.

Remark 10.11. In particular, it uses:

- the curvature relation eq. (7.4),
- the reconstruction relation eq. (7.3),
- and the cant evolution relation eq. (7.7).

10.2.3. Characteristics

Remark 10.12. The geometry layer is:

- intrinsic,
- semantic,
- continuous,
- analytical,
- and independent of a particular metric backend.

Remark 10.13. It defines alignment semantics rather than physically realized track geometry.

10.3. CRS Layer

10.3.1. Role

Remark 10.14. The CRS layer defines external coordinate reference systems and their relations to the Earth.

Remark 10.15. Typical coordinate systems include:

- engineering CRS,
- projected CRS,
- geodetic coordinates,
- local construction coordinate systems,
- and ellipsoid-based representations.

10.3.2. Coordinate Transformation

Remark 10.16. CRS transformations are represented by the coordinate transformation operator $\mathcal{C}$, introduced in theorem 9.27.

Remark 10.17. The CRS layer operates within world space:

$$\mathcal{C} : \Omega_{\text{world}} \rightarrow \Omega_{\text{world}}.$$

10.3.3. Fundamental Observation

Proposition 10.18. *A railway alignment is generally not invariant under CRS transformations.*

Remark 10.19. In particular:

- straight lines,
- curvature,
- transition behaviour,
- stationing-related length interpretation,
- and cant-related spatial interpretation

may change under projection or CRS-dependent embedding.

Remark 10.20. The CRS layer therefore affects spatial interpretation, but it is not identical to intrinsic railway projection.

10.4. Metric Realization Layer

10.4.1. Role

Remark 10.21. The metric realization layer defines how semantic alignment structures are interpreted geometrically within a selected metric model.

Remark 10.22. It is governed conceptually by the metric realization operator $\mathcal{G}$, introduced in theorem 11.22.

10.4.2. Core Principle

Principle 10.23 (Metric realization principle). Semantic alignment logic is separated from metric realization behaviour.

Remark 10.24. Thus:

- semantic curvature laws,
- transition structure,

- sparse alignment logic,
- and stationing semantics

remain conceptually stable even when metric realization changes.

10.4.3. Metric Variants

Remark 10.25. Typical metric realization modes include:

- engineering-Euclidean realization,
- projection-aware realization,
- ellipsoid-aware realization,
- and future differential-geometric realization.

10.4.4. Operator Boundary

Remark 10.26. The metric realization layer forms the operator boundary between:

- semantic alignment logic,
- and metric-dependent geometric evaluation.

Remark 10.27. This architecture is introduced formally in chapter 11.

10.5. Physical Realization Layer

10.5.1. Role

Remark 10.28. The physical realization layer models the transformation from analytical target geometry to physically realized railway geometry.

10.5.2. Core Operator

Remark 10.29. The physical realization layer is governed by the realization operator

$$\mathcal{R} : \mathcal{X}_{\text{target}} \rightarrow \mathcal{X}_{\text{real}},$$

introduced in theorem 9.55.

10.5.3. Characteristics

Remark 10.30. The physical realization layer includes:

- construction constraints,
- machine behaviour,

- discretization,
- interpolation,
- structural response,
- maintenance processes,
- and measurement feedback.

Remark 10.31. Physical realization acts approximately as a low-pass filter on geometry and curvature.

Remark 10.32. Within AIM, physical realization is interpreted as an explicit operator layer rather than as a secondary engineering detail.

Remark 10.33. This layer is developed in chapter 43.

10.6. Runtime Layer

10.6.1. Role

Remark 10.34. The runtime layer transforms stored railway representations into operationally evaluable geometry.

Remark 10.35. Runtime evaluation is governed by the evaluation operator

$$\mathcal{E} : \mathcal{X} \times \Omega_{\text{track}} \rightarrow \mathcal{Y},$$

introduced in theorem 9.38.

10.6.2. Runtime Services

Remark 10.36. Typical runtime services include:

- world-track projection,
- frame evaluation,
- metric-aware geometry evaluation,
- visualization,
- BIM placement,
- spatial queries,
- sensor integration,
- and dynamic quantity evaluation.

10.6.3. Runtime Characteristics

Remark 10.37. The runtime layer is:

- state-dependent,
- scale-dependent,
- metric-aware,
- operator-based,
- computationally adaptive,
- and runtime-evaluated.

Remark 10.38. Many runtime quantities are evaluated dynamically rather than stored explicitly.

Remark 10.39. The runtime layer therefore acts as the operational execution layer of the AIM framework.

Remark 10.40. The runtime architecture is developed further in chapter 36.

10.7. Vehicle Layer

10.7.1. Role

Remark 10.41. The physically relevant object is not merely the alignment centerline, but the railway vehicle moving within space.

10.7.2. Operational Geometry

Remark 10.42. The vehicle layer includes:

- vehicle envelopes,
- bogie motion,
- wheelset orientation,
- center-of-gravity motion,
- operational clearances,
- and platform interaction.

10.7.3. Runtime Dependency

Remark 10.43. Vehicle-space interpretation depends on runtime-evaluated pose3 states, frames, cant, profile, and metric context.

Remark 10.44. Thus, the vehicle layer is not merely an application layer, but an operational interpretation layer built on top of runtime services.

10.7.4. Motivation

Remark 10.45. Classical railway alignment systems often approximate operational geometry using simplified clearance assumptions.

Remark 10.46. Within AIM, operational vehicle-space geometry may instead be modelled explicitly.

10.8. Optimization Layer

10.8.1. Role

Remark 10.47. The optimization layer searches for railway states satisfying:

- geometric,
- metric,
- dynamic,
- operational,
- realization,
- and regulatory constraints.

10.8.2. Realization-Aware Optimization

Remark 10.48. Within AIM, optimization may be evaluated on realized states:

$$J(\mathcal{R}(x(u))).$$

Remark 10.49. This relation is part of the canonical optimization structure introduced in eq. (7.24).

10.8.3. Hybrid Nature

Remark 10.50. Optimization combines:

- analytical models,
- sparse representations,
- discrete constraints,
- runtime evaluation,
- realization operators,
- metric-dependent geometry evaluation,
- and operational vehicle-space constraints.

10.9. Layer Interactions

Remark 10.51. The system layers interact continuously.

Remark 10.52. For example:

- intrinsic geometry feeds metric realization,
- CRS influences metric interpretation and world embedding,
- metric realization affects runtime geometry,
- physical realization modifies target states into realized states,
- runtime supports projection, visualization, and optimization,
- optimization modifies target geometry,
- vehicle-space interpretation depends on pose3, frames, and runtime,
- and physical measurements should generally reference realized geometry.

10.10. Canonical Transformation Structure

Definition 10.53 (Canonical transformation structure). A simplified conceptual AIM structure is:

intrinsic geometry $\rightarrow$ metric realization $\rightarrow$ runtime evaluation $\rightarrow$ vehicle-space interpretation.

Remark 10.54. Physical realization is a separate transformation:

target state $\rightarrow$ realized state.

Remark 10.55. CRS embedding influences metric realization and world-space representation, but it is not identical to intrinsic world-track projection.

Remark 10.56. Optimization closes the loop by modifying target geometry based on runtime-evaluated and realization-aware objectives.

10.11. Scale Dependence

Remark 10.57. Different layers dominate at different spatial and operational scales.

Remark 10.58. For example:

- CRS effects dominate at large spatial scales,
- metric realization dominates at projection transitions,
- physical realization effects dominate at construction scales,

- runtime costs dominate in large operational systems,
- and vehicle-space effects dominate near platforms, switches, and clearance-critical situations.

10.12. Canonical Layer Separation

Principle 10.59 (Canonical separation principle). Intrinsic geometry, CRS embedding, metric realization, physical realization, runtime interpretation, operational behaviour, and optimization should not be merged into a single implicit model.

Remark 10.60. Many inconsistencies in classical railway systems arise from implicit mixing of fundamentally different conceptual layers.

10.13. Key Insight

Proposition 10.61. *Railway alignment modelling is not a single geometric problem, but a multi-layer interaction system.*

Proposition 10.62. *The operational behaviour of railway systems emerges from interactions between:*

- *intrinsic geometry,*
- *CRS embedding,*
- *metric realization,*
- *physical realization,*
- *runtime evaluation,*
- *vehicle-space interpretation,*
- *and optimization.*

10.14. Connection to AIM

Summary 10.63. The AIM framework separates railway alignment modelling into interacting system layers.

This separation enables:

- rigorous intrinsic geometry,
- CRS-aware embedding,
- metric-aware realization,

10. *System Layers*

- physical realization-aware modelling,
- scalable runtime systems,
- explicit vehicle-space interpretation,
- and realization-aware optimization.

Together, these layers form the conceptual architecture underlying the AIM framework.

11. Metric Realization

Motivation

Remark 11.1. Classical railway alignment systems often assume that geometry, distance, curvature, projection, and realization are all defined within one implicit Euclidean coordinate model.

Remark 11.2. This assumption is sufficient for many engineering applications, but it becomes problematic when:

- large-scale coordinate systems are involved,
- Earth curvature becomes relevant,
- projection distortions appear,
- multiple CRS layers interact,
- or alignment semantics must remain independent from metric realization.

Remark 11.3. Within AIM, a strict distinction is therefore introduced between:

- alignment semantics,
- CRS embedding,
- and metric realization.

Remark 11.4. This separation enables:

- stable alignment semantics,
- interchangeable metric models,
- realization-aware geometry,
- projection-aware interpretation,
- and future differential-geometric extensions.

11.1. Core Principle

Principle 11.5 (Alignment semantics principle). The semantic structure of an alignment is independent of the metric space in which it is realized.

Remark 11.6. For example:

- straight elements,
- constant-curvature arcs,
- transition elements,
- topology,
- and switch logic

remain semantically stable even when the underlying metric model changes.

Principle 11.7 (Metric realization principle). Geometric realization depends on the metric model used for evaluation.

Remark 11.8. Thus:

- distances,
- curvature,
- projection,
- interpolation,
- frame construction,
- and geodesic behaviour

may differ between metric realizations.

Principle 11.9 (Canonical separation principle). Alignment semantics and metric realization must remain conceptually separable.

11.2. Semantic Alignment Layer

11.2.1. Intrinsic Semantics

Remark 11.10. The semantic alignment layer describes:

- alignment topology,
- curvature laws,
- transition structure,

11. Metric Realization

- stationing semantics,
- continuity conditions,
- and sparse alignment structure.

Remark 11.11. This layer is intentionally metric-independent.

11.2.2. Sparse Representation

Remark 11.12. Within AIM, the sparse alignment model stores:

- fixed curvature elements,
- transition elements,
- lookup-based curvature evolution,
- sparse reconstruction parameters,
- and associated semantic metadata.

Remark 11.13. The sparse representation therefore acts as a semantic alignment description rather than a finalized geometric embedding.

11.3. CRS Embedding versus Metric Realization

11.3.1. Conceptual Distinction

Remark 11.14. Within AIM, CRS embedding and metric realization are related but distinct concepts.

Remark 11.15. The CRS layer defines:

- coordinate reference systems,
- projection rules,
- Earth-related embedding,
- and coordinate transformations.

Remark 11.16. The metric realization layer defines:

- distance interpretation,
- curvature evaluation,
- frame behaviour,
- transport rules,
- and geometric reconstruction behaviour.

11.3.2. Coordinate Transformation

Remark 11.17. Coordinate transformations are represented by:

$$\mathcal{C} : \Omega_{\text{world}} \rightarrow \Omega_{\text{world}}.$$

Remark 11.18. Coordinate transformation alone does not define intrinsic railway geometry.

11.3.3. Projection Dependence

Proposition 11.19. *A straight line in one projected CRS is generally not identical to a straight line in another projection space.*

Remark 11.20. Similarly:

- curvature,
- transition behaviour,
- stationing interpretation,
- and lateral offsets

may become projection-dependent.

11.4. Metric Realization Spaces

11.4.1. General Interpretation

Definition 11.21 (Metric realization space). A metric realization space defines:

- distance interpretation,
- directional interpretation,
- curvature evaluation,
- frame transport,
- projection behaviour,
- and geometric reconstruction rules.

11.4.2. Metric Realization Operator

Definition 11.22 (Metric realization operator). Metric realization is represented abstractly by:

$$\mathcal{G} : \mathcal{X} \rightarrow \mathcal{X}_{\text{metric}}.$$

Remark 11.23. The operator $\mathcal{G}$ defines how semantic railway states are evaluated geometrically within a selected metric model.

11.4.3. Operator Boundary

Principle 11.24 (Metric operator boundary). Alignment logic should not directly assume a specific metric model.

Remark 11.25. Instead, alignment operations interact with geometry through metric operators and runtime evaluation services.

Remark 11.26. This creates a strict separation between:

- semantic alignment structure,
- and metric realization behaviour.

11.5. Engineering Metric

11.5.1. Role

Remark 11.27. The dominant practical realization mode within AIM is the engineering metric.

11.5.2. Characteristics

Remark 11.28. The engineering metric assumes:

- locally Euclidean geometry,
- Cartesian coordinates,
- fast evaluation,
- simple frame construction,
- and numerically stable reconstruction.

11.5.3. Importance

Principle 11.29 (Engineering priority principle). The engineering metric remains the primary operational mode of AIM and must not suffer unnecessary computational overhead.

Remark 11.30. Advanced metric concepts must therefore remain compatible with highly efficient engineering-scale evaluation.

11.6. Projection-Aware Metric

11.6.1. Motivation

Remark 11.31. Projected coordinate systems introduce:

- scale distortions,

- directional distortions,
- projection-dependent curvature effects,
- and coordinate-dependent metric behaviour.

11.6.2. Interpretation

Remark 11.32. Within AIM, projection-aware realization interprets alignment geometry through projection-dependent metric evaluation.

Remark 11.33. Thus, runtime geometry depends not only on intrinsic alignment semantics but also on projection context.

11.6.3. Operational Consequences

Remark 11.34. Projection-aware realization affects:

- world-track projection,
- nearest-point search,
- stationing interpretation,
- local frame construction,
- and geometric reconstruction.

11.7. Ellipsoid-Based Metric

11.7.1. Motivation

Remark 11.35. At sufficiently large spatial scales, Earth curvature becomes relevant for railway alignment interpretation.

11.7.2. Geodesic Interpretation

Remark 11.36. Within ellipsoid-based geometry:

- Euclidean straightness,
- geodesic straightness,
- and engineering straightness

are no longer identical concepts.

11.7.3. Curvature Interpretation

Remark 11.37. On curved surfaces, curvature decomposes into:

$$\kappa^2 = \kappa_g^2 + \kappa_n^2,$$

where:

- κ_g denotes geodesic curvature,
- κ_n denotes surface-normal curvature.

Remark 11.38. For railway dynamics, geodesic curvature is generally the physically relevant quantity.

11.7.4. Frame Interpretation

Remark 11.39. Within ellipsoid-aware geometry, frame construction may require:

- tangent transport,
- surface-normal evaluation,
- and connection-dependent orientation rules.

11.8. Future Differential-Geometric Layer

11.8.1. Motivation

Remark 11.40. Future railway alignment systems may require:

- intrinsic surface geometry,
- Earth-attached frames,
- geodesic transition interpretation,
- and fully differential-geometric reconstruction methods.

11.8.2. Operator-Based Geometry

Remark 11.41. Within such a framework, geometric operations would no longer be based primarily on Cartesian coordinate algebra.

Remark 11.42. Instead, geometry would emerge from:

- local operators,
- metric tensors,
- connection rules,

- covariant derivatives,
- and differential-geometric transport.

11.9. Runtime Dependency

11.9.1. Runtime Evaluation

Remark 11.43. Metric realization affects runtime services such as:

- projection,
- stationing,
- frame construction,
- nearest-point search,
- pose3 evaluation,
- and geometric reconstruction.

11.9.2. Metric-Aware Runtime

Remark 11.44. Runtime systems should therefore evaluate geometry through metric-aware operator services rather than through hardcoded Euclidean assumptions.

11.9.3. pose3 Dependency

Remark 11.45. Runtime pose3 evaluation depends jointly on:

- semantic alignment states,
- metric realization,
- CRS embedding,
- frame operators,
- profile interpretation,
- and cant interpretation.

Remark 11.46. Thus, pose3 is not a purely intrinsic object, but a runtime-generated metric interpretation of semantic alignment structure.

11. Metric Realization

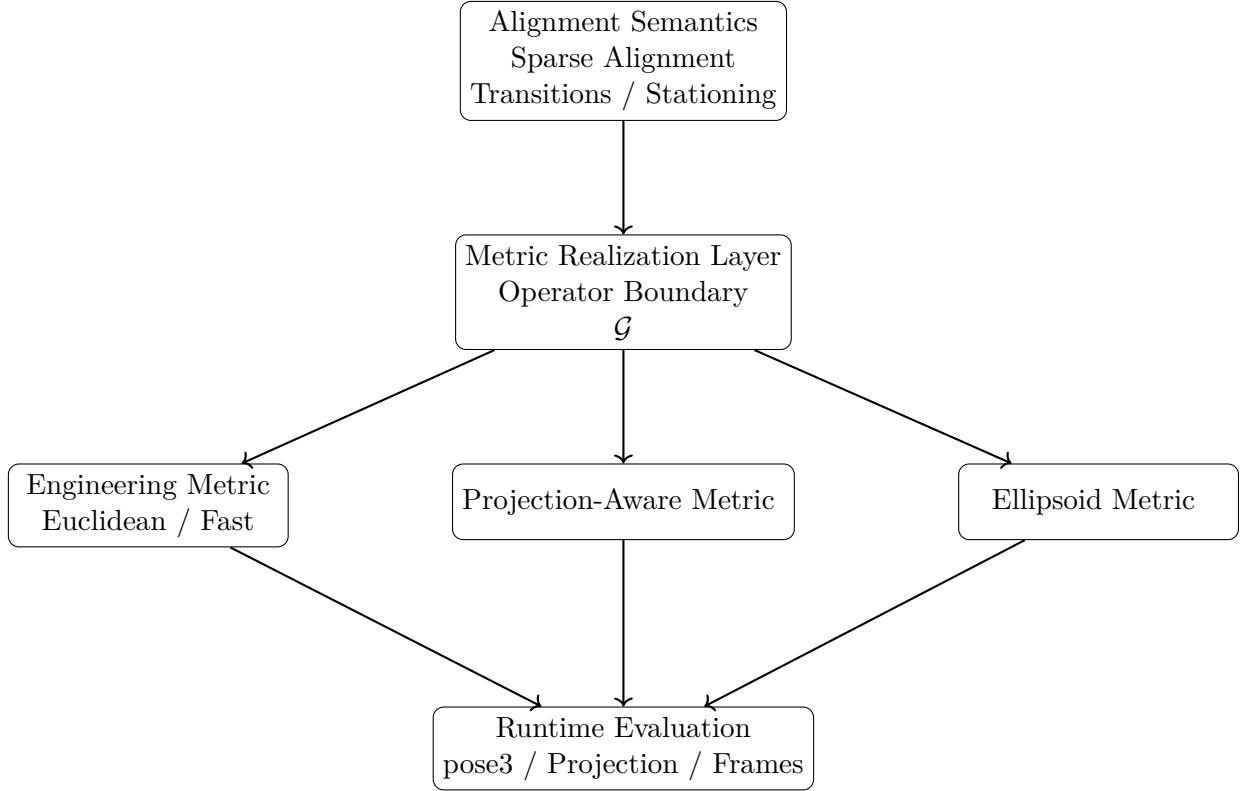


Abbildung 11.1.: AIM separates semantic alignment structure from metric realization. Different metric backends may realize the same semantic alignment model, while runtime services operate on the resulting metric interpretation.

11.10. Conceptual Architecture

11.11. Key Insight

Proposition 11.47. *Railway alignment semantics and metric realization are fundamentally different conceptual layers.*

Proposition 11.48. *The same semantic alignment structure may be realized within different metric spaces.*

Proposition 11.49. *Metric realization therefore acts as an operator layer between semantic alignment logic and spatial geometry evaluation.*

Proposition 11.50. *Runtime geometry emerges from interaction between:*

- *semantic alignment structure,*
- *metric realization,*
- *CRS embedding,*
- *and runtime evaluation operators.*

11.12. Connection to AIM

Summary 11.51. Within AIM, alignment semantics are intentionally separated from metric realization.

This enables:

- stable sparse alignment representations,
- interchangeable metric models,
- realization-aware geometry,
- projection-aware evaluation,
- runtime-dependent pose3 interpretation,
- and future differential-geometric extensions.

The metric realization layer therefore forms the operator boundary between semantic alignment logic and operational geometry.

12. Ontology and Conceptual Structure

Motivation

Remark 12.1. Railway alignment modelling combines concepts originating from:

- geometry,
- mechanics,
- surveying,
- construction,
- vehicle dynamics,
- and operational railway engineering.

Remark 12.2. Many inconsistencies in classical systems arise because these concepts are mixed implicitly without a rigorous conceptual structure.

Remark 12.3. Within AIM, a distinction is made between:

- mathematical objects,
- physical objects,
- operational objects,
- and runtime representations.

12.1. Core Ontological Distinction

Definition 12.4 (Ontological layers). The AIM ontology distinguishes between:

1. abstract geometry,
2. realized physical geometry,
3. operational railway behaviour,
4. and computational runtime representation.

Remark 12.5. These layers are related, but not identical.

12.2. Alignment as Abstract Geometry

12.2.1. Alignment Concept

Definition 12.6 (Alignment). An alignment is an intrinsic geometric object primarily defined through:

$$\kappa(s), \quad \phi(s), \quad \text{pose}(s).$$

Remark 12.7. Within AIM, an alignment is not merely a spatial curve.

Remark 12.8. Instead, it represents a structured operational geometry along arc-length.

12.2.2. Intrinsic Nature

Remark 12.9. The alignment is:

- intrinsic,
- arc-length-based,
- and independent of global coordinates.

12.3. Target Geometry vs Realized Geometry

12.3.1. Target Geometry

Definition 12.10 (Target geometry). The target geometry is the intended analytical railway state defined by the design process.

12.3.2. Realized Geometry

Definition 12.11 (Realized geometry). The realized geometry is the physically existing railway geometry after construction and maintenance.

12.3.3. Fundamental Distinction

Proposition 12.12. *In general:*

$$\text{pose3}_{\text{real}} \neq \text{pose3}_{\text{target}}.$$

Remark 12.13. This distinction is fundamental within AIM.

12.4. Geometry vs Vehicle Space

12.4.1. Classical Simplification

Remark 12.14. Classical railway alignment systems often reduce the railway problem to a centerline geometry problem.

12.4.2. Operational Reality

Remark 12.15. The operationally relevant object, however, is the railway vehicle moving within space.

Remark 12.16. Important operational quantities include:

- center-of-gravity motion,
- vehicle envelope,
- bogie rotation,
- wheelset orientation,
- and platform interaction.

12.4.3. Consequence

Proposition 12.17. *The railway alignment is operationally meaningful only together with its vehicle-space interpretation.*

12.5. Pose, Frame, and State

12.5.1. Pose

Definition 12.18 (Pose). A pose describes a local geometric state attached to arc-length position s .

12.5.2. Frame

Definition 12.19 (Frame). A frame defines a local coordinate interpretation of the pose.

12.5.3. State

Definition 12.20 (State). A state combines:

- geometry,
- orientation,
- and operational interpretation.

12.5.4. Runtime Nature

Remark 12.21. Within AIM, states are runtime-evaluated rather than purely static data objects.

12.6. World Space vs Track Space

12.6.1. Track Space

Definition 12.22 (Track space). Track space is the intrinsic coordinate system attached to the alignment.

12.6.2. World Space

Definition 12.23 (World space). World space is a global spatial coordinate system defined through a CRS.

12.6.3. Transformation

Remark 12.24. The transformation:

$$\text{world} \leftrightarrow \text{track}$$

is nonlinear and context-dependent.

12.7. CRS and Earth Geometry

12.7.1. Projection Dependence

Remark 12.25. Railway geometry depends fundamentally on the underlying coordinate reference system.

12.7.2. Non-Invariance

Proposition 12.26. *Railway alignments are generally not invariant under CRS changes.*

12.7.3. Ellipsoid Reality

Remark 12.27. The physically realized railway exists on the Earth surface rather than within a projected Euclidean plane.

12.8. Runtime Ontology

12.8.1. Runtime Services

Remark 12.28. Many AIM objects exist operationally as runtime services rather than stored static entities.

12.8.2. Examples

Remark 12.29. Examples include:

- world-track projection,
- frame evaluation,
- local coordinate systems,
- and vehicle-space interpretation.

12.9. Realization Ontology

12.9.1. Construction Process

Remark 12.30. The physically realized railway emerges from:

- measurement,
- machine operation,
- maintenance,
- and mechanical response.

12.9.2. Iterative Nature

Remark 12.31. Realization is therefore not a static mapping, but an iterative physical process.

12.10. Optimization Ontology

12.10.1. Classical Optimization

Remark 12.32. Classical optimization often minimizes purely analytical target states.

12.10.2. AIM Optimization

Remark 12.33. Within AIM, optimization operates on:

- realized states,
- operational behaviour,
- and runtime-evaluated constraints.

12.10.3. Inverse Nature

Remark 12.34. Many railway optimization problems are inverse realization problems.

12.11. Sparse vs Runtime Reality

12.11.1. Sparse Representation

Definition 12.35 (Sparse representation). A sparse representation stores only the essential alignment-defining parameters.

12.11.2. Runtime Expansion

Remark 12.36. Operational runtime systems reconstruct:

- geometry,
- frames,
- vehicle-space behaviour,
- and projections

from sparse intrinsic data.

12.12. Fundamental AIM View

Proposition 12.37. *A railway alignment is not merely a geometric curve.*

Remark 12.38. It is a multi-layer operational system connecting:

- intrinsic geometry,
- realization,
- Earth geometry,
- vehicle-space behaviour,
- runtime services,
- and optimization.

12.13. Key Insight

Proposition 12.39. *Many limitations of classical railway systems arise from implicit collapsing of fundamentally different ontological layers into a single geometric abstraction.*

12.14. Connection to AIM

Summary 12.40. The AIM ontology separates:

- target geometry,
- realized geometry,
- operational vehicle behaviour,
- runtime representations,
- and Earth-related coordinate systems.

This separation forms the conceptual foundation for realization-aware, vehicle-aware, and Earth-aware railway alignment modelling.

13. Curvature on the Ellipsoid

Motivation

Remark 13.1. Railway alignment modelling is traditionally performed in projected coordinate systems such as Gauss–Krüger or UTM.

Remark 13.2. Within these systems, curvature, arc-length, and derived dynamic quantities are evaluated in a planar approximation of the Earth.

Remark 13.3. However, such projections introduce geometric distortions and therefore do not exactly represent the physical geometry on which the railway exists.

Remark 13.4. This motivates a formulation of alignment geometry directly on the reference surface of the Earth, typically modelled as an ellipsoid.

13.1. Curve on a Surface

Definition 13.5 (Reference ellipsoid). Let $\mathcal{E} \subset \mathbb{R}^3$ denote a reference ellipsoid representing the Earth.

Definition 13.6 (Alignment on the ellipsoid). An alignment is defined as a curve

$$\gamma : [0, L] \rightarrow \mathcal{E}$$

parameterized by arc-length s .

Remark 13.7. The parameterization satisfies

$$\|\gamma'(s)\| = 1,$$

so that s represents physical distance along the curve.

13.2. Tangent and Normal Decomposition

Definition 13.8 (Tangent vector). The tangent vector is defined as

$$\mathbf{t}(s) = \gamma'(s).$$

Definition 13.9 (Surface normal). Let $\mathbf{n}_{\mathcal{E}}(s)$ denote the unit normal vector of the ellipsoid at $\gamma(s)$.

13. Curvature on the Ellipsoid

Remark 13.10. The second derivative $\gamma''(s)$ can be decomposed into:

- a component tangent to the surface,
- a component normal to the surface.

Definition 13.11 (Decomposition).

$$\gamma''(s) = \underbrace{\nabla_s \mathbf{t}(s)}_{\text{tangential}} + \underbrace{(\gamma''(s) \cdot \mathbf{n}_{\mathcal{E}}) \mathbf{n}_{\mathcal{E}}}_{\text{normal}}.$$

Remark 13.12. Here, $\nabla_s \mathbf{t}$ denotes the covariant derivative of the tangent vector along the curve on the surface.

13.3. Geodesic Curvature

Definition 13.13 (Geodesic curvature). The intrinsic curvature of the alignment on the ellipsoid is defined as

$$\kappa_g(s) = \|\nabla_s \mathbf{t}(s)\|.$$

Remark 13.14. Geodesic curvature measures the change of direction within the surface.

Remark 13.15. It is the surface-intrinsic counterpart of planar curvature and is the relevant curvature notion for alignment geometry constrained to the reference surface.

13.4. Normal Curvature

Definition 13.16 (Normal curvature). The normal curvature is defined as the normal component of the second derivative:

$$\kappa_n(s) = |\gamma''(s) \cdot \mathbf{n}_{\mathcal{E}}(s)|.$$

Remark 13.17. Normal curvature is induced by the curvature of the ellipsoid itself and does not describe the bending of the alignment within the surface.

13.5. Total Curvature

Proposition 13.18. *The total spatial curvature satisfies:*

$$\kappa^2(s) = \kappa_g^2(s) + \kappa_n^2(s).$$

Remark 13.19. Thus, the spatial curvature decomposes into an intrinsic surface contribution and an extrinsic surface-induced contribution.

13.6. Railway-Relevant Curvature

Proposition 13.20. *For railway alignment modelling on a reference surface, the railway-relevant curvature is the geodesic curvature:*

$$\kappa_{\text{rail}}(s) = \kappa_g(s).$$

Remark 13.21. Railway vehicles move along the surface, and dynamic quantities such as lateral acceleration depend on curvature within the surface rather than on the embedding curvature of the Earth.

13.7. Relation to Planar Models

Remark 13.22. In planar geometry, curvature is defined as

$$\kappa(s) = \theta'(s).$$

Remark 13.23. On a curved surface, the heading-angle derivative is replaced by the covariant change of the tangent direction:

$$\kappa_g(s) = \|\nabla_s \mathbf{t}(s)\|.$$

Remark 13.24. Thus, classical planar formulations are a special case of the intrinsic surface formulation.

13.8. Special Case: Geodesics

Definition 13.25 (Geodesic). A curve satisfies

$$\kappa_g(s) = 0$$

if and only if it is a geodesic on the surface.

Remark 13.26. Geodesics represent straightest paths on the ellipsoid in the intrinsic surface sense.

Remark 13.27. They are generally not straight in projected coordinate systems.

Remark 13.28. Railway alignment design is not equivalent to geodesic design. A railway alignment must satisfy engineering, dynamic, operational, and realization constraints.

13.9. Computational Formulation

Remark 13.29. In practical computations, geodesic curvature can be evaluated by projecting the second derivative onto the tangent plane of the surface:

$$\kappa_g = \|\gamma'' - (\gamma'' \cdot \mathbf{n}_\mathcal{E})\mathbf{n}_\mathcal{E}\|.$$

Remark 13.30. This formulation separates curvature within the surface from curvature caused by the embedding of the surface in three-dimensional space.

13.10. Implications for Engineering

Proposition 13.31. *Curvature evaluated in projected coordinate systems differs from intrinsic curvature on the ellipsoid.*

Remark 13.32. Consequently:

- arc-length is only approximately preserved,
- curvature is distorted by projection,
- dynamic quantities are evaluated on an approximate geometry.

Proposition 13.33. *Evaluating curvature intrinsically yields more physically consistent results.*

13.11. Vision

Remark 13.34. A consistent modelling framework may formulate alignment geometry directly on the ellipsoid and use projections only for visualization and data exchange.

Remark 13.35. This enables:

- physically meaningful curvature,
- accurate dynamic evaluation,
- improved robustness in optimization,
- and increased safety in alignment design.

Remark 13.36. This is a long-term research direction rather than a prerequisite for practical AIM implementations.

13.12. Connection to AIM

Summary 13.37. Curvature on the ellipsoid must be understood as an intrinsic quantity, measured within the surface rather than in ambient space.

For railway alignment on a reference surface, the relevant curvature is the geodesic curvature.

This distinction is essential for physically consistent modelling and provides a foundation for extending the AIM framework beyond projection-based engineering geometry.

14. Pose3 on the Ellipsoid

Motivation

Remark 14.1. The planar pose3 model describes railway state by position, direction, curvature, and cant in a projected engineering coordinate system.

Remark 14.2. For physically consistent modelling on the Earth, this state must be defined on the reference ellipsoid rather than in a planar projection.

14.1. Surface-Based Pose

Definition 14.3 (Surface pose). Let $\mathcal{E}$ be a reference ellipsoid and let

$$\gamma : [0, L] \rightarrow \mathcal{E}$$

be an arc-length parameterized alignment.

A surface pose is given by

$$\text{pose}_{\mathcal{E}}(s) = (\gamma(s), \mathbf{t}(s), \mathbf{n}_{\mathcal{E}}(s)),$$

where $\mathbf{t}(s) = \gamma'(s)$ is the tangent vector and $\mathbf{n}_{\mathcal{E}}(s)$ is the ellipsoid normal.

14.2. Railway Surface Frame

Definition 14.4 (Surface railway frame). The local railway frame is defined by

$$\mathbf{t}(s), \quad \mathbf{b}(s) = \mathbf{n}_{\mathcal{E}}(s) \times \mathbf{t}(s), \quad \mathbf{n}_{\mathcal{E}}(s).$$

Remark 14.5. Here, $\mathbf{t}$ is the longitudinal direction, $\mathbf{b}$ is the lateral direction within the surface, and $\mathbf{n}_{\mathcal{E}}$ is the local surface normal.

14.3. Cant as Rotation of the Cross-Section

Definition 14.6 (Cant on the ellipsoid). Cant is represented by a rotation of the cross-section around the tangent vector $\mathbf{t}(s)$:

$$R_{\mathbf{t}}(\phi(s)).$$

14. Pose3 on the Ellipsoid

Remark 14.7. This rotation acts on the lateral-normal plane spanned by

$$\mathbf{b}(s), \quad \mathbf{n}_{\mathcal{E}}(s).$$

Definition 14.8 (Canted lateral direction). The canted lateral direction is

$$\mathbf{b}_{\phi}(s) = R_{\mathbf{t}}(\phi(s)) \mathbf{b}(s).$$

Definition 14.9 (Canted normal direction). The canted normal direction is

$$\mathbf{n}_{\phi}(s) = R_{\mathbf{t}}(\phi(s)) \mathbf{n}_{\mathcal{E}}(s).$$

14.4. Gravity Direction

Remark 14.10. On the ellipsoid, gravity is not identical to the ellipsoid normal in full physical precision.

Definition 14.11 (Gravity vector). Let

$$\mathbf{g}(s)$$

denote the local gravity vector at $\gamma(s)$.

Remark 14.12. In a simplified model, one may approximate

$$\mathbf{g}(s) \parallel -\mathbf{n}_{\mathcal{E}}(s).$$

Remark 14.13. A refined model distinguishes between ellipsoid normal, geoid normal, and local gravity direction.

14.5. Curvature and Lateral Acceleration

Definition 14.14 (Railway curvature on the ellipsoid). The railway-relevant curvature is the geodesic curvature

$$\kappa_{\text{rail}}(s) = \kappa_g(s).$$

Remark 14.15. This curvature measures direction change within the reference surface and replaces the planar relation $\kappa = \theta'$.

Definition 14.16 (Surface lateral acceleration). For speed v , the curvature-induced lateral acceleration is

$$a_{\kappa}(s) = v^2 \kappa_g(s).$$

14.6. Effective Acceleration with Cant

Remark 14.17. The effective lateral acceleration is obtained by projecting both curvature-induced acceleration and gravity into the canted lateral direction.

Definition 14.18 (Effective lateral acceleration). A geometrically consistent formulation is

$$a_{\text{eff}}(s) = v^2 \kappa_g(s) - \mathbf{g}(s) \cdot \mathbf{b}_\phi(s).$$

Remark 14.19. In the planar approximation with

$$\mathbf{g} \parallel -\mathbf{n}_\mathcal{E},$$

this reduces to the familiar expression

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

14.7. Pose3 State on the Ellipsoid

Definition 14.20 (Ellipsoidal pose3). The pose3 state on the ellipsoid is

$$\text{pose3}_\mathcal{E}(s) = (\gamma(s), \mathbf{t}(s), \mathbf{n}_\mathcal{E}(s), \kappa_g(s), \phi(s)).$$

Remark 14.21. Compared to planar pose3, the heading angle $\theta(s)$ is replaced by a tangent vector field on the surface.

14.8. Implication for AIM

Summary 14.22. Pose3 on the ellipsoid replaces planar heading-based geometry by a surface-based moving frame.

Cant becomes a rotation of the track cross-section around the tangent vector, while curvature is represented by geodesic curvature.

This provides a physically more consistent foundation for evaluating railway dynamics, especially where projection distortions become relevant.

Teil III.

Geometry and Curvature

Geometry

What do we already know?

Remark 14.23. The previous part established the mathematical foundation of AIM.

In particular, it introduced the distinction between intrinsic and derived quantities, the importance of arc-length based descriptions, and the operator-based viewpoint adopted throughout this work.

Remark 14.24. However, a railway alignment has not yet been given a geometric form.

At this stage, the mathematical framework exists, but the geometric entity that it is intended to describe remains undefined.

What is still missing?

Remark 14.25. Traditional railway engineering typically describes alignments through coordinates, radii, tangents, circular arcs, and transition elements.

Remark 14.26. While such descriptions are operationally useful, they do not reveal the intrinsic geometric structure of the alignment itself.

Remark 14.27. AIM therefore seeks a representation that is independent of particular coordinate systems, storage formats, and implementation conventions.

Remark 14.28. The missing ingredient is an intrinsic geometric description from which all geometric realizations can be reconstructed.

What comes next?

Remark 14.29. Within AIM, geometry is interpreted not primarily as a collection of geometric primitives, but as controlled curvature evolution along arc-length.

Remark 14.30. The central geometric quantity therefore becomes

$$\kappa(s),$$

rather than position itself.

Remark 14.31. From this viewpoint:

- geometry emerges through curvature integration,

- transition curves become curvature-transition laws,
- sparse alignment models become compact curvature programs,
- and geometric reconstruction becomes a natural consequence of the intrinsic description.

Remark 14.32. This interpretation establishes the foundation for realization, runtime evaluation, dynamics, and optimization developed in later parts of this thesis.

Relation to the AIM Roadmap

Remark 14.33. Within the AIM roadmap, the present part corresponds to the transition from mathematical foundations toward curvature space.

Remark 14.34. It introduces the intrinsic geometric language from which the later state descriptions, runtime models, realization concepts, and optimization methods are derived.

Proposition 14.35. *Within AIM, railway geometry is fundamentally interpreted as curvature evolution along intrinsic arc-length.*

15. Curvature Space

Motivation

Remark 15.1. Classical railway alignment theory is usually formulated in position space.

Remark 15.2. Within AIM, the primary geometric object is not the position curve itself, but the curvature evolution along arc-length.

Remark 15.3. This shifts the interpretation of railway geometry from coordinate-based description toward intrinsic curvature structure.

Remark 15.4. The concept of curvature space provides the mathematical setting for:

- transition curves,
- sparse reconstruction,
- realization analysis,
- runtime evaluation,
- and alignment optimization.

Principle 15.5 (Curvature-space principle). Within AIM, railway alignment geometry is interpreted primarily as a problem in curvature space.

15.1. Curvature as Primary Geometry

Definition 15.6 (Curvature law). A curvature law is a function:

$$\kappa : I \rightarrow \mathbb{R},$$

defined on an arc-length interval $I \subset \mathbb{R}$.

Remark 15.7. Within AIM, curvature is interpreted as the primary intrinsic geometric quantity.

Remark 15.8. Geometry is reconstructed from curvature through integration.

Principle 15.9 (Curvature-first principle). Within AIM, geometry is generated from curvature rather than curvature being derived from geometry.

15.2. Curvature Space

Definition 15.10 (Curvature space). Curvature space denotes the space of admissible curvature laws:

$$\mathcal{K} = \{\kappa : I \rightarrow \mathbb{R}\}.$$

Remark 15.11. The symbol $\mathcal{K}$ does not denote one fixed function class. It denotes the conceptual space in which different admissibility, smoothness, realization, and optimization requirements may be imposed.

Remark 15.12. Different subclasses of curvature space correspond to different geometric and operational properties.

Remark 15.13. Examples include:

- constant curvature,
- piecewise constant curvature,
- continuous curvature,
- differentiable curvature,
- bounded curvature,
- transition curvature,
- and hybrid curvature laws.

OPEN: Later extensions of curvature space include admissible curvature laws, curvature spectra, realization-stable curvature, curvature energy, curvature bandwidth, transition-family subspaces, Vienna-type reinterpretations, and operator deformations.

15.3. Intrinsic Reconstruction

Definition 15.14 (Intrinsic reconstruction). Given a curvature law:

$$\kappa(s),$$

the tangent direction satisfies:

$$\theta'(s) = \kappa(s),$$

and geometry is reconstructed through:

$$\gamma'(s) = \mathbf{t}(s).$$

Remark 15.15. Curvature therefore acts as the differential generator of geometry.

Remark 15.16. The position curve γ is not primary. It is the result of integrating the intrinsic curvature state together with initial pose data.

15.4. Constant Curvature

Definition 15.17 (Constant curvature). A constant curvature law satisfies:

$$\kappa(s) = \kappa_0.$$

Remark 15.18. Special cases are:

- $\kappa_0 = 0$: straight line,
- $\kappa_0 \neq 0$: circular arc.

Remark 15.19. Constant-curvature elements form the fixed parts of the sparse alignment model.

15.5. Transition Curvature

Definition 15.20 (Transition curvature). A transition curvature law connects different curvature states continuously or smoothly.

Remark 15.21. Transition curves are therefore interpreted as curvature-transition processes rather than merely geometric interpolation curves.

Remark 15.22. Typical transition quantities include:

$$\kappa'(s), \quad \kappa''(s), \quad \kappa'''(s).$$

Remark 15.23. In AIM, transition families are interpreted as structured subspaces or parameterized paths within curvature space.

15.6. Curvature Continuity

Definition 15.24 (Curvature continuity). A curvature law is:

- C^0 -continuous if $\kappa(s)$ is continuous,
- C^1 -continuous if $\kappa'(s)$ is continuous,
- C^2 -continuous if $\kappa''(s)$ is continuous.

Remark 15.25. Higher continuity improves:

- dynamic smoothness,
- realization stability,
- numerical robustness,
- and passenger comfort.

Remark 15.26. Continuity of position is not sufficient for high-quality railway geometry. The decisive structure lies in the continuity and behaviour of curvature and its derivatives.

15.7. Sparse Curvature Representation

Definition 15.27 (Sparse curvature representation). A sparse representation describes curvature using:

- fixed curvature elements,
- transition elements,
- and compact parameter sets.

Remark 15.28. Within AIM, sparse alignments are interpreted as compact curvature programs.

Remark 15.29. The sparse model stores:

- curvature structure,
- transition families,
- and reconstruction rules,

rather than dense geometric point clouds.

Remark 15.30. This makes sparse alignment suitable for:

- reconstruction,
- import normalization,
- runtime evaluation,
- realization comparison,
- and optimization.

15.8. Hybrid Curvature Representation

Definition 15.31 (Hybrid curvature model). A hybrid curvature model combines:

- parametric curvature structure,
- sampled correction data,
- and local refinement.

Remark 15.32. Hybrid representations allow:

- compact storage,
- efficient reconstruction,

15. Curvature Space

- realization-aware correction,
- measured-state integration,
- and scalable runtime evaluation.

Remark 15.33. Hybrid curvature models are especially relevant when measured or realized geometry cannot be represented adequately by a purely analytical target model.

15.9. Normalized Curvature Space

Definition 15.34 (Normalized curvature law). A normalized curvature law is defined on:

$$w \in [0, 1].$$

Remark 15.35. Normalized representations separate:

- geometric shape,
- from physical scaling.

Remark 15.36. This enables:

- reusable transition families,
- lookup-based transitions,
- family interpolation,
- and normalized optimization.

Remark 15.37. Physical curvature laws are obtained from normalized laws by scaling in length and curvature amplitude.

15.10. Curvature Derivatives

Definition 15.38 (Curvature derivatives). Important higher-order quantities include:

$$\kappa'(s), \quad \kappa''(s), \quad \kappa'''(s).$$

Remark 15.39. These quantities influence:

- jerk,
- realization behaviour,
- dynamic smoothness,
- vehicle response,

- and transition quality.

Remark 15.40. Curvature derivatives provide the bridge from intrinsic geometry to runtime dynamics.

15.11. Frequency Interpretation

Remark 15.41. Curvature laws may also be interpreted spectrally.

Remark 15.42. Low-frequency curvature components determine global geometric structure, whereas high-frequency components influence:

- realization sensitivity,
- numerical instability,
- measurement sensitivity,
- and dynamic roughness.

Proposition 15.43 (Spectral realization principle). *Realization acts approximately as a low-pass filter on curvature space.*

Remark 15.44. This spectral view connects curvature space directly to construction, maintenance, and physical realization.

15.12. Curvature Space and Runtime

Remark 15.45. Runtime evaluation operates primarily on curvature structures rather than directly on dense geometry.

Remark 15.46. Projection, reconstruction, frame evaluation, pose3 evaluation, and dynamics are derived from runtime evaluation of curvature-based states.

Remark 15.47. Thus, curvature space is not only a design space. It is also the computational state space underlying runtime reconstruction.

15.13. Curvature Space and Realization

Remark 15.48. Realization transforms target curvature into physically realized curvature.

Remark 15.49. This may be written abstractly as:

$$\kappa_{\text{real}} = \mathcal{R}_{\kappa}(\kappa_{\text{target}}).$$

Remark 15.50. The realization operator may attenuate, deform, or locally modify curvature evolution.

Remark 15.51. Consequently, the physically relevant design object is not only the target curvature law, but also its expected realized curvature state.

15.14. Curvature Space and Optimization

Remark 15.52. Optimization within AIM primarily acts in curvature space.

Remark 15.53. Typical optimization variables include:

- transition parameters,
- curvature amplitudes,
- transition lengths,
- curvature anchors,
- and sparse curvature corrections.

Remark 15.54. AIM optimization may therefore be interpreted as the search for admissible, realizable, and operationally suitable paths in curvature space.

15.15. Canonical Interpretation

Proposition 15.55. *Within AIM, railway alignment geometry is fundamentally a curvature-space problem.*

Proposition 15.56. *Position-space geometry is interpreted as reconstructed geometry derived from curvature evolution.*

Proposition 15.57. *Transition curves are interpreted as controlled transitions within curvature space.*

Proposition 15.58. *Realization, runtime evaluation, and optimization act on curvature-space structures before they act on visualized position geometry.*

15.16. Connection to AIM

Summary 15.59. Curvature space forms the intrinsic geometric foundation of AIM.

Within this framework:

- curvature acts as primary geometry,
- sparse models become curvature programs,
- transition curves become curvature transitions,
- runtime systems operate on intrinsic curvature structure,
- realization transforms curvature states,
- and optimization searches within curvature-space structures.

This interpretation unifies geometry, realization, runtime evaluation, measurement interpretation, and optimization within one intrinsic mathematical framework.

16. Curvature Classes

Motivation

Remark 16.1. Not all curvature laws possess the same geometric, dynamic, numerical, or physical properties.

Remark 16.2. Within AIM, curvature laws are therefore classified according to:

- continuity,
- differentiability,
- boundedness,
- transition behaviour,
- realization behaviour,
- runtime stability,
- and operational suitability.

Remark 16.3. This classification forms the basis for:

- transition evaluation,
- runtime robustness,
- realization analysis,
- admissibility checks,
- and optimization strategies.

Principle 16.4 (Curvature classification principle). Within AIM, railway alignment elements are classified primarily by their curvature behaviour rather than by their visual position-space shape.

16.1. Basic Curvature Classes

16.1.1. Constant Curvature

Definition 16.5 (Constant curvature). A curvature law is constant if:

$$\kappa(s) = \kappa_0.$$

Remark 16.6. Special cases are:

- $\kappa_0 = 0$: straight line,
- $\kappa_0 \neq 0$: circular arc.

Remark 16.7. Constant curvature forms the fixed-element basis of classical railway geometry.

16.1.2. Piecewise Constant Curvature

Definition 16.8 (Piecewise constant curvature). A curvature law is piecewise constant if:

$$\kappa(s) = \kappa_i$$

on subintervals I_i .

Remark 16.9. This corresponds to classical tangent–arc railway geometry.

Remark 16.10. Piecewise constant curvature is geometrically simple, but curvature jumps are dynamically and operationally problematic.

16.2. Continuity Classes

16.2.1. C^0 Curvature

Definition 16.11 (C^0 -continuous curvature). A curvature law belongs to C^0 if:

$$\kappa(s)$$

is continuous.

Remark 16.12. Continuous curvature eliminates curvature jumps.

16.2.2. C^1 Curvature

Definition 16.13 (C^1 -continuous curvature). A curvature law belongs to C^1 if:

$$\kappa'(s)$$

exists and is continuous.

Remark 16.14. This improves dynamic smoothness, realization behaviour, and numerical robustness.

16.2.3. C^2 Curvature

Definition 16.15 (C^2 -continuous curvature). A curvature law belongs to C^2 if:

$$\kappa''(s)$$

exists and is continuous.

Remark 16.16. Higher continuity reduces excitation effects, numerical instability, and unwanted high-frequency behaviour.

Remark 16.17. In AIM, continuity class is not merely a mathematical property. It is an indicator for dynamic quality, runtime stability, and realization robustness.

16.3. Boundedness Classes

16.3.1. Bounded Curvature

Definition 16.18 (Bounded curvature). A curvature law is bounded if:

$$|\kappa(s)| \leq \kappa_{\max}.$$

Remark 16.19. Bounded curvature corresponds to a minimum admissible radius.

16.3.2. Bounded Curvature Gradient

Definition 16.20 (Bounded curvature gradient). A curvature law has bounded curvature gradient if:

$$|\kappa'(s)| \leq \eta_{\max}.$$

Remark 16.21. This is closely related to jerk limitations in railway dynamics.

16.3.3. Bounded Higher Derivatives

Definition 16.22 (Bounded higher derivatives). Higher-order boundedness conditions include:

$$|\kappa''(s)| < \infty, \quad |\kappa'''(s)| < \infty.$$

Remark 16.23. Higher-order boundedness becomes relevant for realization behaviour, vehicle response, and transition-family comparison.

16.4. Transition Classes

16.4.1. Monotone Transitions

Definition 16.24 (Monotone transition). A transition curvature law is monotone if:

$$\kappa'(s)$$

does not change sign.

Remark 16.25. Monotone transitions represent direct curvature change between two curvature states.

16.4.2. Symmetric Transitions

Definition 16.26 (Symmetric transition). A normalized transition law is symmetric if its curvature evolution is symmetric with respect to the midpoint of the transition interval.

Remark 16.27. For normalized transition coordinates $w \in [0, 1]$, this may be expressed by a symmetry relation of the transition shape function.

Remark 16.28. Symmetry is not an absolute requirement, but it provides a useful classification property.

16.4.3. Asymmetric Transitions

Definition 16.29 (Asymmetric transition). A transition law is asymmetric if symmetry is intentionally violated.

Remark 16.30. Asymmetric transitions may appear in:

- constrained engineering situations,
- realization-aware optimization,
- vehicle-dynamics optimization,
- and topology-constrained alignments.

16.5. Analytical Classes

16.5.1. Polynomial Curvature Laws

Definition 16.31 (Polynomial curvature). A polynomial curvature law satisfies:

$$\kappa(s) = \sum_{i=0}^n a_i s^i.$$

Remark 16.32. Polynomial curvature laws are useful because derivatives and integrals can often be handled systematically.

16.5.2. Trigonometric Curvature Laws

Definition 16.33 (Trigonometric curvature). A trigonometric curvature law contains sinusoidal or cosine-based components.

Remark 16.34. Trigonometric curvature laws are useful for smooth transition behaviour and frequency-oriented interpretation.

16.5.3. Lookup-Based Curvature Laws

Definition 16.35 (Lookup-based curvature). A lookup-based curvature law is defined by sampled or tabulated representations together with interpolation rules.

Remark 16.36. Lookup-based laws allow practical inclusion of transition families or realization effects that are not conveniently represented by a simple closed formula.

16.5.4. Hybrid Curvature Laws

Definition 16.37 (Hybrid curvature law). A hybrid curvature law combines:

- parametric structure,
- sampled corrections,
- measured data,
- and local refinement.

Remark 16.38. Hybrid laws are especially important for measured and realized railway states.

16.6. Sparse Curvature Classes

Definition 16.39 (Sparse curvature representation). A sparse curvature model represents geometry through:

- fixed curvature elements,
- transition elements,
- and compact parameter sets.

Remark 16.40. Sparse classes emphasize:

- compactness,
- runtime efficiency,
- reconstruction stability,

- and semantic clarity.

Remark 16.41. Within AIM, sparse curvature classes are not merely storage formats. They define executable curvature programs.

16.7. Operational Classes

16.7.1. Dynamically Smooth Curvature

Definition 16.42 (Dynamically smooth curvature). A curvature law is dynamically smooth if:

- jerk remains bounded,
- cant-rate remains admissible,
- curvature variation is controlled,
- and high-frequency excitation is limited.

16.7.2. Realization-Stable Curvature

Definition 16.43 (Realization-stable curvature). A curvature law is realization-stable if:

- realization preserves its dominant structure,
- construction processes do not introduce instability,
- and its low-frequency behaviour remains identifiable after physical realization.

Remark 16.44. Realization-stable curvature is a physical property, not merely an analytical smoothness property.

16.7.3. Runtime-Stable Curvature

Definition 16.45 (Runtime-stable curvature). A curvature law is runtime-stable if:

- projection remains numerically stable,
- frame evaluation remains continuous,
- pose reconstruction behaves robustly,
- and local inverse problems remain well-conditioned.

16.8. Frequency Classes

16.8.1. Low-Frequency Curvature

Definition 16.46 (Low-frequency curvature). Low-frequency curvature components determine large-scale geometric behaviour.

Remark 16.47. These components are usually preserved by construction and maintenance processes.

16.8.2. High-Frequency Curvature

Definition 16.48 (High-frequency curvature). High-frequency curvature components influence:

- realization sensitivity,
- dynamic roughness,
- measurement sensitivity,
- and numerical instability.

Proposition 16.49 (Spectral realization principle). *High-frequency curvature components are attenuated by realization.*

Remark 16.50. This frequency-oriented classification connects curvature classes with the realization filter interpretation.

16.9. Admissibility

Definition 16.51 (Admissible curvature law). A curvature law is admissible if it satisfies:

- geometric constraints,
- realization constraints,
- operational constraints,
- dynamic constraints,
- and numerical stability conditions.

Remark 16.52. Admissibility depends on:

- vehicle dynamics,
- realization processes,
- standards,

- topology,
- runtime requirements,
- and intended operational use.

Remark 16.53. Thus, admissibility is context-dependent. A curvature law may be analytically valid but operationally unsuitable or physically unstable under realization.

16.10. Classification versus Quality

Remark 16.54. Classification does not automatically imply quality.

Remark 16.55. A highly smooth curvature law may still be unsuitable if it is:

- difficult to realize,
- dynamically unfavourable,
- numerically unstable,
- incompatible with topology,
- or unsuitable for the intended operation.

Remark 16.56. Within AIM, curvature quality is evaluated by the interaction of class, context, realization, and runtime behaviour.

16.11. Canonical Interpretation

Proposition 16.57. *Different curvature classes correspond to fundamentally different geometric, operational, runtime, and realization behaviours.*

Proposition 16.58. *Within AIM, railway alignment quality is primarily characterized in curvature space rather than position space.*

Proposition 16.59. *Transition families are interpreted as structured subsets of curvature classes.*

Proposition 16.60. *Admissibility is not a purely geometric property, but a combined geometric, dynamic, runtime, and realization property.*

16.12. Connection to AIM

Summary 16.61. Curvature classes provide the structural classification framework of AIM geometry.

They connect:

16. Curvature Classes

- continuity,
- differentiability,
- boundedness,
- transition behaviour,
- realization behaviour,
- runtime stability,
- dynamic smoothness,
- and optimization suitability.

Within AIM, transition curves are therefore interpreted not merely as geometric entities, but as members of specific curvature classes with distinct operational properties.

This creates the bridge from intrinsic geometry to classification, realization analysis, runtime evaluation, and optimization.

17. Curvature Transitions

Motivation

Remark 17.1. Railway alignments rarely consist of isolated constant-curvature segments.

Remark 17.2. Instead, operational geometry requires smooth transitions between different curvature states.

Remark 17.3. Within AIM, transition curves are interpreted fundamentally as curvature-transition processes.

Remark 17.4. This shifts the interpretation:

- from position interpolation,
- to controlled curvature evolution.

Principle 17.5 (Curvature-transition principle). Within AIM, a transition curve is primarily a law of curvature evolution, not a coordinate-space interpolation object.

17.1. Transition Principle

Definition 17.6 (Curvature transition). A curvature transition is a curvature law:

$$\kappa(s)$$

connecting two curvature states:

$$\kappa_A \rightarrow \kappa_B.$$

Remark 17.7. The transition determines:

- geometric smoothness,
- dynamic behaviour,
- realization stability,
- runtime robustness,
- and operational comfort.

Principle 17.8 (Transition principle). Within AIM, transition geometry is governed primarily by curvature evolution rather than by coordinate interpolation.

17.2. Normalized Transition Space

Definition 17.9 (Normalized transition parameter). Transitions are commonly normalized to:

$$w \in [0, 1].$$

Definition 17.10 (Normalized curvature law). A normalized transition law satisfies:

$$\hat{\kappa}(w) \in [0, 1].$$

Remark 17.11. Normalization separates:

- intrinsic transition shape,
- from physical scaling.

Remark 17.12. This enables:

- reusable transition families,
- family interpolation,
- lookup-based evaluation,
- normalized optimization,
- and compact sparse representation.

Remark 17.13. A physical transition is obtained by scaling normalized curvature and normalized length to the actual curvature interval and physical arc-length.

17.3. Transition Families

Definition 17.14 (Transition family). A transition family is a class of normalized curvature laws sharing common structural properties.

Remark 17.15. Examples include:

- clothoids,
- polynomial transitions,
- trigonometric transitions,
- Bloss transitions,
- Vienna transitions,
- lookup-based transitions,
- and hybrid transition families.

Remark 17.16. Within AIM, transition families are interpreted as structured regions or parameterized paths within normalized curvature space.

17.4. Clothoid Transition

Definition 17.17 (Clothoid). The clothoid is defined by linear curvature evolution:

$$\kappa(s) = as + b.$$

Remark 17.18. In normalized form, the clothoid corresponds to:

$$\hat{\kappa}(w) = w.$$

Remark 17.19. The clothoid provides constant curvature gradient.

Remark 17.20. Historically, the clothoid became dominant because:

- it is simple,
- analytically tractable,
- compatible with classical railway practice,
- and dynamically smoother than curvature jumps.

Remark 17.21. Within AIM, the clothoid is not treated as the universal transition truth, but as one important member of curvature-transition space.

17.5. Polynomial Transitions

Definition 17.22 (Polynomial transition). A polynomial transition satisfies:

$$\kappa(s) = \sum_{i=0}^n a_i s^i.$$

Remark 17.23. Polynomial transitions allow:

- higher continuity,
- endpoint constraints,
- derivative constraints,
- and optimized dynamic behaviour.

Remark 17.24. Polynomial transitions are especially useful when smoothness conditions are imposed directly on curvature derivatives.

17.6. Trigonometric Transitions

Definition 17.25 (Trigonometric transition). A trigonometric transition contains sinusoidal or cosine-based curvature components.

Remark 17.26. These transitions often provide smoother higher-order behaviour than low-degree polynomial transitions.

Remark 17.27. Trigonometric transitions are naturally connected to spectral interpretation of curvature space.

17.7. Half-Wave Interpretation

Definition 17.28 (Half-wave transition). A half-wave transition is a normalized curvature segment connecting:

$$0 \rightarrow 1$$

or

$$1 \rightarrow 0.$$

Remark 17.29. Within AIM, many transition families are interpreted compositionally as:

$$\text{halfWave}_1 \rightarrow \text{core} \rightarrow \text{halfWave}_2.$$

Remark 17.30. This enables:

- modular transition construction,
- asymmetric transitions,
- family interpolation,
- and transition deformation.

Remark 17.31. The half-wave interpretation provides a common language for comparing classical transition families and newly constructed transition laws.

17.8. Curvature Anchors

Definition 17.32 (Curvature anchors). Normalized transition families may define anchor values:

$$0, \quad c_1, \quad c_2, \quad 1.$$

Remark 17.33. Anchors describe:

- transition partitioning,
- continuity conditions,

- normalization structure,
- and family composition.

Remark 17.34. Curvature anchors make it possible to compare transition families by their internal curvature structure rather than by their external position-space appearance.

17.9. Transition Continuity

Definition 17.35 (Transition continuity). Transition quality depends on continuity of:

$$\kappa(s), \quad \kappa'(s), \quad \kappa''(s).$$

Remark 17.36. Higher continuity improves:

- jerk behaviour,
- realization smoothness,
- runtime stability,
- vehicle response,
- and passenger comfort.

Remark 17.37. Continuity in position space alone is insufficient for evaluating transition quality.

17.10. Sparse Transition Representation

Definition 17.38 (Sparse transition). A sparse transition representation stores:

- transition family identity,
- normalization parameters,
- curvature anchors,
- scaling parameters,
- and reconstruction rules.

Remark 17.39. Within AIM, sparse transitions are interpreted as compact curvature operators rather than dense geometric curves.

Remark 17.40. A sparse transition therefore stores how curvature shall evolve, not a sampled polyline approximation of the resulting geometry.

17.11. Lookup-Based Transitions

Definition 17.41 (Lookup-based transition). A lookup-based transition uses:

- sampled normalized curvature,
- interpolation rules,
- derivative reconstruction,
- and runtime evaluation.

Remark 17.42. This enables:

- arbitrary transition families,
- experimentally designed transitions,
- realization-adapted transitions,
- vehicle-specific transitions,
- and runtime-efficient evaluation.

Remark 17.43. Lookup-based transitions are not a fallback for missing theory. They are a practical representation of transition laws in normalized curvature space.

17.12. Transition Derivatives

Definition 17.44 (Transition derivatives). Important quantities include:

$$\kappa'(s), \quad \kappa''(s), \quad \kappa'''(s).$$

Remark 17.45. These derivatives influence:

- jerk,
- dynamic smoothness,
- realization behaviour,
- vehicle response,
- and optimization stability.

Remark 17.46. Derivative behaviour is one of the main differences between transition families that may otherwise appear similar in position space.

17.13. Transition Symmetry

Definition 17.47 (Symmetric transition). A normalized transition is symmetric if its curvature evolution is symmetric with respect to the midpoint of the transition interval.

Definition 17.48 (Asymmetric transition). A transition is asymmetric if this symmetry is intentionally violated.

Remark 17.49. Asymmetric transitions may improve:

- realization quality,
- vehicle dynamics,
- topology fitting,
- or geometric constraint satisfaction.

Remark 17.50. Symmetry is therefore a classification property, not a universal requirement.

17.14. Transition Frequency Behaviour

Remark 17.51. Transitions may be interpreted spectrally through their curvature content.

Remark 17.52. High-frequency curvature behaviour influences:

- realization sensitivity,
- machine smoothing,
- measurement sensitivity,
- and dynamic roughness.

Proposition 17.53 (Spectral transition principle). *Transition quality depends not only on curvature continuity, but also on spectral behaviour within curvature space.*

Remark 17.54. This connects transition theory directly to realization-stable curvature and curvature bandwidth.

17.15. Transition Realization

Remark 17.55. The transition that is designed is not necessarily the transition that is physically realized.

Remark 17.56. Realization may modify transition behaviour through:

- sampling,
- machine response,

- structural filtering,
- ballast mechanics,
- measurement feedback,
- and local correction limits.

Remark 17.57. Thus, transition quality must be evaluated both in target curvature space and in expected realized curvature space.

17.16. Transition Optimization

Remark 17.58. Within AIM, optimization acts primarily on:

- transition families,
- normalization structure,
- transition lengths,
- curvature anchors,
- derivative behaviour,
- and curvature evolution.

Remark 17.59. Thus, optimization is interpreted primarily as optimization in curvature-transition space rather than position space.

Remark 17.60. A realization-aware optimization may optimize not only the target transition law κ_{target} , but also the expected realized transition law:

$$\kappa_{\text{real}} = \mathcal{R}_{\kappa}(\kappa_{\text{target}}).$$

17.17. Canonical Interpretation

Proposition 17.61. *Transition curves are fundamentally curvature-transition laws.*

Proposition 17.62. *Within AIM, transition geometry is interpreted intrinsically within curvature space.*

Proposition 17.63. *Sparse transition models represent compact transition operators rather than explicit geometric curves.*

Proposition 17.64. *Transition quality depends on curvature evolution, derivative behaviour, spectral content, runtime stability, and realization behaviour.*

17.18. Connection to AIM

Summary 17.65. Curvature transitions form the dynamic geometric core of AIM.

Within this framework:

- transitions are interpreted as curvature evolution processes,
- sparse models become transition operators,
- normalized transition space enables reusable families,
- half-wave decomposition enables modular transition construction,
- lookup-based laws enable experimental and realization-aware transition design,
- and runtime systems reconstruct geometry dynamically from curvature structure.

This interpretation unifies transition theory, curvature classification, realization behaviour, runtime evaluation, and optimization within one intrinsic curvature framework.

18. Curvature Classification

Motivation

Remark 18.1. Classical railway alignment theory typically classifies transitions by their analytical formulas or historical engineering usage.

Remark 18.2. Within AIM, transition and alignment structures are classified intrinsically within curvature space.

Remark 18.3. This allows classification based on:

- continuity,
- curvature behaviour,
- dynamics,
- realization behaviour,
- spectral structure,
- runtime robustness,
- and operational suitability.

Remark 18.4. Curvature classification therefore becomes a structural interpretation framework rather than merely a catalogue of curve formulas.

Principle 18.5 (Classification as interpretation). Within AIM, classification is not a naming scheme for curve formulas. It is an interpretation layer for curvature behaviour.

18.1. Classification Principle

Principle 18.6 (Intrinsic classification principle). Within AIM, alignments are classified by intrinsic curvature behaviour rather than by reconstructed position geometry.

Remark 18.7. Two geometrically similar curves in position space may possess very different curvature behaviour.

Remark 18.8. Conversely, geometrically different realizations may belong to the same curvature class if their intrinsic curvature evolution behaves equivalently.

Remark 18.9. Classification is therefore performed in curvature space before it is interpreted in position space.

18.2. Continuity Classification

18.2.1. G^0 Geometry

Definition 18.10 (G^0 -continuous geometry). A geometry is G^0 -continuous if position is continuous.

18.2.2. G^1 Geometry

Definition 18.11 (G^1 -continuous geometry). A geometry is G^1 -continuous if tangent direction is continuous.

18.2.3. G^2 Geometry

Definition 18.12 (G^2 -continuous geometry). A geometry is G^2 -continuous if curvature is continuous.

Remark 18.13. Within railway alignment modelling, G^2 -continuity is often the minimal requirement for dynamically acceptable transitions.

18.2.4. G^3 Geometry

Definition 18.14 (G^3 -continuous geometry). A geometry is G^3 -continuous if curvature gradient is continuous.

Remark 18.15. Higher continuity classes improve:

- jerk behaviour,
- realization smoothness,
- runtime stability,
- and vehicle response.

Remark 18.16. Continuity classification bridges classical geometry notation with curvature-space interpretation.

18.3. Curvature Behaviour Classification

18.3.1. Constant Curvature Class

Definition 18.17 (Constant curvature class). A curvature law belongs to the constant class if:

$$\kappa(s) = \kappa_0.$$

18.3.2. Linear Curvature Class

Definition 18.18 (Linear curvature class). A curvature law belongs to the linear class if:

$$\kappa(s) = as + b.$$

Remark 18.19. The classical clothoid belongs to this class.

18.3.3. Polynomial Curvature Class

Definition 18.20 (Polynomial curvature class). A curvature law belongs to the polynomial class if:

$$\kappa(s) = \sum_i a_i s^i.$$

18.3.4. Trigonometric Curvature Class

Definition 18.21 (Trigonometric curvature class). A curvature law belongs to the trigonometric class if it contains sinusoidal or cosine-based components.

18.3.5. Lookup-Based Curvature Class

Definition 18.22 (Lookup-based curvature class). A curvature law belongs to the lookup-based class if it is defined by sampled or tabulated curvature values together with interpolation and reconstruction rules.

18.3.6. Hybrid Curvature Class

Definition 18.23 (Hybrid curvature class). A hybrid curvature law combines:

- parametric structure,
- lookup structure,
- measured data,
- and sampled corrections.

18.4. Symmetry Classification

18.4.1. Symmetric Transitions

Definition 18.24 (Symmetric transition). A normalized transition is symmetric if its curvature evolution is symmetric with respect to the midpoint of the normalized interval.

Remark 18.25. For transition comparison, symmetry should usually be evaluated in normalized transition space rather than in raw physical coordinates.

18.4.2. Antisymmetric Structure

Definition 18.26 (Antisymmetric transition). A transition has antisymmetric structure if its deviation from a reference or midpoint value changes sign under midpoint reflection.

Remark 18.27. Antisymmetry is especially useful when analysing centred transition shape functions or derivative behaviour.

18.4.3. Asymmetric Transitions

Definition 18.28 (Asymmetric transition). A transition is asymmetric if no symmetry condition is imposed.

Remark 18.29. Asymmetry may arise intentionally through:

- realization-aware design,
- vehicle-dynamics optimization,
- topology constraints,
- or local engineering constraints.

18.5. Derivative Classification

18.5.1. Bounded Gradient Class

Definition 18.30 (Bounded gradient class). A curvature law belongs to the bounded-gradient class if:

$$|\kappa'(s)| \leq \eta_{\max}.$$

18.5.2. Bounded Higher-Derivative Class

Definition 18.31 (Bounded higher-derivative class). A curvature law belongs to this class if:

$$|\kappa''(s)| < \infty, \quad |\kappa'''(s)| < \infty.$$

Remark 18.32. These classes influence:

- dynamic smoothness,
- numerical stability,
- realization robustness,
- and optimization conditioning.

18.6. Spectral Classification

18.6.1. Low-Frequency Structures

Definition 18.33 (Low-frequency curvature class). Low-frequency curvature structures are dominated by slowly varying curvature evolution.

Remark 18.34. Low-frequency structure usually determines the large-scale geometric shape of the alignment.

18.6.2. High-Frequency Structures

Definition 18.35 (High-frequency curvature class). High-frequency curvature structures contain rapidly varying or oscillating components.

Remark 18.36. High-frequency structures tend to:

- amplify realization sensitivity,
- increase dynamic roughness,
- increase measurement sensitivity,
- and reduce numerical stability.

Proposition 18.37 (Spectral realization principle). *Realization suppresses high-frequency curvature components.*

Remark 18.38. Spectral classification therefore links transition theory with physical realization behaviour.

18.7. Realization Classification

18.7.1. Realization-Stable Class

Definition 18.39 (Realization-stable class). A transition belongs to the realization-stable class if:

- dominant curvature structure is preserved under realization,
- low-frequency behaviour remains identifiable,
- and realization does not introduce instability.

18.7.2. Realization-Sensitive Class

Definition 18.40 (Realization-sensitive class). A transition belongs to the realization-sensitive class if small realization perturbations strongly alter curvature behaviour.

Remark 18.41. Realization-sensitive transitions may be mathematically valid but operationally undesirable.

18.8. Runtime Classification

18.8.1. Projection-Stable Class

Definition 18.42 (Projection-stable class). A curvature structure is projection-stable if:

- world-to-track projection remains numerically robust,
- local inverse problems remain well-conditioned,
- and ambiguity remains controllable.

18.8.2. Runtime-Smooth Class

Definition 18.43 (Runtime-smooth class). A curvature structure is runtime-smooth if:

- frame evaluation remains continuous,
- pose reconstruction remains robust,
- visualization remains stable,
- and runtime operators behave predictably.

18.9. Operational Classification

18.9.1. Comfort-Oriented Class

Definition 18.44 (Comfort-oriented class). A transition belongs to the comfort-oriented class if:

- jerk remains low,
- lateral acceleration evolves smoothly,
- and dynamic excitation is minimized.

18.9.2. Realization-Oriented Class

Definition 18.45 (Realization-oriented class). A transition belongs to the realization-oriented class if:

- construction robustness is prioritized,
- machine realization remains stable,
- maintenance sensitivity is reduced,
- and physical filtering preserves the intended behaviour.

18.9.3. Optimization-Oriented Class

Definition 18.46 (Optimization-oriented class). A transition belongs to the optimization-oriented class if:

- parameterization is numerically stable,
- gradients remain well-conditioned,
- constraints are expressible,
- and sparse representation is efficient.

18.10. Family Classification

Definition 18.47 (Transition family classification). Transition families may be classified according to:

- analytical form,
- continuity order,
- derivative behaviour,
- spectral behaviour,
- realization behaviour,
- symmetry,
- runtime stability,
- and operational suitability.

Remark 18.48. Thus, transition classification becomes multidimensional rather than formula-based.

Remark 18.49. A single transition family may be favourable in one classification axis and unfavourable in another.

18.11. Classification Vectors

Definition 18.50 (Classification vector). A transition may be assigned a classification vector:

$$\chi(T) = (\chi_{\text{cont}}, \chi_{\text{deriv}}, \chi_{\text{spec}}, \chi_{\text{real}}, \chi_{\text{run}}, \chi_{\text{op}}),$$

where the components describe continuity, derivative behaviour, spectral structure, realization behaviour, runtime stability, and operational suitability.

Remark 18.51. The classification vector is not intended as a rigid taxonomy. It is a structured way to compare transition families across multiple relevant dimensions.

18.12. Canonical Interpretation

Proposition 18.52. *Within AIM, curvature classification is fundamentally intrinsic rather than coordinate-based.*

Proposition 18.53. *Transition quality cannot be characterized sufficiently by position geometry alone.*

Proposition 18.54. *Realization, runtime stability, dynamics, and optimization require classification directly in curvature space.*

Proposition 18.55. *Transition taxonomy in AIM is multidimensional and context-dependent.*

18.13. Connection to AIM

Summary 18.56. Curvature classification provides the structural interpretation framework of AIM geometry.

Within this framework:

- transitions are classified intrinsically,
- formula names are secondary to curvature behaviour,
- realization behaviour becomes analyzable,
- runtime robustness becomes class-dependent,
- operational suitability becomes comparable,
- and optimization acts directly on curvature structures.

This establishes curvature space as the primary classification domain of railway alignment geometry and provides the basis for a taxonomy of transition families.

Teil IV.

Railway State Model

State

What do we already know?

Remark 18.57. The previous part established the intrinsic geometry of railway alignments.

Remark 18.58. Within AIM, geometry was shown to emerge from curvature evolution along arc-length, rather than from stored coordinates or predefined geometric primitives.

Remark 18.59. Curvature space therefore provides a complete intrinsic description of railway geometry.

Remark 18.60. However, geometry alone does not yet constitute a railway state.

What is still missing?

Remark 18.61. A geometric description specifies what the alignment is.

Remark 18.62. It does not yet specify how the alignment is represented, reconstructed, interpreted, or evaluated during operation.

Remark 18.63. Engineering applications require more than geometry alone.

Remark 18.64. Position, direction, profile, cant, spatial orientation, admissible motions, and runtime interpretation all require additional state variables.

Remark 18.65. A bridge therefore remains between intrinsic geometry and operational railway behaviour.

Remark 18.66. The missing element is a coherent state description.

What comes next?

Remark 18.67. This part introduces the railway state model underlying AIM.

Remark 18.68. The first state layer is *pose2*, the intrinsic reconstruction state generated from curvature.

Remark 18.69. It provides the initial-value structure required to reconstruct position and direction along the alignment.

Remark 18.70. The second state layer is *pose3*, the spatial railway state obtained through profile, cant, frame interpretation, and metric realization.

Remark 18.71. Importantly, *pose3* describes the railway itself.

Remark 18.72. Vehicle states are introduced only later as states evolving relative to pose3.

Remark 18.73. The chapters of this part establish the variables, frames, operators, and transformations required to connect intrinsic geometry with operational railway interpretation.

Relation to the AIM Roadmap

Remark 18.74. Within the AIM roadmap, the present part corresponds to the transition from curvature space toward railway states.

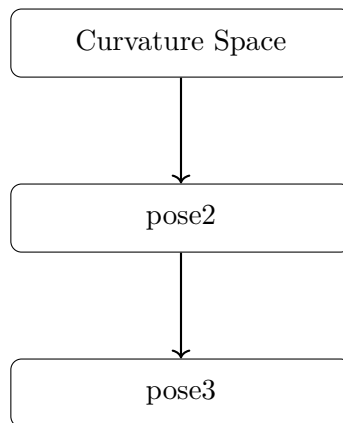


Abbildung 18.1.: State-related section of the AIM roadmap.

Remark 18.75. The progression from curvature space to pose2 and further to pose3 forms the central bridge between intrinsic geometry and all later runtime, realization, and optimization concepts.

Remark 18.76. The state model developed here serves as the foundation for vehicle space, runtime evaluation, realization-aware modelling, and optimization in subsequent parts of this thesis.

Principle 18.77 (Railway-State Principle). Within AIM, railway alignment is interpreted through state evolution rather than through coordinate geometry alone.

Summary 18.78. Geometry explains the structure of the railway.

Summary 18.79. The state model explains how this structure is represented, reconstructed, interpreted, and evaluated.

Summary 18.80. The transition from geometry to state therefore marks the transition from intrinsic shape to operational meaning.

19. Pose2

Motivation

Remark 19.1. In railway alignment modelling, the geometric state of the track at a position along its length must be represented in a minimal, stable, and intrinsic form.

Remark 19.2. While full spatial railway modelling involves cant, profile, realization, runtime services, and operational interpretation, many fundamental geometric properties can already be expressed within a planar intrinsic framework.

Remark 19.3. The concept of *pose2* provides this minimal intrinsic representation.

Remark 19.4. Within AIM, *pose2* is not merely a geometric helper structure. It forms the persistent intrinsic reconstruction kernel of the alignment model.

19.1. Definition of pose2

Definition 19.5 (*pose2*). A planar pose state at arc-length position s is defined as

$$\text{pose2}(s) = (\gamma(s), \theta(s)).$$

Remark 19.6. The *pose2* state describes:

- intrinsic planar position,
- and intrinsic tangent orientation.

Remark 19.7. The canonical *pose2* system is defined in eq. (7.6).

19.2. Curvature-Driven Reconstruction

Principle 19.8 (Curvature-driven reconstruction). Within AIM, planar geometry is reconstructed intrinsically from curvature evolution.

Remark 19.9. The intrinsic reconstruction chain follows directly from eqs. (7.3) to (7.5).

Remark 19.10. Thus, geometry is not treated as primary. Instead, position and orientation emerge through integration of curvature.

19. Pose2

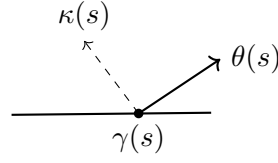


Abbildung 19.1.: Planar geometry emerges through intrinsic curvature-driven reconstruction. Within AIM, curvature is treated as the primary geometric quantity.

19.3. Initial Conditions

Definition 19.11 (Initial pose2 state). A pose2 trajectory is uniquely determined by:

$$\gamma(0), \quad \theta(0),$$

together with the curvature law

$$\kappa(s).$$

Remark 19.12. The curvature law acts as the geometric control function of planar state evolution.

19.4. Pose2 as Persistent Kernel

Proposition 19.13. *Within AIM, pose2 forms the persistent intrinsic reconstruction kernel of the alignment model.*

Remark 19.14. Pose2 is sufficient for persistent planar reconstruction of intrinsic alignment geometry.

Remark 19.15. Higher-level geometric and operational quantities are derived from pose2 rather than stored redundantly.

19.5. Connection to Sparse Alignment

Remark 19.16. Within the sparse alignment model, cGeom elements are interpreted as local curvature operators contributing to reconstruction of the global pose2 trajectory.

Remark 19.17. Fixed and transition elements therefore do not primarily store geometry itself, but rules governing geometric evolution.

Remark 19.18. The sparse alignment model can thus be interpreted as a compact integration program for reconstructing pose2 trajectories.

19.6. Intrinsic Interpretation

Remark 19.19. pose2 separates:

- intrinsic geometric behaviour,
- from derived geometric quantities.

Remark 19.20. Curvature acts as intrinsic control quantity, whereas position and orientation emerge through reconstruction.

Remark 19.21. This separation forms one of the central structural principles of the AIM framework.

19.7. Relation to pose3

Remark 19.22. Within AIM, pose2 forms the persistent intrinsic reconstruction kernel, whereas pose3 typically represents an extended runtime-evaluated spatial state.

Remark 19.23. pose3 is derived dynamically from:

- pose2 reconstruction,
- cant evaluation,
- profile evaluation,
- frame construction,
- and runtime coordinate services.

Remark 19.24. This avoids redundancy and synchronization instability between planar, cant, profile, and runtime representations.

19.8. Canonical Interpretation

Proposition 19.25. *Within AIM, intrinsic railway geometry is fundamentally curvature-driven.*

Proposition 19.26. *pose2 represents the canonical persistent intrinsic geometry state from which higher-level runtime states are derived.*

19.9. Connection to AIM

Summary 19.27. pose2 is the fundamental intrinsic planar state representation of railway alignment.

Within AIM, pose2 forms the persistent reconstruction kernel from which planar geometry is generated through curvature-driven integration.

Higher-dimensional runtime states such as pose3 are derived dynamically from this kernel together with cant, profile, frame, and runtime evaluation services.

20. Pose3, Cant, and Spatial Railway State

Motivation

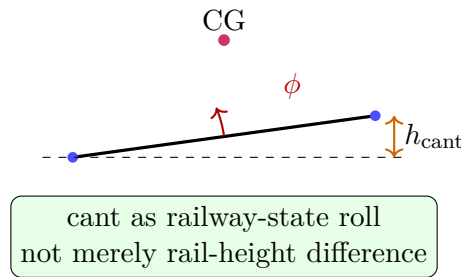


Abbildung 20.1.: Cant interpretation as railway-state roll rather than merely rail-height difference.

Remark 20.1. Planar railway alignment models describe intrinsic geometry using the pose2 state introduced in theorem 19.5.

Remark 20.2. Operational railway systems are spatial systems involving:

- profile geometry,
- cant,
- local frame orientation,
- vehicle-space interpretation,
- and dynamically relevant quantities.

Remark 20.3. This motivates an extended spatial railway state referred to as *pose3*.

Principle 20.4 (Runtime pose3 principle). Within AIM, pose3 is interpreted primarily as a runtime-evaluated spatial railway state rather than as a persistent geometric storage object.

Principle 20.5 (pose3 storage separation). pose3 is not persisted as independent geometry. The persistent model stores only the quantities from which pose3 can be reconstructed:

- intrinsic pose2 geometry,
- profile information,
- cant information,

- gauge and convention data,
- and metric/runtime context.

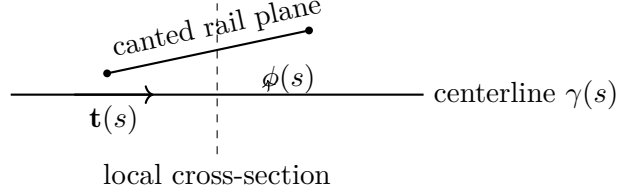


Abbildung 20.2.: pose3 extends planar intrinsic geometry by profile and cant-dependent spatial interpretation.

20.1. Relation to pose2

Remark 20.6. The canonical pose2 system is defined in eq. (7.6).

Remark 20.7. Within AIM, pose2 forms the persistent intrinsic reconstruction kernel, whereas pose3 extends this state by runtime-evaluated spatial and operational interpretation.

Remark 20.8. pose3 depends on:

- pose2 reconstruction,
- profile evaluation,
- cant evaluation,
- frame construction,
- runtime metric services,
- and runtime coordinate services.

Remark 20.9. Conceptually:

- pose2 answers: “where is the intrinsic alignment and how does it turn?”
- profile answers: “how does the alignment rise?”
- cant answers: “how is the rail plane rotated?”
- pose3 answers: “what operational spatial railway state results at runtime?”

20.2. Cant Representation

20.2.1. Cant as Persistent Engineering Quantity

Definition 20.10 (Cant height difference). Within AIM, persistent cant is represented as rail height difference:

$$u(s).$$

Remark 20.11. This follows standard railway engineering practice, where cant is typically specified as cross-level difference between rails.

20.2.2. Derived Roll Angle

Definition 20.12 (Derived roll angle). Given a gauge value g_{gauge} , the corresponding roll angle is:

$$\phi(s) = \arctan\left(\frac{u(s)}{g_{\text{gauge}}}\right). \quad (20.1)$$

Remark 20.13. The roll angle is a derived runtime quantity rather than a persistent primary variable.

Remark 20.14. Within AIM, cant is interpreted geometrically as frame rotation, independent of its engineering realization as rail height difference.

20.2.3. Cant Evolution

Remark 20.15. Cant evolution is governed by the canonical relation eq. (7.7).

Remark 20.16. Cant-rate influences:

- dynamic comfort,
- load transfer,
- vehicle response,
- and operational smoothness.

20.3. Cant Reference Conventions

Definition 20.17 (Cant reference convention). A cant convention specifies:

- the reference axis of rotation,
- the sign convention of cant,
- the relation between rail height difference and roll angle,
- and the induced centerline displacement.

Remark 20.18. Railway source data may define cant relative to:

- the centerline,
- one rail,
- a construction axis,
- or another engineering reference.

Remark 20.19. Within AIM, the intrinsic alignment centerline acts as canonical reference geometry.

Remark 20.20. Rail-based cant definitions must therefore be transformed consistently during import, normalization, or runtime evaluation.

20.4. Pose3 Frames

Definition 20.21 (Uncanted geometric frame). The uncanted geometric frame is denoted by:

$$F_0(s).$$

Remark 20.22. $F_0(s)$ is constructed from the reconstructed spatial alignment without cant rotation.

Definition 20.23 (Railway frame). The railway frame is denoted by:

$$F_{\text{rail}}(s).$$

Remark 20.24. $F_{\text{rail}}(s)$ is obtained by cant rotation of the geometric frame $F_0(s)$.

Definition 20.25 (Vehicle frame). A vehicle-dependent runtime frame may additionally be defined as:

$$F_{\text{veh}}(s).$$

Remark 20.26. Vehicle frames may depend on:

- suspension behaviour,
- center-of-gravity location,
- dynamic response,
- and operational conditions.

20.5. Runtime pose3 State

Definition 20.27 (Runtime pose3). The runtime spatial railway state is the station-dependent evaluated state:

$$\text{pose3}(s) = (\Gamma(s), u(s), F_{\text{rail}}(s)). \quad (20.2)$$

Remark 20.28. Here:

- $\Gamma(s)$ denotes the reconstructed spatial reference trajectory,
- $F_{\text{rail}}(s)$ denotes the cant-aware railway frame,
- $\phi(s)$ denotes the derived cant rotation.

Remark 20.29. The railway frame is evaluated by the frame operator $\mathcal{F}$ introduced in theorem 9.41.

Remark 20.30. Within projected engineering geometry, the spatial trajectory $\Gamma(s)$ is assembled dynamically from:

- planar pose2 reconstruction,
- profile evaluation,
- metric interpretation,
- and runtime coordinate services.

20.6. Persistent Model versus Runtime State

Remark 20.31. The persistent sparse model stores:

- intrinsic curvature geometry,
- cant bands,
- profile bands,
- and sparse reconstruction parameters.

Remark 20.32. Spatial pose3 states are reconstructed dynamically by runtime services.

Remark 20.33. This avoids:

- redundant spatial storage,
- synchronization instability,
- and unnecessary duplication of derived geometry.

Remark 20.34. The runtime interpretation may vary depending on:

- metric realization,
- CRS interpretation,
- level of detail,
- vehicle-space interpretation,
- and runtime evaluation strategy.

20.7. Reconstruction Principle

Proposition 20.35 (pose3 reconstruction principle). *Given:*

$$\kappa(s), \quad u(s), \quad h(s),$$

together with initial pose2 data, gauge information, convention data, and runtime metric context, the runtime pose3 state is determined through evaluation.

Remark 20.36. Thus, pose3 is reconstructed from synchronized runtime evaluation of:

- intrinsic geometry,
- profile,
- cant,
- frame services,
- metric services,
- and coordinate services.

20.8. Dynamic Quantities

20.8.1. Effective Lateral Acceleration

Remark 20.37. The effective lateral acceleration is defined canonically in eq. (7.10).

Remark 20.38. This quantity expresses the operational coupling between curvature and cant.

20.8.2. Jerk

Remark 20.39. The jerk relation is defined in eq. (7.12).

Remark 20.40. Jerk depends jointly on curvature evolution and cant evolution.

Remark 20.41. This is one of the primary reasons why curvature and cant cannot be treated independently in high-quality railway alignment design.

20.9. Cant and Curvature Coupling

Remark 20.42. Curvature and cant are operationally coupled quantities.

Remark 20.43. The equilibrium relation is defined in eq. (7.11).

Remark 20.44. Within AIM, curvature and cant remain persistently separated as distinct bands, whereas their operational interaction is evaluated dynamically.

20.10. Vehicle-Space Interpretation

Remark 20.45. pose3 enables interpretation of:

- vehicle envelopes,
- center-of-gravity motion,
- bogie orientation,
- wheelset behaviour,
- and platform interaction.

Remark 20.46. The physically relevant object is therefore not merely the track centerline, but the operational vehicle-space state generated from the alignment.

Remark 20.47. Vehicle-space interpretation may depend on runtime-generated vehicle frames $F_{\text{veh}}(s)$.

20.11. Towards Schwerpunkt-Trassierung

Remark 20.48. The pose3 framework allows reinterpretation of railway alignment as design of a spatial operational state trajectory.

Remark 20.49. Schwerpunkt-Trassierung does not optimize the rail centerline alone.

Instead, it optimizes vehicle-relevant operational trajectories induced by:

- curvature evolution,
- cant evolution,
- profile evolution,
- frame interpretation,
- and vehicle dynamics.

Remark 20.50. In this interpretation, the railway alignment acts as a generator of runtime vehicle-space states.

OPEN: Vehicle-specific runtime envelopes based on center-of-gravity-dependent cant interpretation.

RESEARCH: Formal definition of Schwerpunkt-based railway alignment optimization.

20.12. Relation to Standards

Remark 20.51. Existing railway standards typically prescribe limits on:

- curvature,
- cant,
- cant-rate,
- jerk,
- and operational acceleration.

Remark 20.52. Within AIM, such rules are interpreted as constraints acting on runtime pose3 states and their derivatives.

20.13. Canonical Interpretation

Proposition 20.53. *Within AIM, pose3 is interpreted as a runtime-evaluated operational spatial railway state.*

Proposition 20.54. *Persistent alignment storage and runtime spatial interpretation are treated as distinct conceptual layers.*

Proposition 20.55. *Curvature, profile, cant, frame evaluation, metric interpretation, and runtime services jointly generate operational railway geometry.*

Proposition 20.56. *The physically relevant railway object is not merely the geometric centerline, but the induced runtime vehicle-space state.*

20.14. Connection to AIM

Summary 20.57. pose3 extends intrinsic planar geometry into operational spatial railway state evaluation.

Within AIM, pose3 is reconstructed dynamically from:

- pose2 geometry,
- cant bands,
- profile bands,

20. Pose3, Cant, and Spatial Railway State

- frame evaluation,
- runtime services,
- metric interpretation,
- and coordinate context.

This provides:

- realization-aware spatial interpretation,
- dynamic operational geometry,
- runtime vehicle-space evaluation,
- and a foundation for advanced alignment optimization concepts.

Within AIM, pose3 is therefore not interpreted as a persistent geometry object, but as a dynamically evaluated operational railway state.

—

Proposition 20.58 (Track state principle). *pose3 describes the railway state.*

Vehicle states are derived through vehicle-space operators.

21. Pose3 Frames and Spatial Reference Systems

Motivation

Remark 21.1. The pose3 state describes railway geometry together with orientation, cant, and spatial interpretation.

Remark 21.2. The state variables alone do not fully determine how the alignment is interpreted in space. For this, local reference frames are required.

Remark 21.3. Within AIM, frames form the operational bridge between:

- intrinsic geometry,
- cant,
- dynamics,
- realization,
- world-track projection,
- and vehicle-space interpretation.

21.1. Pose3 and Frames

Remark 21.4. The canonical pose3 state is defined in eq. (7.1).

Remark 21.5. A pose3 frame provides the local spatial interpretation of this state. It is derived from the alignment and evaluated along arc-length.

21.2. Tangent Direction

Definition 21.6 (Tangent vector). The tangent vector is:

$$\mathbf{t}(s) = \gamma'(s).$$

Remark 21.7. For arc-length parameterization,

$$\|\mathbf{t}(s)\| = 1.$$

Remark 21.8. The tangent direction defines the local forward direction of motion.

21.3. Lateral and Binormal Directions

Definition 21.9 (Lateral direction). The lateral direction is denoted by:

$$\mathbf{n}(s),$$

with

$$\mathbf{n}(s) \cdot \mathbf{t}(s) = 0.$$

Definition 21.10 (Binormal direction). The binormal direction is:

$$\mathbf{b}(s) = \mathbf{t}(s) \times \mathbf{n}(s).$$

Remark 21.11. The lateral and binormal directions define the local cross-section orientation of the railway frame.

21.4. Pose3 Frame

Definition 21.12 (Pose3 frame). The local pose3 frame is:

$$F(s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s)).$$

Remark 21.13. The pose3 frame defines the local operational coordinate system of the railway alignment.

Remark 21.14. It is a runtime-evaluated structure rather than an independently stored geometric object.

21.5. Cant Rotation

Definition 21.15 (Cant rotation). Cant is interpreted as a rotation around the tangent direction:

$$R_\phi(s).$$

Definition 21.16 (Cant-rotated frame). The cant-rotated frame is:

$$F_\phi(s) = (\mathbf{t}(s), \mathbf{n}_\phi(s), \mathbf{b}_\phi(s)).$$

Remark 21.17. The cant-rotated frame determines:

- rail elevations,
- cross-level interpretation,

- vehicle orientation,
- effective gravity direction,
- and platform geometry.

21.6. Rail Plane

Remark 21.18. Gauge, rail elevation, and cross-level geometry are defined relative to the cant-rotated frame.

Remark 21.19. The rail plane is therefore not globally horizontal, but attached to the local pose3 frame and its cant rotation.

21.7. Frenet and Railway Frames

Remark 21.20. Classical Frenet frames are not fully adequate for railway alignment modelling.

Remark 21.21. In particular, Frenet frames become unstable in regions where

$$\kappa(s) \rightarrow 0.$$

Remark 21.22. Railway modelling therefore requires frame formulations that remain stable in straight and low-curvature sections.

Remark 21.23. In practice, railway frames are often transported continuously along the alignment rather than recomputed purely from differential geometry.

21.8. Surface and Ellipsoid Frames

Remark 21.24. When alignments are defined on the Earth surface, local frames must respect ellipsoid geometry.

Definition 21.25 (Surface-attached frame). A surface-attached frame consists of:

$$(\mathbf{t}, \mathbf{n}, \mathbf{n}_E),$$

where $\mathbf{n}_E$ denotes the ellipsoid surface normal.

Remark 21.26. In this setting:

- curvature splits into geodesic and normal curvature,
- gravity direction depends on Earth geometry,
- and world-to-track projection becomes surface-dependent.

21.9. Vehicle-Space Interpretation

Remark 21.27. The physically relevant object is not always the track centerline alone, but the motion of the railway vehicle within space.

Remark 21.28. Pose3 frames provide the spatial reference basis for:

- center-of-gravity motion,
- vehicle envelopes,
- bogie rotation,
- wheelset orientation,
- and platform clearance behaviour.

Remark 21.29. Thus, vehicle-space interpretation is derived from alignment states and their associated frames.

21.10. Connection to World–Track Transformation

Remark 21.30. World–track projection relies directly on the local frame.

Remark 21.31. In particular:

- tangent directions define longitudinal coordinates,
- lateral directions define offsets,
- and cant modifies the operational spatial interpretation.

Remark 21.32. This connects pose3 frames to the canonical world–track relation in eq. (7.14).

21.11. Frame Continuity

Remark 21.33. Frame continuity is essential for:

- smooth visualization,
- stable dynamics,
- cant interpretation,
- vehicle-space evaluation,
- and numerical robustness.

Remark 21.34. Discontinuous frame flips may lead to incorrect cant orientation, unstable vehicle interpretation, and invalid projection behaviour.

21.12. Key Insight

Proposition 21.35. *Pose3 frames are operational runtime structures rather than purely geometric differential objects.*

Remark 21.36. They connect intrinsic railway geometry with realization, dynamics, vehicle behaviour, and spatial interpretation.

21.13. Connection to AIM

Summary 21.37. Pose3 frames provide the spatial reference structure underlying railway alignment modelling.

Within AIM, they form the operational bridge between:

- curvature-based geometry,
- cant,
- realization,
- world-track projection,
- vehicle-space interpretation,
- and runtime system behaviour.

22. Dynamics and Railway State Interpretation

Motivation

Remark 22.1. Geometric alignment alone is not sufficient to describe railway behaviour. The motion of a vehicle depends on both curvature and cant, as well as on operational parameters such as speed.

Remark 22.2. This motivates a dynamic interpretation of alignment in which geometric quantities are translated into physical effects. Such a perspective is consistent with both railway design standards and modern railway vehicle dynamics literature, where curvature, cant, and vehicle response are treated as strongly coupled aspects of the system [6, 4, 16, 49].

22.1. Kinematic Basis

Definition 22.3 (Velocity). Let $v > 0$ denote the velocity of the vehicle along the alignment.

Definition 22.4 (Curvature-induced acceleration). For a planar curve with curvature $\kappa(s)$, the lateral acceleration is

$$a_\kappa(s) = v^2 \kappa(s).$$

Remark 22.5. This is the classical centripetal acceleration associated with curved motion and forms the basic link between geometry and dynamics.

22.2. Cant and Effective Acceleration

Definition 22.6 (Cant angle). Let $\phi(s)$ denote the local roll angle induced by cant.

Definition 22.7 (Gravity component). The gravitational acceleration projected onto the lateral direction is

$$a_g(s) = g \sin \phi(s),$$

where g denotes gravitational acceleration.

Definition 22.8 (Effective lateral acceleration). The effective lateral acceleration acting on the vehicle is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 22.9. This expression captures the fundamental coupling between curvature and cant in railway engineering, where cant is introduced precisely to compensate part of the curvature-induced lateral acceleration [24, 6].

Remark 22.10. From a vehicle-dynamics perspective, this coupling is one of the central mechanisms governing running behaviour in curves [16, 49].

22.3. Equilibrium Condition

Definition 22.11 (Equilibrium cant). A state of equilibrium is achieved when

$$v^2 \kappa(s) = g \sin \phi(s).$$

Remark 22.12. In this case, the effective lateral acceleration vanishes:

$$a_{\text{eff}}(s) = 0.$$

Remark 22.13. This condition corresponds to the classical notion of balanced track, where centrifugal and gravitational effects cancel. The same general idea underlies standard cant and cant-deficiency formulations in railway design [6, 24].

22.4. Cant Deficiency and Excess

Definition 22.14 (Cant deficiency). Cant deficiency is defined as

$$\Delta(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 22.15. Positive values correspond to insufficient cant, negative values to excess cant.

Remark 22.16. Cant deficiency is a central design parameter in railway engineering, as it directly influences passenger comfort and safety [6, 4].

Remark 22.17. In more detailed vehicle-based interpretations, cant deficiency is also closely related to guiding forces, wheel unloading, and curving behaviour [16, 32].

22.5. Dynamic Smoothness

Definition 22.18 (Jerk). The rate of change of effective acceleration is given by

$$j(s) = \frac{d}{dt} a_{\text{eff}}(s).$$

Remark 22.19. Using $dt = \frac{1}{v} ds$, one obtains

$$j(s) = v \frac{d}{ds} a_{\text{eff}}(s).$$

Proposition 22.20. *The jerk is given by*

$$j(s) = v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Remark 22.21. This shows that both curvature variation and cant variation contribute to dynamic effects. Such higher-order quantities are directly relevant for comfort-oriented transition design [34, 27, 28].

Remark 22.22. In railway vehicle dynamics, smoothness criteria of this kind are important because abrupt changes in curvature or cant excite the vehicle–track system beyond what is visible from geometry alone [25, 16].

22.6. Time Parameterization

Remark 22.23. The alignment is parameterized by arc length s , while dynamic quantities evolve in time t .

Definition 22.24 (Arc-length/time relation). Assuming constant speed $v > 0$, we have

$$\frac{ds}{dt} = v.$$

Proposition 22.25. *For any function $f(s)$, the time derivative is given by*

$$\frac{df}{dt} = v \frac{df}{ds}.$$

Remark 22.26. This relation allows all spatial derivatives to be translated into temporal dynamics and provides the mathematical bridge between arc-length-based alignment modelling and time-based vehicle response.

22.7. Dynamic System Formulation

Definition 22.27 (State variables). The railway state is described by

$$x(s), y(s), \theta(s), \phi(s).$$

Proposition 22.28 (Kinematic system). *The system satisfies*

$$\begin{aligned} \frac{dx}{ds} &= \cos \theta(s), & \frac{dy}{ds} &= \sin \theta(s), \\ \frac{d\theta}{ds} &= \kappa(s). \end{aligned}$$

Remark 22.29. These equations correspond to the geometric reconstruction of the curve.

Definition 22.30 (Extended dynamic state). The extended state includes cant:

$$(\gamma(s), \theta(s), \phi(s)).$$

Remark 22.31. The functions $\kappa(s)$ and $\phi(s)$ act as control variables.

Remark 22.32. This formulation is compatible with reduced-order vehicle–track interpretations in which track geometry provides structured inputs to a dynamic system rather than merely static shape data [16, 43].

22.8. Time-Based Dynamics

Proposition 22.33. *In time parameterization, the heading angle satisfies*

$$\frac{d\theta}{dt} = v\kappa(s).$$

Proposition 22.34. *The lateral acceleration is given by*

$$a_\kappa(t) = v^2\kappa(s).$$

Remark 22.35. This formulation makes explicit that a purely geometric quantity, namely curvature, acquires direct physical meaning once speed is introduced.

22.9. Higher-Order Dynamics

Definition 22.36 (Jerk). The jerk is defined as

$$j(t) = \frac{d}{dt}a_{\text{eff}}(t).$$

Proposition 22.37. *The jerk is given by*

$$j(t) = v^3\kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Beweis. Using

$$a_{\text{eff}} = v^2\kappa(s) - g \sin \phi(s),$$

we obtain

$$\frac{d}{dt}a_{\text{eff}} = v^2 \frac{d\kappa}{dt} - g \cos \phi(s) \frac{d\phi}{dt}.$$

With

$$\frac{d}{dt} = v \frac{d}{ds},$$

this yields

$$j(t) = v^3\kappa'(s) - gv \cos \phi(s) \phi'(s).$$

□

Remark 22.38. This explicit expression shows why transition design cannot be reduced to curvature alone if cant development is part of the system model.

22.10. Control-Theoretic Interpretation

Definition 22.39 (Control system). The railway alignment can be interpreted as a control system with

$$\text{state: } (\gamma, \theta, \phi), \quad \text{controls: } (\kappa, \phi').$$

Remark 22.40. The curvature $\kappa(s)$ controls the geometric evolution, while $\phi(s)$ controls the dynamic balancing of forces.

Remark 22.41. This formulation establishes a direct connection between alignment design and optimal control theory, and is consistent with the broader optimization-oriented interpretation of railway alignment proposed by Kufver [31, 30].

22.11. Coupled Dynamics

Remark 22.42. The effective acceleration depends on both curvature and cant:

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi.$$

Remark 22.43. Thus, curvature and cant must be treated as a coupled system rather than independent design variables.

Proposition 22.44. *Perfect equilibrium requires*

$$v^2 \kappa(s) = g \sin \phi(s).$$

Remark 22.45. This coupling is also central from the vehicle–track interaction point of view, since the dynamic response depends on the joint evolution of track geometry, cant, and operating speed [25, 32].

22.12. Coupled State Interpretation

Remark 22.46. The expressions above show that railway behaviour depends on the coupled state variables

$$(\kappa(s), \phi(s)).$$

Remark 22.47. A purely geometric interpretation based on $\kappa(s)$ alone is therefore insufficient for describing the physical behaviour of the system. This is one of the core reasons why a state-based railway model is preferable to a purely planar alignment description [4, 6, 16].

22.13. Extension to Pose3

Remark 22.48. Within the pose3 framework, the railway state is described by

$$(x(s), y(s), z(s), \theta(s), \phi(s)).$$

Remark 22.49. This representation naturally integrates:

- geometry (position and direction),
- vertical profile,
- cant (through ϕ),
- and dynamic interpretation.

Remark 22.50. The extension from planar pose to spatial railway state is therefore not a cosmetic modification, but a structural requirement for coupling alignment description to engineering dynamics.

22.14. Relation to Optimization

Remark 22.51. The dynamic quantities introduced above provide natural objective terms for alignment optimization.

Remark 22.52. Typical optimization criteria include:

- minimizing $\int a_{\text{eff}}(s)^2 ds$,
- minimizing $\int j(s)^2 ds$,
- limiting peak values of cant deficiency.

Such objective terms arise naturally once alignment is interpreted as a coupled geometric–dynamic system rather than as static geometry alone [30, 16].

22.15. Engineering Interpretation

Remark 22.53. In practical railway design, acceptable ranges for cant, cant deficiency, and jerk are prescribed by standards and operational guidelines.

Remark 22.54. The mathematical formulation above provides a unified framework for expressing these rules in analytic form [6, 24, 48].

Remark 22.55. At a more detailed level, such limits are closely related to the running–safety and vehicle–track compatibility questions discussed in railway dynamics literature [48, 16, 49].

22.16. Connection to Schwerpunkt-Trassierung

Remark 22.56. The combined dependence on curvature and cant supports a Schwerpunkt-oriented view of alignment.

Remark 22.57. Rather than describing only a geometric path, alignment is interpreted as a controlled dynamic state along the track.

Remark 22.58. This opens the way toward formulations in which vehicle-relevant derived motions, rather than geometry alone, become the true design object.

22.17. Summary

Summary 22.59. Railway alignment must be interpreted dynamically rather than purely geometrically.

The coupled variables $\kappa(s)$ and $\phi(s)$ determine lateral acceleration, comfort, and operational feasibility.

This provides the foundation for extending alignment modelling toward state-based and optimization-based frameworks.

23. Admissibility States

Motivation

Remark 23.1. Operational railway design is not governed by exact equality conditions alone, but by admissible ranges of runtime states.

Remark 23.2. Admissibility therefore belongs to state interpretation rather than only to engineering rule checking.

23.1. Admissibility State

Definition 23.3 (Admissibility state). An admissibility state is a runtime evaluation state describing whether operational quantities remain within allowed envelopes.

Remark 23.4. Admissibility states may include bounds on effective acceleration, jerk, cant deficiency, roll evolution, velocity, and realization quality.

23.2. Operational Envelopes

Definition 23.5 (Operational envelope state). An operational envelope state describes admissible regions in runtime state space.

23.3. Connection to AIM

Summary 23.6. Admissibility states connect runtime dynamics, engineering rules, realization-aware behaviour, and optimization constraints.

24. World–Track Coordinate Transformation

Motivation

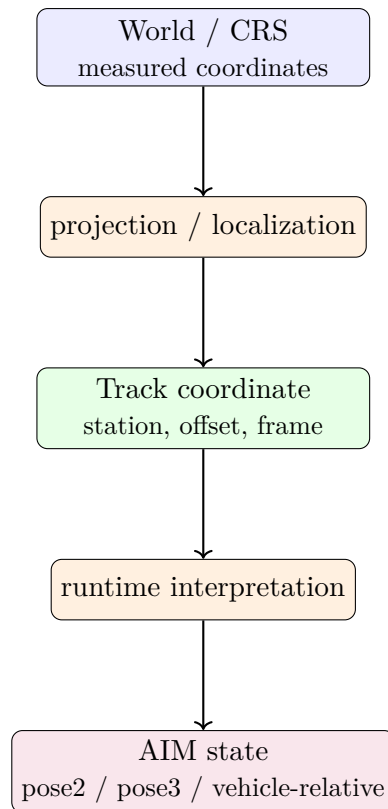


Abbildung 24.1.: World-to-track transformation as runtime interpretation from measured coordinates toward AIM states.

Remark 24.1. Railway alignments are formulated intrinsically using arc-length-based coordinates.

Remark 24.2. Real-world data is typically represented in external coordinate reference systems such as:

- WGS84,
- UTM,
- Gauss–Krüger,
- DBRef,

24. World–Track Coordinate Transformation

- or local engineering CRS definitions.

Remark 24.3. A transformation between intrinsic alignment coordinates and external world coordinates is therefore required.

Remark 24.4. Within AIM, this transformation is interpreted as a runtime evaluation service connecting:

- intrinsic alignment states,
- measured data,
- visualization,
- realization,
- topology,
- and operational geometry.

Principle 24.5 (Intrinsic-space primacy). Within AIM, intrinsic railway geometry is primary. World-space coordinates are interpreted as external embeddings generated through runtime evaluation.

24.1. Track Coordinate System

Definition 24.6 (Track space). The track space is the intrinsic railway coordinate domain:

$$\Omega_{\text{track}} = \{(s, d, h)\}.$$

It represents positions relative to an alignment rather than positions in an external coordinate reference system.

Definition 24.7 (Track coordinate system). The intrinsic track coordinate system consists of:

$$(s, d, h),$$

where:

- s denotes intrinsic arc-length,
- d denotes lateral offset,
- h denotes vertical offset.

Remark 24.8. Track coordinates belong to the intrinsic track-space domain Ω_{track} , introduced in theorem 24.6.

Remark 24.9. In planar situations, only:

$$(s, d)$$

is required.

Remark 24.10. Track coordinates are intrinsic alignment coordinates and must not be confused with projected world coordinates.

24.1.1. Intrinsic Geometry versus External Coordinates

Remark 24.11. Intrinsic geometry is independent of any external coordinate reference system.

Remark 24.12. The same intrinsic alignment may be embedded into:

- projected engineering planes,
- ellipsoidal world models,
- local realization frames,
- or visualization coordinate systems.

Remark 24.13. Thus, intrinsic alignment geometry and external CRS representation form distinct conceptual layers.

24.1.2. Arc-Length and Stationing

Remark 24.14. Intrinsic arc-length is not identical to engineering stationing.

Remark 24.15. Operational kilometre systems may contain:

- station equations,
- kilometre jumps,
- overlapping station ranges,
- topology-dependent references,
- and operational branch relations.

Remark 24.16. Stationing systems are operational reference structures layered on top of intrinsic geometry.

Remark 24.17. Consequently, the mapping:

$$s \leftrightarrow \text{km}$$

is generally neither linear nor globally unique.

24.2. Track-to-World Transformation

Definition 24.18 (Track-to-world mapping). The forward transformation is:

$$x(s, d) = \gamma(s) + d \mathbf{n}(s). \quad (24.1)$$

Remark 24.19. This relation specializes the canonical track-to-world operator $\mathcal{P}_{\text{tw}}$ introduced in theorem 9.22.

24.2.1. Planar Tangent and Normal Directions

Remark 24.20. In the planar case:

$$\mathbf{t}(s) = (\cos \theta(s), \sin \theta(s)),$$

and:

$$\mathbf{n}(s) = (-\sin \theta(s), \cos \theta(s)).$$

24.2.2. Spatial Runtime Evaluation

Remark 24.21. In spatial runtime evaluation, the mapping depends additionally on:

- profile,
- cant,
- frame construction,
- metric interpretation,
- and runtime realization context.

Remark 24.22. Consequently, the world embedding:

$$x(s, d, h)$$

is generally generated dynamically from runtime pose3 evaluation.

24.2.3. Reference-Line Interpretation

Remark 24.23. The lateral coordinate d depends on the chosen reference interpretation.

Remark 24.24. Possible reference structures include:

- alignment centerline,
- track center,
- rail axis,

24. World–Track Coordinate Transformation

- gauge reference,
- or operational offset definitions.

OPEN: Formalize reference-line conventions for:

- multi-track systems,
- cant-dependent offsets,
- and vehicle-space interpretation.

24.3. World-to-Track Transformation

Definition 24.25 (World-to-track projection). The inverse transformation is:

$$(s^*, d^*) = \arg \min_{s, d} \|x - (\gamma(s) + d \mathbf{n}(s))\|. \quad (24.2)$$

Remark 24.26. This relation specializes the canonical world-to-track operator $\mathcal{P}_{\text{wt}}$ introduced in theorem 9.24.

Remark 24.27. The term projection is used operationally and does not necessarily imply a unique orthogonal projection in the classical geometric sense.

Remark 24.28. Within AIM, world-to-track projection is interpreted as an inverse runtime evaluation problem.

24.4. Coordinate Transformation versus Projection

Remark 24.29. Coordinate transformation and world-to-track projection are distinct operations within AIM.

Remark 24.30. Coordinate transformations map between world-space representations:

$$\mathcal{C} : \Omega_{\text{world}} \rightarrow \Omega_{\text{world}},$$

as introduced in theorem 9.27.

Remark 24.31. World-to-track projection maps between:

$$\Omega_{\text{world}} \quad \text{and} \quad \Omega_{\text{track}}.$$

Principle 24.32 (Projection distinction principle). Coordinate transformation must not be confused with intrinsic world-to-track projection.

Remark 24.33. CRS transformations operate entirely within external coordinate representations, whereas projection accesses intrinsic railway geometry.

24.5. Numerical Evaluation

Remark 24.34. The inverse transformation is generally nonlinear.

Remark 24.35. Typical numerical approaches include:

- nearest-point search,
- local Newton iteration,
- segment-wise projection,
- topology-aware lookup,
- and hybrid runtime strategies.

Remark 24.36. Projection quality depends strongly on:

- curvature behaviour,
- topology,
- initialization,
- runtime context,
- and metric interpretation.

24.6. Ambiguity and Context Dependence

Remark 24.37. World-to-track mapping is generally not globally unique.

Remark 24.38. Ambiguities may occur in:

- loops,
- switches and crossings,
- parallel tracks,
- closely spaced alignments,
- station areas,
- and topology-rich railway networks.

Remark 24.39. Multiple intrinsic states may correspond to nearly identical world-space locations.

Remark 24.40. Consequently, projection is fundamentally context dependent.

Remark 24.41. Runtime projection may require:

- track identity,
- route identity,
- topology context,
- expected station range,
- operational direction,
- runtime interaction history,
- vehicle context,
- or network-state constraints.

Principle 24.42 (Projection ambiguity principle). World-to-track projection cannot generally be interpreted as a purely local geometric operation.

24.7. Topology and Operational References

Remark 24.43. Intrinsic geometry alone is insufficient for operational railway referencing.

Remark 24.44. Operational interpretation additionally depends on:

- topology,
- route membership,
- kilometre references,
- operational direction,
- and network semantics.

Remark 24.45. Thus, a complete projection context may require:

$$(x, \mathcal{T}, \mathcal{R}, \mathcal{O}),$$

where:

- x denotes world-space input,
- $\mathcal{T}$ denotes topology context,
- $\mathcal{R}$ denotes route/reference context,
- $\mathcal{O}$ denotes operational context.

24.8. Relation to Curvature

Remark 24.46. Projection stability depends directly on curvature behaviour.

Remark 24.47. High curvature increases projection sensitivity, whereas low curvature typically yields stable projection behaviour.

Remark 24.48. Errors in intrinsic coordinates propagate nonlinearly into world space.

Remark 24.49. Projection behaviour near transition regions may depend strongly on higher-order curvature behaviour.

24.9. Runtime Dependency

Remark 24.50. Within AIM, world–track transformation depends on runtime evaluation of the underlying alignment representation.

Remark 24.51. In particular:

- cGeom defines intrinsic geometry,
- pose2 defines tangent evolution,
- pose3 defines runtime spatial interpretation,
- metric services define realized embedding,
- CRS services define external representation,
- and runtime operators evaluate resulting states.

Remark 24.52. Projection therefore depends on the currently active runtime evaluation pipeline.

Remark 24.53. Different runtime configurations may yield different projection results, for example:

- projected versus ellipsoidal evaluation,
- target versus realized geometry,
- low-resolution versus high-resolution runtime,
- or simplified versus vehicle-aware interpretation.

Principle 24.54 (Runtime projection principle). World–track transformation is a runtime evaluation problem operating on alignment states rather than a static coordinate utility.

24.10. LOD and Hybrid Evaluation

Remark 24.55. Exact world-to-track projection is computationally expensive.

Remark 24.56. Large-scale systems therefore require:

- LOD-dependent evaluation,
- approximate projection,
- topology-aware filtering,
- and hybrid runtime architectures.

Proposition 24.57 (Selective projection principle). *Exact projection should only be evaluated where high precision is required.*

Remark 24.58. A hybrid strategy combines:

- precomputed samples for coarse lookup,
- topology-aware candidate filtering,
- cached runtime states,
- and local parametric refinement for precision.

Remark 24.59. Hybrid projection architectures are particularly important for:

- large railway networks,
- interactive visualization,
- realtime vehicle-space evaluation,
- and multi-resolution runtime systems.

24.11. Connection to CRS and Realization

Remark 24.60. The complete external transformation chain is:

$$\text{CRS} \rightarrow \Omega_{\text{world}} \rightarrow \Omega_{\text{track}}.$$

Remark 24.61. For high-precision applications, the alignment itself may be defined on a reference ellipsoid rather than in a projected plane.

Remark 24.62. Projection may also depend on realized geometry:

$$\gamma_{\text{real}}(s) \neq \gamma_{\text{target}}(s).$$

24. World–Track Coordinate Transformation

Remark 24.63. Physical measurements should therefore reference realized rather than purely analytical geometry.

Remark 24.64. The distinction between:

intrinsic geometry and external embedding

is fundamental within AIM.

24.12. Key Insight

Proposition 24.65. *World–track transformation is an inverse runtime service rather than a static coordinate conversion.*

Proposition 24.66. *Within AIM, intrinsic alignment space is primary, whereas world coordinates are interpreted as external embeddings of alignment states.*

Proposition 24.67. *Projection is generally:*

- *nonlinear,*
- *ambiguous,*
- *topology-dependent,*
- *runtime-dependent,*
- *and context-sensitive.*

24.13. Connection to AIM

Summary 24.68. World–track transformation connects intrinsic railway geometry with external coordinate systems, topology, runtime evaluation, and measured reality.

Its nonlinear, ambiguous, topology-dependent, and scale-dependent nature makes it a central runtime component of the AIM framework.

Within AIM, projection is therefore interpreted not as static coordinate conversion, but as contextual runtime reconstruction of intrinsic railway states from external observations.

Efficient handling consequently requires:

- hybrid runtime architectures,
- topology-aware projection strategies,
- LOD-dependent evaluation,
- sparse-state reconstruction,
- and metric-aware runtime services.

Teil V.

Dynamic Railway State

Dynamics

What do we already know?

Remark 24.69. The previous part introduced the railway state model underlying AIM.

Remark 24.70. In particular, the states `pose2` and `pose3` were established as intrinsic descriptions of railway geometry and spatial railway configuration.

Remark 24.71. These states provide a description of the railway itself.

However, they do not yet describe how railway systems evolve during operation.

What is still missing?

Remark 24.72. A state description specifies what exists at a given location along an alignment.

It does not yet explain how operational quantities emerge from that state.

Remark 24.73. Railway engineering ultimately concerns moving systems.

Vehicles travel through space, passengers experience acceleration, cant influences dynamic balance, and operational constraints evolve along trajectories.

Remark 24.74. The missing element is therefore not another geometric description, but a framework for state evolution and operational interpretation.

Remark 24.75. A distinction must consequently be made between railway states and the states generated from them during operation.

What comes next?

Remark 24.76. This part extends the railway state model toward operational runtime interpretation.

Remark 24.77. The central observation is that railway states and vehicle states are not identical.

Remark 24.78. Within AIM, `pose3` describes the railway itself.

Vehicle states, center-of-gravity states, balance states, comfort states, and optimization states are generated from railway states through runtime interpretation.

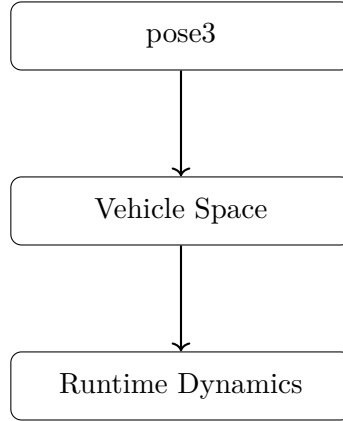
Remark 24.79. The chapters of this part move from state taxonomy and runtime operators toward vehicle space, runtime dynamics, Schwerpunkt interpretation, dynamic balance, Vienna reinterpretation, realization-aware dynamics, and optimization in state space.

Remark 24.80. This part does not introduce a full multibody vehicle model.

Instead, it establishes the structural framework into which such models may later be coupled.

Relation to the AIM Roadmap

Remark 24.81. Within the AIM roadmap, the present part corresponds to the transition from railway states toward operational state evolution.



Remark 24.82. The progression

$$\text{pose3} \rightarrow \text{Vehicle Space} \rightarrow \text{Runtime Dynamics}$$

marks the transition from describing the railway itself toward describing processes evolving relative to the railway.

Remark 24.83. This transition prepares the introduction of realization, optimization, and process-oriented information modelling developed in later parts of this thesis.

Principle 24.84 (Dynamic-State Interpretation Principle). Within AIM, dynamic railway behaviour is interpreted as operator-based runtime state evolution generated from railway states rather than as a direct property of coordinate geometry alone.

Summary 24.85. Geometry explains shape.

States explain configuration.

Dynamics explain evolution.

The present part establishes the operational layer that connects railway states with vehicle behaviour, runtime interpretation, and future optimization concepts.

25. Conceptual Boundaries in Dynamic State Modelling

25.1. Purpose

Remark 25.1. The dynamic chapters of this thesis are closely related. To avoid conceptual overlap, each chapter is assigned a precise responsibility.

Summary 25.2. Each chapter defines one conceptual layer, uses previously established layers, and must not redefine concepts that belong elsewhere.

25.2. Vehicle Space

Definition 25.3 (Responsibility of the vehicle-space chapter). The chapter on vehicle space defines the vehicle state as a state evolving relative to the railway state.

Remark 25.4. This chapter uses:

- pose3,
- railway frames,
- cant,
- profile,
- and the distinction between track state and vehicle state.

Remark 25.5. This chapter must not redefine:

- pose2,
- pose3,
- curvature space,
- runtime state,
- realization,
- or optimization state.

25.3. Runtime Dynamics

Definition 25.6 (Responsibility of the runtime-dynamics chapter). The chapter on runtime dynamics defines the evolution of operational states along or over the railway state.

Remark 25.7. This chapter uses:

- vehicle state,
- pose3,
- speed,
- curvature,
- cant,
- and runtime operators.

Remark 25.8. This chapter must not redefine:

- vehicle space,
- pose3,
- realization,
- dynamic balance,
- or optimization.

25.4. Dynamic Balance

Definition 25.9 (Responsibility of the dynamic-balance chapter). The chapter on dynamic balance defines equilibrium, cant deficiency, cant excess, and admissible balance relations between curvature, speed, and cant.

Remark 25.10. This chapter uses:

- runtime dynamics,
- vehicle-relative quantities,
- $\kappa(s)$,
- $v(s)$,
- $\phi(s)$,
- and admissibility concepts.

Remark 25.11. This chapter must not redefine:

- vehicle state,
- runtime state,
- pose3,
- realization,
- or optimization state.

25.5. Realization-Aware Dynamics

Definition 25.12 (Responsibility of the realization-aware-dynamics chapter). The chapter on realization-aware dynamics defines how realization operators affect dynamic interpretation.

Remark 25.13. This chapter uses:

- designed railway states,
- realized railway states,
- runtime dynamics,
- dynamic balance,
- measurement and construction effects,
- and realization operators.

Remark 25.14. This chapter must not redefine:

- realization itself,
- vehicle state,
- runtime dynamics,
- dynamic balance,
- sparse modelling,
- or optimization.

25.6. Boundary Principle

Principle 25.15 (Single-definition principle). Each conceptual object should be defined in exactly one chapter and used by later chapters through reference.

Summary 25.16. Vehicle space defines the vehicle-relative state.

Runtime dynamics defines state evolution.

Dynamic balance defines equilibrium and admissibility relations.

Realization-aware dynamics defines how realization modifies dynamic interpretation.

None of these chapters should redefine pose3, curvature space, sparse modelling, or optimization.

26. State Taxonomy

Motivation

Remark 26.1. Within AIM, the term *state* appears in multiple contexts:

- geometric reconstruction,
- runtime evaluation,
- realization,
- operational dynamics,
- vehicle interpretation,
- and optimization.

Remark 26.2. Without a consistent taxonomy, different state concepts may become ambiguous or unintentionally conflated.

Remark 26.3. This chapter therefore introduces a structural classification of state concepts within AIM.

Principle 26.4 (State-layer principle). Within AIM, state concepts belong to distinct interpretation layers and must not be identified implicitly.

26.1. General State Interpretation

Definition 26.5 (State). A state is a structured description of a system configuration relative to a specified interpretation layer.

Remark 26.6. The same physical railway system may therefore give rise to different states depending on whether it is interpreted geometrically, operationally, dynamically, through realization, or through vehicle interaction.

26.2. Intrinsic Geometric States

26.2.1. pose2

Definition 26.7 (pose2 state). The pose2 state is the intrinsic planar reconstruction state generated from:

$$\kappa(s).$$

Remark 26.8. pose2 represents:

- intrinsic geometry,
- planar tangent evolution,
- and reconstruction state.

Remark 26.9. pose2 is fundamentally an intrinsic geometric state.

26.2.2. pose3

Definition 26.10 (pose3 state). The pose3 state is the runtime-evaluated spatial railway state generated from:

- pose2 reconstruction,
- profile evaluation,
- cant interpretation,
- frame evaluation,
- and runtime coordinate interpretation.

Remark 26.11. pose3 describes the railway runtime state rather than the vehicle itself.

Remark 26.12. pose3 therefore belongs to the runtime railway-state layer.

26.3. Runtime States

Definition 26.13 (Runtime state). A runtime state is a dynamically evaluated state generated during runtime interpretation.

Remark 26.14. Runtime states depend on:

- runtime operators,
- realization,
- coordinate interpretation,
- frame construction,
- and operational context.

Remark 26.15. Examples include:

- pose3,
- runtime frames,
- projection states,
- and operational interpretation states.

26.4. Realized States

Definition 26.16 (Realized state). A realized state is a runtime state generated from physically realized rather than purely analytical geometry.

Remark 26.17. Realized states depend on:

- construction,
- measurement,
- maintenance,
- realization filtering,
- and operational degradation.

Remark 26.18. Typical examples include:

$$\text{pose3}_{\text{real}}, \quad \kappa_{\text{real}}, \quad \phi_{\text{real}}.$$

26.5. Operational States

Definition 26.19 (Operational state). An operational state is a runtime state relevant for railway operation, comfort, admissibility, or dynamic interpretation.

Remark 26.20. Operational states may include:

- effective acceleration,
- jerk,
- roll evolution,
- cant deficiency,
- excess cant,
- operational balance,
- and balance trajectories.

Remark 26.21. Operational states are derived from runtime railway states rather than stored directly as intrinsic geometry.

26.6. Vehicle States

Definition 26.22 (Vehicle state). A vehicle state is a runtime-relative operational state describing the vehicle relative to the railway runtime state.

Remark 26.23. Vehicle states depend on:

- railway runtime state,
- operational velocity,
- vehicle interpretation,
- and dynamic response.

Remark 26.24. Thus:

$$\text{vehicleState} = \mathcal{V}(\text{pose3}, v, \dots).$$

Remark 26.25. Vehicle states are therefore not intrinsic railway states.

26.7. Center-of-Gravity States

Definition 26.26 (Center-of-gravity state). A center-of-gravity state is an operational abstraction describing the runtime trajectory of the vehicle center of gravity relative to the railway runtime state.

Remark 26.27. Center-of-gravity states are primarily operational interpretation states rather than full mechanical vehicle models.

Remark 26.28. These states are particularly relevant for:

- comfort interpretation,
- balance interpretation,
- Schwerpunkt-Trassierung,
- and runtime operational optimization.

26.8. Dynamic States

Definition 26.29 (Dynamic state). A dynamic state is a runtime state participating in operational state evolution.

Remark 26.30. Dynamic states may evolve according to:

$$x'(s) = f(x, u).$$

Remark 26.31. Dynamic states may include:

- acceleration,
- jerk,
- roll evolution,
- and operational balance quantities.

26.9. Optimization States

Definition 26.32 (Optimization state). An optimization state is the state representation used as the object of an optimization problem.

Remark 26.33. Within AIM, optimization states are preferably runtime-state trajectories rather than isolated geometric parameters.

Remark 26.34. A typical optimization state may be written as:

$$x_{\text{opt}}(s) = x_{\text{real}}(s),$$

where $x_{\text{real}}(s)$ denotes a realized operational runtime state trajectory.

26.10. Relative States

Definition 26.35 (Relative state). A relative state is a state evaluated relative to another runtime state or runtime reference frame.

Remark 26.36. Within AIM, many operational quantities are fundamentally relative states.

Remark 26.37. Examples include:

- vehicle states relative to track states,
- wheelset states relative to local frames,
- center-of-gravity trajectories relative to realized track geometry,
- and operational balance states relative to admissibility envelopes.

26.11. State Transformations

Remark 26.38. Different state classes are connected through runtime operators. The principal operator structure is introduced in chapter 27.

Remark 26.39. Examples include:

$$\mathcal{R}, \quad \mathcal{D}, \quad \mathcal{O}, \quad \mathcal{V}.$$

Remark 26.40. Typical transformations include:

- intrinsic reconstruction,
- realization filtering,
- runtime evaluation,
- operational evaluation,
- vehicle interpretation,
- and optimization-state construction.

26.12. State Hierarchy

Remark 26.41. The AIM state hierarchy may be interpreted schematically as:

$$\kappa(s) \rightarrow \text{pose2} \rightarrow \text{pose3} \rightarrow x(s) \rightarrow x_{\text{vehicle}} \rightarrow x_{\text{CG}} \rightarrow x_{\text{opt}}.$$

Remark 26.42. Realization operators act throughout this hierarchy:

$$\mathcal{R} : x \mapsto x_{\text{real}}.$$

Remark 26.43. This hierarchy is not a rigid implementation sequence. It is a conceptual ordering of interpretation layers.

26.13. Canonical Interpretation

Proposition 26.44. *Within AIM, state concepts belong to distinct interpretation layers.*

Proposition 26.45. *pose3 describes the railway runtime state rather than the vehicle itself.*

Proposition 26.46. *Vehicle states are relative runtime interpretation states generated from interaction with the railway runtime state.*

Proposition 26.47. *Operational and dynamic states are generated through runtime evaluation rather than through static geometric storage.*

Proposition 26.48. *Optimization states are preferably realized runtime-state trajectories rather than isolated geometric parameters.*

26. State Taxonomy

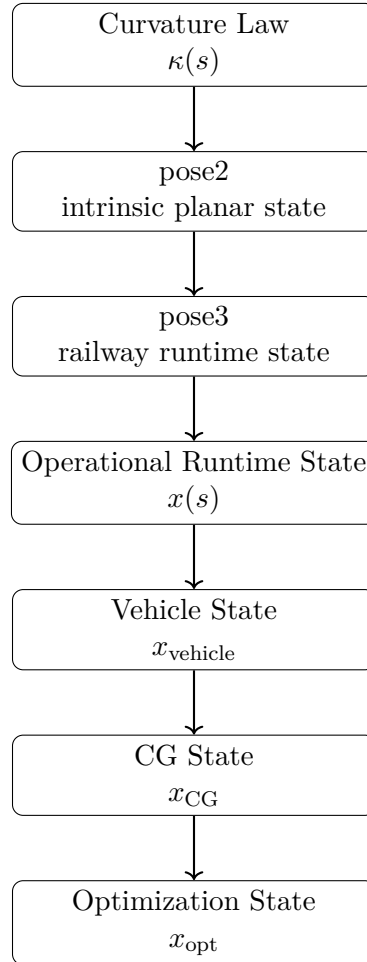


Abbildung 26.1.: AIM state hierarchy from intrinsic curvature to optimization state.

26.14. Connection to AIM

Summary 26.49. State taxonomy provides the structural interpretation framework for AIM runtime architecture.

Within this framework:

- pose2 represents intrinsic reconstruction state,
- pose3 represents railway runtime state,
- realized states represent physical railway realization,
- operational states describe runtime behaviour,
- vehicle states describe relative operational interpretation,
- center-of-gravity states describe operational abstractions,
- dynamic states describe runtime state evolution,

26. *State Taxonomy*

- and optimization states describe the objects of trajectory optimization.

This taxonomy prevents conceptual ambiguity between:

- geometry,
- runtime,
- realization,
- dynamics,
- vehicle interpretation,
- and optimization.

27. Runtime Operators

Motivation

Remark 27.1. The preceding chapters introduced several runtime-dependent concepts:

- railway runtime states,
- realized states,
- operational states,
- vehicle states,
- and optimization states.

Remark 27.2. These state layers are connected by operators rather than by direct identity.

Remark 27.3. This chapter introduces the main runtime operators used within AIM.

Principle 27.4 (Operator-layer principle). Within AIM, runtime interpretation is organized through explicit operators connecting distinct state layers.

27.1. Operator Overview

Remark 27.5. The central runtime operators are:

$$\mathcal{R}, \quad \mathcal{D}, \quad \mathcal{O}, \quad \mathcal{V}.$$

Remark 27.6. They may be interpreted schematically as:

$$\text{pose3}_{\text{target}} \xrightarrow{\mathcal{R}} \text{pose3}_{\text{real}} \xrightarrow{\mathcal{D}} x_{\text{op}} \xrightarrow{\mathcal{V}} x_{\text{vehicle}} \xrightarrow{\mathcal{O}} x_{\text{opt}}.$$

Remark 27.7. This diagram is conceptual. Individual implementations may combine, approximate, or partially evaluate these operators.

27.2. Realization Operator

Definition 27.8 (Realization operator). The realization operator maps target states to realized states:

$$\mathcal{R} : x_{\text{target}} \mapsto x_{\text{real}}.$$

27. Runtime Operators

Remark 27.9. For pose3 states this becomes:

$$\mathcal{R}_{\text{pose3}} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

Remark 27.10. The realization operator includes:

- construction effects,
- measurement effects,
- maintenance effects,
- filtering effects,
- and physical realization constraints.

Remark 27.11. Within AIM, $\mathcal{R}$ is not a correction after design. It is a central operator in the runtime interpretation chain.

27.3. Dynamics Operator

Definition 27.12 (Dynamics operator). The dynamics operator maps railway runtime states to operational dynamic state trajectories:

$$\mathcal{D} : \text{pose3} \mapsto x_{\text{op}}.$$

Remark 27.13. The generated operational state x_{op} may include:

- effective acceleration,
- jerk,
- roll response,
- cant deficiency,
- balance trajectories,
- and admissibility quantities.

Remark 27.14. In realization-aware interpretation:

$$x_{\text{op,real}} = \mathcal{D} \left(\mathcal{R}_{\text{pose3}} \left(\text{pose3}_{\text{target}} \right) \right).$$

Remark 27.15. The dynamics operator is an operational interpretation operator. It is not necessarily a complete multibody simulation model.

27.4. Operational Operator

Definition 27.16 (Operational operator). The operational operator maps runtime dynamic states to operational evaluation states:

$$\mathcal{O} : x_{\text{op}} \mapsto x_{\text{eval}}.$$

Remark 27.17. The operational operator may evaluate:

- admissibility,
- standards compliance,
- comfort indicators,
- operational envelopes,
- and optimization objectives.

Remark 27.18. For example:

$$x_{\text{eval}} = \mathcal{O}(x_{\text{op}}, \mathcal{A}_{\text{adm}}),$$

where $\mathcal{A}_{\text{adm}}$ denotes admissibility constraints.

Remark 27.19. The operator $\mathcal{O}$ separates generation of operational states from evaluation of their quality or admissibility.

27.5. Vehicle Interpretation Operator

Definition 27.20 (Vehicle interpretation operator). The vehicle interpretation operator maps railway runtime states and operational context to vehicle-relative states:

$$\mathcal{V} : (\text{pose3}, v, \mathcal{C}) \mapsto x_{\text{vehicle}}.$$

Remark 27.21. Here $\mathcal{C}$ denotes operational context such as vehicle type, load state, abstraction level, or interpretation model.

Remark 27.22. The vehicle operator may generate:

- vehicle-relative states,
- wheelset interpretation states,
- bogie interpretation states,
- center-of-gravity trajectories,
- and comfort-relevant abstractions.

Remark 27.23. Within AIM, $\mathcal{V}$ is not automatically a full mechanical vehicle model. It is an interpretation operator whose detail level may vary.

27.6. Operator Composition

Remark 27.24. The strength of the operator interpretation lies in composition.

Remark 27.25. A realization-aware operational state may be written as:

$$x_{\text{op,real}} = (\mathcal{D} \circ \mathcal{R}_{\text{pose3}}) (\text{pose3}_{\text{target}}).$$

Remark 27.26. A vehicle-relative operational state may be written as:

$$x_{\text{vehicle}} = \mathcal{V} \left(\mathcal{R}_{\text{pose3}} (\text{pose3}_{\text{target}}), v, \mathcal{C} \right).$$

Remark 27.27. A realization-aware optimization state may be written as:

$$x_{\text{opt}} = \mathcal{O} \left(\mathcal{D} \left(\mathcal{R}_{\text{pose3}} (\text{pose3}_{\text{target}}) \right) \right).$$

27.7. Runtime versus Mechanics

Remark 27.28. The runtime operators introduced here are not intended to replace specialized physical simulation models.

Remark 27.29. In particular, AIM does not require immediate commitment to:

- full vehicle dynamics,
- multibody simulation,
- wheel–rail contact mechanics,
- finite element models,
- or Simpack-like simulation environments.

Remark 27.30. Such models may later be coupled through appropriate operator implementations.

Principle 27.31 (Runtime abstraction principle). Runtime operators define interpretation layers first; detailed mechanics may be added later as specialized implementations.

27.8. Operator Roles

Definition 27.32 (Operator role). An operator role specifies the conceptual transformation performed by an operator, independent of a particular numerical implementation.

Remark 27.33. This distinction allows AIM to remain:

- structurally clear,
- implementation-flexible,

- runtime-oriented,
- and extensible toward higher physical fidelity.

Remark 27.34. For example, $\mathcal{V}$ may initially be a simple operational abstraction and later be replaced or refined by a detailed vehicle model without changing the conceptual state architecture.

27.9. Canonical Interpretation

Proposition 27.35. *Within AIM, runtime interpretation is operator-based.*

Proposition 27.36. *The realization operator $\mathcal{R}$ maps target states to realized states.*

Proposition 27.37. *The dynamics operator $\mathcal{D}$ maps railway runtime states to operational dynamic state trajectories.*

Proposition 27.38. *The vehicle operator $\mathcal{V}$ maps railway runtime states to vehicle-relative interpretation states.*

Proposition 27.39. *The operational operator $\mathcal{O}$ maps operational states to evaluation, admissibility, or optimization states.*

27.10. Connection to AIM

Summary 27.40. Runtime operators provide the structural connection between AIM state layers.

Within this framework:

- $\mathcal{R}$ represents realization,
- $\mathcal{D}$ represents runtime dynamic state generation,
- $\mathcal{V}$ represents vehicle-relative interpretation,
- and $\mathcal{O}$ represents operational evaluation.

Together, these operators connect:

- intrinsic geometry,
- pose3 runtime states,
- realized railway states,
- operational dynamics,
- vehicle-space interpretation,

27. Runtime Operators

- admissibility,
- and optimization.

This keeps AIM structurally, intrinsically, runtime-oriented, and operator-based without prematurely introducing full mechanical simulation models.

28. Vehicle Space and Relative Railway States

28.1. Introduction

Motivation:

- pose3 describes the railway.
- railway operation involves vehicles.
- vehicle behaviour cannot be identified with railway state.
- a separate vehicle-space interpretation is required.

Principle:

Principle 28.1 (Track–Vehicle Distinction). Within AIM, railway states and vehicle states are fundamentally distinct. Vehicle states are interpreted relative to railway states.

28.2. Track State versus Vehicle State

Proposition 28.2 (State hierarchy). *Railway states exist independently of vehicles.*

Vehicle states require railway states as reference states.

Therefore:

$$\text{vehicleState} = \mathcal{V}(\text{pose3}, \dots)$$

but not vice versa.

Definition 28.3 (Track State). The track state is the runtime-evaluated railway state represented by pose3 and its associated railway reference frame.

Definition 28.4 (Vehicle State). A vehicle state is an operational state interpreted relative to a track state.

$$\text{vehicleState} = \mathcal{V}(\text{pose3}, \dots) \tag{28.1}$$

OPENDiscussion.

Insert Figure: Track State $\rightarrow$ Vehicle State.

28.3. Vehicle Space

Definition 28.5 (Vehicle Space). Vehicle space denotes the collection of operational states that are interpreted relative to a railway state.

OPENDiscussion.

28.4. Relative Vehicle States

Definition 28.6 (Relative Vehicle State). A relative vehicle state is a vehicle state expressed in a local track-attached reference frame.

Examples:

- lateral displacement
- yaw deviation
- roll deviation
- wheel unloading

OPENDiscussion.

28.5. Track and Vehicle Frames

Definition 28.7 (Track Frame). Local railway-attached frame generated from pose3.

Definition 28.8 (Vehicle Frame). Local frame attached to a vehicle body or subsystem.

OPENDiscussion.

$$F_{\text{vehicle}} \neq F_{\text{track}} \tag{28.2}$$

28.6. Operational Abstractions

28.6.1. Wheelset State

Definition 28.9. Wheelset state.

OPENSShort discussion.

28.6.2. Bogie State

Definition 28.10. Bogie state.

OPENSShort discussion.

28.6.3. Center-of-Gravity State

Definition 28.11. Center-of-gravity state.

OPENShort discussion.

Remark 28.12. The center-of-gravity state is not identical to the vehicle state.

A vehicle may contain multiple relevant operational states, whereas the center-of-gravity state describes the motion of a selected mass reference point.

Remark 28.13. Within AIM, future optimization approaches may operate on center-of-gravity trajectories rather than directly on geometric centerlines.

28.7. Canonical Interpretation

Proposition 28.14. *pose3 describes the railway rather than the vehicle.*

Proposition 28.15. *Vehicle states are relative interpretation states.*

Proposition 28.16. *Vehicle space is generated from railway states.*

28.8. Summary

Vehicle space introduces the operational entities that are interpreted relative to railway states.

It defines the ontology of vehicle-related states but does not yet describe runtime evolution, dynamic balance, realization effects, or optimization.

29. Operational Velocity

Motivation

Remark 29.1. Railway alignment geometry is naturally parameterized by arc length, whereas operational behaviour is experienced through time.

Remark 29.2. Operational velocity is therefore the bridge between intrinsic geometry and runtime dynamics.

Principle 29.3 (Velocity-runtime principle). Within AIM, velocity is interpreted as a runtime quantity coupling arc-length-based railway geometry to time-based operational behaviour.

29.1. Arc Length and Time

Definition 29.4 (Velocity relation). For an operational velocity $v(s)$, arc-length and time derivatives are related by:

$$\frac{d}{dt} = v \frac{d}{ds}.$$

Remark 29.5. This relation is fundamental for translating curvature evolution into operational runtime effects.

29.2. Design Velocity and Operational Velocity

Definition 29.6 (Design velocity). Design velocity denotes the velocity assumption used during alignment design.

Definition 29.7 (Operational velocity). Operational velocity denotes the realized or intended runtime velocity during operation.

Remark 29.8. Design velocity and operational velocity need not be identical.

29.3. Velocity Envelopes

Definition 29.9 (Velocity envelope). A velocity envelope defines admissible velocity ranges along an alignment.

Remark 29.10. Velocity envelopes interact with curvature, cant, comfort, admissibility, and operational constraints.

29.4. Connection to AIM

Summary 29.11. Operational velocity connects arc-length-based geometry with time-dependent runtime behaviour.

Within AIM, velocity is not treated as an external scalar only, but as a runtime coupling quantity connecting curvature, cant, dynamics, admissibility, and optimization.

30. Runtime Dynamics

Motivation

Railway alignments are not merely stored geometric objects. Within AIM, they are interpreted as state-generating structures.

The preceding chapters introduced railway states and vehicle-relative interpretation. This chapter now introduces the operational layer that describes how runtime states evolve.

Geometry defines what the railway is.

Runtime dynamics define which operational states emerge from it.

Principle 30.1 (Runtime-dynamics principle). Within AIM, operational railway dynamics are interpreted as runtime state evolution generated from railway runtime states.

30.1. Dynamic Runtime States

Definition 30.2 (Dynamic runtime state). A dynamic runtime state is an operational state generated from railway runtime states during runtime interpretation.

Dynamic runtime states do not belong to the intrinsic alignment model. They are derived states.

They are generated from railway state, operational context, and runtime interpretation.

Conceptually, a dynamic runtime state may be written as

$$x(s) = \mathcal{D}(\text{pose3}(s), \mathcal{C}),$$

where $\text{pose3}(s)$ denotes the railway runtime state, $\mathcal{C}$ denotes operational context, and $x(s)$ denotes the resulting operational state.

30.2. Runtime Dynamics Operator

Definition 30.3 (Runtime dynamics operator). The runtime dynamics operator is the operator

$$\mathcal{D}$$

that maps railway runtime states and operational context to dynamic runtime states.

The operator $\mathcal{D}$ represents the dynamic interpretation layer of AIM.

It does not prescribe a particular vehicle model. Instead, it describes where such models, simplified operational relations, admissibility checks, and runtime evaluations are coupled to the railway state.

Thus,

$$\mathcal{D} : (\text{pose3}, \mathcal{C}) \mapsto x$$

expresses the transition from railway state to operational state.

30.3. Arc-Length State Evolution

Definition 30.4 (Runtime state evolution). Runtime state evolution describes the change of a dynamic runtime state along the alignment parameter.

Within AIM, the natural evolution parameter is arc-length.

An abstract runtime evolution law may be written as

$$x'(s) = f(x(s), u(s)),$$

where $x(s)$ denotes the dynamic runtime state, $u(s)$ denotes operational input, and f denotes the evolution law.

If a velocity function $v(s)$ is known, time-based and arc-length-based derivatives are connected by

$$\frac{d}{dt} = v(s) \frac{d}{ds}.$$

This relation links intrinsic railway geometry to operational runtime interpretation.

30.4. Geometry and Runtime Behaviour

Intrinsic geometry describes the structure of the railway.

Runtime dynamics describe the operational behaviour generated from that structure.

The distinction is essential.

Curvature, profile, cant, and frame reconstruction define railway state. Dynamic runtime states are derived from this state through operational interpretation.

Proposition 30.5. *Dynamic runtime states are derived from railway geometry, but they are not themselves intrinsic geometric quantities.*

30.5. Curvature as Runtime Generator

Runtime dynamics originate from intrinsic railway structure.

30. Runtime Dynamics

In particular, curvature evolution acts as one of the primary generators of operational behaviour:

$$\kappa(s) \longrightarrow \mathcal{D} \longrightarrow x(s).$$

The quantities

$$\kappa(s), \quad \kappa'(s), \quad \kappa''(s)$$

are therefore not only geometric descriptors. They influence the dynamic runtime states generated from the railway.

This observation is one of the central motivations of AIM: operational behaviour is rooted in intrinsic curvature structure.

30.6. Runtime Dynamics and Vehicle Models

Runtime dynamics in AIM are not identical to full vehicle mechanics.

A complete vehicle-dynamics simulation may be attached to the AIM runtime layer, but it is not required for defining the layer itself.

The role of runtime dynamics is to provide the operational state-evolution interface between railway state and dynamic interpretation.

Principle 30.6 (Mechanics separation principle). Runtime dynamics in AIM form an operational state-evolution layer rather than a complete vehicle-dynamics simulation layer.

30.7. Canonical Interpretation

Proposition 30.7. *Within AIM, railway dynamics are interpreted as runtime state evolution.*

Proposition 30.8. *Dynamic runtime states are generated from railway runtime states.*

Proposition 30.9. *The runtime dynamics operator $\mathcal{D}$ maps railway runtime states to operational state trajectories.*

Proposition 30.10. *Runtime dynamics belong to the operational interpretation layer and not to intrinsic geometry itself.*

Proposition 30.11. *The AIM runtime-dynamics framework provides operational state evolution rather than complete vehicle mechanics.*

30.8. Connection to AIM

Summary 30.12. Runtime dynamics extend AIM from railway-state reconstruction toward operational state evolution.

Within this framework:

30. Runtime Dynamics

- `pose3` provides railway runtime states,
- $\mathcal{D}$ generates dynamic runtime states,
- state evolution is naturally formulated over arc-length,
- curvature evolution acts as a generator of runtime behaviour,
- geometry and dynamics remain conceptually separated,
- and detailed vehicle mechanics may be coupled through dedicated runtime models.

Runtime dynamics therefore establish the bridge between intrinsic geometry, railway state, vehicle-space interpretation, realization-aware dynamics, and optimization.

31. Schwerpunkt States and Operational Balance

Motivation

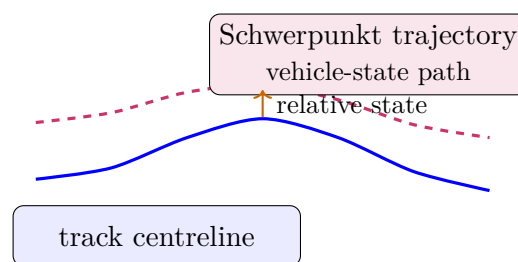


Abbildung 31.1.: Schwerpunkt trajectory as vehicle-state path relative to the railway state.

Remark 31.1. Classical railway alignment theory primarily interprets transition design through geometric smoothness.

Remark 31.2. Within AIM, however, the operationally relevant object is not merely the track geometry itself, but the dynamically evolving vehicle state generated relative to the railway state.

Remark 31.3. This distinction builds on the state taxonomy introduced in chapter 26.

Remark 31.4. This shifts the interpretation of railway alignment from:

curve construction

toward:

trajectory shaping of operational vehicle states.

Remark 31.5. The center-of-gravity interpretation plays a particularly important role within this framework.

Principle 31.6 (Schwerpunkt principle). Within AIM, Schwerpunkt-Trassierung is interpreted as operational trajectory shaping of realized runtime vehicle states.

Principle 31.7 (Schwerpunkt is not geometry). Schwerpunkt-Trassierung optimizes operational runtime trajectories, not geometric curves.

31.1. Center-of-Gravity State

Definition 31.8 (Center-of-gravity state). The center-of-gravity state describes the operational runtime state of the vehicle center of gravity relative to the railway state.

Remark 31.9. The center-of-gravity state depends on:

- realized railway geometry,
- cant evolution,
- operational speed,
- vehicle configuration,
- suspension behaviour,
- and runtime frame interpretation.

Remark 31.10. Within AIM, the center-of-gravity state is interpreted as a runtime state generated from

$$\text{pose3}_{\text{real}}.$$

Remark 31.11. The center-of-gravity state is an operational abstraction. It is not a complete mechanical vehicle model.

31.2. Center-of-Gravity Trajectory

Definition 31.12 (Center-of-gravity trajectory). The center-of-gravity trajectory is the operational trajectory

$$x_{\text{CG}}(s)$$

traced by the vehicle center of gravity within runtime space.

Remark 31.13. This trajectory is generally not identical to

$$\gamma(s).$$

Remark 31.14. Instead, it emerges from:

- curvature evolution,
- cant evolution,
- roll response,
- vehicle geometry,
- operational velocity,

- and runtime dynamics.

Remark 31.15. The operational trajectory may therefore differ substantially from the pure geometric centerline.

Remark 31.16. Within Schwerpunkt interpretation, $x_{CG}(s)$ is one of the central operational state trajectories generated by railway geometry.

31.3. Operational Balance

Definition 31.17 (Operational balance). Operational balance denotes the dynamic relationship between:

- curvature-induced acceleration,
- cant-induced compensation,
- and vehicle response.

Remark 31.18. A dynamically balanced state approximately satisfies

$$v^2 \kappa(s) \approx g \sin \phi(s).$$

Remark 31.19. In practice, operational balance is not exact but evolves continuously within admissible runtime ranges.

Remark 31.20. Operational balance is therefore interpreted as a runtime state relation, not as a purely static equilibrium formula.

31.4. Balance Trajectories

Definition 31.21 (Balance trajectory). A balance trajectory is the runtime evolution of operational equilibrium or imbalance states along the railway alignment.

Remark 31.22. Balance trajectories may be interpreted through:

- effective acceleration,
- roll evolution,
- cant deficiency,
- and center-of-gravity response.

Remark 31.23. Thus, operational balance is interpreted dynamically rather than as a single static equilibrium condition.

Remark 31.24. Within AIM, balance trajectories may become a central class of operational runtime trajectories.

31.5. Effective Acceleration Interpretation

Definition 31.25 (Effective operational acceleration). The effective operational acceleration is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 31.26. This quantity measures the operational imbalance experienced within the vehicle state.

Remark 31.27. Within Schwerpunkt interpretation, effective acceleration is more relevant than curvature alone.

Remark 31.28. Operational smoothness therefore depends on the trajectory of

$$a_{\text{eff}}(s),$$

rather than merely on geometric continuity.

31.6. Roll Evolution

Definition 31.29 (Roll evolution). Roll evolution denotes the runtime evolution of vehicle roll state relative to the railway frame.

Remark 31.30. Roll evolution depends jointly on:

- cant evolution,
- suspension response,
- center-of-gravity height,
- and operational velocity.

Remark 31.31. Within operational railway dynamics, abrupt roll evolution may generate:

- discomfort,
- load transfer,
- wheel unloading,
- and dynamic instability.

31.7. Comfort Interpretation

Definition 31.32 (Operational comfort). Operational comfort denotes the dynamic smoothness experienced by the vehicle state during runtime evolution.

Remark 31.33. Comfort depends on:

- effective acceleration,
- jerk,
- roll evolution,
- vibration behaviour,
- center-of-gravity trajectory,
- and realization smoothness.

Remark 31.34. Comfort is therefore not purely geometric. It emerges from runtime dynamic state evolution.

31.8. Trajectory Shaping

Definition 31.35 (Operational trajectory shaping). Operational trajectory shaping denotes the design of runtime vehicle state evolution through curvature and cant control.

Remark 31.36. Within this interpretation, railway alignment design becomes:

- shaping of effective acceleration trajectories,
- shaping of roll evolution,
- shaping of operational balance,
- and shaping of center-of-gravity motion.

Remark 31.37. Thus, railway alignment design is elevated from:

curve construction

to:

operational state design.

31.9. Schwerpunkt Interpretation

Remark 31.38. Within AIM, Schwerpunkt-Trassierung is not interpreted as a particular curve formula or historical transition family.

Remark 31.39. Instead, it denotes:

runtime-oriented shaping of operational vehicle-state trajectories.

Remark 31.40. The central engineering question therefore becomes:

Which operational state trajectory is generated?

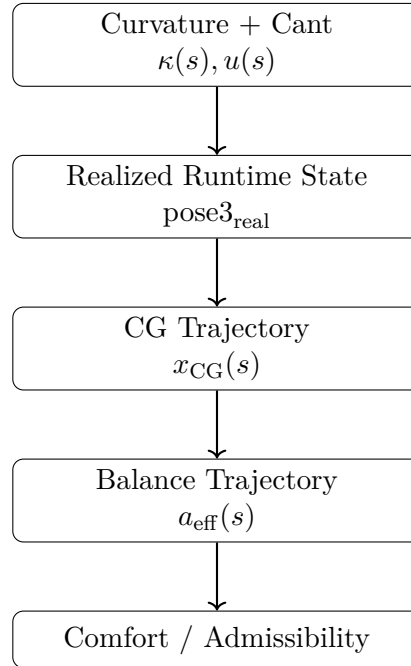


Abbildung 31.2.: Schwerpunkt interpretation: alignment design becomes shaping of operational state trajectories.

rather than:

Which curve formula is used?

Remark 31.41. The decisive object is not the visual curve, but the operational runtime trajectory generated from realized geometry.

31.10. Schwerpunkt and Realization

Remark 31.42. Operational trajectory shaping depends fundamentally on realized rather than purely analytical geometry.

Remark 31.43. Thus,

$$x_{\text{CG,real}}(s)$$

is operationally more relevant than

$$x_{\text{CG,target}}(s).$$

Remark 31.44. Realization therefore directly influences:

- comfort,
- roll smoothness,
- operational balance,

- center-of-gravity motion,
- and dynamic admissibility.

31.11. Schwerpunkt and Curvature Space

Remark 31.45. Operational state trajectories are generated fundamentally through curvature evolution.

Remark 31.46. Thus, curvature space acts simultaneously as:

- geometric space,
- runtime space,
- and operational state-generation space.

Remark 31.47. Different transition families may therefore generate fundamentally different operational state trajectories even when their position-space geometry appears similar.

31.12. Towards Dynamic State Design

Remark 31.48. The Schwerpunkt interpretation transforms railway alignment theory into a theory of operational runtime-state evolution.

Remark 31.49. This prepares the conceptual transition toward:

- realization-aware dynamics,
- Vienna reinterpretation,
- operational admissibility,
- and runtime-state optimization.

Remark 31.50. The relevant runtime operators are introduced in chapter 27.

31.13. Canonical Interpretation

Proposition 31.51. *Within AIM, Schwerpunkt-Trassierung is interpreted as shaping of operational runtime vehicle states.*

Proposition 31.52. *Schwerpunkt-Trassierung optimizes operational runtime trajectories, not geometric curves.*

Proposition 31.53. *The operationally relevant object is not merely the railway curve, but the realized trajectory of vehicle states relative to the railway frame.*

Proposition 31.54. *Operational smoothness depends primarily on runtime state evolution rather than on coordinate geometry alone.*

31.14. Connection to AIM

Summary 31.55. *Schwerpunkt* states extend railway alignment interpretation from geometry toward operational runtime-state design.

Within AIM:

- center-of-gravity trajectories become operationally relevant,
- $x_{CG}(s)$ becomes an operational state trajectory,
- balance trajectories replace purely geometric interpretation,
- comfort emerges from runtime evolution,
- effective acceleration becomes a state variable,
- and railway alignment becomes operational trajectory shaping.

This provides the conceptual bridge toward:

- Vienna reinterpretation,
- realization-aware dynamics,
- operational admissibility,
- and runtime-state optimization.

32. Dynamic Balance

Motivation

Operational railway motion is governed by the interaction between curvature, cant, and velocity.

A railway vehicle travelling through a curve experiences lateral acceleration generated by curvature. Cant partially compensates this acceleration by rotating the local vehicle reference frame.

The resulting balance between both effects is one of the central quantities of railway operation.

Within AIM, dynamic balance is interpreted as a runtime-state relation derived from railway and vehicle states.

Principle 32.1 (Dynamic-balance principle). Dynamic balance emerges from the interaction between curvature, cant, and operational velocity.

32.1. Balance Relation

Definition 32.2 (Balance relation). A dynamically balanced state approximately satisfies

$$v^2 \kappa(s) = g \sin \phi(s).$$

The balance relation expresses equilibrium between curvature-induced lateral acceleration and cant-induced compensation.

A perfectly balanced state produces no residual lateral acceleration in the local vehicle frame.

32.2. Effective Lateral Acceleration

Definition 32.3 (Effective lateral acceleration). The effective lateral acceleration is

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi.$$

The quantity a_{eff} measures the residual imbalance remaining after cant compensation.

It is one of the most important operational quantities in railway engineering because it directly relates geometry to passenger perception.

32. Dynamic Balance

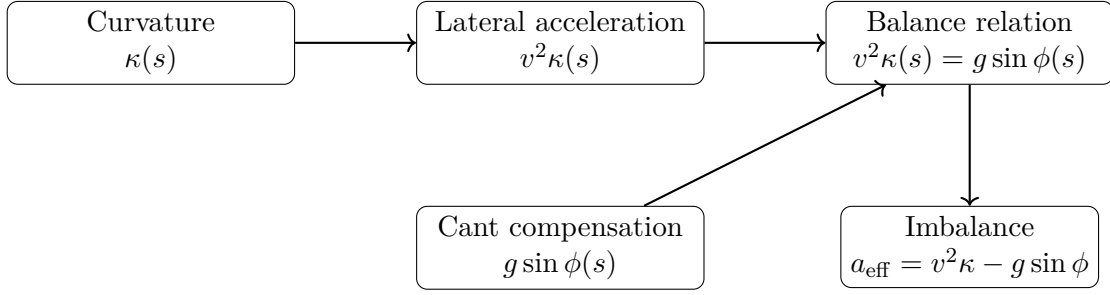


Abbildung 32.1.: Balance relation between curvature, cant, and velocity.

32.3. Cant Deficiency and Excess Cant

Definition 32.4 (Cant deficiency). Cant deficiency occurs whenever

$$a_{\text{eff}} > 0.$$

In this case the available cant is insufficient for the current operating velocity.

Definition 32.5 (Excess cant). Excess cant occurs whenever

$$a_{\text{eff}} < 0.$$

In this case the cant exceeds the amount required for dynamic balance.

Both situations represent operational imbalance states.

32.4. Balance Trajectories

Definition 32.6 (Balance trajectory). A balance trajectory is the evolution of

$$a_{\text{eff}}(s)$$

along the alignment.

Balance trajectories are generated jointly by

- curvature evolution,
- cant evolution,
- and velocity evolution.

Operational smoothness depends not only on the magnitude of imbalance but also on how imbalance changes along the route.

32.5. Operational Interpretation

Dynamic balance forms the connection between

- railway geometry,
- railway state,
- vehicle-space interpretation,
- and passenger comfort.

It therefore occupies a central position between the geometric and dynamic layers of AIM.

Balance is not a geometric property.

Balance is an operational state generated from geometry.

32.6. Canonical Interpretation

Proposition 32.7. *Within AIM, dynamic balance is interpreted as a runtime-derived operational state.*

Proposition 32.8. *Effective lateral acceleration measures the degree of operational imbalance.*

Proposition 32.9. *Cant deficiency and excess cant are complementary imbalance states.*

Proposition 32.10. *Balance trajectories describe the evolution of operational balance along the alignment.*

32.7. Connection to AIM

Summary 32.11. Dynamic balance provides the fundamental coupling between curvature, cant, and operational velocity.

Within AIM,

- balance is interpreted as an operational state,
- effective acceleration quantifies imbalance,
- cant deficiency and excess cant describe deviation from equilibrium,
- and balance trajectories characterize operational behaviour along the alignment.

This chapter establishes the balance concepts required for Schwerpunkt-Trassierung, realization-aware dynamics, admissibility evaluation, and optimization.

33. Vienna Reinterpretation and Dynamic State Shaping

Motivation

Remark 33.1. Classical transition theory usually interprets transition curves as special geometric constructions.

Remark 33.2. Within AIM, however, transition structures are interpreted primarily through their generated runtime behaviour.

Remark 33.3. This changes the interpretation of Vienna-style transition concepts fundamentally.

Remark 33.4. Within this framework, Vienna is not interpreted as:

- a specific curve formula,
- a historical construction rule,
- or a particular interpolation method,

but rather as:

a strategy for shaping operational runtime states.

Principle 33.5 (Vienna reinterpretation principle). Within AIM, Vienna-type transitions are interpreted as dynamic state-shaping strategies operating in curvature-runtime space.

Principle 33.6 (No special-curve principle). Vienna reinterpretation does not elevate one special curve formula. It interprets transition structure by its generated operational runtime behaviour.

33.1. From Curves to Runtime States

Remark 33.7. Classical railway alignment theory focuses primarily on:

$$\gamma(s).$$

Remark 33.8. Within AIM, however, the operationally relevant quantity is:

$$\text{pose3}_{\text{real}}(s),$$

33. Vienna Reinterpretation and Dynamic State Shaping

together with the generated vehicle-space trajectory.

Remark 33.9. Thus, transition interpretation shifts:

- from coordinate geometry,
- toward runtime operational state evolution.

Remark 33.10. The central engineering question therefore becomes:

Which operational runtime trajectory is generated?

rather than:

Which geometric curve formula is used?

33.2. Vienna as Runtime-State Strategy

Definition 33.11 (Vienna-type state shaping). A Vienna-type transition strategy is a curvature-evolution strategy designed to shape operational runtime-state trajectories.

Remark 33.12. Relevant runtime quantities include:

- effective acceleration,
- jerk,
- roll evolution,
- operational balance,
- and center-of-gravity motion.

Remark 33.13. Thus, Vienna interpretation operates fundamentally in:

- curvature space,
- runtime space,
- and operational state space.

Remark 33.14. The term *Vienna* is therefore used here as an interpretive strategy, not as a commitment to one fixed analytical transition law.

33.3. Half-Wave Interpretation

Definition 33.15 (Half-wave interpretation). Half-wave interpretation denotes a modular description of transition evolution through normalized curvature segments.

33. Vienna Reinterpretation and Dynamic State Shaping

Remark 33.16. Within AIM, many transition structures may be interpreted compositionally as:

$$\text{halfWave}_1 \rightarrow \text{core} \rightarrow \text{halfWave}_2.$$

Remark 33.17. This decomposition is a modelling aid. It should not be interpreted as a new special-curve dogma.

Remark 33.18. The half-wave interpretation allows:

- modular runtime shaping,
- asymmetric runtime behaviour,
- spectral interpretation,
- and realization-aware transition design.

Remark 33.19. The half-wave structure is relevant only insofar as it affects generated runtime behaviour.

33.4. Dynamic State Shaping

Definition 33.20 (Dynamic state shaping). Dynamic state shaping denotes controlled generation of operational runtime trajectories through curvature evolution.

Remark 33.21. Within AIM, transition geometry is interpreted fundamentally as:

runtime-state shaping through curvature control.

Remark 33.22. Relevant shaped quantities include:

- effective acceleration,
- jerk,
- cant deficiency,
- roll evolution,
- and center-of-gravity trajectories.

33.5. Center-of-Gravity Interpretation

Remark 33.23. Within Vienna reinterpretation, the operationally relevant object is not merely the rail geometry itself.

Remark 33.24. Instead, emphasis shifts toward:

$$x_{\text{CG}}(s),$$

the runtime trajectory of the vehicle center of gravity.

33. Vienna Reinterpretation and Dynamic State Shaping

Remark 33.25. Different transition families may produce:

- similar geometric centerlines,
- but fundamentally different center-of-gravity trajectories.

Remark 33.26. Thus, Vienna reinterpretation naturally connects transition theory with:

- Schwerpunkt-Trassierung,
- operational balance,
- and runtime comfort interpretation.

33.6. Spectral Interpretation

Remark 33.27. Transition laws may be interpreted spectrally through their curvature content.

Remark 33.28. Low-frequency curvature behaviour primarily influences:

- global operational balance,
- smooth acceleration evolution,
- and large-scale runtime behaviour.

Remark 33.29. High-frequency curvature behaviour tends to influence:

- realization sensitivity,
- machine smoothing,
- dynamic roughness,
- measurement sensitivity,
- and operational excitation.

Proposition 33.30 (Spectral smoothing principle). *Operationally smooth runtime trajectories correspond to spectrally controlled curvature evolution.*

Remark 33.31. This interpretation shifts transition evaluation away from geometric appearance and toward spectral runtime behaviour.

33.7. Realization Filtering

Remark 33.32. Realization acts approximately as a low-pass filter on curvature space.

Remark 33.33. Thus, highly oscillatory transition structures may collapse into similar realized operational states after construction and maintenance.

Remark 33.34. Within AIM, Vienna reinterpretation therefore includes:

- realization filtering,
- machine smoothing,
- measurement reconstruction,
- and runtime operational robustness.

Remark 33.35. A transition law is operationally meaningful only if its intended runtime behaviour survives realization.

Principle 33.36 (Realization-survival principle). A transition strategy is operationally relevant only to the extent that its intended runtime behaviour survives realization filtering.

Remark 33.37. A possible Vienna-type hypothesis is therefore that certain transition strategies may preserve their intended operational behaviour more robustly under realization filtering than others.

33.8. Runtime Smoothness

Definition 33.38 (Runtime smoothness). Runtime smoothness denotes smooth operational evolution of runtime states generated through curvature and cant evolution.

Remark 33.39. Runtime smoothness includes:

- smooth acceleration evolution,
- smooth roll evolution,
- smooth operational balance,
- and smooth center-of-gravity trajectories.

Remark 33.40. Within AIM, runtime smoothness is operationally more relevant than purely geometric smoothness.

Remark 33.41. Spectral smoothing provides one possible interpretation of runtime smoothness in curvature space.

33.9. Vienna and Curvature Space

Remark 33.42. Vienna reinterpretation operates fundamentally within curvature space.

Remark 33.43. Different curvature structures generate:

- different operational balance trajectories,
- different roll behaviour,
- different jerk evolution,
- different spectral content,
- and different realization behaviour.

Remark 33.44. Thus, transition quality cannot be characterized sufficiently through position geometry alone.

33.10. Vienna and Runtime Dynamics

Remark 33.45. Vienna reinterpretation is fundamentally a runtime-dynamics concept.

Remark 33.46. It couples:

- curvature evolution,
- runtime-state evolution,
- operational balance,
- spectral smoothing,
- realization filtering,
- and vehicle-space interpretation.

Remark 33.47. The transition curve therefore acts as:

a runtime-state generator.

33.11. Operational Interpretation

Remark 33.48. Within AIM, the operational railway problem is interpreted as:

generation of admissible runtime operational trajectories.

Remark 33.49. This transforms railway alignment theory from:

- geometric interpolation,

33. Vienna Reinterpretation and Dynamic State Shaping

- toward operational runtime-state design.

Remark 33.50. Vienna reinterpretation therefore acts as a bridge between:

- transition theory,
- runtime dynamics,
- realization,
- and operational optimization.

33.12. Towards Runtime-State Optimization

Remark 33.51. The reinterpretation developed here prepares the transition toward:

- realization-aware optimization,
- operational admissibility optimization,
- runtime-state shaping,
- and Schwerpunkt-based alignment design.

Remark 33.52. Optimization therefore acts not merely on:

$$\gamma(s),$$

but on:

$$\text{pose3}_{\text{real}}(s),$$

and the generated operational vehicle-state trajectories.

33.13. Canonical Interpretation

Proposition 33.53. *Within AIM, Vienna-type transitions are interpreted as runtime-state shaping strategies rather than as isolated geometric curve formulas.*

Proposition 33.54. *Operational transition quality depends fundamentally on generated runtime-state trajectories.*

Proposition 33.55. *Realization filtering and spectral behaviour are central components of transition interpretation.*

Proposition 33.56. *The operationally relevant object is the realized runtime trajectory of vehicle states rather than the abstract geometric curve alone.*

Proposition 33.57. *The half-wave interpretation is a modular modelling aid, not a special-curve doctrine.*

33.14. Connection to AIM

Summary 33.58. Vienna reinterpretation transforms transition theory into a theory of runtime-state shaping.

Within AIM:

- transitions generate operational runtime trajectories,
- half-wave structures provide modular interpretation rather than special-curve dogma,
- realization filtering determines operational robustness,
- spectral smoothing influences comfort and admissibility,
- and center-of-gravity trajectories become primary operational objects.

This establishes the conceptual bridge between:

- curvature space,
- runtime dynamics,
- realization-aware interpretation,
- Schwerpunkt-Trassierung,
- and operational state optimization.

34. Realization-Aware Runtime Dynamics

Motivation

Operational railway behaviour is commonly derived from analytical target geometry.

Within AIM, this interpretation is incomplete.

Vehicles do not interact with target geometry. They interact with realized railway states reconstructed and evaluated during runtime.

The relevant chain is therefore

$$\text{pose3}_{\text{target}} \rightarrow \mathcal{R} \rightarrow \text{pose3}_{\text{real}} \rightarrow \mathcal{D} \rightarrow \text{runtime}.$$

Dynamic behaviour emerges from realized railway states rather than from idealized design geometry alone.

Principle 34.1 (Realization-aware dynamics principle). Within AIM, operational railway dynamics are generated from realized runtime states rather than directly from analytical target geometry.

34.1. Realized Runtime States

Definition 34.2 (Realized runtime state). A realized runtime state is a railway state evaluated from realized rather than idealized geometry.

Within AIM, the operationally relevant state is

$$\text{pose3}_{\text{real}}(s).$$

This state is obtained through realization, reconstruction, and runtime evaluation.

The analytical target state remains important, but only through the realized state that ultimately governs operational behaviour.

34.2. Realization Filtering

Realization alters the states from which operational behaviour emerges.

Construction tolerances, maintenance history, measurement uncertainty, and reconstruction procedures modify the effective railway state available during operation.

Consequently, realization acts as a filter between design intent and runtime behaviour.

Proposition 34.3 (Realization-filter principle). *Operational behaviour is generated from realized states rather than directly from analytical target states.*

A design concept is operationally relevant only if its intended runtime behaviour survives realization.

34.3. Realization-Aware Admissibility

Definition 34.4 (Realization-aware admissibility). A railway state is realization-aware admissible if its realized runtime behaviour remains inside the relevant operational envelope.

Admissibility therefore depends not only on design geometry but also on the realized state that emerges after construction, reconstruction, and runtime evaluation.

Operational admissibility is consequently a runtime property rather than a purely geometric property.

34.4. Runtime Envelopes

Definition 34.5 (Runtime envelope). A runtime envelope is the admissible region of realized operational states.

Runtime envelopes define the operational limits within which realized states are considered acceptable.

Examples include comfort limits, dynamic admissibility limits, and other project-specific operational constraints.

Runtime envelopes provide the connection between realized states and engineering evaluation.

34.5. Realization-Aware State Design

The realization-aware viewpoint changes the engineering objective.

Railway alignment design is no longer interpreted merely as the design of analytical geometry.

Instead, it becomes the design of realized operational states.

The central engineering question becomes:

Which realized operational state should emerge during operation?

This shifts attention from target geometry toward the behaviour ultimately experienced in reality.

34.6. Canonical Interpretation

Proposition 34.6. *Operational railway behaviour emerges from realized runtime states rather than directly from analytical target geometry.*

Proposition 34.7. *Realization acts as a state-generating operator between design intent and operation.*

Proposition 34.8. *Admissibility is fundamentally a realization-aware property.*

Proposition 34.9. *Within AIM, railway alignment design is interpreted as realization-aware state design.*

34.7. Connection to AIM

Summary 34.10. Realization-aware dynamics establish the connection between realization and operational behaviour.

Within AIM:

- operational behaviour is generated from realized states,
- realization filters design intent,
- admissibility becomes realization-aware,
- runtime envelopes define operational limits,
- and railway alignment becomes realization-aware state design.

This chapter therefore provides the conceptual bridge between realization, runtime interpretation, engineering admissibility, and optimization.

35. Optimization in Runtime State Space

Motivation

Remark 35.1. Classical railway alignment optimization is usually formulated as a geometric optimization problem.

Remark 35.2. Typical optimization variables include:

- points,
- radii,
- tangent lengths,
- and geometric transition parameters.

Remark 35.3. Within AIM, however, operational railway behaviour emerges from runtime state evolution rather than from geometry alone.

Remark 35.4. Consequently, the relevant optimization object is not merely geometric shape, but:

runtime operational state trajectories.

Principle 35.5 (State-space optimization principle). Within AIM, railway alignment optimization is interpreted fundamentally as optimization of realized runtime-state trajectories.

Principle 35.6 (Trajectory-shaping principle). Railway optimization in AIM is trajectory shaping in runtime state space.

35.1. From Geometry to Runtime States

Remark 35.7. Geometric railway alignments generate:

$$\text{pose3}(s),$$

which in turn generates:

- operational balance,
- acceleration evolution,
- jerk evolution,

35. Optimization in Runtime State Space

- roll evolution,
- and vehicle-state trajectories.

Remark 35.8. Thus, geometry acts indirectly through:

$$\kappa(s), \quad u(s), \quad \phi(s),$$

to generate operational runtime states.

Remark 35.9. Optimization therefore acts fundamentally on:

$$x(s),$$

where $x(s)$ denotes the evolving operational runtime state.

Remark 35.10. The geometry remains important, but it is interpreted as a generator of runtime-state trajectories rather than as the final optimization object.

35.2. Runtime State Trajectories

Definition 35.11 (Runtime state trajectory). A runtime state trajectory is the operational evolution:

$$x(s)$$

generated along the railway alignment.

Remark 35.12. The runtime state may include:

- effective acceleration,
- jerk,
- roll state,
- cant deficiency,
- operational balance,
- and center-of-gravity behaviour.

Remark 35.13. The operationally relevant optimization object is therefore:

$$x(s),$$

rather than merely:

$$\gamma(s).$$

35.3. State Evolution

Definition 35.14 (Runtime state evolution). Runtime state evolution may be written abstractly as:

$$x'(s) = f(x(s), u(s)),$$

where:

- $x(s)$ denotes the runtime state,
- and $u(s)$ denotes control variables.

Remark 35.15. Typical control variables include:

- curvature evolution,
- cant evolution,
- transition structure,
- operational velocity,
- and sparse correction parameters.

Remark 35.16. Within AIM, geometry therefore acts as:

a runtime-state generator.

35.4. Operational State Variables

Remark 35.17. Operational state variables may include:

$$x(s) = (a_{\text{eff}}, j, \phi, \phi', \Delta_{\text{cant}}, x_{\text{CG}}, \dots).$$

Remark 35.18. The exact state structure depends on:

- operational interpretation,
- realization level,
- vehicle-space abstraction,
- and optimization objectives.

35.5. Realized Runtime States

Remark 35.19. Operational railway systems interact with:

$$\text{pose3}_{\text{real}},$$

rather than purely analytical target geometry.

35. Optimization in Runtime State Space

Remark 35.20. Thus, optimization should operate fundamentally on:

$$x_{\text{real}}(s),$$

generated through:

$$\mathcal{R}_{\text{pose3}}.$$

Remark 35.21. This transforms optimization into:

$$J = J(x_{\text{real}}(s)).$$

Remark 35.22. The central optimization object is therefore a realized operational runtime-state trajectory.

35.6. Optimization Functionals

Definition 35.23 (Runtime-state functional). A runtime-state optimization functional may be written as:

$$J[x] = \int_0^L F(x(s), x'(s), u(s)) \, ds.$$

Remark 35.24. The functional $J[x]$ evaluates the quality of an operational runtime-state trajectory.

Remark 35.25. Typical optimization targets include:

- minimizing jerk,
- minimizing effective acceleration variation,
- minimizing roll excitation,
- minimizing realization sensitivity,
- satisfying operational envelopes,
- and maximizing operational smoothness.

Remark 35.26. This shifts optimization away from isolated geometric residuals such as radius errors, tangent errors, or point-list errors toward functional evaluation of runtime-state trajectories.

35.7. Trajectory Shaping

Definition 35.27 (Operational trajectory shaping). Operational trajectory shaping denotes optimization of runtime-state evolution through curvature and cant control.

Remark 35.28. Within AIM, optimization therefore acts primarily on:

35. Optimization in Runtime State Space

- acceleration trajectories,
- roll trajectories,
- balance trajectories,
- admissibility trajectories,
- and center-of-gravity trajectories.

Remark 35.29. This interpretation extends classical railway optimization toward:

runtime-state trajectory optimization.

Remark 35.30. Trajectory shaping is therefore not an auxiliary concept, but one of the central optimization interpretations of AIM.

35.8. Curvature Space Optimization

Remark 35.31. Runtime-state trajectories are generated fundamentally from curvature evolution.

Remark 35.32. Thus, optimization acts intrinsically within:

$$\mathcal{K},$$

the curvature-space domain.

Remark 35.33. Typical optimization variables include:

- transition families,
- curvature amplitudes,
- transition lengths,
- normalization structure,
- and sparse curvature corrections.

Remark 35.34. Curvature-space optimization is therefore the upstream design mechanism for downstream runtime-state trajectory shaping.

35.9. Sparse Runtime Optimization

Definition 35.35 (Sparse runtime optimization). Sparse runtime optimization uses compact curvature representations to optimize runtime-state trajectories efficiently.

Remark 35.36. Within AIM, sparse models act as:

compact runtime-state generators.

Remark 35.37. This enables:

- scalable optimization,
- efficient runtime evaluation,
- realization-aware optimization,
- and robust parameter control.

35.10. Operational Admissibility

Definition 35.38 (Operationally admissible trajectory). A runtime-state trajectory is operationally admissible if:

- effective acceleration remains bounded,
- jerk remains admissible,
- roll evolution remains smooth,
- realization remains stable,
- and operational envelopes remain satisfied.

Remark 35.39. Admissibility therefore acts directly within runtime-state space.

Remark 35.40. Operational admissibility provides the constraint structure for runtime-state trajectory optimization.

35.11. Realization-Aware Optimization

Remark 35.41. Optimization should operate on:

$$x_{\text{real}}(s),$$

rather than on purely analytical target states.

Remark 35.42. Thus:

$$J = J\left(\mathcal{D}(\mathcal{R}_{\text{pose3}}(\text{pose3}_{\text{target}}))\right).$$

Remark 35.43. Optimization therefore couples:

- geometry,
- realization,
- runtime reconstruction,
- operational dynamics,
- and admissibility envelopes.

35.12. Center-of-Gravity Optimization

Remark 35.44. Within advanced runtime-state interpretation, optimization may act directly on:

$$x_{CG}(s),$$

the center-of-gravity trajectory.

Remark 35.45. Thus, optimization becomes:

- balance shaping,
- comfort shaping,
- roll shaping,
- and operational trajectory shaping.

Remark 35.46. This interpretation connects directly to:

- Schwerpunkt-Trassierung,
- Vienna reinterpretation,
- and realization-aware runtime dynamics.

35.13. Optimization as Runtime-State Design

Remark 35.47. Within AIM, railway optimization is interpreted fundamentally as:

design of admissible realized runtime-state trajectories.

Remark 35.48. The central engineering question therefore becomes:

Which operational state trajectory should emerge?

rather than:

Which geometric curve should be constructed?

Remark 35.49. Geometry, realization, and runtime dynamics therefore become coupled components of one operational design framework.

35.14. AXTRAN Reinterpretation

Remark 35.50. Classical optimization systems such as AXTRAN already moved partially toward operational optimization beyond purely geometric fitting.

Remark 35.51. AIM generalizes classical alignment optimization toward runtime-state trajectory optimization.

Remark 35.52. This reinterpretation should not be read as a rejection of classical optimization systems, but as an extension of their geometric core toward realization-aware operational state design.

35.15. Canonical Interpretation

Proposition 35.53. *Within AIM, railway optimization acts fundamentally on runtime-state trajectories rather than on coordinate geometry alone.*

Proposition 35.54. *Geometry acts as a generator of operational runtime states.*

Proposition 35.55. *Operational admissibility is fundamentally a runtime-state property.*

Proposition 35.56. *Realization-aware runtime-state optimization forms the core of advanced railway alignment design.*

Proposition 35.57. *Trajectory shaping is the central optimization interpretation of AIM runtime-state design.*

35.16. Connection to AIM

Summary 35.58. Optimization in AIM operates fundamentally in runtime-state space.

Within this framework:

- geometry generates runtime states,
- curvature space generates operational trajectories,
- realization filters runtime behaviour,
- admissibility acts in operational state space,
- optimization functionals evaluate $J[x]$,
- and optimization becomes trajectory shaping.

This interpretation unifies:

- curvature evolution,
- runtime dynamics,
- realization,
- Schwerpunkt-Trassierung,
- Vienna reinterpretation,
- operational admissibility,
- and operational optimization.

Railway alignment therefore becomes:

realization-aware runtime-state trajectory design.

Teil VI.

Runtime and Evaluation

Runtime

What do we already know?

Remark 35.59. The previous part established that railway states do not remain static descriptions.

They generate operational state evolution.

Remark 35.60. In particular, vehicle states, balance states, comfort states, and dynamic quantities were interpreted as states evolving relative to the railway state.

What is still missing?

Remark 35.61. State evolution alone is not yet an executable engineering system.

Remark 35.62. AIM requires runtime services that evaluate, reconstruct, transform, project, cache, refine, and interpret railway states in operational contexts.

What comes next?

Remark 35.63. This part introduces the runtime layer of AIM.

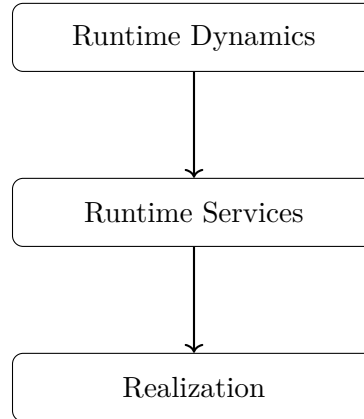
Remark 35.64. Runtime is the operational evaluation layer that transforms stored alignment information and railway states into usable engineering quantities.

Remark 35.65. Runtime services connect:

- sparse alignment semantics,
- metric-aware geometry,
- frame construction,
- projection services,
- realization-aware states,
- dynamic quantities,
- and vehicle-space interpretation.

Relation to the AIM Roadmap

Remark 35.66. Within the AIM roadmap, the present part corresponds to the transition from dynamic state evolution toward operational runtime execution.



Remark 35.67. The progression

Runtime Dynamics $\rightarrow$ Runtime Services $\rightarrow$ Realization

marks the transition from state evolution toward operational execution and evaluation.

Principle 35.68 (Runtime Layer Principle). Within AIM, operational quantities should be generated by runtime operators from underlying railway states rather than stored redundantly as independent truth.

Summary 35.69. Dynamics explain state evolution.

Runtime explains operational execution.

The runtime layer therefore provides the bridge between mathematical state models and interactive, realization-aware railway engineering systems.

36. Runtime Services

Motivation

Remark 36.1. Railway alignment systems are not static collections of stored geometry.

Remark 36.2. Within AIM, geometry, frames, projections, realization effects, dynamic quantities, and vehicle-space behaviour are evaluated during runtime.

Remark 36.3. The runtime layer forms the operational bridge between:

- stored alignment semantics,
- metric-aware geometric evaluation,
- visualization,
- optimization,
- and interaction with real-world data.

Remark 36.4. Runtime therefore acts as the operational execution layer of the AIM framework.

36.1. Runtime Interpretation

Definition 36.5 (Runtime service). A runtime service is an operator-based evaluation process acting on alignment states during operational use.

Remark 36.6. Runtime services evaluate:

- geometry,
- projections,
- frames,
- realization-dependent states,
- dynamic quantities,
- and operational vehicle-space behaviour.

Principle 36.7 (Runtime evaluation principle). Operational quantities should be evaluated from underlying state descriptions rather than stored redundantly.

Remark 36.8. Within AIM, runtime does not merely denote software execution. It denotes dynamic operational evaluation of geometric, physical, and spatial railway quantities.

36.2. Runtime State Evaluation

36.2.1. Evaluation Operator

Remark 36.9. Runtime evaluation is governed by the evaluation operator introduced in theorem 9.38.

Definition 36.10 (Runtime evaluation). Runtime evaluation is represented by:

$$\mathcal{E} : \mathcal{X} \times \Omega_{\text{track}} \rightarrow \mathcal{Y}.$$

Remark 36.11. Here, $\mathcal{X}$ denotes railway state space, Ω_{track} denotes intrinsic track space, and $\mathcal{Y}$ denotes runtime-evaluated quantities.

Remark 36.12. Runtime evaluation may operate on:

- target states,
- realized states,
- measured states,
- or temporary interaction states.

36.2.2. Typical Runtime Quantities

Remark 36.13. Typical runtime quantities include:

- world coordinates,
- tangent directions,
- local frames,
- curvature,
- cant,
- geodesic curvature,
- dynamic quantities,
- and vehicle-space geometry.

36.3. Persistent State versus Runtime State

Remark 36.14. Within AIM, the persistent model intentionally stores sparse and semantically stable alignment information.

Remark 36.15. Operational geometry is reconstructed dynamically during runtime.

Principle 36.16 (Runtime reconstruction principle). Runtime geometry should be reconstructed from sparse alignment semantics rather than stored redundantly.

Remark 36.17. This avoids:

- synchronization instability,
- redundant geometry storage,
- and unnecessary realization inconsistencies.

36.4. pose3 as Runtime State

Remark 36.18. The minimal intrinsic pose3 notation is introduced in theorem 6.39.

Remark 36.19. Within runtime evaluation, pose3 is interpreted as a spatial operational state generated from:

- intrinsic pose2 reconstruction,
- profile evaluation,
- cant evaluation,
- metric realization,
- and frame construction.

Definition 36.20 (Runtime pose3 evaluation). Runtime pose3 evaluation may be represented schematically as:

$$\text{pose3}_{\text{run}}(s) = \mathcal{E}_{\text{pose3}}(x, s).$$

Remark 36.21. The runtime pose3 state is therefore not persistent truth. It is an evaluated operational state.

36.5. Lazy Evaluation

Definition 36.22 (Lazy evaluation). Lazy evaluation computes quantities only when explicitly required.

Remark 36.23. This avoids unnecessary evaluation of expensive geometric operations.

Proposition 36.24 (Selective evaluation principle). *Exact evaluation should be performed only where precision is required.*

Remark 36.25. This principle becomes essential for:

- world-to-track projection,
- large-scale visualization,

- optimization,
- interactive runtime systems,
- and vehicle-space evaluation.

36.6. Projection Services

Remark 36.26. Runtime projection services are governed by the projection operators introduced in:

- theorem 9.22,
- and theorem 9.24.

36.6.1. Track-to-World Evaluation

Definition 36.27 (Track-to-world runtime service). The track-to-world runtime service evaluates:

$$\mathcal{P}_{\text{tw}} : \Omega_{\text{track}} \rightarrow \Omega_{\text{world}}.$$

36.6.2. World-to-Track Evaluation

Definition 36.28 (World-to-track runtime service). The inverse runtime projection service evaluates:

$$\mathcal{P}_{\text{wt}} : \Omega_{\text{world}} \rightarrow \Omega_{\text{track}}.$$

Remark 36.29. The inverse projection is:

- nonlinear,
- potentially ambiguous,
- metric-dependent,
- context-dependent,
- and computationally expensive.

Remark 36.30. Projection must not be confused with CRS coordinate transformation. The distinction is established in theorem 9.27.

36.7. Frame Services

Remark 36.31. Runtime frame construction is governed by the frame operator introduced in theorem 9.41.

Definition 36.32 (Frame service). The frame service evaluates:

$$\mathcal{F}(x_{\text{metric}}, s) = (\mathbf{t}(s), \mathbf{n}(s), \mathbf{b}(s)).$$

Remark 36.33. Frame services support:

- cant interpretation,
- pose3 evaluation,
- operational geometry,
- vehicle-space interpretation,
- visualization,
- and runtime dynamics.

Remark 36.34. Frame evaluation may depend on metric realization. For example, projected engineering frames and ellipsoid-attached frames need not coincide.

36.8. Dynamic Runtime Quantities

36.8.1. Operational Dynamics

Remark 36.35. Many operational railway quantities are runtime-derived rather than persistent alignment quantities.

36.8.2. Effective Lateral Acceleration

Definition 36.36 (Effective lateral acceleration).

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi.$$

Remark 36.37. This is the simplified engineering relation introduced canonically in eq. (7.10).

36.8.3. Jerk

Definition 36.38 (Runtime jerk).

$$j = v^3 \kappa' - g v \cos \phi \phi'.$$

Remark 36.39. This is the runtime form of the canonical jerk relation in eq. (7.12).

Remark 36.40. These quantities emerge from:

- curvature evolution,
- cant evolution,

- runtime frame evaluation,
- metric realization,
- and vehicle-state interpretation.

36.9. Hybrid Runtime Evaluation

Principle 36.41 (Hybrid runtime principle). Efficient runtime systems combine:

- approximate coarse evaluation,
- cached representations,
- and exact local refinement.

Remark 36.42. Typical hybrid strategies combine:

- sampled lookup structures,
- sparse parametric models,
- local numerical refinement,
- and context-dependent precision selection.

Remark 36.43. Hybrid runtime evaluation is especially important for world-to-track projection, frame construction, and large-scale visualization.

36.10. Level-of-Detail Runtime

Definition 36.44 (LOD-dependent evaluation). Runtime evaluation depends on:

- scale,
- precision requirements,
- visibility,
- operational context,
- interaction state,
- and runtime cost.

Remark 36.45. Examples include:

- coarse map rendering,
- exact local projection,

- simplified distant geometry,
- high-precision construction-scale evaluation,
- and precise nearby vehicle interaction.

Remark 36.46. LOD-dependent runtime evaluation is one of the central scalability mechanisms of AIM.

36.11. Caching

Definition 36.47 (Runtime cache). A runtime cache stores previously evaluated quantities to avoid repeated computation.

Remark 36.48. Caching may be applied to:

- projection results,
- local frames,
- sampled geometry,
- dynamic quantities,
- nearest-point queries,
- and metric-aware reconstruction results.

Remark 36.49. Caching introduces a trade-off between:

- computational cost,
- memory usage,
- consistency,
- invalidation complexity,
- and numerical freshness.

36.12. Metric-Aware Runtime

Remark 36.50. Runtime evaluation depends on the metric realization layer introduced in chapter 11.

Remark 36.51. Different metric realizations may affect:

- distance interpretation,
- curvature evaluation,

- projection behaviour,
- frame construction,
- stationing,
- and pose3 evaluation.

Principle 36.52 (Metric-aware runtime principle). Runtime systems should evaluate geometry through metric-aware operator services rather than through hardcoded Euclidean assumptions.

Remark 36.53. Metric-aware runtime preserves fast Euclidean engineering evaluation as the default case while allowing projection-aware and ellipsoid-aware extensions.

36.13. Runtime Pipelines

36.13.1. Conceptual Composition

Definition 36.54 (Runtime operator pipeline). A runtime pipeline is a composed operator chain such as:

$$\mathcal{E}_{\text{run}} \sim \mathcal{P} \circ \mathcal{F} \circ \mathcal{G}.$$

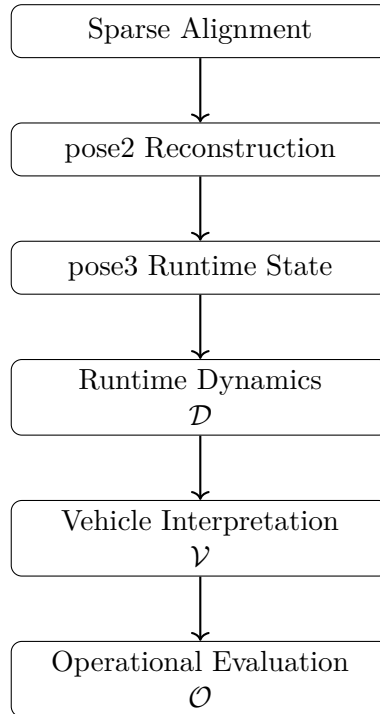


Abbildung 36.1.: Conceptual runtime evaluation pipeline.

Remark 36.55. This pipeline expresses that runtime evaluation may depend on:

- metric realization,

- frame evaluation,
- and projection services.

36.13.2. Realized Runtime Evaluation

Definition 36.56 (Realized runtime pipeline). When realized geometry is required, runtime evaluation may conceptually depend on:

$$\mathcal{E}_{\text{real}} \sim \mathcal{P} \circ \mathcal{F} \circ \mathcal{G} \circ \mathcal{R}.$$

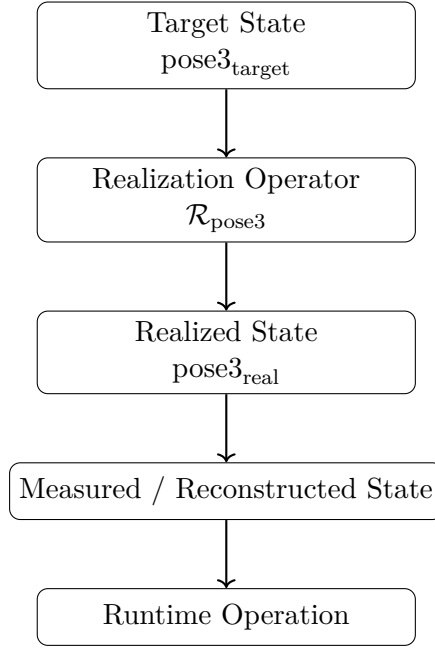


Abbildung 36.2.: Realization-aware runtime evaluation pipeline.

Remark 36.57. This expresses the dependency on physical realization, but it should not be read as a rigid software pipeline.

36.13.3. Runtime Pipeline Variants

Remark 36.58. Different runtime services may activate different operator chains. For example:

- visualization may use coarse metric evaluation,
- projection may require local refinement,
- optimization may require realized-state evaluation,
- and vehicle-space analysis may require frame and cant evaluation.

36.14. Runtime and Realization

Remark 36.59. Runtime systems should distinguish:

- target geometry,
- realized geometry,
- measured geometry,
- and temporary interaction geometry.

Remark 36.60. Operational services often operate on realized states:

$$\mathcal{X}_{\text{real}}.$$

Remark 36.61. Runtime evaluation and physical realization are coupled, but they are not identical. Runtime evaluates states; realization transforms target states into realized states.

36.15. Runtime and Vehicle Space

Remark 36.62. Vehicle-space interpretation requires runtime evaluation of:

- cant,
- local frames,
- vehicle envelopes,
- platform interaction,
- bogie motion,
- center-of-gravity motion,
- and operational clearances.

Remark 36.63. These quantities cannot be represented sufficiently by static centerline geometry alone.

Remark 36.64. Vehicle-space interpretation therefore depends on runtime pose3 evaluation rather than on stored centerline geometry alone.

36.16. Conceptual Runtime Architecture

36.17. Canonical Interpretation

Proposition 36.65. *Within AIM, runtime services are operator-based evaluation systems acting on alignment states.*

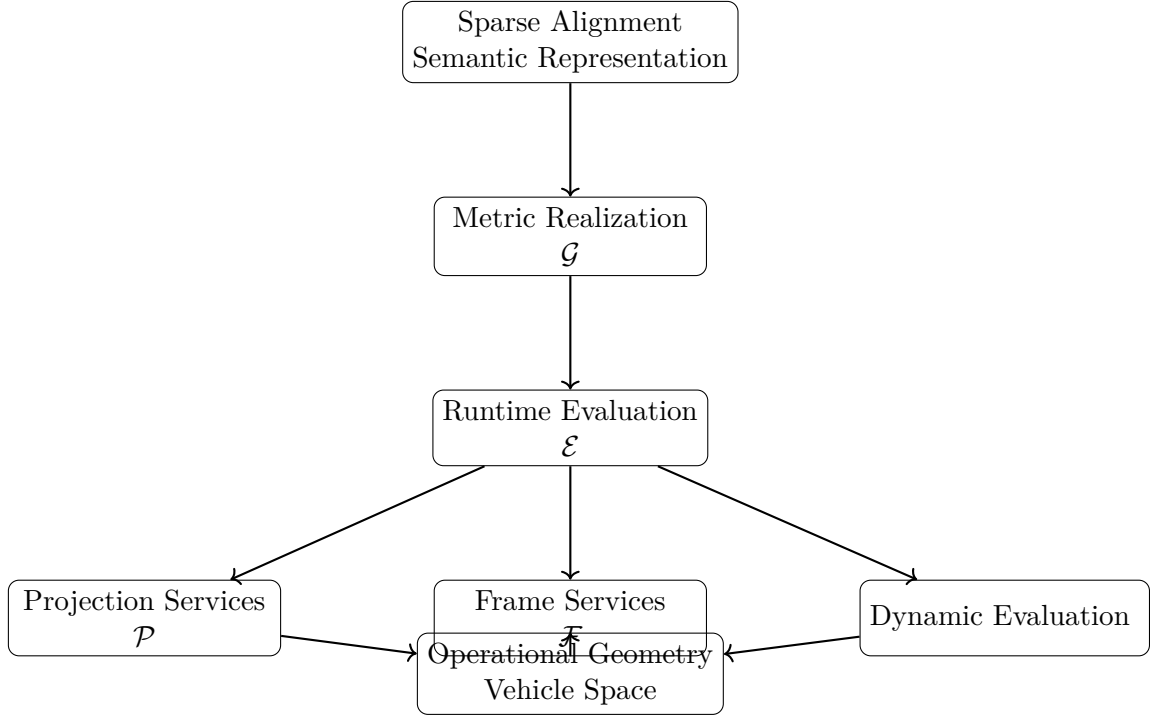


Abbildung 36.3.: Runtime services transform sparse alignment semantics through metric realization into operationally interpretable geometry and vehicle-space behaviour.

Proposition 36.66. *Runtime geometry is interpreted as dynamically evaluated operational geometry rather than static stored geometry.*

Proposition 36.67. *Projection, metric realization, frame interpretation, dynamics, and realization-aware evaluation form one coupled runtime architecture.*

36.18. Connection to AIM

Summary 36.68. The AIM runtime architecture transforms sparse alignment semantics into operationally interpretable geometry.

Runtime services provide:

- projection,
- frame evaluation,
- pose3 evaluation,
- dynamic evaluation,
- realization-aware geometry,
- metric-aware evaluation,

- and vehicle-space interpretation.

Thus, runtime evaluation forms the operational execution layer of the AIM framework.

AIM is therefore not merely a stored alignment model, but an operational evaluation system.

Teil VII.

Reality and Data

Reality

What do we already know?

Remark 36.69. The previous parts established a complete mathematical description of railway alignment geometry, railway states, runtime interpretation, and operational evaluation.

Remark 36.70. Within AIM, geometry emerges from intrinsic curvature evolution, railway states describe spatial configuration, and runtime services generate operational quantities through evaluation operators.

Remark 36.71. These constructions provide a coherent idealized railway model.

What is still missing?

Remark 36.72. Real railway systems do not exist purely within mathematical models.

Remark 36.73. Railway alignments are embedded in coordinate systems, measurement processes, construction workflows, maintenance histories, and operational environments.

Remark 36.74. Measurements are imperfect.

Coordinate systems may be ambiguous.

Constructed infrastructure deviates from design intent.

Observed geometry may differ from modeled geometry.

Remark 36.75. The missing element is therefore the interaction between the ideal alignment model and physical reality.

What comes next?

Remark 36.76. This part introduces the reality layer of AIM.

Remark 36.77. The chapters that follow investigate:

- coordinate reference systems,
- measurement and uncertainty,
- stationing systems,
- topology and special railway elements,
- construction processes,

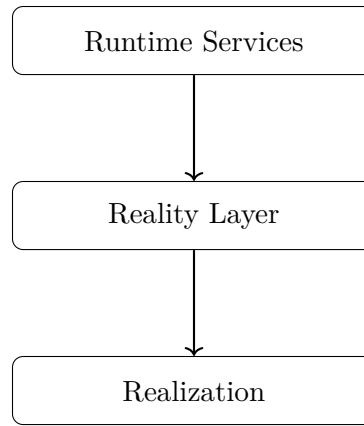
- and realized railway geometry.

Remark 36.78. Particular emphasis is placed on the distinction between *designed* and *realized* railway states.

Remark 36.79. This distinction ultimately leads to the concept of realization-aware alignment modelling, one of the central ideas underlying AIM.

Relation to the AIM Roadmap

Remark 36.80. Within the AIM roadmap, the present part corresponds to the transition from ideal runtime evaluation toward interaction with the physical world.



Remark 36.81. The progression

Runtime Services $\rightarrow$ Reality Layer $\rightarrow$ Realization

marks the transition from ideal state evaluation toward interaction with measured and constructed railway systems.

Principle 36.82 (Reality Principle). Within AIM, railway models and physical railway systems must be treated as related but distinct entities connected through measurement, construction, realization, and interpretation processes.

Summary 36.83. Geometry explains shape.

States explain configuration.

Runtime explains evaluation.

Reality explains how mathematical railway models interact with the physical world.

The present part therefore establishes the bridge between ideal railway states and realized railway infrastructure.

37. Measurement Data and Reality

Motivation

Remark 37.1. In practical applications, alignment geometry is rarely available in a clean, analytic form. Instead, it is derived from heterogeneous data sources with varying levels of precision, structure, and semantic completeness.

Remark 37.2. Therefore, the modelling process must explicitly distinguish between

- observed data,
 - interpreted structure,
 - and reconstructed geometry.
-

37.1. Sources of Data

Definition 37.3 (Data source). A data source is any origin of geometric or alignment-related information.

Remark 37.4. Typical sources include:

- survey measurements,
 - legacy engineering data,
 - container formats such as **LandXML**,
 - railway-specific formats,
 - point clouds and derived datasets.
-

37.2. Nature of Real-World Data

Remark 37.5. Real-world data exhibits several characteristic properties:

- **Incompleteness:** Not all required information is present.

- **Noise:** Measurements are subject to error.
- **Ambiguity:** Multiple interpretations may be possible.
- **Inconsistency:** Different parts of the data may contradict each other.

Remark 37.6. As a consequence, data cannot be directly equated with geometry.

37.3. Coordinate Systems

Remark 37.7. Geometric data is always embedded in a coordinate reference system (CRS).

Definition 37.8 (Coordinate reference system). A coordinate reference system defines how numerical coordinates relate to positions in physical space.

Remark 37.9. In infrastructure applications, multiple systems may coexist, including:

- global systems (e.g. WGS84),
- projected systems (e.g. UTM, Gauss–Krüger),
- domain-specific systems (e.g. railway reference systems).

OPEN: Formal integration of CRS handling into the alignment model.

RESEARCH: Transformation consistency across heterogeneous data sources.

37.4. Interpretation Layer

Definition 37.10 (Interpretation). Interpretation is the process of assigning structure and meaning to raw data.

Remark 37.11. Typical interpretation tasks include:

- identifying alignment segments,
- reconstructing curvature behaviour,
- associating auxiliary data such as profile or cant,
- resolving ambiguities in topology.

Remark 37.12. Interpretation is inherently non-unique and often requires domain knowledge.

OPEN: Formal separation between data and interpretation.

37.5. From Data to Sparse Model

Remark 37.13. The sparse alignment model provides a target structure for data interpretation.

Definition 37.14 (Reconstruction from data). Reconstruction is the process of deriving a sparse alignment model from a given data source.

Remark 37.15. This process typically involves:

- segmentation of data,
- classification of element types,
- estimation of curvature laws,
- assignment of initial conditions.

TODO: Formal pipeline from raw data to sparse alignment.

OPEN: Handling of incomplete or conflicting input data.

37.6. Container Formats

Remark 37.16. Many real-world datasets are provided in structured container formats.

Remark 37.17. Examples include:

- LandXML,
- railway-specific XML formats,
- spreadsheet-based engineering datasets.

Remark 37.18. Such formats often mix geometric, semantic, and metadata in ways that are not directly aligned with the sparse model.

RESEARCH: Mapping of container formats to sparse representation.

TODO: Definition of intermediate formats (e.g. landFAT).

37.7. Uncertainty and Error

Remark 37.19. All real-world data is subject to uncertainty.

Definition 37.20 (Measurement uncertainty). Measurement uncertainty describes the deviation between observed data and the true physical quantity.

Remark 37.21. Uncertainty propagates through:

- interpretation,
- reconstruction,
- numerical integration.

RESEARCH: Propagation of uncertainty through the reconstruction pipeline.

OPEN: Integration of uncertainty into the sparse model.

37.8. Topological vs Geometric Information

Remark 37.22. Data may contain both geometric and topological information.

Definition 37.23 (Geometric information). Geometric information describes positions, directions, and curvature.

Definition 37.24 (Topological information). Topological information describes connectivity and relationships between elements.

Remark 37.25. In many datasets, these two aspects are interwoven and not cleanly separated.

OPEN: Separation and integration of topology and geometry in the model.

37.9. Engineering Reality

Remark 37.26. Engineering data is often shaped by practical constraints rather than mathematical clarity.

Remark 37.27. Typical phenomena include:

- piecewise definitions without explicit continuity guarantees,
- implicit assumptions,
- domain-specific conventions,
- historical artefacts in legacy data.

RESEARCH: Systematic classification of real-world alignment data quality.

37.10. Role within the Overall Framework

Remark 37.28. The data layer connects the abstract modelling framework to real-world applications.

Remark 37.29. It forms the entry point for:

- import pipelines,
 - validation procedures,
 - reconstruction workflows.
-

37.11. Summary

Summary 37.30. Real-world alignment data is heterogeneous, noisy, and often ambiguous.

A clear separation between raw data, interpretation, and reconstructed geometry is essential.

The sparse alignment model provides a structured target for reconstruction, while numerical methods and domain knowledge are required to bridge the gap between data and geometry.

38. Coordinate Systems and Spatial Reference

Motivation

Remark 38.1. Railway alignment data is inherently spatial. Its interpretation depends fundamentally on the coordinate system in which it is expressed.

Remark 38.2. In practice, railway projects involve multiple coordinate reference systems (CRS), often used simultaneously.

Remark 38.3. Failure to handle coordinate systems consistently leads to:

- geometric inconsistencies,
- loss of accuracy,
- and incorrect integration of data sources.

CRS is not Metadata

In many software systems, the coordinate reference system (CRS) is treated as metadata.

It is stored, displayed, and occasionally converted — but rarely considered part of the model itself.

In railway alignment modelling, this view is fundamentally wrong.

The CRS determines:

- *how distances are measured,*
- *how curvature is interpreted,*
- *and how different datasets relate to each other.*

A shift of a few centimeters in coordinates may appear negligible.

A change in curvature is not.

If coordinate systems are inconsistent:

- *alignments no longer match,*

38. Coordinate Systems and Spatial Reference

- *curvature estimates become unstable,*
- *and optimization results lose physical meaning.*

The problem is not numerical.

It is structural.

A railway alignment does not exist independently of its spatial reference.

The coordinate system is part of the model.

38.1. Coordinate Reference Systems

Definition 38.4 (Coordinate reference system). A coordinate reference system (CRS) is a mathematical framework that defines how spatial coordinates relate to positions in the real world.

Remark 38.5. A CRS typically consists of:

- a reference surface (e.g. ellipsoid),
- a coordinate system (e.g. Cartesian or projected),
- and a transformation to real-world locations.

38.2. Global and Local Systems

Remark 38.6. Railway data commonly uses both global and local coordinate systems:

- global systems (e.g. WGS84),
- projected systems (e.g. UTM, Gauss–Krüger),
- engineering systems (e.g. track-based coordinates).

Remark 38.7. Each system serves a different purpose:

- global systems enable interoperability,
- projected systems provide metric consistency,
- local systems support engineering workflows.

38.3. Engineering Coordinate Systems

Remark 38.8. In railway engineering, alignments are often described in track-relative coordinate systems.

Definition 38.9 (Stationing). Let s denote the arc-length parameter along an alignment. This parameter is referred to as *stationing*.

Remark 38.10. Stationing defines a one-dimensional coordinate system along the track, independent of the global spatial reference.

Remark 38.11. Additional local coordinates may include:

- lateral offsets,
- vertical offsets,
- and local reference frames attached to the track.

38.4. Multiple CRS in Practice

Remark 38.12. Real-world projects frequently involve multiple CRS simultaneously.

Example 38.13. Typical combinations include:

- survey data in local engineering coordinates,
- GIS data in WGS84,
- design data in projected systems (UTM or GK).

Remark 38.14. These systems are often not perfectly consistent, due to:

- measurement errors,
- transformation approximations,
- and legacy definitions.

38.5. Transformations

Definition 38.15 (Coordinate transformation). A coordinate transformation is a mapping

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

that converts coordinates from one CRS to another.

Remark 38.16. Transformations may include:

- translations,
- rotations,
- scaling,

- and projection changes.

Remark 38.17. In practice, transformations are often defined by:

- reference points,
- parameter sets,
- or standardized transformation models.

38.6. Accuracy and Consistency

Remark 38.18. Coordinate transformations introduce uncertainties.

Remark 38.19. Even small inconsistencies can lead to:

- visible misalignment of datasets,
- incorrect curvature estimation,
- and unstable numerical reconstruction.

Remark 38.20. Therefore, a modelling framework must:

- track the CRS of all data,
- apply transformations explicitly,
- and avoid implicit assumptions.

38.7. CRS in Alignment Modelling

Remark 38.21. In a curvature-based alignment model, geometry is reconstructed from functions such as $\kappa(s)$.

Remark 38.22. This reconstruction requires a consistent spatial reference:

- initial position,
- initial orientation,
- and coordinate system definition.

Remark 38.23. Thus, CRS information is not auxiliary, but an essential component of the alignment model.

38.8. Challenges

Remark 38.24. The integration of CRS into alignment modelling raises several challenges:

- handling heterogeneous input data,
- maintaining numerical stability,
- supporting large coordinate values,
- and enabling local computations with high precision.

Remark 38.25. In particular, large global coordinates may lead to numerical issues in floating-point computations.

38.9. Local Coordinate Frames

Remark 38.26. To address numerical issues, local coordinate frames are often used.

Definition 38.27 (Local frame). A local frame is a coordinate system defined relative to a reference point, typically near the area of interest.

Remark 38.28. Using local frames:

- improves numerical stability,
- simplifies geometric computations,
- and allows efficient visualization.

38.10. Towards a Coordinate-Aware Model

Remark 38.29. A modern alignment modelling framework must be coordinate-aware.

Remark 38.30. This includes:

- explicit CRS representation,
- consistent transformation handling,
- integration with geometry reconstruction,
- and support for local and global coordinates.

38.11. Connection to the AIM Framework

Summary 38.31. Coordinate systems are a fundamental aspect of railway alignment data.

A consistent treatment of CRS is essential for integrating real-world data, ensuring numerical stability, and enabling reliable modelling.

Within the AIM framework, CRS is treated as an explicit component of the alignment state, rather than as external metadata.

38.12. Alignment Modelling Beyond Map Projections

38.12.1. Motivation

Remark 38.32. Conventional railway alignment design is typically performed in planar coordinate systems derived from map projections, such as Gauss–Krüger, UTM, or Mercator.

Remark 38.33. These projections introduce distortions in length, angle, and area, which are controlled but fundamentally unavoidable.

Remark 38.34. For large-scale infrastructure or high-precision applications, these distortions become conceptually and computationally relevant.

38.12.2. Intrinsic Surface Formulation

Remark 38.35. An alternative perspective is to formulate alignment geometry directly on the Earth’s reference surface, typically modeled as an ellipsoid.

Definition 38.36 (Surface-based alignment). An alignment is defined as a curve

$$\gamma : s \mapsto \mathcal{E},$$

where $\mathcal{E}$ denotes the reference ellipsoid.

Remark 38.37. In this formulation, arc-length is measured intrinsically on the surface, and curvature is defined with respect to the surface geometry.

38.12.3. Geometric Consequences

Remark 38.38. Curves on the ellipsoid are no longer planar:

- geodesics replace straight lines,
- curvature must be interpreted in a surface-intrinsic sense,
- normal vectors depend on the local surface geometry.

Remark 38.39. Thus, the classical decomposition into planar geometry and vertical profile becomes insufficient.

38.12.4. Relation to AIM

Remark 38.40. The AIM framework is inherently curvature-based and therefore not tied to a specific embedding space.

Remark 38.41. In particular, the representation

$$\kappa(s)$$

remains valid on curved surfaces, provided curvature is defined with respect to the surface metric.

38.12.5. Practical Considerations

Remark 38.42. Despite its conceptual appeal, a full ellipsoidal formulation introduces significant complexity:

- nonlinear surface geometry,
- increased computational cost,
- limited compatibility with existing engineering workflows.

Remark 38.43. For this reason, planar projections remain the dominant approach in practice.

38.12.6. Alignment as Curve on a Surface

Remark 38.44. The Earth is modeled as a reference surface $\mathcal{E}$, typically an ellipsoid.

Definition 38.45 (Surface alignment). An alignment can be interpreted as a curve

$$\gamma : s \mapsto \mathcal{E},$$

embedded in the surface.

Remark 38.46. In this setting, curvature decomposes into:

- intrinsic curvature (within the surface),
- normal curvature (with respect to embedding).

Remark 38.47. Straight lines in planar projections correspond to geodesics on the surface, which are not necessarily suitable as railway alignments.

Remark 38.48. Thus, alignment design is not equivalent to geodesic design.

38.12.7. Vision

Summary 38.49. AIM opens the possibility of alignment modelling directly on the Earth's reference surface.

While current engineering practice relies on planar projections, future systems may operate intrinsically on the ellipsoid, avoiding projection distortions and enabling globally consistent modelling.

This perspective represents a long-term research direction rather than an immediate engineering requirement.

Remark 38.50. In this sense, map projections can be understood as implementation choices rather than fundamental modelling constraints.

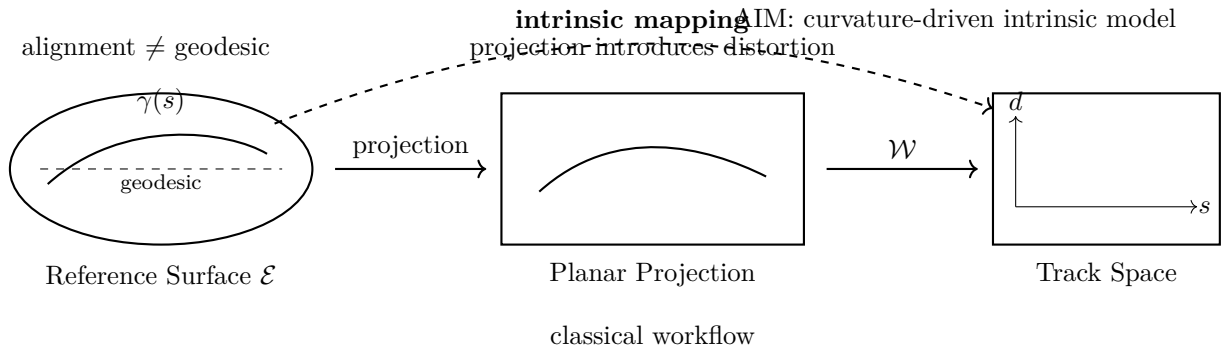


Abbildung 38.1.: Alignment modelling across geometric representations. On the reference surface, alignments are curves that generally differ from geodesics. Classical workflows project geometry to a plane before further processing, introducing distortions. The AIM perspective enables direct intrinsic modelling and mapping to track coordinates, independent of projection.

38.13. Physical Validity and Coordinate Systems

38.13.1. Current Practice

Remark 38.51. In current engineering workflows, railway alignment design and validation are typically performed in projected coordinate systems (e.g. Gauss–Krüger, UTM).

Remark 38.52. All geometric quantities, including curvature, distances, and derived dynamic measures, are evaluated within these planar representations.

Remark 38.53. However, such coordinate systems are approximations of the Earth’s geometry and introduce distortions:

- metric distortion (scale),
- angular distortion,
- non-uniform curvature representation.

38.13.2. Fundamental Limitation

Proposition 38.54. *Physical quantities derived in a projected coordinate system are not strictly consistent with the underlying geometry of the Earth.*

Remark 38.55. In particular:

- arc-length is only approximately preserved,
- curvature is distorted by projection,
- dynamic quantities (e.g. acceleration, jerk) are evaluated on an approximate geometry.

Remark 38.56. Thus, current alignment validation is performed in an *engineering approximation space*, not in the physical space in which the railway actually exists.

38.13.3. Ellipsoid-Based Perspective

Remark 38.57. A more physically consistent approach is to formulate alignment geometry directly on a reference ellipsoid.

Remark 38.58. In this setting:

- arc-length is defined intrinsically,
- curvature is defined with respect to the surface geometry,
- transformations to world coordinates are exact (up to numerical precision).

38.13.4. Differential-Geometric Formulation

Definition 38.59 (Intrinsic alignment on a surface). Let $\mathcal{E}$ denote a reference ellipsoid. An alignment is defined as a curve

$$\gamma : s \mapsto \mathcal{E},$$

with curvature defined intrinsically via differential geometry.

Remark 38.60. In this formulation:

- curvature is a geometric invariant,
- arc-length is physically meaningful,
- projection artifacts are eliminated.

38.13.5. Implication for Dynamics

Remark 38.61. Dynamic quantities such as lateral acceleration and jerk depend on curvature and arc-length derivatives.

Remark 38.62. If these are evaluated in a distorted coordinate system, the resulting values are only approximate.

Proposition 38.63. *Evaluating dynamics on an intrinsic (ellipsoid-based) geometry yields more physically accurate and reliable results.*

38.13.6. Safety Perspective

Remark 38.64. Railway alignment design directly affects safety-critical quantities:

- lateral acceleration,
- cant deficiency,
- jerk.

Remark 38.65. Errors introduced by coordinate distortions may be small individually, but can accumulate over long distances or in high-speed scenarios.

Proposition 38.66. *A geometrically consistent modelling framework improves the robustness and safety of alignment design.*

38.13.7. Vision

Remark 38.67. The long-term vision is to move from projection-based engineering geometry to intrinsic, surface-based modelling.

Remark 38.68. This implies:

- alignment design on the ellipsoid,
- differential-geometric formulation of curvature and dynamics,
- projection used only for visualization and data exchange.

Summary 38.69. Current railway alignment validation is performed in projected coordinate systems and is therefore only approximately physical.

A shift toward ellipsoid-based or differential-geometric modelling enables:

- physically consistent curvature,
- accurate dynamic evaluation,
- increased robustness,
- and ultimately safer alignment design.

This represents a fundamental extension of current engineering practice within the AIM framework.

Remark 38.70. Curvature evaluated in projected coordinate systems differs from intrinsic curvature on the ellipsoid.

See chapter 13.

39. Kilometre Line and Stationing

Motivation

Remark 39.1. Railway infrastructure is not only described geometrically, but also by stationing.

Remark 39.2. The kilometre line defines the longitudinal reference used for addressing positions along a railway route.

Remark 39.3. It must not be confused with the physical track centerline.

39.1. Definition

Definition 39.4 (Kilometre line). A kilometre line is a longitudinal reference alignment that assigns a station value to positions along a railway route.

Remark 39.5. The kilometre line may coincide with a track centerline, but this is not required.

39.2. Relation to Track Alignments

Remark 39.6. Several track alignments may refer to the same kilometre line.

Remark 39.7. This is common in double-track lines, junctions, station areas, and switching zones.

Remark 39.8. In single-track cases, the kilometre line and the track centerline may be identical.

39.3. Outer Stationing

Remark 39.9. When the physical track alignment differs from the kilometre line, positions along the track must be related back to the kilometre reference.

Definition 39.10 (Outer stationing). Outer stationing describes the mapping between a track alignment and an external kilometre reference.

39.4. Topological Role

Remark 39.11. The kilometre line acts as a topological and administrative reference rather than merely a geometric object.

Remark 39.12. It provides continuity across track changes, parallel tracks, and operationally relevant segments.

39.5. Implication for AIM

Summary 39.13. Within AIM, the kilometre line is treated as a reference structure separate from physical track alignments.

This distinction is essential for modelling multi-track situations, switches, stationing jumps, and real-world railway project data.

40. Kilometre Line, Stationing, and Spatial Reference

Motivation

Remark 40.1. Railway infrastructure is not only described by geometry, but also by stationing.

Remark 40.2. A point on a railway route is usually not addressed by global coordinates, but by a kilometre value along a reference line.

Remark 40.3. Thus, railway modelling must distinguish between:

- geometric location,
- spatial reference system,
- and operational addressing.

40.1. Three Reference Layers

Definition 40.4 (Spatial reference layer). The spatial reference layer defines where geometry lives. It may be based on:

- a planar projection,
- a local engineering coordinate system,
- or, in a long-term view, a reference ellipsoid.

Definition 40.5 (Geometric alignment layer). The geometric alignment layer describes the physical or designed track geometry, typically through curvature, pose, profile, and cant.

Definition 40.6 (Stationing layer). The stationing layer assigns address values to positions along a railway route.

Summary 40.7. Spatial reference answers: where is it? Geometry answers: what shape is it? Stationing answers: how is it addressed?

40.2. Kilometre Line

Definition 40.8 (Kilometre line). A kilometre line is a longitudinal reference structure used to assign station values to positions within a railway route or project.

Remark 40.9. The kilometre line may coincide with a physical track centerline, but this is not required.

Remark 40.10. It is therefore incorrect to treat the kilometre line simply as another track alignment.

40.3. Relation to Track Geometry

Remark 40.11. Several physical track alignments may refer to the same kilometre line.

Remark 40.12. This occurs in:

- double-track lines,
- station areas,
- junctions,
- switches and crossings,
- temporary or construction-stage alignments.

Remark 40.13. Conversely, in simple single-track situations, the kilometre line and track centerline may be identical.

40.4. Outer Stationing

Definition 40.14 (Outer stationing). Outer stationing describes the mapping between a physical track alignment and an external kilometre reference.

Remark 40.15. Outer stationing is required whenever the track geometry and the kilometre reference do not coincide.

Remark 40.16. In such cases, a point on the track has at least two relevant longitudinal coordinates:

- its intrinsic track coordinate s_{track} ,
- its assigned kilometre value s_{km} .

40.5. Relation to Ellipsoid and CRS

Remark 40.17. The kilometre line is independent of the coordinate reference system in which the geometry is represented.

Remark 40.18. A kilometre value is not a projected coordinate. It is an addressing value along a railway-specific reference structure.

Remark 40.19. Thus, even if geometry is formulated on a reference ellipsoid, in a Gauss–Krüger projection, or in a local engineering CRS, the kilometre line remains a separate semantic layer.

40.6. Reference Separation Principle

Proposition 40.20. *A railway project requires separation of at least three reference concepts:*

$$\textit{spatial reference} \neq \textit{geometric alignment} \neq \textit{stationing reference}.$$

Remark 40.21. Confusing these layers leads to systematic modelling errors, especially in multi-track and junction situations.

40.7. Implication for AIM

Remark 40.22. Within AIM, the kilometre line is treated as a reference object, not as the primary physical track geometry.

Remark 40.23. Track alignments may be assigned to kilometre lines by explicit stationing relations.

Remark 40.24. This allows AIM to model cases where:

- one kilometre line serves multiple tracks,
- a track has no independent kilometre line,
- stationing jumps occur,
- geometric and operational continuity differ.

40.8. Relation to the Seven-Line Model

Remark 40.25. The kilometre line plays a central role in the seven-line import and sorting model.

Remark 40.26. A simplified structure is:

$$\text{km-line} \rightarrow \begin{cases} \text{right track} \rightarrow (\text{profile, cant}), \\ \text{left track} \rightarrow (\text{profile, cant}), \\ \text{outer stationing.} \end{cases}$$

Remark 40.27. This expresses that the kilometre line provides the addressing context, while the tracks provide the physical geometry and engineering state.

40.9. Key Insight

Proposition 40.28. *The kilometre line is not primarily a geometric curve, but an addressing structure for railway infrastructure.*

Summary 40.29. The km-line, the geometric alignment, and the spatial reference system serve different roles.

The km-line addresses railway infrastructure.

The alignment describes physical geometry.

The CRS or ellipsoid defines the spatial embedding.

Keeping these layers separate is essential for robust railway modelling, especially in multi-track, station, and junction contexts.

41. Measurement and Imperfect Data

Motivation

Remark 41.1. Railway alignment modelling is not performed in an ideal mathematical environment, but based on measured and legacy data.

Remark 41.2. Such data is inherently imperfect. It contains noise, inconsistencies, and incomplete information.

Remark 41.3. A realistic modelling framework must therefore account explicitly for data imperfections.

41.1. Sources of Data

Remark 41.4. Typical sources of railway alignment data include:

- geodetic surveys,
- track measurement systems,
- legacy design files,
- and extracted data from infrastructure models.

Remark 41.5. These sources differ significantly in:

- accuracy,
- resolution,
- completeness,
- and coordinate reference systems.

41.2. Types of Imperfections

41.2.1. Measurement Noise

Definition 41.6 (Noise). Let $p_i \in \mathbb{R}^2$ denote measured points. Noise is the deviation between measured and true positions.

Remark 41.7. Noise arises from:

- instrument precision limits,
- environmental effects,
- and data processing steps.

41.2.2. Systematic Errors

Definition 41.8 (Systematic error). A systematic error is a consistent deviation affecting a dataset.

Remark 41.9. Examples include:

- incorrect CRS parameters,
- transformation offsets,
- calibration errors.

41.2.3. Incompleteness

Remark 41.10. Data may be incomplete:

- missing segments,
- missing metadata,
- or missing structural information.

41.2.4. Topological Ambiguity

Remark 41.11. In complex railway systems, data may not explicitly encode topology.

Remark 41.12. This leads to ambiguities such as:

- unclear connectivity of track segments,
- duplicated or overlapping elements,
- inconsistent ordering of points.

41.3. Representation of Measured Geometry

Definition 41.13 (Polyline representation). Measured alignments are often given as a sequence of points

$$P = (p_0, p_1, \dots, p_N), \quad p_i \in \mathbb{R}^2.$$

Remark 41.14. This representation is:

- simple,

- widely used,
- but not directly suitable for analysis or optimization.

Remark 41.15. In particular, curvature is not explicitly given, but must be inferred.

41.4. From Data to Model

Remark 41.16. A key task in alignment modelling is the transformation:

$$\text{raw data} \longrightarrow \text{structured model}.$$

Remark 41.17. This involves:

- filtering noise,
- reconstructing geometry,
- identifying structural elements,
- and assigning semantic meaning.

41.5. Curvature Estimation

Definition 41.18 (Discrete curvature). Given three consecutive points, curvature may be approximated by

$$\kappa_i \approx \frac{\theta_{i+1} - \theta_i}{\Delta s_i},$$

where θ_i denotes the local direction.

Remark 41.19. This approximation is sensitive to noise and sampling density.

Remark 41.20. Therefore, direct use of raw curvature estimates is often unstable.

41.6. Smoothing and Filtering

Remark 41.21. To obtain usable curvature information, smoothing is required.

Definition 41.22 (Smoothing operator). A smoothing operator transforms noisy data into a more regular form:

$$\kappa_i \mapsto \tilde{\kappa}_i.$$

Remark 41.23. Common techniques include:

- moving averages,
- Gaussian filtering,
- spline fitting,
- and regularization methods.

41.7. Model Reconstruction

Remark 41.24. The goal of reconstruction is to obtain a structured alignment model from imperfect data.

Remark 41.25. This includes:

- determining curvature functions,
- identifying transition regions,
- and constructing a consistent representation.

Remark 41.26. Reconstruction is not unique and depends on modelling assumptions.

41.8. Uncertainty

Definition 41.27 (Uncertainty). Uncertainty describes the lack of exact knowledge about the true alignment.

Remark 41.28. Uncertainty arises from:

- measurement errors,
- incomplete data,
- and modelling assumptions.

Remark 41.29. A robust modelling framework should:

- tolerate uncertainty,
- avoid overfitting,
- and provide stable results.

41.9. Impact on Optimization

Remark 41.30. Imperfect data directly affects optimization:

- objective functions may reflect noise,
- constraints may be violated by measurement errors,
- and solutions may become unstable.

Remark 41.31. Therefore, optimization must account for uncertainty, for example by:

- regularization,
- robust objective functions,
- or probabilistic models.

41.10. Engineering Perspective

Remark 41.32. In practice, engineers routinely work with imperfect data.

Remark 41.33. This includes:

- interpreting measurement results,
- reconciling inconsistent datasets,
- and making design decisions under uncertainty.

Remark 41.34. A modelling framework should support these tasks rather than assume ideal data.

41.11. Towards Data-Aware Modelling

Remark 41.35. A modern alignment model must be data-aware.

Remark 41.36. This includes:

- explicit handling of noise,
- robust reconstruction methods,
- integration with coordinate systems,
- and compatibility with real-world workflows.

41.12. Connection to the AIM Framework

Summary 41.37. Railway alignment data is inherently imperfect.

A realistic modelling framework must therefore incorporate measurement noise, incomplete data, and uncertainty.

Within the AIM framework, data is not assumed to be ideal, but is treated as an integral part of the modelling process.

42. Topology and Special Elements

Motivation

Remark 42.1. Railway systems are not isolated curves, but interconnected networks.

Remark 42.2. While mathematical models often focus on single alignments, real-world railway infrastructure consists of:

- branching tracks,
- merging lines,
- parallel alignments,
- and complex switching structures.

Remark 42.3. A complete modelling framework must therefore go beyond single curves and incorporate topology.

42.1. From Curves to Networks

Definition 42.4 (Alignment). An alignment is a parametrized curve

$$\gamma : [0, L] \rightarrow \mathbb{R}^2.$$

Definition 42.5 (Railway network). A railway network is a collection of alignments together with a set of connectivity relations.

Remark 42.6. The transition from a single alignment to a network introduces topological structure in addition to geometry.

42.2. Topological Relations

Remark 42.7. Alignments in a network may be related by:

- continuation (end-to-start connections),
- branching (one alignment splits into several),
- merging (multiple alignments join),

- parallel relations (tracks running side by side).

Definition 42.8 (Connection point). A connection point is a location where two or more alignments meet.

Remark 42.9. At connection points, geometric and dynamic consistency conditions must be satisfied.

42.3. Switches and Turnouts

Remark 42.10. Switches (turnouts) are fundamental elements of railway networks.

Definition 42.11 (Switch). A switch is a structure that allows a vehicle to move from one alignment to another.

Remark 42.12. Switches introduce:

- branching geometry,
- nontrivial curvature transitions,
- and constraints on feasible trajectories.

Remark 42.13. Unlike simple transitions, switches cannot be described by a single curvature function.

42.4. Geometric Complexity

Remark 42.14. Special elements often exhibit geometric properties that differ from standard alignment elements:

- discontinuities in curvature,
- piecewise-defined geometry,
- and locally non-smooth behaviour.

Remark 42.15. These properties must be handled explicitly in a modelling framework.

42.5. Topological Ambiguity in Data

Remark 42.16. Real-world data often does not explicitly encode topology.

Remark 42.17. Typical issues include:

- missing connectivity information,
- duplicated segments,
- and inconsistent ordering of elements.

Remark 42.18. Therefore, topology must often be inferred from geometric and contextual information.

42.6. Representation of Networks

Definition 42.19 (Graph representation). A railway network can be represented as a graph:

- nodes represent connection points,
- edges represent alignments.

Remark 42.20. This abstraction separates:

- topology (graph structure),
- and geometry (alignment shape).

42.7. Coupling of Topology and Geometry

Remark 42.21. Although topology and geometry can be separated conceptually, they are strongly coupled in practice.

Remark 42.22. For example:

- curvature constraints depend on connectivity,
- feasible transitions depend on network structure,
- and optimization must consider multiple interacting alignments.

42.8. Special Engineering Cases

Remark 42.23. Railway systems include a variety of special elements, such as:

- switches and crossings,
- sidings and yards,
- station layouts,
- and maintenance tracks.

Remark 42.24. These elements often violate assumptions of standard alignment models:

- unique parameterization by arc length,
- smooth curvature evolution,
- and single-track continuity.

42.9. Implications for Modelling

Remark 42.25. The presence of topology and special elements implies that:

- alignment models must support multiple connected curves,
- curvature-based descriptions must be extended,
- and reconstruction algorithms must handle branching structures.

Remark 42.26. In particular, optimization problems must be formulated on networks rather than individual curves.

42.10. Towards Topology-Aware Modelling

Remark 42.27. A topology-aware modelling framework must include:

- explicit representation of connectivity,
- integration of geometry and topology,
- support for special elements,
- and compatibility with real-world data.

42.11. Connection to the AIM Framework

Summary 42.28. Railway systems are inherently network-based and include complex special elements such as switches.

A complete modelling framework must therefore extend beyond single alignments and incorporate topology.

Within the AIM framework, topology is treated as an integral component of the alignment model, enabling consistent representation and optimization of railway networks.

43. Construction and Realization

Motivation

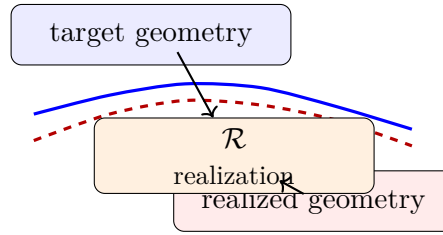


Abbildung 43.1.: Target geometry and realized geometry connected by the realization operator.

Remark 43.1. Theoretical railway alignments are defined as continuous geometric and curvature objects.

Remark 43.2. In practice, tracks are not constructed as continuous mathematical curves. They are realized through discrete construction and maintenance processes, local measurements, machine operations, correction limits, and structural response of the track.

Proposition 43.3. *The curve that is constructed is not identical to the curve that is designed.*

Remark 43.4. A realistic modelling framework must therefore include an explicit realization layer between analytical design and built geometry.

Principle 43.5 (Realization principle). Within AIM, realization is treated as an operator layer transforming target alignment states into physically achievable railway states.

43.1. Realized Alignment

Definition 43.6 (Realized alignment). The realized alignment is the physically constructed railway geometry, denoted by

$$\gamma_{\text{real}}(s).$$

Remark 43.7. It may be represented continuously, but it is usually generated from discrete construction, measurement, and maintenance data.

43. Construction and Realization

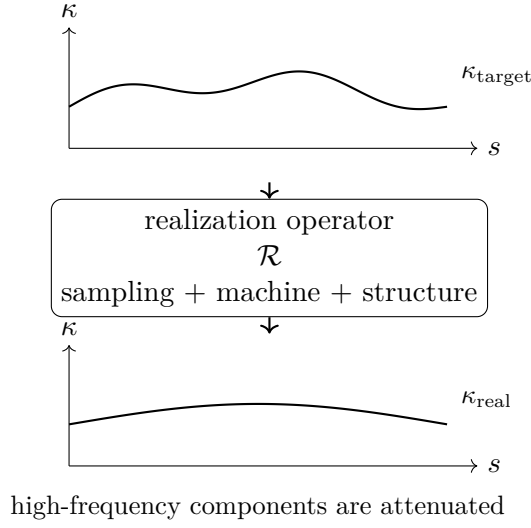


Abbildung 43.2.: Realization acts as a filtering process: target curvature may contain high-frequency components that are attenuated by sampling, machine action, and structural response.

Definition 43.8 (Realized curvature). The realized curvature is:

$$\kappa_{\text{real}}(s) = \frac{d\theta_{\text{real}}}{ds}.$$

Remark 43.9. The realized track should not be judged only by positional agreement, but also by preservation of curvature evolution.

Remark 43.10. Within AIM, realized geometry may be interpreted as:

$$x_{\text{real}} = \mathcal{R}(x_{\text{target}}),$$

where $\mathcal{R}$ denotes the realization operator.

43.2. Realization Error

Definition 43.11 (Curvature realization error). A curvature-based realization error may be written as:

$$E_{\kappa} = \int_0^L (\kappa_{\text{real}} - \kappa_{\text{target}})^2 ds.$$

Definition 43.12 (Curvature derivative error). A higher-order realization error may be written as:

$$E_{\kappa'} = \int_0^L (\kappa'_{\text{real}} - \kappa'_{\text{target}})^2 ds.$$

Remark 43.13. These measures reflect that different transition families may be close in position space while differing significantly in curvature evolution and dynamic interpretation.

Remark 43.14. Realization error may therefore be evaluated in:

- position space,
- tangent space,
- curvature space,
- curvature-derivative space,
- and runtime pose3 space.

43.3. Sampling and Control Points

Remark 43.15. Construction and maintenance operate through discrete control data.

Definition 43.16 (Control points). Let

$$0 = s_0 < s_1 < \cdots < s_N = L$$

be a strictly increasing sequence of arc-length positions. These positions are called control points.

Remark 43.17. The realized alignment may be reconstructed from these points using:

- piecewise linear interpolation,
- piecewise polynomial interpolation,
- spline-based reconstruction,
- machine-specific reconstruction rules,
- or sparse alignment reconstruction.

Proposition 43.18 (Adaptive sampling principle). *Sampling density should increase in regions of strong curvature variation:*

$$\rho(s) \sim f(|\kappa'(s)|, |\kappa''(s)|),$$

for a suitable increasing function f .

Remark 43.19. Nearly constant-curvature regions require fewer control points, whereas transition zones require denser control.

43.4. Adjustment as Iterative Process

Remark 43.20. In railway construction and maintenance, track geometry is adjusted through a sequence of localized operations, for example by tamping machines.

43. Construction and Realization

Remark 43.21. The canonical adjustment operator is denoted by $\mathcal{A}$, as introduced in theorem 9.53.

Definition 43.22 (Iterative realization). Starting from an initial alignment γ_0 , the iterative process is:

$$\gamma_{k+1} = \mathcal{A}_{\gamma_{\text{target}}}(\gamma_k).$$

Remark 43.23. The operator depends on local measurement data, machine parameters, correction limits, and physical properties of the track.

Definition 43.24 (Realized alignment as limit). If the sequence converges, the realized alignment is:

$$\gamma_{\text{real}} = \lim_{k \rightarrow \infty} \gamma_k.$$

Proposition 43.25 (Fixed point interpretation). *If the iterative realization converges, then:*

$$\gamma_{\text{real}} = \mathcal{A}_{\gamma_{\text{target}}}(\gamma_{\text{real}}).$$

Remark 43.26. In practice, convergence cannot be assumed a priori, since it depends on measurement feedback, machine limits, and local track conditions.

43.5. Locality and Correction Limits

Definition 43.27 (Local correction). A guided adjustment may be written as:

$$\mathcal{A}_{\gamma_{\text{target}}}(\gamma) = \gamma + \Delta(\gamma, \gamma_{\text{target}}),$$

where Δ is a local correction operator.

Remark 43.28. Typically, corrections are bounded:

$$\|\Delta(\gamma, \gamma_{\text{target}})\| \leq \Delta_{\text{max}}.$$

Proposition 43.29 (Locality). *The adjustment operator is local in the sense that:*

$$(\mathcal{A}_{\gamma_{\text{target}}} \gamma)(s)$$

depends only on γ and γ_{target} in a neighbourhood of s .

Remark 43.30. This reflects the finite working range of construction and maintenance equipment.

43.6. Kernel-Based Realization Model

Definition 43.31 (Kernel-based adjustment). A linearized model of the adjustment process is:

$$(\mathcal{A}_{\gamma_{\text{target}}} \gamma)(s) = \gamma(s) + \int_I K(s, \sigma) (\gamma_{\text{target}}(\sigma) - \gamma(\sigma)) d\sigma,$$

43. Construction and Realization

where $K(s, \sigma)$ is a localized influence kernel.

Remark 43.32. The kernel represents the spatial influence of local construction or maintenance action.

Proposition 43.33 (Localization). *There exists a characteristic length $\ell > 0$ such that:*

$$K(s, \sigma) \approx 0 \quad \text{for } |s - \sigma| > \ell.$$

Definition 43.34 (Discrete adjustment model). For control points s_i , the kernel model becomes:

$$\gamma_i^{(k+1)} = \gamma_i^{(k)} + \sum_j K_{ij}(\gamma_j^{\text{target}} - \gamma_j^{(k)}).$$

Remark 43.35. This form is compatible with sparse numerical implementation.

Remark 43.36. The matrix K_{ij} should not be interpreted merely as numerical smoothing. It represents physical influence: machine action, track-bed response, sleeper spacing, stiffness, local constraints, and measurement feedback.

43.7. Filter Interpretation

Proposition 43.37. *The realization process acts approximately as a low-pass filter on curvature.*

Remark 43.38. High-frequency curvature components are attenuated during construction and maintenance, while low-frequency structure is preserved.

Remark 43.39. This explains why transition curves with similar low-order behaviour may become nearly indistinguishable in position space after realization, while still differing in curvature evolution and dynamic interpretation.

Remark 43.40. In frequency-oriented notation, realization may be idealized as:

$$\hat{\kappa}_{\text{real}}(\omega) \approx H_{\mathcal{R}}(\omega) \hat{\kappa}_{\text{target}}(\omega),$$

where $H_{\mathcal{R}}$ denotes a realization transfer function.

Remark 43.41. For typical physical realization processes, $H_{\mathcal{R}}$ attenuates high-frequency components.

Principle 43.42 (Realization filter principle). Construction and maintenance do not merely approximate target geometry; they transform it through a physical filter induced by sampling, machinery, structure, and correction limits.

43.8. Relation to the Realization Operator

Remark 43.43. In the AIM operator framework, realization is represented by:

$$\mathcal{R} : \mathcal{X}_{\text{target}} \rightarrow \mathcal{X}_{\text{real}},$$

as introduced in theorem 9.55.

Remark 43.44. The construction process described in this chapter provides one interpretation of how such an operator may arise physically.

Remark 43.45. The simplified realization decomposition:

$$\mathcal{R} \approx \mathcal{M} \circ \mathcal{I} \circ \mathcal{S}$$

separates sampling, reconstruction, and mechanical response.

Remark 43.46. Here:

- $\mathcal{S}$ denotes sampling or control-point extraction,
- $\mathcal{I}$ denotes interpolation or reconstruction,
- $\mathcal{M}$ denotes mechanical and structural response.

Remark 43.47. More generally, $\mathcal{R}$ may depend on:

- machinery,
- measurement process,
- construction method,
- track structure,
- maintenance history,
- local constraints,
- and environmental conditions.

Remark 43.48. The realization workflow is summarized in fig. 36.2.

43.9. Sparse Compatibility

Remark 43.49. The realization layer must remain compatible with sparse alignment representation.

Remark 43.50. Within AIM, this means that realization may operate on:

- sparse target parameters,
- sampled runtime states,
- reconstructed curvature functions,
- or realized sparse approximations.

Principle 43.51 (Sparse realization compatibility). Realization-aware modelling must not destroy the sparse structure of the alignment model.

Instead, realized geometry should be interpretable either as a runtime state, a sampled state, or a sparse-compatible approximation.

Remark 43.52. A realized alignment may therefore be represented as:

- measured sample data,
- a reconstructed dense curve,
- a sparse re-fit,
- or a hybrid measured–sparse object.

43.10. Realization-Aware Optimization

Remark 43.53. Classical design optimizes the target curve.

Remark 43.54. Realization-aware design optimizes the target curve with respect to its expected realized outcome.

Definition 43.55 (Realization-aware objective). A realization-aware optimization objective may be written as:

$$\min_{x_{\text{target}}} J(\mathcal{R}(x_{\text{target}})).$$

Remark 43.56. Here, the design variable is the analytical target, while the evaluated object is the realized railway state.

Remark 43.57. This distinction is essential whenever construction, maintenance, machine response, or structural filtering significantly affect the final geometry.

Proposition 43.58 (Realization-aware design principle). *The best target alignment is not necessarily the analytically smoothest alignment, but the target whose realized state best satisfies the operational objective.*

43.11. Design Implication

Remark 43.59. The designed transition curve is not simply the curve that is built. It is a target that produces a realized behaviour after construction and maintenance.

Remark 43.60. Thus, alignment design implicitly includes assumptions about construction, machine behaviour, and structural response.

Proposition 43.61. *The relevant physical design space is the image of the realization operator, not the full space of analytical curvature laws.*

Remark 43.62. This connects construction and realization to realization-aware optimization.

Remark 43.63. For AIM, this means that transition classification, sparse alignment modeling, runtime pose3 evaluation, and optimization must be interpreted not only in analytical target space, but also in realized physical state space.

43.12. Connection to AIM

Summary 43.64. Construction and realization describe the transformation from analytical target alignment to physically realized railway geometry.

Within AIM, this transformation is treated as an explicit operator layer:

$$x_{\text{real}} = \mathcal{R}(x_{\text{target}}).$$

The realization layer explains why physical railway geometry must be understood through construction processes, measurement feedback, local correction limits, machine behaviour, sparse reconstruction, and mechanical response rather than through analytical geometry alone.

Realization is therefore not an implementation detail. It is a central mathematical and physical layer connecting:

- target geometry,
- sparse alignment representation,
- measured reality,
- runtime state evaluation,
- and realization-aware optimization.

44. Realized Pose3 and Dynamic Realization

Motivation

Remark 44.1. Classical alignment theory typically assumes that the designed geometry is identical to the realized geometry.

Remark 44.2. In reality, railway tracks are produced through iterative and physically constrained realization processes.

Remark 44.3. As a consequence, not only the geometric position of the track differs from the theoretical target alignment, but also its orientation, curvature, cant, and dynamic behaviour.

Remark 44.4. Thus, realization affects the full spatial railway state rather than geometry alone.

Remark 44.5. This motivates the introduction of a realized pose3 state:

$$\text{pose3}_{\text{real}}(s).$$

Principle 44.6 (Realized-state principle). Within AIM, the physically relevant railway state is the realized runtime state, not merely the analytical target state.

44.1. Target and Realized States

Definition 44.7 (Target pose3). The target railway state is defined as

$$\text{pose3}_{\text{target}}(s) = (\Gamma_{\text{target}}(s), F_{\text{rail,target}}(s), \phi_{\text{target}}(s)).$$

Definition 44.8 (Realized pose3). The realized railway state is defined as

$$\text{pose3}_{\text{real}}(s) = (\Gamma_{\text{real}}(s), F_{\text{rail,real}}(s), \phi_{\text{real}}(s)).$$

Remark 44.9. The realized state is the physically constructed, measurable, and operationally effective state of the railway track.

Remark 44.10. In practice,

$$\text{pose3}_{\text{real}} \neq \text{pose3}_{\text{target}}.$$

Remark 44.11. The difference is not limited to position. It affects tangent direction, curvature, cant, frame orientation, and derived dynamic quantities.

44.2. Measured States

Definition 44.12 (Measured pose3 state). A measured pose3 state is an observation-derived approximation of the realized railway state:

$$\text{pose3}_{\text{meas}}(s_i) \approx \text{pose3}_{\text{real}}(s_i).$$

Remark 44.13. Measured pose3 states are usually available only at discrete stations, timestamps, or sensor positions.

Remark 44.14. Measurement may provide direct or indirect information about:

- position,
- tangent direction,
- curvature,
- cant,
- gauge,
- vertical profile,
- acceleration,
- and local frame orientation.

Remark 44.15. Measured states are not identical to realized states. They include sensor noise, sampling effects, calibration errors, filtering effects, and interpretation assumptions.

Principle 44.16 (Measurement separation principle). AIM distinguishes target state, realized state, and measured state as separate conceptual layers.

44.3. Sources of Realization Deviations

Remark 44.17. Deviations between target and realized state arise from:

- measurement uncertainty,
- machine limitations,
- discrete control points,
- structural elasticity,
- ballast behaviour,
- rail bending,
- settlement processes,

- maintenance history,
- and operational degradation.

Remark 44.18. Thus, realization is not a purely geometric transformation, but a mechanical and operational process.

44.4. Realized Geometry

Definition 44.19 (Realized spatial trajectory). The realized spatial trajectory is denoted by:

$$\Gamma_{\text{real}}(s).$$

Remark 44.20. $\Gamma_{\text{real}}(s)$ represents the physically realized railway reference trajectory after construction, maintenance, settlement, and operational loading.

Remark 44.21. It may be represented as:

- discrete measured samples,
- a reconstructed dense curve,
- a sparse re-fit,
- or a hybrid measured–analytical object.

Remark 44.22. Within AIM, realized geometry remains compatible with sparse modelling by being interpreted either as reconstructed runtime state or as a sparse-compatible approximation.

44.5. Runtime Reconstruction

Remark 44.23. Realized pose3 is generally not stored directly as a complete continuous object.

Remark 44.24. Instead, it is reconstructed at runtime from:

- measured samples,
- sparse target geometry,
- realization models,
- frame operators,
- cant reconstruction,
- profile reconstruction,
- and metric context.

Definition 44.25 (Runtime realization reconstruction). Runtime reconstruction of realized pose3 may be written abstractly as:

$$\text{pose3}_{\text{real}}(s) = \mathcal{E}_{\text{real}}(s, x_{\text{target}}, m, \mathcal{R}, \mathcal{C}),$$

where m denotes measured data, $\mathcal{R}$ denotes realization model information, and $\mathcal{C}$ denotes coordinate and metric context.

Remark 44.26. The operator $\mathcal{E}_{\text{real}}$ is a runtime evaluation operator rather than a persistent storage object.

44.6. Realized Curvature and Cant

Definition 44.27 (Realized curvature). The realized curvature is:

$$\kappa_{\text{real}}(s) = \frac{d\theta_{\text{real}}}{ds}.$$

Definition 44.28 (Realized cant). The realized cant angle is:

$$\phi_{\text{real}}(s).$$

Remark 44.29. Both quantities differ from their target values due to realization effects.

Remark 44.30. In particular,

$$\kappa_{\text{real}} \neq \kappa_{\text{target}}, \quad \phi_{\text{real}} \neq \phi_{\text{target}}.$$

Remark 44.31. Measured curvature and measured cant are further approximations:

$$\kappa_{\text{meas}} \approx \kappa_{\text{real}}, \quad \phi_{\text{meas}} \approx \phi_{\text{real}}.$$

44.7. Cant Realization

Remark 44.32. Cant realization is particularly sensitive because cant is not only a geometric quantity but also a dynamically relevant operational state.

Remark 44.33. Cant realization is affected by:

- tamping accuracy,
- rail fastening systems,
- ballast stiffness,
- torsion limits,
- local support conditions,

- measurement procedures,
- and sensor interpretation.

Remark 44.34. In practice, cant ramps are realized only approximately and may exhibit local irregularities.

Remark 44.35. This is especially relevant in:

- transition zones,
- switches and crossings,
- station areas,
- overlapping transition regions,
- and locally constrained construction zones.

44.8. Measurement Ambiguity

Remark 44.36. Measured railway data does not automatically define a unique realized pose3 state.

Remark 44.37. Ambiguities may arise from:

- uncertain station assignment,
- CRS uncertainty,
- projection ambiguity,
- sensor offset from the reference line,
- vehicle motion,
- filtering by onboard systems,
- incomplete cant or profile information,
- and topology ambiguity in multi-track environments.

Remark 44.38. Therefore, measurement interpretation is itself a runtime reconstruction problem.

Principle 44.39 (Measured-state ambiguity). Measured data must be interpreted as evidence for realized pose3, not as realized pose3 itself.

44.9. Sensor Integration

Remark 44.40. Different sensors observe different projections of the realized railway state.

Remark 44.41. Examples include:

- surveying points,
- track geometry cars,
- inertial measurement units,
- GNSS receivers,
- laser scanning,
- photogrammetry,
- gauge and cant measurement systems,
- and vehicle-mounted acceleration sensors.

Remark 44.42. Sensor integration requires mapping heterogeneous observations into a common runtime state interpretation.

Definition 44.43 (Sensor observation model). A sensor observation may be written as:

$$y_i = \mathcal{H}_i(\text{pose3}_{\text{real}}(s_i)) + \varepsilon_i,$$

where $\mathcal{H}_i$ denotes the sensor observation operator and ε_i denotes measurement error.

Remark 44.44. The reconstruction task is therefore an inverse problem: estimate the realized pose3 state from heterogeneous noisy observations.

44.10. Coupling of Curvature and Cant

Remark 44.45. Curvature and cant are dynamically coupled quantities.

Remark 44.46. Therefore, realization errors in curvature and cant cannot be treated independently.

Remark 44.47. The physically relevant quantity is the realized dynamic state:

$$(\kappa_{\text{real}}, \phi_{\text{real}}).$$

Definition 44.48 (Realized equilibrium condition). A realized track is dynamically balanced if:

$$v^2 \kappa_{\text{real}}(s) = g \sin \phi_{\text{real}}(s).$$

Remark 44.49. In practice, deviations from equilibrium are unavoidable and constrained only within admissible limits.

44.11. Dynamic Realization Error

Definition 44.50 (Realized effective lateral acceleration). The realized effective lateral acceleration is:

$$a_{\text{eff,real}}(s) = v^2 \kappa_{\text{real}}(s) - g \sin \phi_{\text{real}}(s).$$

Remark 44.51. This quantity is operationally more relevant than geometric deviation alone.

Definition 44.52 (Realized jerk). The realized jerk is:

$$j_{\text{real}}(s) = v^3 \kappa'_{\text{real}}(s) - gv \cos \phi_{\text{real}}(s) \phi'_{\text{real}}(s).$$

Remark 44.53. Passenger comfort depends directly on the realized jerk rather than on the analytical target functions.

44.12. Realization as State Transformation

Definition 44.54 (Pose3 realization operator). The realization process may be written as:

$$\mathcal{R}_{\text{pose3}} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

Remark 44.55. This operator includes:

- geometric realization,
- cant realization,
- profile realization,
- mechanical filtering,
- measurement feedback,
- and operational constraints.

Remark 44.56. Thus, realization acts on the full railway state rather than only on position space.

Remark 44.57. In practice, $\mathcal{R}_{\text{pose3}}$ is not directly known. It is approximated through construction knowledge, measurement data, machine models, and runtime reconstruction.

44.13. Filtering Interpretation

Proposition 44.58. *The realization operator acts as a low-pass filter on:*

- *curvature,*
- *cant,*

- *profile irregularities,*
- *and higher-order derivatives.*

Remark 44.59. High-frequency components are attenuated through:

- machine limitations,
- structural smoothing,
- ballast mechanics,
- vehicle-track interaction,
- and measurement discretization.

Remark 44.60. Therefore, highly detailed analytical target states may collapse into similar realized states.

44.14. Operational Interpretation

Remark 44.61. Railway operation interacts with the realized state rather than with the theoretical target geometry.

Remark 44.62. Consequently, the relevant engineering question is not:

“Was the theoretical geometry built exactly?”

but rather:

“Does the realized state produce the intended dynamic behaviour?”

Remark 44.63. This makes realized pose3 the natural bridge between construction, measurement, runtime evaluation, and operation.

44.15. Connection to Optimization

Remark 44.64. Optimization should therefore operate on realized states:

$$J = J(\text{pose3}_{\text{real}}) = J(\mathcal{R}_{\text{pose3}}(\text{pose3}_{\text{target}})).$$

Remark 44.65. This extends classical alignment optimization toward realization-aware optimization.

Remark 44.66. When measured states are available, optimization may additionally use:

$$J = J(\text{pose3}_{\text{real}}, \text{pose3}_{\text{meas}}),$$

to calibrate realization models or identify maintenance needs.

44.16. Connection to AIM

Summary 44.67. Real railway tracks are not analytical curves but realized physical states.

The realization process acts on the complete pose3 state, including geometry, curvature, profile, frame orientation, and cant.

Measured railway data provides evidence for this realized state, but it does not define it uniquely. Therefore, AIM distinguishes target pose3, realized pose3, and measured pose3 as separate layers.

Within the AIM framework, realized pose3 forms the bridge between:

- analytical geometry,
- runtime reconstruction,
- sensor integration,
- construction and maintenance,
- physical realization,
- and operational dynamics.

Thus, railway alignment design must be understood as the design of a realizable dynamic state rather than merely a geometric target curve.

Teil VIII.

Alignment Modeling

Modeling

What do we already know?

Remark 44.68. The previous parts established the conceptual foundations of AIM.

Remark 44.69. Intrinsic geometry, railway states, runtime evaluation, realization, and reality-aware interpretation have been developed as coherent theoretical constructs.

Remark 44.70. These concepts define what railway alignments are, how they evolve, and how they interact with the physical world.

Remark 44.71. However, concepts alone do not yet define how alignment information is represented within an operational information model.

What is still missing?

Remark 44.72. An information model requires explicit internal structures.

Remark 44.73. Geometric concepts must be transformed into representations that can be stored, exchanged, reconstructed, analyzed, and optimized.

Remark 44.74. Traditional railway systems frequently rely on coordinate lists, geometric primitives, or implementation-specific data structures.

Remark 44.75. While such representations may be sufficient for storage purposes, they often obscure the intrinsic structure of the alignment and limit subsequent analysis.

Remark 44.76. The missing element is therefore a modeling framework capable of capturing railway semantics while remaining compact, interpretable, and computationally efficient.

What comes next?

Remark 44.77. This part introduces the modeling layer of AIM.

Remark 44.78. The central objective is the construction of sparse, semantic alignment models.

Remark 44.79. Within AIM, an alignment is not primarily represented as sampled coordinates.

Instead, it is represented through a compact collection of geometric states, transition descriptions, and semantic relationships from which geometry can be reconstructed whenever required.

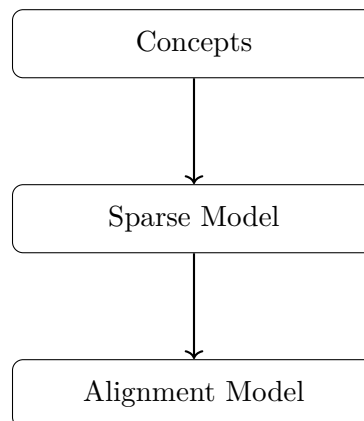
Remark 44.80. The chapters that follow introduce:

- sparse alignment representations,
- transition modeling,
- transition classification,
- higher-level alignment semantics,
- and Schwerpunkt-based modeling concepts.

Remark 44.81. These structures form the operational core of alignment-based information modelling.

Relation to the AIM Roadmap

Remark 44.82. Within the AIM roadmap, the present part corresponds to the transition from conceptual railway descriptions toward explicit alignment models.



Remark 44.83. The progression

Concepts $\rightarrow$ Sparse Model $\rightarrow$ Alignment Model

marks the transition from theoretical railway descriptions toward operational information structures.

Remark 44.84. These models subsequently become the basis for runtime services, analysis pipelines, optimization procedures, and engineering applications.

Principle 44.85 (Sparse Modeling Principle). Within AIM, railway alignments should be represented through compact semantic structures from which geometric, operational, and realization-aware information can be reconstructed when required.

Summary 44.86. Concepts explain railway structure.

Reality explains interaction with the physical world.

Modeling explains how railway knowledge is represented.

The present part therefore establishes the sparse and semantic information structures that form the core of alignment-based information modelling.

45. Sparse Alignment Model

Motivation

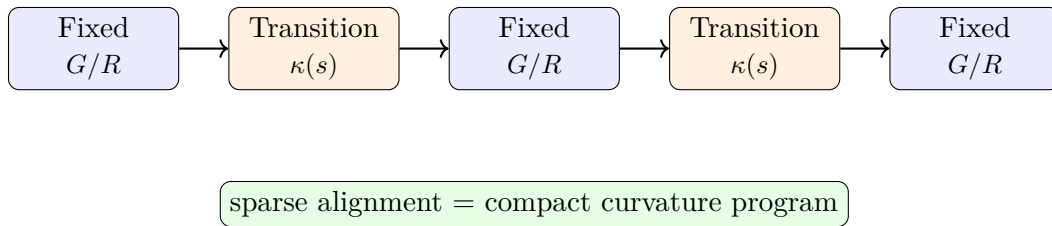


Abbildung 45.1.: Sparse alignment as an alternating sequence of fixed and transition elements.

Remark 45.1. A continuous curvature law is mathematically natural, but it is not by itself an operational representation for editing, exchange, validation, or reconstruction from heterogeneous engineering data. For this reason, a sparse model is introduced as a structured, piecewise-defined representation of an alignment.

Remark 45.2. The sparse model is not intended as a denial of continuous geometry. On the contrary, it serves as an intermediate layer between continuous curvature theory and practical modelling tasks such as import, computation, optimization, and interoperability.

Remark 45.3. Dense geometric representations such as point clouds or sampled polylines preserve positional information, but largely destroy the structural semantics of the alignment itself.

45.1. General Idea

Definition 45.4 (Sparse alignment model). A sparse alignment model is a structured longitudinal representation consisting of synchronized alignment bands together with sufficient reconstruction information.

Remark 45.5. The adjective *sparse* refers here to the fact that the alignment is not stored as a dense point cloud, but as a compact sequence of semantically meaningful elements.

Remark 45.6. Within AIM, the sparse model is not interpreted as a static 3D geometry, but as a persistent structural kernel from which geometry and dynamics are derived.

45.2. Element Types

Definition 45.7 (Alignment element). An alignment element is a segment of an alignment with a prescribed local curvature behaviour on a parameter interval.

Definition 45.8 (Fixed element). A fixed element is an alignment element with constant curvature

$$\kappa(s) = \kappa_0.$$

Typical special cases are straight segments with $\kappa_0 = 0$ and circular arcs with $\kappa_0 \neq 0$.

Definition 45.9 (Transition element). A transition element is an alignment element with non-constant curvature, i.e.

$$\kappa(s) = f(s)$$

for a suitable curvature law f .

Remark 45.10. Within AIM, transition elements are primarily interpreted through their curvature evolution rather than through explicit geometric shape.

Remark 45.11. At this level of abstraction, no commitment is yet made to a specific family of transition laws. This includes classical clothoids as well as polynomial, lookup-based, or hybrid constructions.

TODO: Formal classification of transition families.

45.3. Minimal Parameterization

Definition 45.12 (Arc length of an element). Each element e_i is associated with an arc length

$$L_i > 0.$$

Definition 45.13 (Local curvature law). Each element e_i is associated with a curvature law

$$\kappa_i : [0, L_i] \rightarrow \mathbb{R}.$$

Remark 45.14. For fixed elements, κ_i is constant. For transition elements, κ_i varies along the local parameter.

Definition 45.15 (Sparse alignment). A sparse alignment is an ordered tuple

$$A = (e_1, e_2, \dots, e_n)$$

together with sufficient initial data for geometric reconstruction.

45.4. Initial Data and Pose

Definition 45.16 (Planar start pose). A planar start pose is a tuple

$$P_0 = (x_0, y_0, \theta_0)$$

with position $(x_0, y_0) \in \mathbb{R}^2$ and initial direction $\theta_0 \in \mathbb{R}$.

Proposition 45.17. *The planar alignment is reconstructed by successive integration of the curvature law:*

$$\kappa(s) \rightarrow \theta(s) \rightarrow \gamma(s).$$

Remark 45.18. Together with the ordered element sequence and the local curvature laws, the start pose determines the alignment geometry by successive integration.

Definition 45.19 (Reconstructable sparse alignment). A sparse alignment is called reconstructable if its element sequence and its initial data suffice to determine a unique planar alignment up to the intended modelling conventions.

45.5. Sparse Longitudinal Band Structure

Principle 45.20 (Sparse longitudinal band principle). Railway alignments are represented as synchronized longitudinal bands rather than as monolithic geometric objects.

Remark 45.21. Within the AIM framework, the persistent alignment model is intentionally kept sparse and decomposed into synchronized longitudinal bands.

45.5.1. cGeom

Definition 45.22 (Core geometric band). The central geometric band is denoted by

$$\text{cGeom}.$$

It represents the curvature-driven planar alignment core and consists of:

- fixed elements,
- transition elements,
- pose2-based reconstruction anchors.

Remark 45.23. The cGeom band forms the primary integration core of the alignment.

45.5.2. Cant Band

Definition 45.24 (Cant band). Cant is represented as an independent longitudinal band correlated with fixed alignment elements.

45. Sparse Alignment Model

Remark 45.25. The cant band:

- does not define an independent stationing system,
- stores rail height differences rather than roll angles,
- follows the same normalized curvature family as the associated cGeom element,
- may contain local station offsets at element boundaries.

Remark 45.26. Since cant evolution follows the same normalized curvature family as the associated cGeom element, no independent geometric evolution model is required for the cant band itself.

Remark 45.27. The corresponding roll angle is derived via:

$$\phi(s) = \arctan\left(\frac{u(s)}{g}\right),$$

where:

- $u(s)$: rail height difference,
- g : gauge.

45.5.3. Profile Band

Definition 45.28 (Profile band). Vertical geometry is represented independently through a profile band.

Remark 45.29. Thus, cGeom, cant, and profile are treated as synchronized but distinct bands sharing a common arc-length domain.

45.5.4. Runtime State

Proposition 45.30. *Pose3 is not treated as persistent truth within the sparse model.*

Remark 45.31. Persisting pose3 directly would introduce redundancy and synchronization instabilities between geometry, cant, and profile representations.

Remark 45.32. Instead, pose3 is derived dynamically from:

- cGeom,
- cant,
- profile,
- and runtime evaluation services.

Remark 45.33. Consequently, the primary intelligence of the AIM framework resides not in the storage format itself, but in the mathematical operators and evaluation services acting on the sparse model.

45.6. Concatenation

Definition 45.34 (Element concatenation). Two consecutive elements are concatenated if the end state of the first element provides the start state of the second element.

Definition 45.35 (Positional continuity). A sparse alignment is positionally continuous if consecutive elements meet in a common endpoint.

Definition 45.36 (Directional continuity). A sparse alignment is directionally continuous if consecutive elements share the same tangent direction at their common boundary.

Remark 45.37. In most engineering contexts, positional and directional continuity are minimal requirements. Higher continuity requirements may depend on the chosen transition families and on the intended application.

TODO: Formalize continuity classes such as C^0 , C^1 , and curvature-related continuity conditions.

45.7. Normal Form and Alternation

Remark 45.38. In practical sparse representations, a frequent pattern is the alternation between fixed elements and transition elements. This corresponds closely to engineering intuition and to many traditional alignment descriptions.

OPEN: Should strict alternation between fixed and transition elements be formulated as a mathematical property, as a preferred normal form, or merely as an implementation convention?

Remark 45.39. It is plausible to regard alternation as a normal form rather than a fundamental axiom. Multiple neighbouring fixed elements may be merged, and certain degenerate transition cases may collapse into fixed elements.

45.8. Normalized Representation of Elements

Definition 45.40 (Normalized local parameter). A normalized local parameter is a variable

$$u \in [0, 1]$$

used to represent an element independently of its physical length.

Definition 45.41 (Normalized curvature law). A normalized curvature law is a law defined on the unit interval, typically of the form

$$\hat{\kappa} : [0, 1] \rightarrow \mathbb{R}.$$

Remark 45.42. Normalization separates shape from scale. It allows the same transition family to be reused for different lengths and different curvature magnitudes.

Remark 45.43. Normalized representations form the basis for reusable transition registries, family classification, and implementation-independent comparison of alignment elements.

45.9. Structural Role of the Sparse Model

Remark 45.44. The sparse model occupies an intermediate conceptual level:

- below the purely abstract curvature-theoretic foundation,
- but above dense numerical sampling, imported point lists, or format-specific container structures.

Remark 45.45. The sparse model is intentionally not a dense geometric storage format, nor a direct copy of external exchange containers.

Remark 45.46. This intermediate status makes the sparse model suitable as

- an internal modelling layer,
- a reconstruction target from imported data,
- a source for numerical evaluation,
- and a bridge toward standards and BIM-related representations.

45.10. Relation to Imported and Standardized Data

Remark 45.47. Existing engineering data rarely arrives directly in sparse form. Instead, it is typically given through container formats, survey data, legacy descriptions, or derived coordinate sequences.

Remark 45.48. Such data may contain:

- inconsistent stationing,
- incomplete cant information,
- broken coordinate reference systems,
- discontinuities,
- partial geometry fragments,
- or non-synchronized alignment components.

Remark 45.49. The sparse model therefore should not be identified with any one exchange standard. It is more appropriate to view it as a compact, internal representation that can be derived from, and mapped back to, multiple external formats.

RESEARCH: Dock sparse model to railML alignment structures. **RESEARCH:** Dock sparse model to LandXML alignment representations. **OPEN:** Define a minimal common abstraction layer for standards mapping. **OPEN:** Clarify the relation between topology-oriented standards and purely geometric container models.

45.11. Relation to BIM and BIMalignment

Remark 45.50. A sparse alignment model may support BIM-oriented workflows, but it should not merely imitate BIM container logic. Its primary responsibility is to represent alignment structure in a mathematically and operationally coherent way.

RESEARCH: Formulate a BIM-compatible alignment core without losing geometric precision. **OPEN:** Position the sparse model between generic BIM, infrastructure BIM, and explicit BIMalignment concepts. **TODO:** State explicitly where the manuscript aims to push BIMalignment rather than only satisfy BIM conformity.

45.12. Summary

Summary 45.51. The sparse alignment model is a compact, structured representation of an alignment by means of synchronized longitudinal bands together with reconstruction data and concatenation rules.

Within AIM, the sparse alignment model is not interpreted as a static 3D geometry, but as a persistent structural kernel from which geometry, runtime state, and dynamic interpretation are derived.

It serves as a bridge between continuous curvature theory and practical engineering operations such as import, validation, editing, numerical evaluation, optimization, and interoperability.

46. Transition Functions

Motivation

Remark 46.1. Transitions are the decisive element type for curvature-driven alignment modelling. While fixed elements describe geometric states of constant curvature, transitions describe the controlled change between such states.

Remark 46.2. In railway engineering, transitions are essential for ensuring smooth vehicle motion, limiting dynamic forces, and satisfying comfort and safety requirements, as discussed in both engineering-oriented studies and review literature [26, 4].

Remark 46.3. Within AIM, transitions are not primarily interpreted through explicit geometric shape, but through the evolution of curvature.

46.1. Definition by Curvature Evolution

Principle 46.4 (Curvature evolution principle). Within AIM, transitions are classified primarily by their curvature evolution rather than by explicit geometric shape.

Definition 46.5 (Transition). A transition is an alignment element whose curvature varies over its parameter interval.

Definition 46.6 (Transition curvature law). Let $L > 0$. A transition curvature law is a function

$$\kappa : [0, L] \rightarrow \mathbb{R}$$

with prescribed boundary values

$$\kappa(0) = \kappa_0, \quad \kappa(L) = \kappa_1.$$

Remark 46.7. This formulation directly follows the curvature-based description of planar curves [8].

46.2. Normalized Representation

Definition 46.8 (Normalized transition law). A normalized transition is defined by

$$\hat{\kappa} : [0, 1] \rightarrow \mathbb{R}$$

with

$$\hat{\kappa}(0) = 0, \quad \hat{\kappa}(1) = 1.$$

Remark 46.9. Normalization separates the geometric shape of a transition from its scale and allows systematic comparison of transition families.

Remark 46.10. Normalization separates intrinsic transition behaviour from physical scale and therefore enables reusable transition registries and family-based modelling.

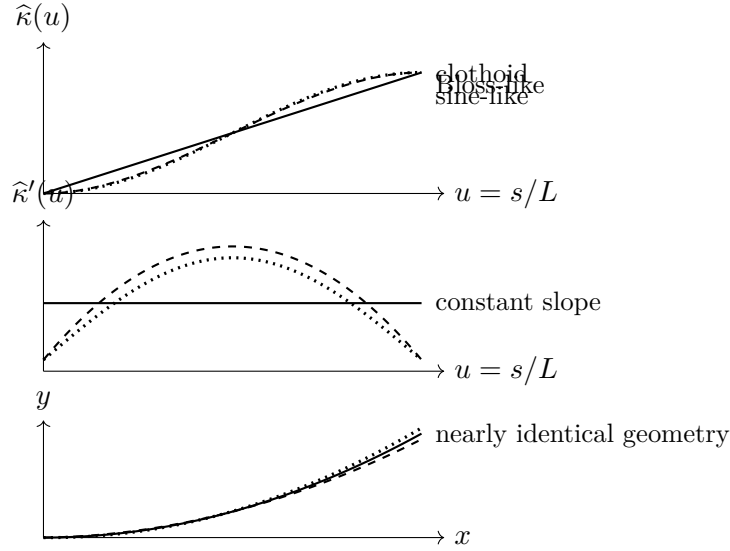


Abbildung 46.1.: Transition families may produce very similar geometric paths while differing strongly in curvature evolution and curvature derivatives. This is why transition design must be evaluated in curvature and dynamic terms, not only in position space.

Remark 46.11. Transitions that appear nearly identical in position space may differ substantially in curvature derivatives and therefore in dynamic behaviour.

46.3. Scaling

Proposition 46.12. Let $\hat{\kappa}$ be a normalized transition law. Then a physical transition is given by

$$\kappa(s) = \kappa_0 + (\kappa_1 - \kappa_0) \hat{\kappa}\left(\frac{s}{L}\right).$$

Remark 46.13. This separates shape from scale and provides the basis for reusable transition families and parameterized design.

46.4. Regularity and Smoothness

Definition 46.14 (Regularity class). A transition is of class C^k if its curvature function is k -times continuously differentiable.

Remark 46.15. In engineering practice, higher-order properties such as curvature derivatives are directly related to ride comfort and dynamic loading, as shown in both theoretical and engineering studies [34, 27].

OPEN: Formal role of κ' and κ'' in engineering rules.

46.5. Transition Families

Definition 46.16 (Transition family). A transition family is a class of curvature laws sharing a common functional structure.

46.5.1. Classical Families

Remark 46.17. The classical transition curve in railway engineering is the clothoid, characterized by linear curvature evolution:

$$\kappa(s) \propto s.$$

Remark 46.18. Historically, this line of development can be traced through early work by Glover [9], Horsburgh [15], and Higgins [13].

46.5.2. Polynomial and Alternative Families

Remark 46.19. Polynomial transition curves provide additional flexibility in shaping curvature evolution and its derivatives.

Remark 46.20. Such approaches are associated with modern polynomial smoothing research [51] as well as control-oriented constructions proposed by Klauder [19, 22].

TODO: Formal definition of polynomial transition families.

46.5.3. Geometric Families

Remark 46.21. Some transition curves are introduced through geometric or implicit relations rather than explicit curvature laws.

Example 46.22. Lemniscate-type or sinusoid-related constructions belong to this class after suitable reparameterization into curvature space [37].

Remark 46.23. These curves require transformation into $\kappa(s)$ representation before they can be integrated into the present framework.

46.5.4. Engineering and Practice-Oriented Families

Remark 46.24. Engineering practice often introduces transition curves through design rules and operational considerations rather than through purely theoretical derivation.

Remark 46.25. The Bloss transition is a notable example of a practical engineering family, originally described by Bloss [3] and further discussed in modern summaries [7].

46.5.5. Modern and Hybrid Families

Remark 46.26. Modern engineering practice considers a variety of transition families with improved dynamic behaviour, smoother endpoint properties, or higher-order control.

Remark 46.27. Examples include smoothed-curvature transitions proposed by Koc [27, 29, 28] and Wiener-Bogen-related approaches [50].

RESEARCH: Systematic classification of modern transition families.

46.6. Decomposition Approaches

Remark 46.28. A transition need not always be treated as a single indivisible object. It may be useful to represent it as a composition of sub-elements.

Remark 46.29. Within AIM, complex transition behaviour may be constructed from composable normalized suboperators such as half-wave and clothoid components.

TODO: Formalize halfWave – clothoid – halfWave decomposition.

OPEN: Is decomposition fundamental or implementation-specific?

46.7. Operator View

Principle 46.30 (Transition operator principle). A transition element acts as a local curvature evolution operator transforming one geometric state into another.

Remark 46.31. Transitions can therefore be interpreted as state evolution operators rather than merely as predefined geometric curve shapes.

RESEARCH: Formal operator algebra of transitions.

46.8. Implementation Perspective

Remark 46.32. In computational systems, transition curves may be represented using:

- analytic expressions,
- polynomial approximations,
- lookup tables,
- compiled hybrid representations.

Remark 46.33. Transition families may be represented through registries containing normalized curvature laws together with derivative and integration operators.

TODO: Registry-based transition system.

OPEN: Trade-off between analytic precision and computational efficiency.

46.9. Relation to Railway Engineering

Remark 46.34. In railway applications, transitions are tightly coupled to operational parameters such as speed and cant, and must satisfy strict engineering constraints [30, 24].

Remark 46.35. The review literature confirms that transition curves must be understood not only geometrically, but also with respect to comfort, maintenance, and dynamic response [4].

46.10. Summary

Summary 46.36. Transitions define the controlled evolution of curvature between geometric states.

Within AIM, they are interpreted as normalized curvature evolution operators rather than merely as predefined geometric curve types.

Their normalized representation enables reusable transition families, operator-based modelling, and runtime reconstruction of alignment geometry and dynamics.

47. Transition Families and Classification

Motivation

Remark 47.1. Transition curves form a central component of alignment design, yet their treatment in the literature remains fragmented across geometric, engineering, and dynamic perspectives.

Remark 47.2. A unified classification is therefore required in order to compare, evaluate, and optimize transition concepts systematically.

—

47.1. Curvature-Based Representation

Proposition 47.3. *Any transition curve can be represented by a curvature function*

$$\kappa : [0, L] \rightarrow \mathbb{R}.$$

Remark 47.4. This representation is independent of the original definition of the curve and provides a common mathematical framework.

—

47.1.1. Normalization

Definition 47.5. A normalized transition is defined on

$$s \in [0, 1]$$

with

$$\kappa(0) = 0, \quad \kappa(1) = 1.$$

Remark 47.6. Normalization removes scale and allows direct comparison of transition families.

—

47.2. Classification by Functional Form

47.2.1. Linear Curvature: Clothoid

Definition 47.7.

$$\kappa(s) = as.$$

Remark 47.8. The clothoid is characterized by constant curvature derivative

$$\kappa'(s) = \text{const.}$$

Remark 47.9. This simplicity explains its widespread adoption in railway engineering.

47.2.2. Polynomial Curvature Families

Definition 47.10.

$$\kappa(s) = \sum_{i=0}^n a_i s^i.$$

Remark 47.11. Polynomial transitions allow explicit control of higher-order derivatives of curvature.

Remark 47.12. Such flexibility enables optimization with respect to dynamic criteria.

Remark 47.13. Klauder-type transitions belong to this class, incorporating dynamic considerations directly into the curvature design.

47.2.3. Geometric Families

Remark 47.14. Some transition curves are defined implicitly or through geometric relations rather than explicit curvature functions.

Remark 47.15. These curves must be transformed into curvature representation to be integrated into the present framework.

47.2.4. Engineering-Based Families

Remark 47.16. Engineering approaches often define transition curves through design rules or empirical considerations.

Remark 47.17. The Bloss transition represents a notable example of such a practical engineering compromise.

47.2.5. Optimized and Hybrid Families

Remark 47.18. Modern approaches introduce transition curves designed explicitly to optimize dynamic performance.

Remark 47.19. These include smoothed curvature transitions and jerk-optimized polynomial curves.

—

47.3. Classification by Higher-Order Behaviour

47.3.1. Curvature Derivatives

Definition 47.20.

$$\kappa'(s), \quad \kappa''(s)$$

denote the first and second derivatives of curvature.

—

Remark 47.21. These quantities determine dynamic behaviour, including acceleration variation and smoothness of force development.

—

47.3.2. Smoothness Classes

Definition 47.22. A transition is of class C^k if κ is k -times continuously differentiable.

—

Remark 47.23. Higher smoothness typically improves dynamic performance and reduces wear.

—

47.4. Comparison of Families

Family	κ	κ'	κ''
Clothoid	linear	constant	0
Polynomial	flexible	flexible	flexible
Engineering	implicit	implicit	implicit
Optimized	designed	controlled	controlled

—

47.5. Interpretation as Control Functions

Proposition 47.24. *Transition curves can be interpreted as control functions acting on system dynamics through curvature evolution.*

Remark 47.25. This interpretation links geometric design directly to dynamic performance.

47.6. Connection to Railway Engineering

Remark 47.26. Railway transition design is governed by dynamic constraints such as acceleration, jerk, and wear.

Remark 47.27. These constraints translate directly into requirements on

$$\kappa(s), \quad \kappa'(s), \quad \kappa''(s).$$

47.7. Main Insight

Theorem 47.28. *Transition curves are not merely geometric connectors, but control functions governing system behaviour.*

47.8. Conclusion

Summary 47.29. Transition curves form a unified class of curvature functions.

Their classification by functional form and higher-order behaviour provides a systematic framework for comparison, optimization, and integration into engineering systems.

This perspective establishes a direct link between geometry and system dynamics.

48. Schwerpunkt-Trassierung

Motivation

Remark 48.1. Classical railway alignment design separates:

- geometric design (alignment),
- cant design,
- dynamic evaluation.

Remark 48.2. This separation leads to suboptimal solutions, as geometry and dynamics are intrinsically coupled. In railway practice, this coupling appears through the interaction of curvature, cant, speed, and vehicle response [24, 6, 4].

Remark 48.3. The concept of *Schwerpunkt-Trassierung* aims to unify these aspects by defining a single design principle based on the motion of a representative point.

Remark 48.4. The idea that railway alignment should be related to the motion of the vehicle centre of gravity rather than merely to the track centerline has also appeared in engineering practice and patent literature [12].

Remark 48.5. This supports the broader AIM interpretation that alignment should be understood as a dynamically relevant spatial state trajectory rather than as a purely geometric curve.

48.1. Conceptual Idea

Definition 48.6 (Reference trajectory). Let

$$\gamma_c : I \rightarrow \mathbb{R}^2$$

denote a reference trajectory representing the motion of a characteristic point of the vehicle system.

Remark 48.7. This point may be interpreted as:

- the center of mass,
- a dynamically relevant reference point,
- or an abstract design trajectory.

Remark 48.8. The conceptual shift consists in evaluating track design through the motion of such a reference point rather than through geometry alone.

48.2. Relation to Track Geometry

Remark 48.9. The physical track centerline $\gamma(s)$ and cant angle $\phi(s)$ induce a spatial configuration of the vehicle.

Definition 48.10 (Induced reference trajectory). Given a pose3 state

$$(\gamma(s), \theta(s), \phi(s)),$$

the induced reference trajectory is

$$\gamma_c(s) = \gamma(s) + d n(s, \phi(s)),$$

where:

- d is a characteristic offset,
- $n(s, \phi)$ is a direction depending on orientation and cant.

Remark 48.11. The exact form of $n(s, \phi)$ depends on the chosen mechanical model.

RESEARCH: Precise definition of offset direction based on vehicle model.

OPEN: Consistent integration of local track frames and global CRS transformations in the Schwerpunkt model.

48.3. Schwerpunkt Principle

Definition 48.12 (Schwerpunkt principle). A railway alignment is said to satisfy the Schwerpunkt principle if the induced reference trajectory $\gamma_c(s)$ possesses prescribed smoothness and dynamical properties.

Remark 48.13. In this formulation, the primary design object is not the track itself, but the induced trajectory of the reference point.

Remark 48.14. This makes the design task closer to vehicle-oriented engineering than to purely geometric construction.

48.4. Variational Formulation

Definition 48.15 (Schwerpunkt functional). Let

$$J_c[\gamma_c] = \int_0^L \|\gamma_c''(s)\|^2 ds.$$

Remark 48.16. This functional penalizes curvature of the reference trajectory and therefore promotes smooth motion.

Definition 48.17 (Schwerpunkt optimization problem). Find

$$(\kappa, \phi)$$

such that the induced trajectory γ_c minimizes

$$J_c[\gamma_c]$$

subject to the pose3 dynamics.

Remark 48.18. This formulation extends the variational interpretation of transition design from curvature laws alone to vehicle-relevant derived motion.

48.5. Coupling to Pose3

Remark 48.19. The trajectory γ_c depends on:

$$\gamma(s), \quad \theta(s), \quad \phi(s).$$

Remark 48.20. Thus, the Schwerpunkt problem becomes a constrained optimization problem in the variables:

$$\kappa(s), \quad \phi(s).$$

Remark 48.21. This places Schwerpunkt-Trassierung naturally within the pose3-based state model introduced earlier and within the broader optimization perspective of railway alignment [31, 30].

48.6. Interpretation

Remark 48.22. The Schwerpunkt formulation shifts the perspective:

- from track geometry
- to vehicle motion.

Remark 48.23. This leads to a unified framework in which:

- geometry,
- cant,
- and dynamics

are optimized simultaneously.

Remark 48.24. Such a viewpoint is consistent with the general observation that railway transition and alignment quality cannot be judged from geometry alone [4, 6].

48.7. Relation to Classical Design

Proposition 48.25. *Classical transition curves correspond to special cases of the Schwerpunkt formulation where the reference trajectory coincides with the track centerline.*

Remark 48.26. In this case, the problem reduces to curvature-based optimization in the plane.

Remark 48.27. Thus, classical alignment design is not contradicted, but embedded as a special case of a more general formulation.

48.8. Advantages

- unified modelling of geometry and dynamics,
- natural integration of cant,
- compatibility with optimization frameworks,
- extensibility to vehicle-specific models.

Remark 48.28. These advantages derive from the fact that the model treats alignment as a state-governed physical system rather than as a static geometric line.

48.9. Open Problems

OPEN: Precise mechanical interpretation of the reference trajectory.

OPEN: Integration with full vehicle dynamics models.

OPEN: Extension to network-level alignment design.

OPEN: Explicit treatment of CRS, local track frames, and coordinate transformations in large-scale alignment systems.

48.10. Connection to the AIM Framework

Summary 48.29. Schwerpunkt-Trassierung provides a conceptual and mathematical extension of classical alignment design.

Instead of prescribing geometric elements, it defines the design problem in terms of a reference trajectory derived from the railway state.

This aligns with the AIM philosophy of treating alignment as a functional object governed by curvature and its derived quantities.

48.11. Vehicle Model

48.11.1. Rigid Body Approximation

Remark 48.30. We model the railway vehicle as a rigid body with a characteristic center of mass.

Definition 48.31 (Center of mass). Let

$$C(s) \in \mathbb{R}^3$$

denote the spatial position of the vehicle center of mass.

Remark 48.32. For planar track geometry, we consider the projection

$$\gamma_c(s) \in \mathbb{R}^2$$

of the center of mass trajectory.

Remark 48.33. This is intentionally a simplified model. Its purpose is not to replace full multibody simulation, but to introduce a mathematically tractable vehicle-related state layer.

48.11.2. Track Coordinate Frame

Definition 48.34 (Frenet frame). Let

$$t(s) = (\cos \theta(s), \sin \theta(s))$$

be the tangent direction and

$$n(s) = (-\sin \theta(s), \cos \theta(s))$$

the normal direction.

Remark 48.35. The pair (t, n) defines a local coordinate system attached to the track.

Remark 48.36. This local frame is sufficient for the present formulation, but future extensions should consistently relate it to a global project CRS.

48.11.3. Cant Rotation

Definition 48.37 (Rotated normal). Let the cant angle $\phi(s)$ define a rotation of the cross-section.

Remark 48.38. In a simplified model, the lateral offset direction remains aligned with $n(s)$, while vertical effects are captured implicitly.

Remark 48.39. This simplification is acceptable at the present conceptual stage, but a full treatment should explicitly include the three-dimensional rotated cross-section and its transformation into the global frame.

48.11.4. Center of Mass Offset

Definition 48.40 (Offset model). Let $d > 0$ denote a characteristic lateral offset. The center of mass trajectory is approximated by

$$\gamma_c(s) = \gamma(s) + d n(s).$$

Remark 48.41. More refined models may include dependence on $\phi(s)$, e.g.

$$\gamma_c(s) = \gamma(s) + d \cos \phi(s) n(s).$$

RESEARCH: Inclusion of full 3D vehicle geometry.

48.12. Dynamics of the Center of Mass

48.12.1. Velocity and Acceleration

Proposition 48.42. *The velocity of the center of mass is*

$$\gamma'_c(s) = t(s) + d n'(s).$$

Remark 48.43. Using

$$n'(s) = -\kappa(s) t(s),$$

we obtain

$$\gamma'_c(s) = (1 - d\kappa(s)) t(s).$$

Proposition 48.44. *The acceleration of the center of mass is proportional to*

$$\gamma''_c(s) = -d\kappa'(s) t(s) + (1 - d\kappa(s))\kappa(s) n(s).$$

Remark 48.45. Thus, both curvature and its derivative influence the motion of the center of mass.

Remark 48.46. This is one of the key conceptual benefits of the formulation: it makes the dynamic relevance of higher-order geometric quantities explicit.

48.12.2. Effective Forces

Definition 48.47 (Lateral acceleration). The lateral acceleration of the center of mass is

$$a_c(s) = v^2(1 - d\kappa(s))\kappa(s).$$

48. Schwerpunkt-Trassierung

Remark 48.48. Including cant yields the effective acceleration

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 48.49. The latter term reflects the same curvature–cant coupling that appears in railway design standards and in dynamic alignment interpretation [24, 6].

48.13. Coupled Design Problem

Definition 48.50 (Vehicle-based objective). Define

$$J_{\text{veh}}[\kappa, \phi] = \int_0^L \|\gamma_c''(s)\|^2 ds.$$

Remark 48.51. This functional directly measures the smoothness of the motion of the center of mass.

Definition 48.52 (Dynamic objective). Define

$$J_{\text{dyn}}[\kappa, \phi] = \int_0^L \left(v^2 \kappa(s) - g \sin \phi(s) \right)^2 ds.$$

Definition 48.53 (Combined vehicle functional). A combined objective is

$$J[\kappa, \phi] = \alpha J_{\text{veh}} + \beta J_{\text{dyn}} + \gamma \int_0^L (\kappa'(s))^2 ds + \delta \int_0^L (\phi'(s))^2 ds.$$

Remark 48.54. This combines geometric smoothing, dynamic balancing, and higher-order regularization within a single optimization framework.

48.14. Schwerpunkt-Trassierung (Formal Definition)

Definition 48.55 (Schwerpunkt alignment). A Schwerpunkt alignment is a pair

$$(\kappa, \phi)$$

that minimizes the functional

$$J[\kappa, \phi]$$

subject to:

- pose3 dynamics,
- boundary conditions,
- engineering constraints.

Remark 48.56. This turns alignment design into the optimization of a vehicle-related state trajectory rather than the mere composition of geometric elements.

48.15. Interpretation

Remark 48.57. In this formulation:

- the track is no longer the primary design object,
- the motion of the vehicle becomes central.

Remark 48.58. Classical alignment design is recovered as a special case where

$$d = 0 \quad \text{and} \quad \phi(s) = 0.$$

48.16. Discussion

Remark 48.59. The presented model is intentionally simplified. However, it already captures:

- coupling between curvature and cant,
- influence of curvature derivatives,
- relation between geometry and dynamics.

Remark 48.60. Its main contribution is therefore conceptual and structural: it defines a mathematically coherent bridge between alignment geometry and vehicle-oriented design thinking.

RESEARCH: Extension to full multibody vehicle models.

RESEARCH: Calibration using measured track data.

OPEN: Integration into practical design workflows.

48.17. Connection to the AIM Framework

Summary 48.61. The combination of pose3 and a vehicle-based model provides a rigorous foundation for Schwerpunkt-Trassierung.

It transforms alignment design into a problem of optimizing the motion of a representative physical system.

This supports the central AIM idea that alignment is not merely a geometric object, but a functional entity governed by curvature and its physical implications.

Teil IX.

System and Pipeline

Analysis

What do we already know?

Remark 48.62. The previous parts established the conceptual, geometric, operational, realization-aware, and modeling foundations of AIM.

Remark 48.63. Railway alignments can now be represented through sparse semantic models, reconstructed through runtime services, interpreted through state operators, and related to physical reality through realization concepts.

Remark 48.64. The individual building blocks of the framework therefore exist.

Remark 48.65. However, a collection of components does not yet constitute a complete system.

What is still missing?

Remark 48.66. Practical railway information modelling requires more than isolated models and operators.

Remark 48.67. Data must move through processing chains.

States must be transformed.

Services must interact.

Failures must be detected.

Results must be generated from intermediate representations.

Remark 48.68. The missing element is therefore the processing architecture that connects individual concepts into a coherent operational framework.

Remark 48.69. AIM must be understood not only as a collection of models, but as a pipeline of interacting transformations.

What comes next?

Remark 48.70. This part introduces the system-level interpretation of AIM.

Remark 48.71. The focus shifts from individual concepts toward the interaction between concepts.

Remark 48.72. The chapters that follow investigate:

- transformation pipelines,

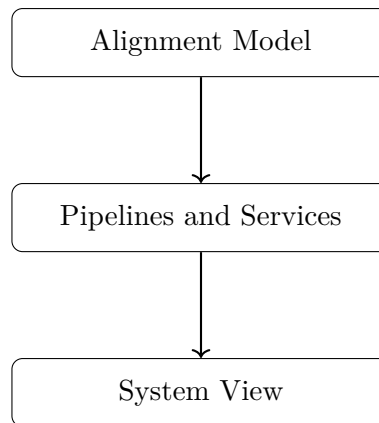
- runtime services,
- operator chains,
- system interactions,
- failure modes,
- and architectural design principles.

Remark 48.73. The objective is to explain how railway information flows through the AIM framework from acquisition to evaluation, realization, and optimization.

Remark 48.74. This perspective ultimately reveals AIM as an operator-based railway information system rather than a mere collection of geometric methods.

Relation to the AIM Roadmap

Remark 48.75. Within the AIM roadmap, the present part corresponds to the transition from alignment models toward processing pipelines and system services.



Remark 48.76. The progression

Alignment Model $\rightarrow$ Pipelines and Services $\rightarrow$ System View

marks the transition from static railway representations toward operational processing architectures.

Remark 48.77. This system perspective prepares the subsequent treatment of optimization, engineering workflows, and practical deployment.

Principle 48.78 (Pipeline Principle). Within AIM, railway information is processed through explicit operator pipelines in which states, models, realizations, and evaluations are connected by well-defined transformations.

Summary 48.79. Modeling explains how railway information is represented.

Analysis explains how railway information is processed.

The present part therefore establishes the system architecture that connects models, operators, services, and transformations into a coherent AIM framework.

48.18. AIM Processing Pipeline

Motivation

Remark 48.80. Railway alignment modelling involves multiple domains:

- global coordinate systems (CRS),
- geometric alignment representation,
- physical realization,
- measurement data,
- optimization.

Remark 48.81. These domains are often treated separately, leading to fragmented models and inconsistent results.

Remark 48.82. Within the AIM framework, these components are unified into a single processing pipeline.

48.18.1. Operator Chain

Definition 48.83 (AIM pipeline). The complete modelling and optimization process is described as a composition of operators:

$$\boxed{\mathcal{P} = \mathcal{W}^{-1} \circ \mathcal{R} \circ \mathcal{M} \circ \mathcal{W} \circ \mathcal{C}}$$

Remark 48.84. The operators are:

- $\mathcal{C}$: CRS transformation,
- $\mathcal{W}$: world-to-track transformation,
- $\mathcal{M}$: alignment model (parametric or hybrid),
- $\mathcal{R}$: realization operator,
- $\mathcal{W}^{-1}$: track-to-world transformation.

—

48.18.2. Detailed Structure

Remark 48.85. The pipeline maps raw data to realized geometry:

CRS data $\xrightarrow{\mathcal{C}}$ world coordinates $\xrightarrow{\mathcal{W}}$ (s, d) $\xrightarrow{\mathcal{M}}$ $\kappa(s)$ $\xrightarrow{\mathcal{R}}$ $\gamma_{\text{real}}(s)$ $\xrightarrow{\mathcal{W}^{-1}}$ world geometry.

—

48.18.3. Hybrid Model Integration

Remark 48.86. The alignment model operator is hybrid:

$$\mathcal{M} : (\mathbf{T}, \delta\kappa) \mapsto \kappa(s).$$

Remark 48.87. Thus, the pipeline operates on both:

- structured parameters,
- local corrections.

—

48.18.4. Inverse Problem

Definition 48.88 (Pipeline inversion). Given observed data x_i , the inverse problem is:

$$\min_{\mathbf{x}} \sum_i \|\mathcal{P}(\mathbf{x})(s_i) - x_i\|^2.$$

Remark 48.89. This corresponds to fitting a model such that the realized geometry matches observations.

—

48.18.5. Optimization Embedding

Remark 48.90. The pipeline is embedded into an optimization problem:

$$\min_{\mathbf{x}} J(\mathcal{P}(\mathbf{x})).$$

Remark 48.91. The objective is evaluated on the realized geometry, not on the parametric model.

—

48.18.6. Jacobian Structure

Remark 48.92. The derivative of the pipeline follows from the chain rule:

$$\frac{\partial \mathcal{P}}{\partial \mathbf{x}} = D\mathcal{W}^{-1} \cdot D\mathcal{R} \cdot D\mathcal{M} \cdot D\mathcal{W}.$$

Remark 48.93. Each component has specific properties:

- DM : sparse and structured,
 - DR : smoothing operator,
 - DW : nonlinear projection,
 - DC : linear transformation.
-

48.18.7. Computational Implications

Remark 48.94. The pipeline exhibits:

- sparsity (model level),
- nonlinearity (projection level),
- smoothing (realization level),
- scaling effects (CRS level).

Remark 48.95. Efficient implementation requires:

- hybrid representations,
 - block-structured solvers,
 - LOD-dependent evaluation.
-

48.18.8. Level-of-Detail Interpretation

Remark 48.96. Different parts of the pipeline dominate at different scales:

- large scale: CRS and world geometry,
 - medium scale: parametric alignment,
 - small scale: corrections and realization.
-

48.18.9. Key Insight

Proposition 48.97. *Alignment modelling is not a single mapping, but a composition of geometric, physical, and computational operators.*

48.18.10. Connection to AIM

Summary 48.98. The AIM framework is fundamentally defined by its operator pipeline.

It integrates coordinate systems, geometry, physics, and optimization into a single coherent model.

This unified perspective enables:

- consistent data integration,
- physically meaningful optimization,
- scalable computation.

Thus, AIM transforms alignment modelling from a collection of isolated methods into a structured computational system.

48.19. AIM Pipeline Overview

48.20. AIM Master Architecture

48.21. Pipeline Story: From Measurement to Alignment

48.21.1. A Single Point

Remark 48.99. Consider a single measured point x in the real world.

Remark 48.100. It may originate from:

- a surveying campaign,
- a track recording vehicle,
- or a design dataset.

Remark 48.101. At this stage, the point exists only in a global coordinate system.

48.21.2. Step 1: CRS Transformation

Remark 48.102. The point is first interpreted within a coordinate reference system:

$$x_{\text{crs}} \xrightarrow{\mathcal{C}} x_{\text{world}}.$$

Remark 48.103. This step ensures that all data is expressed in a consistent spatial frame.

48.21.3. Step 2: Projection onto the Track

Remark 48.104. The point is projected onto the alignment:

$$x_{\text{world}} \xrightarrow{\mathcal{W}} (s, d).$$

Remark 48.105. This is a nonlinear operation:

- the longitudinal position s is determined,
- the lateral deviation d is computed.

Remark 48.106. At this moment, the point becomes part of the intrinsic alignment coordinate system.

48.21.4. Step 3: Interpretation through the Model

Remark 48.107. The alignment model provides a curvature description:

$$(s, d) \xrightarrow{\mathcal{M}} \kappa(s).$$

Remark 48.108. Here, two components interact:

- the parametric structure (design intent),
- local corrections (data-driven adjustments).

Remark 48.109. The point is no longer treated as isolated data, but as part of a structured geometric system.

48.21.5. Step 4: Physical Realization

Remark 48.110. The theoretical model is transformed by physical processes:

$$\kappa(s) \xrightarrow{\mathcal{R}} \gamma_{\text{real}}(s).$$

Remark 48.111. This step reflects:

- construction constraints,
- mechanical behaviour,
- maintenance operations.

Remark 48.112. The resulting geometry is not identical to the theoretical model.

48.21.6. Step 5: Back to World Coordinates

Remark 48.113. The realized alignment is mapped back into world space:

$$\gamma_{\text{real}}(s) \xrightarrow{\mathcal{W}^{-1}} x_{\text{real}}.$$

Remark 48.114. This produces the geometry that is visible and measurable in reality.

48.21.7. Step 6: Comparison and Feedback

Remark 48.115. The original measurement and the realized position are compared:

$$x_{\text{real}} \approx x_{\text{world}}.$$

Remark 48.116. The deviation drives optimization:

- parameters are adjusted,
- corrections are updated,
- the pipeline is evaluated again.

48.21.8. Interpretation

Remark 48.117. The point has passed through multiple representations:

- global coordinates,
- intrinsic alignment coordinates,
- parametric model,
- realized geometry.

Remark 48.118. Each transformation introduces structure, constraints, and interpretation.

48.21.9. Key Insight

Proposition 48.119. *A measured point is not directly part of the alignment.*

It becomes part of the alignment only after passing through the world-to-track transformation and the alignment model.

48.21.10. Connection to AIM

Summary 48.120. The AIM framework transforms raw spatial data into structured alignment information through a sequence of operators.

This process integrates:

- coordinate systems,

- geometric modelling,
- physical realization,
- optimization.

Thus, alignment modelling is not a static representation, but a dynamic process linking measurement, theory, and construction.

48.22. Failure Modes and Limitations of the AIM Pipeline

Motivation

Remark 48.121. Any modelling and optimization framework must account not only for its capabilities, but also for its limitations.

Remark 48.122. Within the AIM pipeline, failure modes arise from the interaction of:

- coordinate systems,
- geometric transformations,
- physical realization,
- numerical optimization.

—

48.22.1. CRS-Related Errors

Remark 48.123. Errors in coordinate reference systems include:

- incorrect CRS identification,
- inconsistent transformations,
- mixed coordinate systems within datasets.

Proposition 48.124. *CRS inconsistencies lead to systematic spatial distortions that cannot be corrected by local alignment optimization.*

—

48.22.2. Projection Errors (World-to-Track)

Remark 48.125. The world-to-track transformation may fail due to:

- ambiguous nearest points,
- high curvature regions,

- sparse or noisy alignment representations.

Proposition 48.126. *Projection errors introduce nonlocal distortions in the intrinsic coordinate s , affecting all downstream computations.*

48.22.3. Model Mis-Specification

Remark 48.127. The parametric model may fail to represent the true alignment due to:

- insufficient transition families,
- incorrect parameterization,
- overly restrictive structure.

Remark 48.128. Hybrid correction terms mitigate but do not eliminate this issue.

48.22.4. Overfitting vs. Underfitting

Remark 48.129. Hybrid models introduce a trade-off:

- underfitting: parametric model too rigid,
- overfitting: correction terms capture noise.

Proposition 48.130. *Regularization is required to balance model fidelity and stability.*

48.22.5. Realization Effects

Remark 48.131. The realization operator introduces:

- smoothing of curvature,
- loss of high-frequency information,
- dependence on construction processes.

Proposition 48.132. *Certain features of the target alignment are not physically realizable and will be systematically suppressed.*

48.22.6. Inverse Problem Instability

Remark 48.133. The inverse realization problem is ill-posed:

- multiple target alignments yield similar realized geometry,
- small changes in data may cause large parameter variations.

Proposition 48.134. *Regularization and prior knowledge are essential for stable solutions.*

48.22.7. Numerical Optimization Issues

Remark 48.135. Optimization may fail due to:

- poor initial guesses,
- nonconvex objective functions,
- ill-conditioned Jacobians.

Remark 48.136. Despite structured sparsity, convergence is not guaranteed.

48.22.8. Block Structure Degeneracy

Remark 48.137. The block structure may become degenerate if:

- coupling terms dominate,
- insufficient data separates variables,
- parameters become redundant.

Proposition 48.138. *Loss of block separability reduces computational efficiency and interpretability.*

48.22.9. Measurement Noise and Bias

Remark 48.139. Measurement data may contain:

- random noise,
- systematic bias,
- missing or inconsistent segments.

Proposition 48.140. *Noise propagates through the pipeline and may be amplified by inverse operations.*

48.22.10. Scale-Dependent Failures

Remark 48.141. Different failure modes dominate at different scales:

- large scale: CRS and projection errors,
 - medium scale: model mis-specification,
 - small scale: realization and measurement noise.
-

48.22.11. Non-Uniqueness of Solutions

Remark 48.142. Different alignment models may produce nearly identical realized geometry.

Proposition 48.143. *The mapping*

$$\kappa_{\text{target}} \mapsto \gamma_{\text{real}}$$

is not injective.

48.22.12. Practical Engineering Limits

Remark 48.144. Certain effects are outside the scope of the model:

- material-dependent behaviour,
- long-term degradation,
- construction variability.

Remark 48.145. These factors must be treated as external constraints.

48.22.13. Key Insight

Proposition 48.146. *The AIM pipeline is inherently approximate and limited by both physical and computational constraints.*

48.22.14. Connection to AIM

Summary 48.147. Failure modes in the AIM pipeline arise from the interaction of geometry, physics, data, and computation.

Understanding these limitations is essential for:

- robust model design,
- stable optimization,
- reliable engineering application.

Thus, failure analysis is not an auxiliary component, but an integral part of the AIM framework.

48.23. Design Principles of Alignment-Based Information Modelling

Motivation

Remark 48.148. The AIM framework integrates geometry, coordinate systems, realization, and optimization into a unified model.

Remark 48.149. From this integration, a set of fundamental design principles emerges.

48.23.1. Principle 1: Curvature-Centric Modelling

Proposition 48.150. *Railway alignments should be modelled primarily through curvature functions $\kappa(s)$, not through point-based geometry.*

Remark 48.151. Curvature provides:

- intrinsic representation,
 - direct link to dynamics,
 - natural integration with optimization.
-

48.23.2. Principle 2: Separation of Representation and Realization

Proposition 48.152. *The theoretical **alignment** model must be separated from its physical realization.*

Remark 48.153. The realization operator introduces:

- smoothing,

- constraints,
- deviations from the ideal model.

Remark 48.154. Ignoring this separation leads to systematic modelling errors.

48.23.3. Principle 3: Hybrid Modelling

Proposition 48.155. *Alignment models should combine parametric structure with local correction terms.*

Remark 48.156. This allows:

- interpretable design intent,
 - robust handling of imperfect data,
 - flexible refinement.
-

48.23.4. Principle 4: Operator-Based Architecture

Proposition 48.157. *Alignment modelling should be formulated as a composition of operators:*

$$\mathcal{P} = \mathcal{W}^{-1} \circ \mathcal{R} \circ \mathcal{M} \circ \mathcal{W} \circ \mathcal{C}.$$

Remark 48.158. Each operator represents a distinct domain:

- coordinate systems,
 - geometric modelling,
 - physical processes,
 - data transformation.
-

48.23.5. Principle 5: Design in Realized Space

Proposition 48.159. *Optimization objectives must be evaluated on realized geometry, not on theoretical models.*

Remark 48.160. The relevant mapping is:

$$\kappa_{\text{target}} \mapsto \gamma_{\text{real}}.$$

Remark 48.161. Design must account for the transformation induced by realization.

48.23.6. Principle 6: Exploitation of Structure

Proposition 48.162. *The intrinsic block and sparsity structure of the problem must be exploited for efficient computation.*

Remark 48.163. This includes:

- locality in arc-length,
 - block decomposition (curvature vs. cant),
 - sparse Jacobians.
-

48.23.7. Principle 7: Scale Awareness

Proposition 48.164. *Different components of the pipeline must be evaluated at appropriate scales.*

Remark 48.165. • CRS dominates at large scale,

- parametric modelling at medium scale,
 - corrections and realization at small scale.
-

48.23.8. Principle 8: Measurement Integration

Proposition 48.166. *Measured data must be integrated through the world-to-track transformation, not directly in world space.*

Remark 48.167. This ensures consistency with the intrinsic alignment representation.

48.23.9. Principle 9: Regularization as Necessity

Proposition 48.168. *Regularization is an essential component of alignment modelling and optimization.*

Remark 48.169. It stabilizes:

- inverse problems,
 - hybrid corrections,
 - numerical optimization.
-

48.23.10. Principle 10: Acceptance of Non-Uniqueness

Proposition 48.170. *Alignment solutions are not unique and must be evaluated based on engineering criteria.*

Remark 48.171. Different models may produce nearly identical realized geometry.

Remark 48.172. Thus, solution quality must be defined through:

- dynamics,
 - constraints,
 - robustness.
-

48.23.11. Summary

Summary 48.173. The AIM framework is governed by a set of design principles that emerge from the interaction of geometry, physics, data, and computation.

These principles define:

- how alignments should be represented,
- how they should be transformed,
- how they should be optimized,
- and how their limitations must be handled.

Together, they provide a coherent foundation for alignment-based information modelling as a unified engineering and computational discipline.

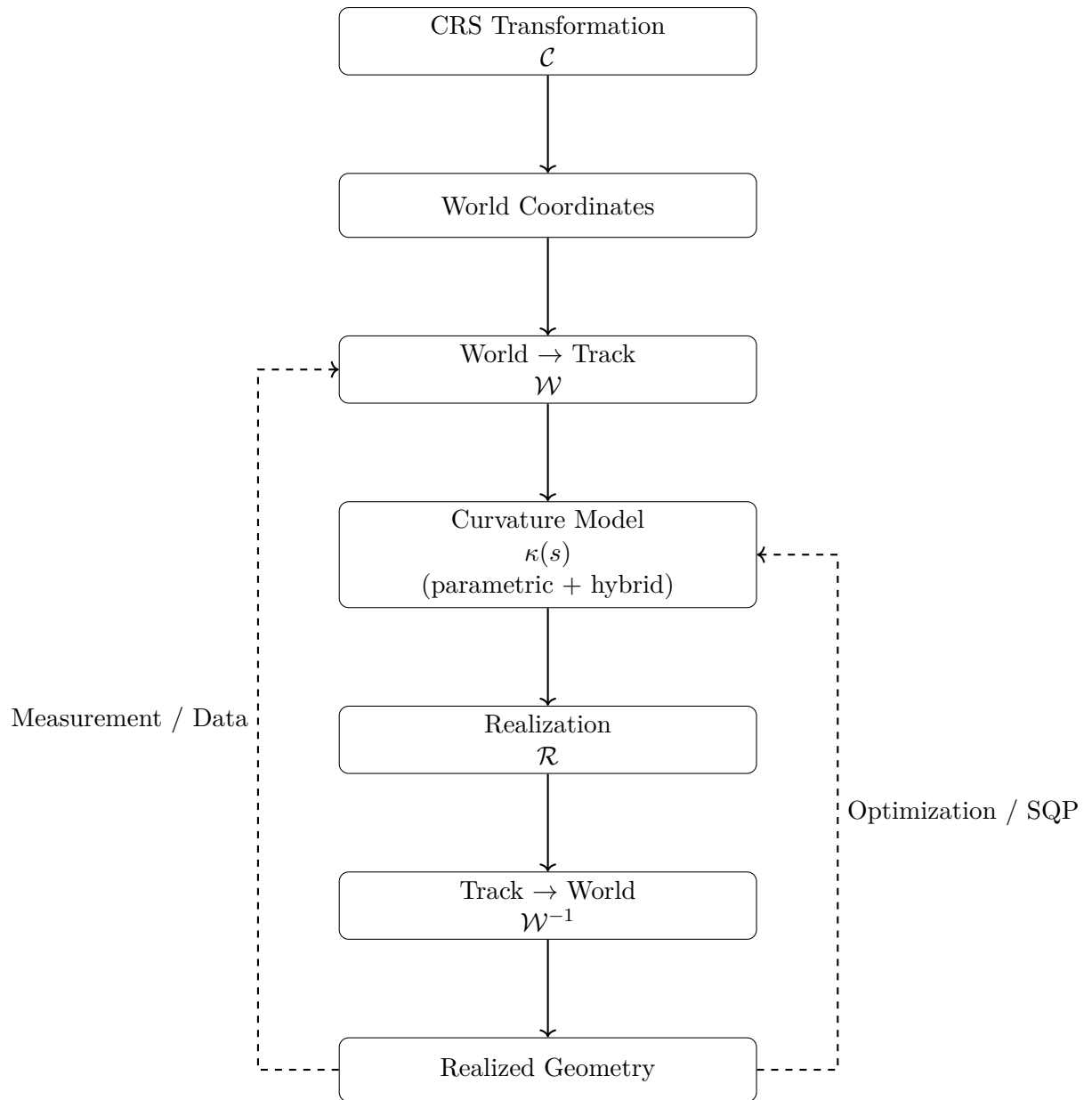


Abbildung 48.1.: AIM processing pipeline as composition of operators linking coordinate systems, alignment modelling, realization, and optimization.

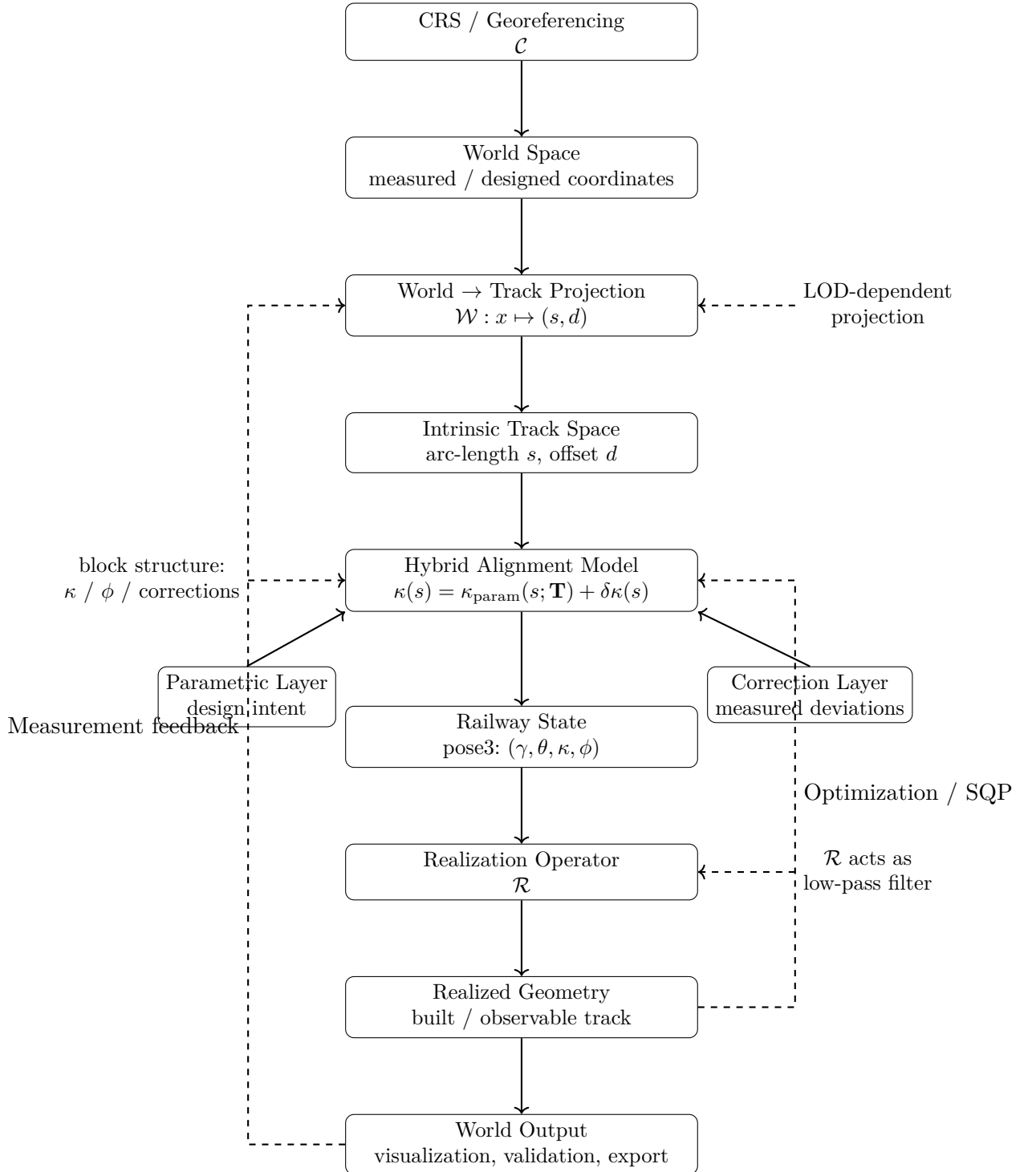


Abbildung 48.2.: Level-2 AIM master architecture. The framework combines CRS handling, nonlinear world-to-track projection, hybrid curvature modelling, pose3 state reconstruction, realization, and optimization feedback. The central modelling object is not the geometric curve itself, but the curvature-driven and realization-aware operator system producing it.

Teil X.

Optimization

Optimization

What do we already know?

Remark 48.174. The previous parts established railway geometry, railway states, runtime evaluation, realization-aware modelling, sparse alignment representations, and processing pipelines.

Remark 48.175. Within AIM, railway alignments can be represented, reconstructed, evaluated, measured, realized, and analyzed through a coherent operator-based framework.

Remark 48.176. The framework is therefore capable of describing railway systems and their interaction with reality.

Remark 48.177. However, description alone is not the ultimate objective of railway engineering.

What is still missing?

Remark 48.178. Engineering design requires improvement.

Remark 48.179. Railway alignments must be adapted to constraints, operational requirements, construction conditions, comfort criteria, realization effects, and economic objectives.

Remark 48.180. Measurements must be interpreted.

Designs must be refined.

Conflicting objectives must be balanced.

Alternative solutions must be evaluated.

Remark 48.181. The missing element is therefore optimization.

Remark 48.182. Optimization transforms railway modelling from descriptive analysis into active design and decision support.

What comes next?

Remark 48.183. This part formulates railway alignment design, reconstruction, and improvement as optimization problems.

Remark 48.184. Within AIM, optimization is not an external component added after modelling.

It emerges naturally from the intrinsic state and curvature representations developed throughout the previous parts.

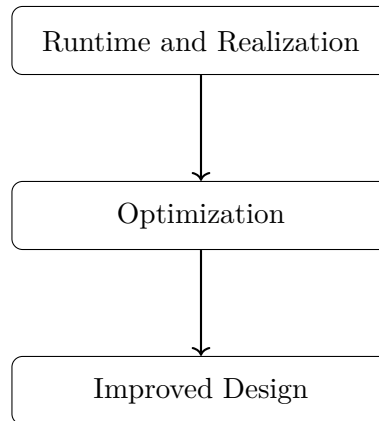
Remark 48.185. The chapters that follow investigate:

- numerical foundations,
- alignment optimization,
- structured residual formulations,
- sparse and block-structured solvers,
- optimizer architectures,
- and realization-aware optimization.

Remark 48.186. Special emphasis is placed on exploiting the analytical structure of railway alignment models in order to obtain efficient and interpretable optimization procedures.

Relation to the AIM Roadmap

Remark 48.187. Within the AIM roadmap, the present part corresponds to the transition from runtime evaluation and realization toward systematic improvement of railway systems.



Remark 48.188. The progression

Runtime and Realization $\rightarrow$ Optimization $\rightarrow$ Improved Design

marks the transition from evaluating railway systems toward actively improving them.

Remark 48.189. Optimization therefore forms the bridge between railway information modelling and railway engineering decision making.

Principle 48.190 (Optimization Principle). Within AIM, railway optimization is interpreted as the systematic improvement of railway states, realizations, and operational behaviour through structured objective functions and operator-based models.

Summary 48.191. Geometry explains shape.

States explain configuration.

Runtime explains evaluation.

Reality explains realization.

Optimization explains improvement.

The present part therefore transforms AIM from a descriptive framework into a framework capable of supporting railway design, reconstruction, and decision making.

49. Numerical Reconstruction

Motivation

Remark 49.1. In the curvature-based framework, geometric objects are not given explicitly as coordinate sets. Instead, they are reconstructed from curvature information by integration.

Remark 49.2. As a consequence, numerical methods are not merely implementation details, but an essential part of the modelling process.

49.1. Reconstruction Principle

Definition 49.3 (Reconstruction from curvature). Let $\kappa(s)$ be a curvature function and let

$$(x_0, y_0, \theta_0)$$

be a start pose. The corresponding curve is obtained by solving

$$\theta'(s) = \kappa(s), \quad \gamma'(s) = (\cos \theta(s), \sin \theta(s)).$$

Remark 49.4. Even if κ is given analytically, closed-form solutions for γ are generally not available.

49.2. Discretization

Definition 49.5 (Discretization). A discretization of the parameter interval $[0, L]$ is a finite sequence

$$0 = s_0 < s_1 < \cdots < s_n = L.$$

Definition 49.6 (Discrete reconstruction). A discrete reconstruction computes approximate values

$$\gamma(s_i), \quad \theta(s_i)$$

at the discretization points.

Remark 49.7. The choice of discretization strongly influences both computational cost and geometric accuracy.

TODO: Formal criteria for discretization strategies (uniform vs adaptive).

49.3. Integration Methods

Remark 49.8. The reconstruction problem can be formulated as an initial value problem for a system of ordinary differential equations.

49.3.1. Basic Scheme

Definition 49.9 (Explicit step). Given (γ_i, θ_i) at s_i , an explicit step computes

$$\theta_{i+1} = \theta_i + \kappa(s_i) \Delta s,$$

$$\gamma_{i+1} = \gamma_i + (\cos \theta_i, \sin \theta_i) \Delta s.$$

Remark 49.10. This corresponds to a first-order method and is generally insufficient for high-precision reconstruction.

49.3.2. Higher-Order Methods

TODO: Runge–Kutta methods for coupled systems.

TODO: Romberg integration for curvature integration.

TODO: Adaptive step size control.

OPEN: Separation of curvature integration and geometric integration.

49.4. Error Analysis

Definition 49.11 (Local error). The local error of a step is the deviation introduced at a single integration step.

Definition 49.12 (Global error). The global error is the accumulated deviation over the entire interval.

Remark 49.13. Errors arise from:

- discretization,
- numerical integration,
- approximation of curvature laws.

RESEARCH: Quantitative bounds for error propagation in curvature-driven reconstruction.

49.5. Sensitivity

Remark 49.14. The reconstruction process is sensitive to perturbations in curvature. Small deviations in κ may lead to significant geometric drift.

RESEARCH: Sensitivity analysis with respect to curvature perturbations.

OPEN: Relation between curvature noise and positional uncertainty.

49.6. Sampling and Representation

Definition 49.15 (Sampling). Sampling refers to the extraction of a finite set of points from a continuous curve representation.

Remark 49.16. Sampling is required for:

- visualization,
- collision detection,
- export to point-based formats.

TODO: Error-controlled sampling strategies.

OPEN: Difference between sampling for visualization and sampling for computation.

49.7. Reconstruction in the Sparse Model

Remark 49.17. In the sparse alignment model, reconstruction is performed element-wise. Each element contributes a local curvature law, which is integrated sequentially.

Definition 49.18 (Sequential reconstruction). Let $(e_1, \dots, e_n)$ be a sparse alignment. The reconstruction proceeds by integrating each element in sequence, using the end state of the previous element as initial state for the next.

OPEN: Handling of numerical drift across element boundaries.

49.8. Numerical Stability

Remark 49.19. Numerical stability is a central concern, especially for long alignments or highly curved geometries.

RESEARCH: Stability criteria for curvature-based integration.

TODO: Techniques for drift correction and re-normalization.

49.9. Connection to Optimization

Remark 49.20. Numerical reconstruction is closely linked to optimization problems. In fitting and design tasks, reconstruction is performed repeatedly within iterative algorithms.

TODO: SQP formulation for alignment fitting.

RESEARCH: Connection to optimal control problems.

49.10. Summary

Summary 49.21. Numerical reconstruction is the process of deriving geometric representations from curvature data by integration.

It introduces discretization, approximation, and error propagation as intrinsic components of the modelling framework.

Therefore, numerical methods are not merely technical tools, but an essential layer connecting mathematical definitions with practical geometry.

49.11. Numerical Reconstruction from Curvature

49.11.1. Problem Statement

Remark 49.22. Given a curvature function

$$\kappa : [0, L] \rightarrow \mathbb{R},$$

and initial conditions

$$\gamma(0), \quad \theta(0),$$

the geometric alignment is obtained by solving:

$$\gamma'(s) = (\cos \theta(s), \sin \theta(s)), \quad \theta'(s) = \kappa(s).$$

Remark 49.23. This defines a nonlinear initial value problem.

49.11.2. Discrete Approximation

Definition 49.24 (Discretization). Let

$$0 = s_0 < s_1 < \cdots < s_N = L$$

be a partition with step sizes Δs_i .

Remark 49.25. We approximate:

$$\kappa_i \approx \kappa(s_i).$$

—

49.11.3. Forward Integration Scheme

Definition 49.26 (Explicit integration). Given (γ_i, θ_i) , define:

$$\theta_{i+1} = \theta_i + \kappa_i \Delta s_i,$$

$$\gamma_{i+1} = \gamma_i + \Delta s_i (\cos \theta_i, \sin \theta_i).$$

Remark 49.27. This corresponds to an explicit Euler scheme.

—

49.11.4. Midpoint Scheme

Definition 49.28 (Midpoint method). Define:

$$\theta_{i+\frac{1}{2}} = \theta_i + \frac{1}{2} \kappa_i \Delta s_i,$$

$$\gamma_{i+1} = \gamma_i + \Delta s_i (\cos \theta_{i+\frac{1}{2}}, \sin \theta_{i+\frac{1}{2}}),$$

$$\theta_{i+1} = \theta_i + \kappa_i \Delta s_i.$$

Remark 49.29. This improves accuracy and stability compared to the explicit scheme.

—

49.11.5. Extension to Pose3

Remark 49.30. Including cant, we obtain:

$$\phi_{i+1} = \phi_i + \omega_i \Delta s_i.$$

Remark 49.31. Thus, the full discrete system is:

$$(\gamma_i, \theta_i, \phi_i) \longrightarrow (\gamma_{i+1}, \theta_{i+1}, \phi_{i+1}).$$

—

49.11.6. Numerical Stability

Remark 49.32. The reconstruction is sensitive to:

- step size Δs ,

- noise in κ_i ,
- and large coordinate values.

Remark 49.33. To ensure stability:

- sufficiently small step sizes must be used,
- curvature should be smoothed,
- and local coordinate frames are recommended.

—

49.11.7. Floating Origin Technique

Remark 49.34. For large coordinates, numerical precision degrades.

Definition 49.35 (Floating origin). A local coordinate system is introduced such that:

$$\gamma_i^{\text{local}} = \gamma_i - \gamma_{\text{ref}}.$$

Remark 49.36. This reduces numerical errors in computation and visualization.

—

49.11.8. Consistency with CRS

Remark 49.37. Reconstruction depends on:

- initial position,
- initial orientation,
- and coordinate system definition.

Remark 49.38. Thus, numerical reconstruction must be performed within a consistent CRS.

—

49.11.9. Interpretation

Summary 49.39. Numerical reconstruction provides the link between curvature-based models and geometric representation.

It transforms abstract functions into concrete spatial geometry, while preserving the structure required for analysis and optimization.

50. Optimization of Alignments

Motivation

Remark 50.1. In practical applications, alignments are not only reconstructed from data, but also designed, modified, and improved.

Remark 50.2. This leads naturally to optimization problems in which alignment parameters are adjusted to satisfy geometric, physical, and regulatory constraints, as already emphasized in railway alignment literature [30, 24].

50.1. General Optimization Problem

Definition 50.3 (Alignment optimization problem). An alignment optimization problem consists of:

- a parameterized alignment model,
- an objective function,
- a set of constraints.

Definition 50.4 (Objective function). An objective function is a mapping

$$J : \mathcal{A} \rightarrow \mathbb{R}$$

that assigns a cost or quality measure to an alignment $\mathcal{A}$.

Remark 50.5. Typical objectives include:

- minimizing curvature variation,
- minimizing deviation from measurement data,
- minimizing construction cost,
- maximizing comfort.

Such objective classes are consistent with both general numerical optimization literature [36] and railway-specific optimization approaches [30].

50.2. Parameterization

Remark 50.6. The choice of parameters is central to the formulation of the optimization problem.

Definition 50.7 (Parameter vector). An alignment is described by a parameter vector

$$\mathbf{x} \in \mathbb{R}^n$$

encoding quantities such as:

- element lengths,
- curvature values,
- transition parameters,
- geometric constraints.

Remark 50.8. The sparse alignment model provides a natural parameterization by element-wise variables.

TODO: Explicit parameter sets for fixed and transition elements.

50.3. Constraints

Definition 50.9 (Constraint). A constraint is a condition of the form

$$g_i(\mathbf{x}) \leq 0, \quad h_j(\mathbf{x}) = 0.$$

Remark 50.10. Constraints arise from:

- geometric continuity,
- engineering design rules,
- physical limitations,
- regulatory requirements.

50.3.1. Geometric Constraints

- continuity of position and direction,
- continuity of curvature.

50.3.2. Engineering Constraints

- minimum radius,
- maximum cant,
- limits on curvature derivatives.

Remark 50.11. Such conditions reflect standard railway design practice and are closely related to design rules discussed in engineering literature and standards [24, 6].

OPEN: Formal representation of engineering rules in constraint form.

50.4. Relation to Numerical Reconstruction

Remark 50.12. Evaluation of the objective function requires reconstruction of the alignment geometry from the parameter vector.

Remark 50.13. Thus, optimization is intrinsically linked to numerical reconstruction, as discussed in the previous chapter. From a numerical perspective, this places alignment optimization within the broader context of differential equations and numerical approximation [11, 41].

50.5. Sequential Quadratic Programming

Remark 50.14. Sequential Quadratic Programming (SQP) is a powerful method for solving nonlinear constrained optimization problems [36].

Definition 50.15 (SQP iteration). An SQP method approximates the original problem by a sequence of quadratic subproblems.

Remark 50.16. Each iteration solves a quadratic approximation of the objective function subject to linearized constraints.

TODO: Explicit formulation of SQP for alignment parameters.

RESEARCH: Adaptation of SQP to sparse alignment structures.

50.6. Alignment Optimization as Control Problem

Remark 50.17. The curvature function $\kappa(s)$ may be interpreted as a control variable in a continuous system.

Definition 50.18 (Control formulation). The alignment problem can be formulated as an optimal control problem with state variables (γ, θ) and control κ .

Remark 50.19. This viewpoint is compatible with the curvature-centred perspective on railway alignment advocated by Kufver [31, 30].

RESEARCH: Connection between alignment optimization and optimal control theory.

50.7. Network-Level Optimization

Remark 50.20. Real-world railway systems involve not only single alignments, but networks with branching and merging structures.

RESEARCH: Extension of optimization from single alignments to networks.

OPEN: Handling of switches, crossings, and parallel tracks.

50.8. Robustness and Uncertainty

Remark 50.21. Optimization must account for uncertainty in input data.

RESEARCH: Robust optimization under uncertain geometry.

OPEN: Trade-off between optimality and robustness.

50.9. Implementation Perspective

Remark 50.22. Practical optimization systems must balance:

- numerical stability,
- computational efficiency,
- model expressiveness.

Remark 50.23. This balance is central both in generic numerical optimization [36] and in railway-specific optimization settings [30].

TODO: Integration of optimization into the modelling workflow.

50.10. Historical Context

Remark 50.24. Classical systems such as AXTRAN have demonstrated the feasibility of optimization-based alignment design.

Remark 50.25. Modern approaches aim to extend these ideas using more flexible models and improved numerical methods, in line with the development from classical railway design methods toward explicit optimization formulations [30].

TODO: Reconstruction and generalization of AXTRAN-like methods.

50.11. Formal Optimization in Curvature Space

50.11.1. Problem Setting

Remark 50.26. Let $L > 0$ be fixed and let

$$\kappa : [0, L] \rightarrow \mathbb{R}$$

50. Optimization of Alignments

denote the curvature function of a transition.

Remark 50.27. We assume the boundary conditions

$$\kappa(0) = 0, \quad \kappa(L) = \kappa_1,$$

and optionally

$$\kappa'(0) = 0, \quad \kappa'(L) = 0.$$

50.11.2. Admissible Set

Definition 50.28 (Admissible transition set).

$$\mathcal{A} = \left\{ \kappa \in H^1(0, L) \mid \kappa(0) = 0, \kappa(L) = \kappa_1 \right\}.$$

50.11.3. Variational Problem

Definition 50.29 (Quadratic curvature-gradient functional).

$$J[\kappa] = \int_0^L (\kappa'(s))^2 ds.$$

Definition 50.30 (Optimization problem). Find $\kappa^* \in \mathcal{A}$ such that

$$J[\kappa^*] = \min_{\kappa \in \mathcal{A}} J[\kappa].$$

50.11.4. Euler–Lagrange Equation

Proposition 50.31. *If κ^* is a minimizer, then*

$$\kappa^{*''}(s) = 0.$$

50.11.5. Interpretation

Proposition 50.32. *The minimizer is*

$$\kappa^*(s) = \frac{\kappa_1}{L} s.$$

Remark 50.33. Thus, the clothoid appears as the solution of a variational problem. This connects classical railway transition design with an optimization view in curvature space and provides a formal counterpart to the historical role of the clothoid in railway engineering [9, 13].

50.11.6. Higher-Order Model

Definition 50.34.

$$J_2[\kappa] = \int_0^L (\kappa''(s))^2 ds.$$

Proposition 50.35. *Minimizers satisfy*

$$\kappa^{(4)}(s) = 0.$$

Remark 50.36. This explains the occurrence of cubic transition laws and links them to modern polynomial transition design [51].

50.11.7. Generalized View

Remark 50.37. General functionals of the form

$$J[\kappa] = \int_0^L F(\kappa, \kappa', \kappa''; v, c) \, ds$$

allow incorporation of dynamic and engineering criteria.

50.11.8. Connection to the Main Thesis

Summary 50.38. Transition design can be formulated as an optimization problem in curvature space.

Classical transition curves arise as solutions of such problems, supporting the interpretation of transitions as curvature control functions.

50.12. Coupled Optimization of Curvature and Cant

50.12.1. Motivation

Remark 50.39. In railway applications, curvature and cant cannot be optimized independently. Both jointly determine the effective lateral acceleration and its rate of change.

Remark 50.40. This motivates a coupled optimization problem in which both

$$\kappa(s) \quad \text{and} \quad \phi(s)$$

are treated as design variables.

Remark 50.41. Such coupling is fully consistent with railway alignment design standards, where curvature, speed, and cant are not independent design quantities [6, 24].

50.12.2. Effective Acceleration Functional

Definition 50.42 (Effective lateral acceleration). The effective lateral acceleration is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \sin \phi(s).$$

Remark 50.43. A natural optimization objective is to reduce the magnitude of unbalanced lateral acceleration.

Definition 50.44 (Acceleration-based functional). Define

$$J_a[\kappa, \phi] = \int_0^L a_{\text{eff}}(s)^2 ds = \int_0^L (v^2 \kappa(s) - g \sin \phi(s))^2 ds.$$

50.12.3. Jerk-Based Functional

Definition 50.45 (Jerk). The jerk is given by

$$j(s) = v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s).$$

Definition 50.46 (Jerk functional). Define

$$J_j[\kappa, \phi] = \int_0^L j(s)^2 ds.$$

That is,

$$J_j[\kappa, \phi] = \int_0^L (v^3 \kappa'(s) - gv \cos \phi(s) \phi'(s))^2 ds.$$

Remark 50.47. This functional penalizes rapid variation of effective acceleration and therefore provides a natural dynamic smoothness criterion. Such criteria are closely related to engineering concerns about comfort and dynamic compatibility [6, 4].

50.12.4. Combined Objective

Definition 50.48 (Coupled objective functional). A general coupled objective may be written as

$$J[\kappa, \phi] = \alpha J_a[\kappa, \phi] + \beta J_j[\kappa, \phi] + \gamma J_\kappa[\kappa] + \delta J_\phi[\phi],$$

where:

$$J_\kappa[\kappa] = \int_0^L (\kappa'(s))^2 ds, \quad J_\phi[\phi] = \int_0^L (\phi'(s))^2 ds.$$

Remark 50.49. The weights

$$\alpha, \beta, \gamma, \delta \geq 0$$

determine the relative importance of equilibrium, comfort, curvature smoothness, and cant smoothness.

50.12.5. Admissible Set

Definition 50.50 (Admissible railway state set). Let

$$\mathcal{B} = \left\{ (\kappa, \phi) \mid \kappa \in H^1(0, L), \phi \in H^1(0, L), \kappa(0) = 0, \kappa(L) = \kappa_1 \right\}.$$

Remark 50.51. Additional boundary conditions may be imposed, for example:

$$\phi(0) = 0, \quad \phi(L) = \phi_1,$$

or derivative constraints such as

$$\kappa'(0) = \kappa'(L) = 0, \quad \phi'(0) = \phi'(L) = 0.$$

50.12.6. Coupled Optimization Problem

Definition 50.52 (Coupled railway optimization problem). Find

$$(\kappa^*, \phi^*) \in \mathcal{B}$$

such that

$$J[\kappa^*, \phi^*] = \min_{(\kappa, \phi) \in \mathcal{B}} J[\kappa, \phi].$$

50.12.7. Interpretation

Remark 50.53. This problem extends classical transition design in two directions:

- from pure geometry to coupled geometry–cant design,
- from curve selection to functional optimization.

Remark 50.54. In this framework, the transition curve is no longer chosen from a catalogue, but emerges as part of an optimal coupled state.

50.12.8. Special Cases

Proposition 50.55. *If $\phi(s) \equiv 0$, the coupled problem reduces to a pure curvature optimization problem.*

Proposition 50.56. *If $a_{\text{eff}}(s) = 0$ for all s , then*

$$v^2 \kappa(s) = g \sin \phi(s),$$

and the design is perfectly balanced.

Remark 50.57. These special cases show that classical equilibrium design appears as a subcase of the coupled optimization framework.

50.12.9. Relation to Optimal Control

Remark 50.58. The coupled optimization problem may be interpreted as an optimal control problem with controls

$$\kappa(s), \quad \phi'(s),$$

state variables

$$x(s), y(s), \theta(s), \phi(s),$$

and cost functional $J[\kappa, \phi]$.

Remark 50.59. This provides a direct bridge from railway alignment design to modern control-theoretic and variational methods.

50.12.10. Connection to the Main Thesis

Summary 50.60. The coupled optimization problem shows that railway alignment design is not a purely geometric task.

Instead, it is the optimization of a coupled state system in which curvature and cant jointly determine dynamic behaviour.

This supports the central thesis of the manuscript that transition curves and cant evolution should be understood as control functions in a curvature-based railway state model.

50.13. Summary

Summary 50.61. Alignment optimization treats geometry as a variable rather than a fixed input.

It combines parameterization, numerical reconstruction, and constrained optimization methods to produce alignments that satisfy both geometric and engineering requirements.

50.14. Discrete Formulation of Alignment Optimization

50.14.1. Discretization of the Domain

Definition 50.62 (Discretization). Let

$$0 = s_0 < s_1 < \cdots < s_N = L$$

be a partition of the interval $[0, L]$.

Remark 50.63. We denote

$$\Delta s_i = s_{i+1} - s_i.$$

50.14.2. Discrete Variables

Definition 50.64 (Discrete curvature). Let

$$\kappa_i \approx \kappa(s_i)$$

denote the curvature values at the nodes.

Definition 50.65 (Discrete cant). Let

$$\phi_i \approx \phi(s_i)$$

50. Optimization of Alignments

denote the cant values.

Definition 50.66 (Parameter vector). The optimization variable is

$$\mathbf{x} = (\kappa_0, \dots, \kappa_N, \phi_0, \dots, \phi_N) \in \mathbb{R}^{2(N+1)}.$$

50.14.3. Discrete Pose Reconstruction

Remark 50.67. The pose3 state is reconstructed by numerical integration.

Definition 50.68 (Discrete integration). Given initial conditions

$$\gamma_0, \theta_0, \phi_0,$$

we compute iteratively:

$$\theta_{i+1} = \theta_i + \kappa_i \Delta s_i,$$

$$\gamma_{i+1} = \gamma_i + \Delta s_i (\cos \theta_i, \sin \theta_i),$$

$$\phi_{i+1} = \phi_i + \omega_i \Delta s_i.$$

Remark 50.69. Here, ω_i may be expressed via differences of ϕ_i . This discrete reconstruction corresponds to standard numerical integration ideas [11, 41].

50.14.4. Finite Differences

Definition 50.70 (First derivative). Approximate derivatives by

$$\kappa'(s_i) \approx \frac{\kappa_{i+1} - \kappa_i}{\Delta s_i}, \quad \phi'(s_i) \approx \frac{\phi_{i+1} - \phi_i}{\Delta s_i}.$$

Definition 50.71 (Second derivative).

$$\kappa''(s_i) \approx \frac{\kappa_{i+1} - 2\kappa_i + \kappa_{i-1}}{(\Delta s_i)^2}.$$

50.14.5. Discrete Objective Functions

Curvature Smoothness

$$J_\kappa(\mathbf{x}) = \sum_{i=0}^{N-1} \left(\frac{\kappa_{i+1} - \kappa_i}{\Delta s_i} \right)^2 \Delta s_i.$$

Cant Smoothness

$$J_\phi(\mathbf{x}) = \sum_{i=0}^{N-1} \left(\frac{\phi_{i+1} - \phi_i}{\Delta s_i} \right)^2 \Delta s_i.$$

Dynamic Objective

$$J_{\text{dyn}}(\mathbf{x}) = \sum_{i=0}^N (v^2 \kappa_i - g \sin \phi_i)^2 \Delta s_i.$$

Schwerpunkt Objective

Remark 50.72. Using finite differences, we approximate

$$\|\gamma_c''(s_i)\|^2$$

by second-order differences of γ_c .

$$J_{\text{veh}}(\mathbf{x}) = \sum_{i=1}^{N-1} \|\gamma_{c,i+1} - 2\gamma_{c,i} + \gamma_{c,i-1}\|^2.$$

50.14.6. Combined Discrete Problem

Definition 50.73 (Discrete objective).

$$J(\mathbf{x}) = \alpha J_{\text{veh}} + \beta J_{\text{dyn}} + \gamma J_{\kappa} + \delta J_{\phi}.$$

50.14.7. Constraints

Definition 50.74 (Equality constraints).

$$\kappa_0 = 0, \quad \kappa_N = \kappa_1.$$

Definition 50.75 (Inequality constraints).

$$|\kappa_i| \leq \kappa_{\max}, \quad |\phi_i| \leq \phi_{\max}.$$

Remark 50.76. Further constraints may include:

- bounds on derivatives,
- continuity conditions,
- alignment anchoring points.

50.14.8. Final Optimization Problem

Definition 50.77 (Discrete optimization problem). Find

$$\mathbf{x}^* \in \mathbb{R}^{2(N+1)}$$

such that

$$J(\mathbf{x}^*) = \min_{\mathbf{x}} J(\mathbf{x})$$

subject to the given constraints.

50.14.9. Structure of the Problem

Remark 50.78. The problem has the structure of a nonlinear least-squares problem.

Remark 50.79. It is therefore well suited for:

- Sequential Quadratic Programming,
- Gauss–Newton methods,
- sparse optimization techniques.

50.14.10. Connection to Implementation

Remark 50.80. The discrete formulation directly corresponds to a computational model:

- κ_i correspond to element parameters,
- ϕ_i correspond to cant samples,
- reconstruction matches numerical evaluation in software.

Summary 50.81. The continuous alignment optimization problem can be reduced to a finite dimensional nonlinear optimization problem.

This provides the direct link between theoretical modelling and practical implementation.

50.15. Sequential Quadratic Programming (SQP)

50.15.1. Problem Structure

Remark 50.82. We consider the discrete optimization problem

$$\min_{\mathbf{x}} J(\mathbf{x}) \quad \text{subject to} \quad g(\mathbf{x}) = 0, \quad h(\mathbf{x}) \leq 0.$$

Definition 50.83 (Parameter vector).

$$\mathbf{x} = (\kappa_0, \dots, \kappa_N, \phi_0, \dots, \phi_N).$$

50.15.2. Residual Formulation

Definition 50.84 (Residual vector).

$$J(\mathbf{x}) = \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|^2.$$

Remark 50.85. Typical residual components:

$$r_i^{(\kappa)} = \sqrt{\gamma} \frac{\kappa_{i+1} - \kappa_i}{\Delta s}, \quad r_i^{(\phi)} = \sqrt{\delta} \frac{\phi_{i+1} - \phi_i}{\Delta s},$$

$$r_i^{(\text{dyn})} = \sqrt{\beta} (v^2 \kappa_i - g \sin \phi_i).$$

50.15.3. Gauss–Newton Step

Definition 50.86 (SQP step). At iteration k , compute $\mathbf{p}_k$ from:

$$(J_r^T J_r) \mathbf{p}_k = -J_r^T \mathbf{r}.$$

Remark 50.87. The Hessian is approximated by:

$$H \approx J_r^T J_r.$$

50.15.4. Sparse Structure

Remark 50.88. The system matrix has band structure:

- tridiagonal coupling for smoothness terms,
- diagonal contributions from dynamic terms.

50.15.5. Constraints Handling

Remark 50.89. Constraints are handled via:

- projection,
- penalty terms,
- Lagrange multipliers.

50.15.6. Iteration Scheme

Algorithm 1 SQP iteration

```

initialize  $\mathbf{x}_0$ 
for  $k = 0, 1, \dots$  do
  compute  $\mathbf{r}(\mathbf{x}_k)$ 
  compute  $J_r(\mathbf{x}_k)$ 
  solve  $(J_r^T J_r) \mathbf{p}_k = -J_r^T \mathbf{r}$ 
  choose step size  $\alpha_k$ 
  update  $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{p}_k$ 
end for
```

50.15.7. Interpretation

Summary 50.90. SQP provides a practical and efficient method for solving the discrete alignment optimization problem.

Its efficiency follows from the sparse and local structure of the underlying model.

50.16. Construction of Initial Guesses

50.16.1. Motivation

Remark 50.91. The performance of SQP methods strongly depends on the choice of the initial parameter vector.

Remark 50.92. In railway applications, initial data is often available in the form of:

- measurement polylines,
- existing alignments,
- partially structured design data.

50.16.2. Input Representation

Definition 50.93 (Input polyline). Let

$$P = (p_0, \dots, p_M), \quad p_i \in \mathbb{R}^2$$

denote a measured or imported polyline.

50.16.3. Arc-Length Parameterization

Definition 50.94. Define cumulative arc-length

$$s_0 = 0, \quad s_{i+1} = s_i + \|p_{i+1} - p_i\|.$$

Remark 50.95. This defines a parameterization of the input data.

50.16.4. Initial Curvature Estimation

Definition 50.96 (Discrete curvature). For three consecutive points, define:

$$\kappa_i \approx \frac{\theta_{i+1} - \theta_i}{\Delta s_i},$$

where

$$\theta_i = \arg(p_{i+1} - p_i).$$

Remark 50.97. This provides a first estimate of curvature from geometric data.

50.16.5. Smoothing

Remark 50.98. Raw curvature estimates are typically noisy.

Definition 50.99 (Smoothing operator). Apply a smoothing filter:

$$\kappa_i^{(0)} = \sum_j w_{ij} \kappa_j.$$

Remark 50.100. Possible choices:

- moving average,
- Gaussian filter,
- low-pass filtering.

50.16.6. Initial Cant Estimation

Remark 50.101. Cant may be initialized based on equilibrium:

$$\phi_i^{(0)} = \arcsin \left(\frac{v^2 \kappa_i^{(0)}}{g} \right).$$

Remark 50.102. Alternatively, cant may be:

- set to zero,
- taken from input data,
- approximated by design rules.

50.16.7. Projection to Discrete Grid

Definition 50.103. Interpolate the values $\kappa_i^{(0)}, \phi_i^{(0)}$ onto the optimization grid:

$$s_0, \dots, s_N.$$

50.16.8. Heuristic Segmentation

Remark 50.104. The input alignment may be segmented into:

- straight sections,
- circular arcs,
- transition regions.

Definition 50.105 (Segment detection). A segment is classified based on curvature:

$$\kappa_i \approx 0 \quad \Rightarrow \text{straight},$$

$$\kappa_i \approx \text{const} \quad \Rightarrow \text{arc}.$$

Remark 50.106. This structure can be used to improve the initial guess and is compatible with railway alignment reconstruction approaches such as those discussed by Kufver [31, 30].

50.16.9. Final Initial Guess

Definition 50.107 (Initial parameter vector). The initial guess is

$$\mathbf{x}_0 = (\kappa_0^{(0)}, \dots, \kappa_N^{(0)}, \phi_0^{(0)}, \dots, \phi_N^{(0)}).$$

50.16.10. Interpretation

Remark 50.108. The initial guess combines:

- measured geometry,
- smoothing,
- physical approximation.

Summary 50.109. A robust initial guess transforms raw geometric input into a structured parameter vector suitable for optimization.

This step is essential for bridging real-world data and numerical methods.

50.17. Analytical Structure of the Optimization Problem

50.17.1. Motivation

Remark 50.110. In classical numerical optimization, Jacobians are often computed by finite differences or automatic differentiation.

Remark 50.111. Within the AIM framework, this is not necessary. The alignment model is inherently differentiable.

50.17.2. Fundamental Observation

Proposition 50.112. *All state variables are obtained via arc-length integration:*

$$\begin{aligned}\theta(s) &= \theta_0 + \int_0^s \kappa(\sigma) \, d\sigma, \\ \gamma(s) &= \gamma_0 + \int_0^s (\cos \theta, \sin \theta) \, d\sigma.\end{aligned}$$

50.17.3. Parameter Types

Definition 50.113. Primary parameter classes:

- curvature parameters dK ,
- length parameters dL ,
- transition parameters dT .

50.17.4. Analytical Derivatives

Proposition 50.114.

$$\frac{\partial \theta(s)}{\partial p} = \int_0^s \frac{\partial \kappa}{\partial p} d\sigma.$$

Proposition 50.115.

$$\frac{\partial \gamma(s)}{\partial p} = \int_0^s \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix} \frac{\partial \theta}{\partial p} d\sigma.$$

Remark 50.116. All derivatives reduce to arc-length integrals.

50.17.5. Key Insight

Proposition 50.117. *The optimization problem admits exact Jacobians without numerical approximation.*

Remark 50.118. This eliminates:

- finite differences,
- numerical instability,
- high computational cost.

50.17.6. Block Structure

Remark 50.119. The parameter vector decomposes as:

$$(\boldsymbol{\kappa}, \boldsymbol{\phi}).$$

Remark 50.120. This induces a block structure:

$$H = \begin{pmatrix} H_{\kappa\kappa} & H_{\kappa\phi} \\ H_{\phi\kappa} & H_{\phi\phi} \end{pmatrix}.$$

50.17.7. Interpretation

Summary 50.121. The structure of the optimization problem is not imposed numerically, but follows directly from the curvature-based modelling approach.

This yields:

- exact derivatives,
- sparse matrices,
- natural block structure.

This is a central advantage of the AIM framework.

50.18. Parametric Representation instead of Sampling

50.18.1. Motivation

Remark 50.122. Classical discretization represents curvature and cant by samples:

$$\kappa_i \approx \kappa(s_i), \quad \phi_i \approx \phi(s_i).$$

Remark 50.123. This leads to high-dimensional optimization problems with weak structure and limited interpretability.

Remark 50.124. Within the AIM framework, curvature and cant are instead represented by parameterized functions.

—

50.18.2. Parametric Curvature Model

Definition 50.125 (Parametric curvature). A transition is described by:

$$\kappa(s) = \kappa_0 + (\kappa_1 - \kappa_0) \hat{\kappa}\left(\frac{s}{L}, \mathbf{T}\right),$$

where:

- L is the element length,
- $\mathbf{T}$ is a vector of transition parameters,
- $\hat{\kappa}$ is a normalized prototype function.

—

50.18.3. Parametric Cant Model

Definition 50.126 (Parametric cant). Cant is represented analogously:

$$\phi(s) = \phi_0 + (\phi_1 - \phi_0) \hat{\phi}\left(\frac{s}{L}, \mathbf{T}_\phi\right).$$

—

50.18.4. Parameter Vector

Definition 50.127 (Reduced parameter vector). The optimization variable becomes:

$$\mathbf{x} = (L, \kappa_0, \kappa_1, \mathbf{T}, \phi_0, \phi_1, \mathbf{T}_\phi).$$

Remark 50.128. The dimension is independent of discretization density.

—

50.18.5. Dimensional Reduction

Remark 50.129. A sampled representation uses $O(N)$ variables.

Remark 50.130. The parametric representation uses:

$$O(1) \text{ parameters per element.}$$

Remark 50.131. This yields a drastic reduction in problem size.

—

50.18.6. Derivative Structure

Proposition 50.132. *Derivatives reduce to:*

$$\frac{\partial \kappa}{\partial p} = (\kappa_1 - \kappa_0) \frac{\partial \hat{\kappa}}{\partial p}.$$

Remark 50.133. Thus, all derivatives are computed analytically from prototype functions.

—

50.18.7. Integration into Reconstruction

Remark 50.134. The parametric representation is evaluated through:

$$\theta(s) = \int \kappa(s) ds, \quad \gamma(s) = \int (\cos \theta, \sin \theta) ds.$$

Remark 50.135. This yields a smooth and consistent geometry without discretization artifacts.

—

50.18.8. Comparison to Sample-Based Methods

	Sample-based	Parametric
dimension	high	low
smoothness	enforced numerically	intrinsic
interpretability	low	high
constraints	many local	few global

—

50.18.9. Interpretation

Remark 50.136. The parametric approach shifts the problem from:

- fitting values

to:

- designing functions.

—

50.18.10. Connection to Transition Families

Remark 50.137. Different transition curves correspond to different prototype functions $\hat{\kappa}$.

Remark 50.138. Thus, the optimization problem includes selection and tuning of transition families.

—

50.18.11. Computational Implications

- smaller optimization problems,
- better conditioning,
- faster convergence,
- direct integration with analytical Jacobians.

—

50.18.12. Connection to Classical Systems

Remark 50.139. This formulation is conceptually close to classical systems such as AXTRAN, which operate on element parameters rather than sampled data.

—

50.18.13. Connection to AIM

Summary 50.140. The parametric representation replaces discrete sampling by structured functions.

It enables a compact, interpretable, and analytically differentiable model, which forms the basis for efficient and robust alignment optimization.

This approach is central to the AIM philosophy.

50.19. Hybrid Parametric–Discrete Model

50.19.1. Motivation

Remark 50.141. Purely parametric models provide low-dimensional structure but may lack flexibility for fitting real-world data.

Remark 50.142. Purely sampled models provide flexibility but lead to large and ill-conditioned optimization problems.

Remark 50.143. A hybrid approach combines both advantages.

—

50.19.2. Hybrid Representation

Definition 50.144 (Hybrid curvature model).

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s),$$

where:

- κ_{param} is a parametric base model,
- $\delta\kappa(s)$ is a correction term.

Definition 50.145 (Hybrid cant model).

$$\phi(s) = \phi_{\text{param}}(s; \mathbf{T}_\phi) + \delta\phi(s).$$

—

50.19.3. Discretization of Correction

Remark 50.146. The correction term is represented discretely:

$$\delta\kappa_i = \delta\kappa(s_i), \quad \delta\phi_i = \delta\phi(s_i).$$

—

50.19.4. Extended Parameter Vector

Definition 50.147.

$$\mathbf{x} = (\mathbf{T}, \mathbf{T}_\phi, \delta\kappa, \delta\phi).$$

—

50.19.5. Regularization of Corrections

Remark 50.148. To avoid overfitting, the correction terms are penalized:

$$J_\delta = \int_0^L (\delta\kappa'(s))^2 ds + \int_0^L (\delta\phi'(s))^2 ds.$$

Remark 50.149. This enforces smooth, small deviations from the parametric model.

—

50.19.6. Residual Structure

Remark 50.150. The residual now consists of:

- parametric contributions,
 - correction contributions,
 - data fitting terms.
-

50.19.7. Block Structure

Remark 50.151. The parameter vector decomposes as:

$$(\mathbf{T}, \delta\boldsymbol{\kappa}) \quad \text{and} \quad (\mathbf{T}_\phi, \delta\boldsymbol{\phi}).$$

Remark 50.152. This yields a multi-level block structure:

- coarse scale (parameters),
 - fine scale (corrections).
-

50.19.8. Interpretation

Remark 50.153. The parametric model captures:

- geometric intent,
- transition structure,
- engineering semantics.

Remark 50.154. The correction term captures:

- measurement noise,
 - local deviations,
 - modelling imperfections.
-

50.19.9. Connection to Measurement Data

Remark 50.155. Measured polylines can be fitted by minimizing:

$$\sum_i \|\gamma(s_i) - p_i\|^2,$$

with the hybrid representation.

Remark 50.156. The parametric part explains the global structure, while the correction absorbs residual errors.

50.19.10. Computational Strategy

- first optimize parametric variables,
 - then refine using correction terms,
 - optionally alternate between both levels.
-

50.19.11. Relation to LOD

Remark 50.157. The hybrid model naturally supports level-of-detail concepts:

- coarse LOD: parametric model only,
 - fine LOD: parametric + corrections.
-

50.19.12. Connection to AIM

Summary 50.158. The hybrid representation combines structured modelling and local flexibility.

It enables robust alignment reconstruction from imperfect data while preserving the interpretability of parametric models.

This approach bridges engineering design and data-driven fitting within the AIM framework.

50.20. Hybrid Representation and Block Structure

50.20.1. Motivation

Remark 50.159. Purely parametric models provide a low-dimensional and interpretable representation of alignments, but lack flexibility when fitting real-world data.

Remark 50.160. Purely sampled models provide maximum flexibility, but lead to high-dimensional and often ill-conditioned optimization problems.

Remark 50.161. A hybrid formulation combines both advantages.

50.20.2. Hybrid Representation

Definition 50.162 (Hybrid curvature model).

$$\kappa(s) = \kappa_{\text{param}}(s; \mathbf{T}) + \delta\kappa(s),$$

where:

- κ_{param} is a structured parametric model,
- $\delta\kappa(s)$ is a correction term.

Definition 50.163 (Hybrid cant model).

$$\phi(s) = \phi_{\text{param}}(s; \mathbf{T}_\phi) + \delta\phi(s).$$

50.20.3. Discretization of Corrections

Remark 50.164. The correction terms are represented discretely:

$$\delta\kappa_i = \delta\kappa(s_i), \quad \delta\phi_i = \delta\phi(s_i).$$

Definition 50.165 (Extended parameter vector).

$$\mathbf{x} = (\mathbf{T}, \mathbf{T}_\phi, \delta\kappa, \delta\phi).$$

50.20.4. Regularization

Remark 50.166. To avoid overfitting, correction terms are regularized:

$$J_\delta = \int_0^L (\delta\kappa'(s))^2 ds + \int_0^L (\delta\phi'(s))^2 ds.$$

Remark 50.167. Thus, the correction captures only smooth, physically meaningful deviations.

50.20.5. Residual Structure

Remark 50.168. The residual vector consists of:

- parametric contributions,
- correction contributions,
- physical constraints,

- data fitting terms.

50.20.6. Block Structure of Variables

Remark 50.169. The parameter vector decomposes naturally as:

$$(\mathbf{T}, \delta\boldsymbol{\kappa}) \quad \text{and} \quad (\mathbf{T}_\phi, \delta\boldsymbol{\phi}).$$

Remark 50.170. This yields a two-level structure:

- coarse scale: parametric variables,
- fine scale: correction variables.

50.20.7. Jacobian Block Structure

Definition 50.171 (Block Jacobian). The Jacobian takes the form:

$$J_r = \begin{pmatrix} J_{\kappa\kappa}^{\text{param}} & J_{\kappa\kappa}^{\text{corr}} & 0 & 0 \\ 0 & 0 & J_{\phi\phi}^{\text{param}} & J_{\phi\phi}^{\text{corr}} \\ J_{\text{dyn},\kappa} & J_{\text{dyn},\kappa}^\delta & J_{\text{dyn},\phi} & J_{\text{dyn},\phi}^\delta \end{pmatrix}.$$

Remark 50.172. Key properties:

- parametric and correction terms are weakly coupled,
- curvature and cant are largely separable,
- coupling occurs mainly through dynamics.

50.20.8. Hierarchical Interpretation

Remark 50.173. The hybrid model introduces a hierarchical structure:

- parametric layer defines global alignment structure,
- correction layer refines local deviations.

Remark 50.174. This reflects engineering practice:

- design defines intent,
- construction introduces deviations.

50.20.9. Computational Strategy

- optimize parametric variables first,
- introduce correction terms for refinement,
- optionally alternate between both levels.

50.20.10. Connection to Level-of-Detail

Remark 50.175. The hybrid representation naturally supports LOD concepts:

- coarse LOD: parametric model only,
- fine LOD: parametric + corrections.

50.20.11. Key Insight

Proposition 50.176. *The hybrid formulation separates structural modelling from data fitting while preserving a sparse and well-conditioned optimization problem.*

50.20.12. Connection to AIM

Summary 50.177. The combination of hybrid modelling and block structure is a central feature of the AIM framework.

It enables:

- interpretable parametric modelling,
- robust handling of imperfect data,
- efficient optimization due to sparsity and locality.

Thus, it bridges the gap between engineering design and data-driven reconstruction.

51. Optimizer Architecture

51.1. Overview

Remark 51.1. The previous chapters introduced the mathematical and physical foundations of alignment modelling and optimization.

Remark 51.2. In this chapter, these concepts are combined into a concrete system architecture that enables the implementation of a full alignment optimization engine.

Remark 51.3. The architecture consists of the following main components:

- data ingestion (import and preprocessing),
 - initial guess construction,
 - constraint library,
 - objective engine,
 - numerical solver,
 - live visualization.
-

51.2. Data Flow

Remark 51.4. The overall data flow can be summarized as:

Input Data $\rightarrow$ Initial Guess $\rightarrow$ Optimization $\rightarrow$ Result

Remark 51.5. In the implemented system, this flow is realized through a sequence of well-defined transformations.

Drop $\rightarrow$ Parser $\rightarrow$ Grabbeltisch $\rightarrow$ Initial Guess $\rightarrow$ SQP $\rightarrow$ Model $\rightarrow$ View

51.3. Optimization Session

Definition 51.6 (Optimization session). An optimization session encapsulates all data and operations required for a single optimization run.

```
class OptimizationSession {
  constructor({ input, constraints, objectives }) {
    this.input = input
    this.constraints = constraints
    this.objectives = objectives
    this.initialGuess = null
    this.result = null
  }

  buildInitialGuess() {}
  runOptimization() {}
  commitToModel(projectModel) {}
}
```

Remark 51.7. This abstraction separates:

- data preparation,
- numerical optimization,
- integration into the system model.

51.4. Initial Guess Construction

Remark 51.8. The initial guess transforms raw geometric data into optimization variables.

```
function buildInitialGuess(polyline) {
  const s = computeArcLength(polyline)

  const theta = computeDirections(polyline)
  const kappa = diff(theta) / ds

  const kappaSmooth = smooth(kappa)

  const phi = kappaSmooth.map(k =>
    Math.asin(clamp(v*v*k/g, -1, 1))
  )
}
```



```

    return { kappa: kappaSmooth, phi }
}

```

Remark 51.9. This step is essential for robust convergence of the optimization algorithm.

51.5. Constraint Library

Definition 51.10 (Constraint). A constraint is a function mapping the current state to a residual.

```

function minRadius(kappa, Rmin) {
    return Math.abs(kappa) - 1/Rmin
}

function parallelConstraint(p1, p2, d) {
    const dx = p1.x - p2.x
    const dy = p1.y - p2.y
    return Math.sqrt(dx*dx + dy*dy) - d
}

```

Remark 51.11. All constraints are expressed uniformly as residuals and integrated into the optimization problem.

51.6. Objective Engine

Definition 51.12 (Objective function). The objective function is constructed as a weighted sum of residual terms.

```

function objectiveJerk(kappa, phi, ds, v, g) {
    const r = []
    for (let i = 0; i < kappa.length - 1; i++) {
        const dk = (kappa[i + 1] - kappa[i]) / ds
        const dphi = (phi[i + 1] - phi[i]) / ds
        const j = v*v*v*dk - g*v*Math.cos(phi[i]) * dphi
        r.push(j)
    }
    return r
}

```

```
function buildObjectiveResiduals(ctx, objectives) {
  const r = []

  for (const obj of objectives) {
    const ri = obj.evaluate(ctx)
    const w = Math.sqrt(obj.weight ?? 1)
    for (const v of ri) r.push(w * v)
  }

  return r
}
```

Remark 51.13. This formulation enables flexible adaptation to different design criteria.

51.7. Residual Assembly

Remark 51.14. The optimization problem is expressed entirely in terms of residuals.

```
function buildResidualVector(ctx, constraints, objectives) {
  return [
    ...buildConstraintResiduals(ctx, constraints),
    ...buildObjectiveResiduals(ctx, objectives)
  ]
}
```

51.8. Numerical Solver

51.8.1. Gauss–Newton / SQP Core

```
async function runSQPInteractive(x0, onStep) {
  let x = x0

  for (let k = 0; k < maxIter; k++) {

    const r = residuals(x)
    const J = jacobian(x)

    const p = solve(J, r)
```

```

    x = add(x, scale(p, alpha))

    onStep({ x, k, r })

    await nextFrame()
  }

  return x
}

```

Remark 51.15. The solver operates on the residual formulation and exploits sparsity.

51.9. Live Visualization

Remark 51.16. A key feature of the system is the live visualization of the optimization process.

```

optSession.runOptimization({
  onStep: ({ x, k }) => {
    const geom = reconstruct(x)

    geoRender.updateAlignment(geom)
    plotKappa(x.kappa)
    plotPhi(x.phi)
  }
})

```

Remark 51.17. This enables interactive exploration and debugging of the optimization.

51.10. Network Extension

Remark 51.18. The architecture extends naturally to networks of alignments.

```

function residuals(network) {

  const r = []

  for (edge of network.edges) {
    r.push(...smoothness(edge))
  }
}

```

```

for (node of network.nodes) {
  r.push(...nodeConstraints(node))
}

return r
}

```

Remark 51.19. This allows simultaneous optimization of multiple connected tracks.

51.11. System Integration

Remark 51.20. The optimization engine is integrated into the user workflow through the interactive selection interface.

```

onUserFinalizeSelection(selection) {

  const alignmentData =
    buildAlignmentFromSelection(selection)

  const optSession = new OptimizationSession({
    alignmentData
  })

  optSession.buildInitialGuess()
  optSession.runOptimization()
}

```

51.12. Summary

Summary 51.21. The presented architecture combines mathematical modelling, numerical optimization, and interactive visualization into a unified system.

By representing all constraints and objectives as residuals, the system achieves a high degree of modularity and extensibility.

This architecture enables the transition from static alignment modelling to dynamic, optimization-driven design of railway infrastructure.

52. Realization-Aware Optimization

Motivation

Remark 52.1. Classical alignment optimization operates directly on analytical target geometry.

Remark 52.2. Real railway tracks, however, differ from their analytical target states.

Remark 52.3. The physically realized alignment is influenced by:

- construction procedures,
- machine behaviour,
- ballast and track mechanics,
- local correction limits,
- discretization,
- and measurement feedback.

Remark 52.4. Thus, optimization should minimize:

$$J(\mathcal{R}(\kappa, \phi)),$$

rather than:

$$J(\kappa, \phi).$$

Remark 52.5. Within AIM, optimization is therefore interpreted as optimization of expected realized operational behaviour rather than of purely analytical geometry.

52.1. Realization Operator

Definition 52.6 (Realization operator). The realization operator is:

$$\mathcal{R} : \text{pose3}_{\text{target}} \mapsto \text{pose3}_{\text{real}}.$$

Remark 52.7. The operator represents the transformation from analytical target state to physically realized railway state.

Remark 52.8. The realization operator may include:

- sampling,
- machine action,
- smoothing,
- local correction kernels,
- and structural response of the track.

52.2. Optimization on Realized States

Remark 52.9. Objective functions should be evaluated on realized states rather than on analytical target states.

Definition 52.10 (Realization-aware optimization). The general optimization problem is:

$$\min_{\kappa, \phi} J(\mathcal{R}(\kappa, \phi)).$$

Remark 52.11. This formulation explicitly integrates realization effects into the optimization process.

52.3. Pose3 Optimization

Remark 52.12. Optimization operates on the coupled pose3 state:

$$(\gamma, \theta, \phi).$$

Remark 52.13. Curvature and cant must therefore be optimized jointly rather than independently.

Remark 52.14. This reflects the dynamic coupling between geometry and vehicle behaviour.

52.4. Dynamic Objectives

52.4.1. Effective Lateral Acceleration

Definition 52.15 (Effective lateral acceleration). The effective lateral acceleration is:

$$a_{\text{eff}} = v^2 \kappa - g \sin \phi.$$

Remark 52.16. This quantity represents the dynamically relevant acceleration acting on the vehicle.

52.4.2. Jerk

Definition 52.17 (Jerk). The jerk is:

$$j = v^3 \kappa' - gv \cos \phi \phi'.$$

Remark 52.18. Jerk constraints strongly influence transition design and passenger comfort.

52.4.3. Typical Objective Components

Remark 52.19. Typical optimization objectives include:

- curvature smoothness,
- cant smoothness,
- jerk minimization,
- cant deficiency control,
- vehicle comfort,
- energy efficiency,
- realization robustness.

52.5. Realization Constraints

Remark 52.20. Optimization must respect realization constraints such as:

- machine limits,
- tamping limits,
- ballast behaviour,
- local correction ranges,
- and measurement resolution.

Remark 52.21. Thus, the feasible design space is constrained not only by geometry and standards, but also by realizability.

52.6. Inverse Realization

Remark 52.22. The optimization problem may be interpreted as an inverse realization problem.

Definition 52.23 (Inverse realization formulation). The objective is:

$$\mathcal{R}(\kappa_{\text{target}}, \phi_{\text{target}}) \approx (\kappa_{\text{desired}}, \phi_{\text{desired}}).$$

Remark 52.24. The analytical target state therefore acts as a pre-image under the realization operator.

52.7. Frequency Interpretation

Remark 52.25. The realization operator acts approximately as a low-pass filter on curvature and cant.

Remark 52.26. High-frequency components are attenuated during realization.

Remark 52.27. Optimization must therefore avoid introducing geometrically meaningless high-frequency corrections that cannot survive realization.

52.8. Sparse and Hybrid Optimization

Remark 52.28. Sparse and hybrid models are particularly suitable for realization-aware optimization.

Remark 52.29. Sparse structures provide:

- numerical stability,
- low parameter dimension,
- and interpretable geometry.

Remark 52.30. Hybrid models additionally permit:

- local corrections,
- realization compensation,
- and adaptive refinement.

52.9. Runtime Optimization

Remark 52.31. Optimization may operate continuously during:

- construction,
- maintenance,
- measurement feedback,
- and operational monitoring.

Remark 52.32. Thus, optimization becomes a runtime process rather than a purely offline planning step.

52.10. Vehicle-Space Interpretation

Remark 52.33. The physically relevant object is not necessarily the track centerline, but the motion of the railway vehicle within space.

Remark 52.34. This includes:

- vehicle center of gravity,
- bogie motion,
- vehicle envelope,
- platform interaction,
- and clearance behaviour.

Remark 52.35. Such interpretations connect realization-aware optimization to Schwerpunkt-oriented alignment concepts [12].

52.11. Ellipsoid-Aware Optimization

Remark 52.36. For large-scale or high-precision systems, optimization may be performed directly on the Earth reference surface.

Remark 52.37. In this case:

- curvature becomes geodesic curvature,
- frames become surface-attached frames,
- and world-to-track projection depends on ellipsoid geometry.

52.12. Key Insight

Proposition 52.38. *The relevant optimization target is not the analytical alignment itself, but the expected realized operational behaviour.*

Remark 52.39. This fundamentally changes the interpretation of railway alignment optimization.

52.13. Connection to AIM

Summary 52.40. Realization-aware optimization extends classical alignment optimization toward physically realizable railway states.

Within AIM, optimization is formulated on:

- pose3 states,

52. *Realization-Aware Optimization*

- realization operators,
- runtime projection structures,
- and dynamically relevant vehicle-space behaviour.

This provides a unified framework linking geometry, realization, dynamics, optimization, and operational interpretation.

Teil XI.

Engineering and Standards

Engineering

What do we already know?

Remark 52.41. The previous parts established the mathematical, geometric, operational, realization-aware, and optimization foundations of AIM.

Remark 52.42. Railway alignments can now be represented, evaluated, analyzed, and improved through structured operator-based methods.

Remark 52.43. Optimization provides mechanisms for generating improved solutions according to specified objectives.

Remark 52.44. However, optimization alone does not determine whether a solution is acceptable from an engineering perspective.

What is still missing?

Remark 52.45. Railway engineering operates within a framework of rules, recommendations, standards, regulations, and operational constraints.

Remark 52.46. Not every mathematically feasible alignment is an admissible railway alignment.

Remark 52.47. Track geometry must satisfy engineering limits.

Operational states must remain safe.

Construction solutions must remain practical.

Design alternatives must comply with applicable standards.

Remark 52.48. The missing element is therefore the engineering constraint layer that connects mathematical models with railway practice.

What comes next?

Remark 52.49. This part introduces engineering constraints as explicit components of the AIM framework.

Remark 52.50. Rather than treating standards as external documents consulted after design, AIM interprets engineering rules as structured constraints that can participate directly in analysis and optimization.

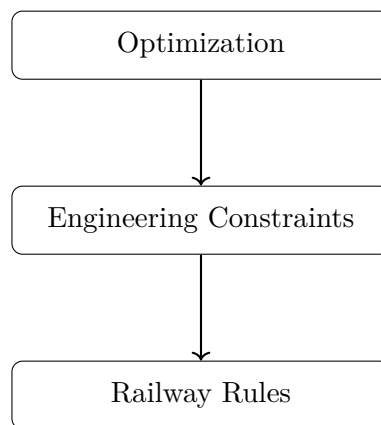
Remark 52.51. The chapters that follow investigate:

- standards and regulations,
- admissibility constraints,
- railway-specific engineering rules,
- and formal representations of operational requirements.

Remark 52.52. The objective is to transform railway standards from passive reference material into active components of railway information modelling.

Relation to the AIM Roadmap

Remark 52.53. Within the AIM roadmap, the present part corresponds to the transition from optimization toward engineering admissibility.



Remark 52.54. The progression

Optimization $\rightarrow$ Engineering Constraints $\rightarrow$ Railway Rules

marks the transition from finding improved solutions toward ensuring that those solutions remain admissible within real railway systems.

Remark 52.55. Engineering therefore acts as the bridge between mathematical optimization and practical railway implementation.

Principle 52.56 (Engineering Constraint Principle). Within AIM, railway standards, regulations, and operational rules should be represented as explicit engineering constraints that can participate directly in modelling, analysis, and optimization.

Summary 52.57. Optimization explains how railway systems can be improved.

Engineering explains which improvements are admissible.

The present part therefore establishes the constraint layer that connects mathematical optimization with practical railway design and operation.

53. Standards and Data Models

Motivation

Remark 53.1. Railway alignment data does not exist in isolation. It is embedded in a landscape of established standards and engineering data models.

Remark 53.2. Relevant formats include:

- LandXML,
- IFC (Industry Foundation Classes),
- railML,
- and various proprietary or legacy formats.

Remark 53.3. The AIM framework does not replace these standards. Instead, it provides a consistent internal representation that can interpret and integrate them.

53.1. General Perspective

Remark 53.4. Most existing standards describe railway geometry in a layered or fragmented way.

- horizontal alignment,
- vertical profile,
- cant (superelevation),
- additional metadata.

Remark 53.5. In contrast, the AIM framework proposes a unified state-based representation:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 53.6. This representation serves as a canonical internal model.

53.2. LandXML

53.2.1. Overview

Remark 53.7. LandXML is one of the most widely used formats for alignment data exchange.

Remark 53.8. It typically represents:

- horizontal geometry (CoordGeom),
- vertical profiles,
- cant definitions.

53.2.2. Characteristics

- structured XML format,
- hierarchical organization,
- multiple alignments per file,
- explicit naming of geometric elements.

53.2.3. Limitations

Remark 53.9. Despite its structure, LandXML exhibits several limitations:

- separation of horizontal and vertical geometry,
- lack of a unified dynamic interpretation,
- ambiguity in coordinate reference systems,
- inconsistent usage across software systems.

53.2.4. AIM Interpretation

Remark 53.10. Within the AIM framework, LandXML is treated as a *fat container*:

- all available data is parsed,
- relevant components are extracted,
- the result is transformed into a unified state model.

Remark 53.11. The conversion step

$$\text{LandXML} \rightarrow \text{AIM (sparse)}$$

is essential for further processing.

—

53.3. IFC

53.3.1. Overview

Remark 53.12. IFC is a general BIM standard designed for interdisciplinary data exchange.

Remark 53.13. In the context of infrastructure, IFC is extended by infrastructure schemas (IFC Rail, IFC Alignment).

53.3.2. Characteristics

- object-oriented data model,
- rich semantic structure,
- integration of geometry and metadata,
- support for relationships and hierarchies.

53.3.3. Challenges

Remark 53.14. For alignment modelling, IFC presents several challenges:

- complex schema structure,
- multiple representation alternatives,
- inconsistent implementation across tools,
- limited support for advanced alignment concepts.

53.3.4. AIM Interpretation

Remark 53.15. The AIM framework interprets IFC data by extracting:

- alignment geometry,
- reference systems,
- semantic relations.

Remark 53.16. AIM acts as a normalization layer that reduces IFC complexity to a computationally usable form.

—

53.4. railML

53.4.1. Overview

Remark 53.17. railML is a domain-specific XML standard for railway systems.

Remark 53.18. It focuses on:

- infrastructure,
- timetable data,
- rolling stock.

53.4.2. Relevance for Alignment

Remark 53.19. railML provides a more explicit representation of railway topology than most other formats.

Remark 53.20. However, detailed geometric modelling is often secondary to network and operational aspects.

53.4.3. AIM Interpretation

Remark 53.21. Within AIM, railML is particularly relevant for:

- network topology,
 - connections between tracks,
 - system-level structure.
-

53.5. Legacy and Proprietary Formats

Remark 53.22. In practice, a large amount of alignment data exists in proprietary or legacy formats.

Remark 53.23. Examples include:

- TRA / GRA (Technet, Vermessung),
- spreadsheet-based formats,
- CAD exports.

53.5.1. Characteristics

- incomplete or implicit structure,
- missing type information,
- reliance on conventions rather than formal schema.

53.5.2. AIM Strategy

Remark 53.24. These formats are interpreted using:

- sniffing and classification,
- format-specific parsers,
- reconstruction of implicit structure.

—

53.6. Canonical Internal Model

Definition 53.25 (AIM core model). The AIM framework uses a canonical internal representation based on:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 53.26. All external formats are mapped to this representation.

53.6.1. Advantages

- unified handling of geometry and dynamics,
- compatibility with numerical optimization,
- independence from specific file formats,
- extensibility toward new standards.

—

53.7. Transformation Pipeline

Remark 53.27. The transformation from external data to the AIM model follows a pipeline structure.

```
external file
→ parser
→ fat representation
→ validation
→ sparse conversion
→ AIM state model
```

Remark 53.28. This pipeline ensures robustness and traceability of data processing.

—

53.8. Relation to BIM

53.8.1. Conceptual Position

Remark 53.29. BIM (Building Information Modelling) aims to integrate geometry, semantics, and lifecycle information.

Remark 53.30. However, alignment modelling is often underrepresented or treated as a secondary aspect.

53.8.2. AIM Contribution

Remark 53.31. The AIM framework contributes to BIM by:

- providing a rigorous alignment model,
- enabling dynamic interpretation,
- supporting optimization-driven design.

Remark 53.32. In this sense, AIM can be interpreted as a specialization of BIM for alignment-centric infrastructure modelling.

53.9. Interoperability

Remark 53.33. A key requirement for practical application is interoperability with existing tools and workflows.

Remark 53.34. The AIM framework supports interoperability through:

- import adapters,
 - export converters,
 - intermediate canonical representation.
-

53.10. Discussion

Remark 53.35. Existing standards focus primarily on data exchange and documentation.

Remark 53.36. The AIM framework complements these standards by introducing:

- a computationally consistent model,
- a dynamic interpretation,
- integration with optimization methods.

Remark 53.37. Thus, AIM bridges the gap between static data representation and active engineering design.

53.11. Summary

Summary 53.38. Railway alignment data is governed by a variety of standards, each with its own strengths and limitations.

The AIM framework provides a unified internal representation that allows these heterogeneous sources to be interpreted consistently.

By combining standard-based data exchange with a state-based, optimization-ready model, AIM enables a transition from static data handling to dynamic and computational alignment design.

54. Constraint Code Architecture

Motivation

Remark 54.1. Railway alignment rules must be represented in a form that is both mathematically meaningful and computationally tractable.

Remark 54.2. Within the AIM framework, this is achieved by expressing all rules as constraint residuals that can be evaluated uniformly inside an optimization loop.

Remark 54.3. The implementation goal is therefore not a collection of isolated rule checks, but a modular constraint library.

54.1. General Principle

Definition 54.4 (Constraint residual). A constraint is represented as a function

$$c(\mathbf{x}) \in \mathbb{R}^m$$

whose value vanishes in the ideal case and whose magnitude measures violation.

Remark 54.5. This unified representation allows the same mathematical object to be used for:

- feasibility checks,
- penalty terms,
- direct inclusion into SQP solvers.

```
function evaluateConstraint(ctx) {  
    return [residual1, residual2, ...]  
}
```

54.2. Constraint Registry

Remark 54.6. A practical implementation should organize constraints in a registry.

```
const ConstraintRegistry = {  
    geometric: {},  
    dynamic: {},  
}
```

```

    topologic: {},
    regulatory: {}
}

```

Remark 54.7. This keeps domain logic extensible and avoids hard-coded monolithic solver structures.

54.3. Constraint Object Model

```

const constraint = {
  id: "min_radius",
  group: "regulatory",
  appliesTo: "edge",
  weight: 1.0,
  evaluate: (ctx) => []
}

```

Remark 54.8. The fields have the following meaning:

- `id`: unique name,
- `group`: semantic class,
- `appliesTo`: edge, node, pair, network,
- `weight`: optional scaling,
- `evaluate`: residual callback.

54.4. Context Object

Remark 54.9. Constraints should not operate directly on raw global arrays. Instead, they receive a normalized context object.

```

const ctx = {
  edge,
  node,
  pair,
  i,
  kappa,
  phi,
  geometry,
  speed,
  params,
}

```

```

    metadata
}

```

Remark 54.10. This provides a stable interface between model, solver, and rule system.

54.5. Geometric Constraints

54.5.1. Position Continuity

```

function continuityConstraint(p1, p2) {
    return [
        p1.x - p2.x,
        p1.y - p2.y
    ]
}

```

Remark 54.11. This enforces C^0 -continuity between adjacent geometric elements or connected alignments.

54.5.2. Direction Continuity

```

function directionConstraint(t1, t2) {
    return [
        t1.dx - t2.dx,
        t1.dy - t2.dy
    ]
}

```

Remark 54.12. This corresponds to tangent continuity and therefore to C^1 -style behaviour in geometric terms.

54.5.3. Curvature Continuity

```

function curvatureConstraint(k1, k2) {
    return [k1 - k2]
}

```

Remark 54.13. This enforces continuity of curvature across boundaries.

54.6. Dynamic Constraints

54.6.1. Equilibrium Constraint

```

function equilibriumConstraint(kappa, phi, v, g) {
    return [v * v * kappa - g * Math.sin(phi)]
}

```

Remark 54.14. This residual measures cant deficiency or excess in analytic form.

54.6.2. Jerk Constraint

```
function jerkConstraint(kappa0, kappa1, phi0, phi1, ds, v, g) {
  const dk = (kappa1 - kappa0) / ds
  const dphi = (phi1 - phi0) / ds
  const j = v * v * v * dk - g * v * Math.cos(phi0) * dphi
  return [j]
}
```

Remark 54.15. This constraint reflects dynamic smoothness and comfort-related limits.

54.7. Regulatory Constraints

54.7.1. Minimum Radius

```
function minRadiusConstraint(kappa, Rmin) {
  return [Math.abs(kappa) - 1 / Rmin]
}
```

Remark 54.16. A non-positive value indicates compliance with the radius limit.

54.7.2. Maximum Cant

```
function maxCantConstraint(phi, phiMax) {
  return [Math.abs(phi) - phiMax]
}
```

54.7.3. Cant Gradient

```
function cantGradientConstraint(phi0, phi1, ds, limit) {
  return [Math.abs((phi1 - phi0) / ds) - limit]
}
```

54.8. Topological Constraints

54.8.1. Switch Node Constraint

```
function switchConstraint(main, branch) {
  return [
    main.x - branch.x,
    main.y - branch.y,
    main.dx - branch.dx,
  ]
}
```



```

    main.dy - branch.dy
  ]
}

```

Remark 54.17. This expresses the basic coupling at branching nodes.

54.8.2. Parallel Track Spacing

```

function parallelConstraint(p1, p2, d) {
  const dx = p1.x - p2.x
  const dy = p1.y - p2.y
  return [Math.sqrt(dx * dx + dy * dy) - d]
}

```

54.9. Constraint Assembly

Remark 54.18. The full residual vector is assembled from all active constraints.

```

function buildConstraintResiduals(ctx, constraints) {
  const r = []

  for (const c of constraints) {
    const ri = c.evaluate(ctx)
    const w = Math.sqrt(c.weight ?? 1)
    for (const v of ri) r.push(w * v)
  }

  return r
}

```

Remark 54.19. This produces a single solver-compatible vector independent of the semantic source of the constraints.

54.10. Factory Functions

Remark 54.20. Constraints should be created through factories rather than by writing anonymous closures throughout the code base.

```

function makeMinRadiusConstraint(Rmin, weight = 1) {
  return {
    id: "min_radius",
    group: "regulatory",
    appliesTo: "edge",
  }
}

```

```

    weight,
    evaluate: ({ kappa, i }) => [
      Math.abs(kappa[i]) - 1 / Rmin
    ]
  }
}

function makeJerkConstraint(v, g, ds, weight = 1) {
  return {
    id: "jerk",
    group: "dynamic",
    appliesTo: "edge",
    weight,
    evaluate: ({ kappa, phi, i }) => {
      const dk = (kappa[i + 1] - kappa[i]) / ds
      const dphi = (phi[i + 1] - phi[i]) / ds
      return [v * v * v * dk - g * v * Math.cos(phi[i]) * dphi]
    }
  }
}

```

54.11. Constraint Presets

Remark 54.21. In practical railway design, different projects require different constraint configurations.

```

const presetHighSpeed = [
  makeMinRadiusConstraint(4000, 10),
  makeJerkConstraint(v, g, ds, 8),
  makeMaxCantConstraint(phiMax, 6)
]

const presetExistingLine = [
  makeMinRadiusConstraint(800, 3),
  makeJerkConstraint(v, g, ds, 2),
  makeParallelConstraint(trackDistance, 5)
]

```

54.12. Integration into the Solver

```

function buildResidualVector(ctx, constraints, objectives) {
  return [

```

```

    ...buildConstraintResiduals(ctx, constraints),
    ...buildObjectiveResiduals(ctx, objectives)
]
}
```

Remark 54.22. This shows the central design principle of the architecture: constraints and objectives share the same residual-based interface.

54.13. Discussion

Remark 54.23. The proposed architecture has several advantages:

- modularity,
- extensibility,
- compatibility with least-squares solvers,
- transparent mapping between engineering rules and code.

Remark 54.24. Instead of scattering rule checks throughout the application, the constraint library concentrates domain logic in a single mathematical layer.

54.14. Summary

Summary 54.25. The constraint library translates railway rules, topology, and dynamic requirements into a unified residual-based architecture.

This provides the technical bridge between theoretical modelling, engineering regulations, and computational optimization.

As a result, the AIM framework becomes not only mathematically coherent, but also directly implementable in software.

55. Railway Design Rules

Motivation

Remark 55.1. Railway alignment design is not solely a geometric or mathematical task. It is governed by engineering rules, safety requirements, and operational constraints.

Remark 55.2. These rules originate from:

- national regulations,
- international standards (e.g. UIC),
- operator-specific guidelines (e.g. Deutsche Bahn),
- empirical engineering practice.

Remark 55.3. Within the AIM framework, such rules are not treated as external constraints only, but are integrated into the modelling and optimization process.

55.1. Categories of Rules

55.1.1. Geometric Rules

- minimum radius,
- continuity of alignment,
- transition length requirements,
- curvature monotonicity in transitions.

55.1.2. Kinematic and Dynamic Rules

- limits on lateral acceleration,
- limits on jerk,
- compatibility of curvature and cant,
- speed-dependent requirements.

55.1.3. Operational Rules

- allowable speeds,
- safety margins,
- maintenance considerations,
- interoperability constraints.

55.1.4. Topological Rules

- constraints at switches and crossings,
 - parallel track spacing,
 - network connectivity conditions.
-

55.2. Geometric Constraints

55.2.1. Minimum Radius

Definition 55.4 (Minimum radius constraint). The curvature must satisfy

$$|\kappa(s)| \leq \kappa_{\max}, \quad \kappa_{\max} = \frac{1}{R_{\min}}.$$

Remark 55.5. The value of $R_{\min}$ depends on:

- design speed,
 - vehicle characteristics,
 - regulatory standards.
-

55.2.2. Continuity Conditions

Remark 55.6. Railway alignments must satisfy at least:

- C^0 continuity (position),
- C^1 continuity (direction),
- typically C^2 continuity (curvature).

Remark 55.7. Higher-order smoothness may be required for high-speed lines.

55.2.3. Transition Length

Definition 55.8 (Minimum transition length). A transition between curvatures must satisfy

$$L \geq L_{\min}(v, \kappa_1, \kappa_2).$$

Remark 55.9. Typical formulations relate transition length to:

- allowable rate of change of lateral acceleration,
- cant development limits.

—

55.3. Dynamic Constraints

55.3.1. Lateral Acceleration

Definition 55.10 (Lateral acceleration). The lateral acceleration is given by

$$a_y(s) = v^2 \kappa(s).$$

Constraint 55.11.

$$|a_y(s)| \leq a_{\max}.$$

—

55.3.2. Cant and Cant Deficiency

Definition 55.12 (Cant angle). Let $\phi(s)$ denote the cant angle.

Definition 55.13 (Effective acceleration). The effective lateral acceleration is

$$a_{\text{eff}}(s) = v^2 \kappa(s) - g \tan(\phi(s)).$$

Constraint 55.14.

$$|a_{\text{eff}}(s)| \leq a_{\text{eff},\max}.$$

—

55.3.3. Jerk

Definition 55.15 (Jerk). The lateral jerk is defined as

$$j(s) = \frac{d}{dt} a_y(s).$$

Remark 55.16. Using s as parameter, one obtains

$$j(s) = v^3 \kappa'(s).$$

Constraint 55.17.

$$|j(s)| \leq j_{\max}.$$

—

55.4. Coupling of Curvature and Cant

Remark 55.18. Curvature and cant must not be treated independently.

Constraint 55.19.

$$|\kappa'(s)| \leq f_1(v, \phi), \quad |\phi'(s)| \leq f_2(v, \kappa).$$

Remark 55.20. These relations express that:

- curvature changes require corresponding cant development,
- cant changes must respect dynamic limits.

—

55.5. Speed-Dependent Constraints

Remark 55.21. All relevant constraints depend on speed.

Definition 55.22 (Speed profile). A speed profile is a function

$$v : [0, L] \rightarrow \mathbb{R}_+.$$

Remark 55.23. Constraints must be evaluated pointwise with respect to $v(s)$.

—

55.6. Constraint Representation in AIM

55.6.1. Abstract Form

Definition 55.24 (Constraint functional). A constraint is represented as a functional

$$C_i[\kappa, \phi, v] \leq 0.$$

Remark 55.25. This allows constraints to be:

- evaluated continuously,
- discretized for numerical optimization,
- combined into a unified constraint system.

—

55.6.2. Discrete Representation

Remark 55.26. In numerical implementation, constraints are evaluated on a grid

$$s_0, \dots, s_N.$$

Definition 55.27 (Discrete constraint).

$$C_i(\mathbf{x}) = \max_k C_i(s_k).$$

—

55.7. Integration into Optimization

Remark 55.28. Constraints enter the optimization problem as:

$$g_i(\mathbf{x}) \leq 0.$$

Remark 55.29. They may be handled via:

- hard constraints (feasibility),
- penalty terms,
- barrier methods.

—

55.8. Relation to Standards

Remark 55.30. The presented constraints correspond to rules found in:

- UIC codes,
- national railway regulations,
- operator guidelines.

Remark 55.31. The AIM framework provides a mathematical representation that allows these rules to be applied consistently across different data sources.

—

55.9. Discussion

Remark 55.32. Traditional workflows treat rules as external checks applied after design.

Remark 55.33. In contrast, AIM integrates rules directly into:

- modelling,
- reconstruction,
- optimization.

Remark 55.34. This leads to a shift from validation to constraint-driven design.

55.10. Summary

Summary 55.35. Railway design rules define the feasible set of alignment solutions.

Within the AIM framework, these rules are formalized as mathematical constraints and integrated into the optimization process.

This allows alignment design to be treated as a constrained optimization problem that respects both geometric and dynamic requirements.

Teil XII.

Applications

Applications

What do we already know?

Remark 55.36. The previous parts established the theoretical foundations of AIM.

Remark 55.37. Geometry, states, runtime evaluation, realization, modelling, optimization, and engineering constraints have been developed as components of a coherent alignment-based information framework.

Remark 55.38. The resulting system provides a rigorous mathematical and engineering basis for railway alignment modelling.

Remark 55.39. However, a framework ultimately derives its value from practical use.

What is still missing?

Remark 55.40. Theoretical concepts alone do not demonstrate practical relevance.

Remark 55.41. A railway information framework must be capable of addressing concrete engineering problems, supporting operational workflows, and providing insight into real applications.

Remark 55.42. The missing element is therefore the application perspective.

Remark 55.43. The question is no longer whether the framework is mathematically consistent, but how it can be employed in practical railway and infrastructure contexts.

What comes next?

Remark 55.44. This part illustrates how the AIM framework can be applied to concrete problems.

Remark 55.45. The focus shifts from framework construction toward framework usage.

Remark 55.46. The chapters that follow investigate:

- railway alignment applications,
- vehicle-space interpretation,
- dynamic engineering scenarios,
- practical examples,

- use-case driven modelling,
- and BIMalignment concepts.

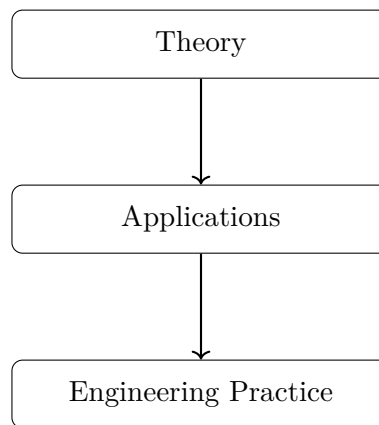
Remark 55.47. These applications are not intended merely as demonstrations.

They illustrate how the concepts developed throughout AIM become operational engineering tools.

Remark 55.48. The purpose is therefore to connect theory with practical railway engineering practice.

Relation to the AIM Roadmap

Remark 55.49. Within the AIM roadmap, the present part corresponds to the transition from engineering theory toward practical deployment.



Remark 55.50. The progression

Theory $\rightarrow$ Applications $\rightarrow$ Engineering Practice

marks the transition from conceptual understanding toward practical engineering value.

Remark 55.51. Applications therefore serve as the validation layer of AIM, showing how abstract concepts become useful engineering instruments.

Principle 55.52 (Application Principle). Within AIM, theoretical concepts achieve their full significance only when they can be transformed into practical workflows, engineering tools, and operational railway applications.

Summary 55.53. Theory explains what is possible.

Engineering explains what is admissible.

Applications explain what is useful.

The present part therefore demonstrates how alignment-based information modelling can be employed to address practical railway and infrastructure problems.

56. Railway Alignment

Motivation

Remark 56.1. Railway alignment is one of the most demanding application domains for curvature-based geometric modelling.

Remark 56.2. In railway engineering, alignment is not merely a planar curve design problem. It is a coupled system involving horizontal geometry, vertical profile, cant, operating speed, dynamic vehicle response, safety, and maintenance.

Remark 56.3. This makes railway alignment a natural and severe test case for alignment-based information modelling.

56.1. Railway Alignment as a Multi-Layer Object

Definition 56.4 (Railway alignment). A railway alignment is the spatial path associated with a railway track, typically represented through coordinated horizontal, vertical, and cant-related descriptions.

Remark 56.5. In practical standards and data models, railway alignment is usually not represented as a single monolithic object, but as an ordered system of segments and auxiliary alignment layers.

56.1.1. Horizontal alignment

Remark 56.6. The horizontal alignment is primarily governed by curvature. In its classical engineering decomposition it consists of straight segments, circular arcs, and transition curves.

56.1.2. Vertical alignment

Remark 56.7. The vertical alignment describes elevation and gradient behaviour along the route. In practical data models it is segmented into line, arc, and transition primitives.

OPEN: Unified mathematical treatment of horizontal and vertical alignment.

56.1.3. Cant alignment

Definition 56.8 (Cant). Cant is the elevation difference between the rails, introduced to compensate lateral acceleration in curves.

Remark 56.9. Cant is not an accessory parameter. It forms an alignment layer of its own, with segment structure, continuity rules, and possible smoothing requirements.

Remark 56.10. Recent railway conceptual models explicitly represent cant as an ordered sequence of segments, including line and transition segments, linked to the horizontal alignment by chainage or equivalent position reference.

OPEN: Integrate cant into a unified geometric and kinematic state model.

56.2. Parameter Coupling

Remark 56.11. Railway alignment is characterized by strong coupling between geometric and operational variables.

- curvature,
- cant,
- speed,
- transition length,
- gradient,
- and boundary conditions imposed by rolling stock and operation

Remark 56.12. These quantities are not independent. Increased speed requires either larger radii, larger cant, longer transition lengths, or a different trade-off between these variables.

Remark 56.13. Likewise, high cant or severe curvature may improve one operational aspect while worsening another, for example when stopped trains or restricted sight distances are involved.

56.3. Why Transition Curves Matter in Railways

Remark 56.14. The transition curve is not merely a geometric connector between a straight line and a circular arc.

Remark 56.15. Its primary role is to avoid abrupt changes in lateral forces, improve ride comfort, reduce dynamic excitation, and limit wear on both vehicle and infrastructure.

Remark 56.16. The state-of-the-art review literature explicitly emphasizes that transition curves are crucial for safe and comfortable railway travel and for reducing maintenance demand.

Remark 56.17. This is also why modern railway research increasingly evaluates transition curves not only geometrically, but in terms of lateral change of acceleration, force development, wear, and vehicle dynamics.

56.4. From Classical Geometry to Dynamic Interpretation

Remark 56.18. Classical alignment design often idealizes the influence of curvature, cant, and transition shape through simplified or quasi-static considerations.

Remark 56.19. However, practical evidence and later engineering work show that the vehicle-track system exhibits dynamic responses which depend strongly on the detailed shape of the transition curve.

Remark 56.20. In particular, non-linear transition curves were introduced to mitigate the abrupt onset and disappearance of the variation of unbalanced lateral acceleration that arises with linear transitions at higher speeds.

Remark 56.21. This gives railway transition design a systems character: the geometric curve acts as a control function for dynamic behaviour.

56.5. Curvature, Cant, and Speed

Remark 56.22. Curvature alone does not determine railway behaviour. The relevant quantity is the combined system of curvature, cant, speed, and their rates of change.

Remark 56.23. Cant and cant variation influence lateral acceleration compensation, while transition shape influences how these effects develop along the track.

Remark 56.24. Practical railway design manuals also show that excessive superelevation, insufficient tangent lengths, or poorly located curves can create operational and derailment risks even when the pure geometry is formally traversable.

OPEN: Formal coupled model of curvature, cant, speed, and dynamic response.

56.6. Operational and Maintenance Constraints

Remark 56.25. Railway alignment is constrained not only by construction feasibility, but also by long-term operation and maintenance.

- minimum radii,
- maximum cant,
- limits on cant change,
- curvature and transition limits,
- sight distance restrictions,
- turnout and crossing constraints,
- wear and maintenance considerations

Remark 56.26. Practical design guides repeatedly emphasize that sharper curvature may be physically negotiable, yet still be undesirable because of higher maintenance cost, lower operating efficiency, or increased derailment risk.

56.7. Freight, Passenger, High-Speed, and Special Systems

Remark 56.27. Railway alignment is not uniform across operational domains. Freight, passenger, high-speed, light rail, and special systems such as maglev may lead to different admissible geometry and different transition preferences.

Remark 56.28. This means that an alignment model must be flexible enough to represent domain-specific constraints without losing mathematical coherence.

OPEN: Taxonomy of railway application classes and their geometric consequences.

56.8. Chainage and Engineering Referencing

Definition 56.29 (Chainage). Chainage is the engineering reference system used to identify positions along an alignment.

Remark 56.30. In practice, alignment data is often anchored more robustly by chainage than by raw Cartesian coordinates alone, especially when the alignment is revised or partially reconstructed.

Remark 56.31. Modern railway information models therefore include dedicated chainage mechanisms in order to preserve engineering references despite local alignment changes.

OPEN: Relation between arc-length parameterization and engineering chainage.

56.9. Railway Alignment in Data Models and BIM

Remark 56.32. Railway alignment is increasingly represented within interoperable data models. These models typically distinguish horizontal alignment, vertical alignment, and cant as related but separate geometric layers.

Remark 56.33. Current IFC-related railway modelling efforts explicitly represent transition curves, vertical transition segments, and cant segments, and already reveal that infrastructure alignment cannot be reduced to a single curve primitive.

Remark 56.34. This strongly supports the view adopted in the present manuscript: alignment should be understood as a structured object rather than as an undifferentiated line.

OPEN: Formal docking of the sparse alignment model to IFC/railway data models.

56.10. Why Railway Alignment is a Privileged Use Case

Remark 56.35. Railway alignment combines

- strict geometric structure,
- strong parameter coupling,
- dynamic sensitivity,
- safety implications,
- heavy maintenance consequences,
- and demanding interoperability requirements.

Remark 56.36. For that reason, railway alignment is not merely one application among many, but an especially revealing use case for a general theory of curvature-based structured curve modelling.

56.11. Outlook

Remark 56.37. A mature railway-alignment framework should therefore support:

- structured geometric representation,
- multiple transition families,
- coupled cant and profile modelling,
- numerical reconstruction,
- dynamic and safety-oriented evaluation,
- and interoperability with engineering standards and BIM models.

Summary 56.38. Railway alignment is a multi-layer object rather than a single planar curve. It couples geometry, cant, speed, vehicle dynamics, operational constraints, maintenance, and engineering referencing.

This makes railway alignment a privileged and highly demanding domain for alignment-based information modelling and provides the main application context for the concepts developed in this manuscript.

57. Vehicle Space and Platform Interaction

Motivation

Remark 57.1. Classical railway alignment modelling is primarily based on the track centerline.

Remark 57.2. In this view, the railway vehicle is treated indirectly through:

- clearance envelopes,
- cant rules,
- safety margins,
- and operational standards.

Remark 57.3. This abstraction has proven remarkably successful and enabled the development of reliable railway systems over many decades.

Remark 57.4. However, the physically relevant object is not the centerline itself, but the spatial motion of the railway vehicle along the track.

Remark 57.5. This becomes particularly visible in:

- tight urban rail systems,
- platform-edge design,
- articulated vehicles,
- narrow tunnel geometries,
- and highly optimized clearance situations.

Remark 57.6. This perspective is closely related to approaches that interpret alignment geometry from the viewpoint of vehicle motion rather than purely from the track centerline.

Such concepts have been explored in advanced railway engineering, including Schwerpunkt-oriented alignment approaches [12].

57.1. Centerline as Reduced Representation

Remark 57.7. The classical alignment model represents the railway track by:

$$\gamma(s), \quad \kappa(s), \quad \phi(s).$$

Remark 57.8. This defines:

- position,
- curvature,
- and cant,

along the alignment.

Remark 57.9. The vehicle itself is not explicitly represented.

Remark 57.10. Instead, its behaviour is incorporated indirectly through engineering rules and geometric tolerances.

Proposition 57.11. *The centerline representation is a reduced-order approximation of the actual spatial motion of railway vehicles.*

Remark 57.12. This approximation is generally sufficient for conventional railway engineering.

57.2. Vehicle-Induced Geometry

Remark 57.13. A railway vehicle does not follow the centerline as a rigid point.

Remark 57.14. Its occupied space depends on:

- bogie geometry,
- wheelbase,
- car-body length,
- articulation,
- cant,
- suspension,
- and dynamic motion.

Remark 57.15. As a consequence:

- vehicle ends may swing outward,
- the center of the car body may move inward,
- and the effective occupied space varies along the track.

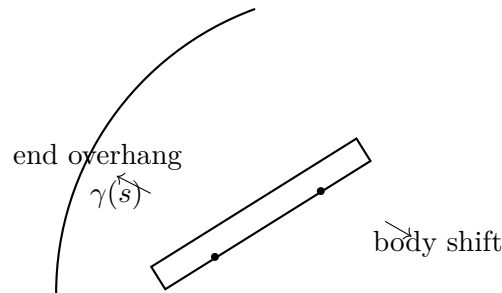


Abbildung 57.1.: Vehicle motion differs from the centerline geometry. Car-body overhang and bogie-constrained motion generate occupied space that cannot be represented by the centerline alone.

57.3. Platform Interaction

Remark 57.16. Platform-edge design is one of the most sensitive examples of vehicle-space interaction.

Remark 57.17. The effective gap between vehicle and platform depends on:

- curvature,
- cant,
- vehicle geometry,
- suspension state,
- and operational conditions.

Remark 57.18. In some urban railway systems, such as light rail and metro systems, vehicle overhang is explicitly considered in platform-edge geometry.

Remark 57.19. This demonstrates that the practically relevant geometry is not merely the track centerline, but the dynamically occupied vehicle space.

57.4. Standards and Conservative Approximation

Remark 57.20. Conventional railway standards often address vehicle motion indirectly through conservative clearance envelopes.

Remark 57.21. This approach increases robustness and simplifies engineering workflows.

Remark 57.22. However, it may also lead to:

- increased platform gaps,
- reduced passenger comfort,
- larger-than-necessary clearance reserves,

- and loss of geometric precision.

Remark 57.23. Thus, many existing systems compensate modelling limitations through additional safety margins.

57.5. Towards Vehicle-Centric Alignment Modelling

Remark 57.24. Future alignment systems may extend the classical centerline model towards vehicle-centric representations.

Remark 57.25. In such a framework, the relevant object would no longer be only:

$$\gamma(s),$$

but the occupied vehicle space evolving along the alignment.

Remark 57.26. Possible extensions include:

- dynamic vehicle envelopes,
- bogie-constrained motion models,
- passenger-space optimization,
- platform interaction models,
- and runtime vehicle-state evaluation.

Remark 57.27. Within the AIM framework, such concepts are regarded as potential future extensions rather than immediate replacement of existing engineering practice.

57.6. Connection to Schwerpunkt-Trassierung

Remark 57.28. The concept of Schwerpunkt-Trassierung already shifts the interpretation of alignment design from purely geometric construction toward the motion of a dynamically relevant state.

Remark 57.29. Vehicle-space modelling can be interpreted as a continuation of this idea.

Remark 57.30. Instead of optimizing only a centerline curve, one may eventually seek to optimize the spatial behaviour of vehicles and passengers along the track.

57.7. Connection to AIM

Summary 57.31. Classical railway alignment modelling represents the railway track by a centerline-based geometric abstraction.

This abstraction is highly successful, but does not explicitly represent the spatial behaviour of railway vehicles.

57. Vehicle Space and Platform Interaction

Practical situations such as platform interaction reveal the limitations of purely centerline-based modelling and motivate future vehicle-centric extensions.

Within the AIM framework, this is regarded as an important long-term research direction.

58. Railway Dynamics versus Aircraft Dynamics

Motivation

Remark 58.1. Aircraft dynamics and railway dynamics are often treated as fundamentally different engineering domains.

Remark 58.2. However, both systems are governed by related physical principles:

- inertia,
- curvature-driven motion,
- center-of-gravity behaviour,
- rotational response,
- and passenger comfort constraints.

Remark 58.3. The essential difference is not dynamics itself, but the degree of kinematic freedom.

58.1. Aircraft Dynamics

58.1.1. Free Spatial Motion

Remark 58.4. Aircraft move within largely unconstrained three-dimensional space.

Remark 58.5. The aircraft continuously adapts:

- roll,
- yaw,
- pitch,
- and trajectory

to dynamic requirements.

58.1.2. Center-of-Gravity Interpretation

Remark 58.6. Aircraft dynamics are strongly centered around:

- center-of-gravity motion,
- stability,
- inertia tensors,
- and aerodynamic force balance.

58.1.3. Dynamic Freedom

Remark 58.7. The aircraft may continuously adapt:

- orientation,
- banking angle,
- and trajectory curvature

during operation.

58.2. Railway Dynamics

58.2.1. Track-Constrained Motion

Remark 58.8. Railway vehicles are fundamentally different because their motion is strongly constrained by track geometry.

Remark 58.9. The railway vehicle cannot freely select:

- curvature,
- roll angle,
- or spatial trajectory.

Remark 58.10. Instead, these quantities are largely prescribed by:

- alignment geometry,
- cant,
- vehicle suspension,
- and operational velocity.

58.3. The Hidden Similarity

Remark 58.11. Despite these differences, both aircraft and railway systems ultimately attempt to manage:

inertial forces

relative to:

vehicle orientation and passenger comfort.

Remark 58.12. In aircraft:

- trajectory adapts to dynamics.

Remark 58.13. In railways:

- geometry and cant must be designed in advance to support future dynamic behaviour.

58.4. Cant as Railway Banking

Remark 58.14. Cant may be interpreted as the railway equivalent of aircraft banking.

Remark 58.15. However, unlike aircraft banking:

- railway cant is infrastructure-bound,
- spatially fixed,
- and shared by all vehicles using the track.

Remark 58.16. This creates a fundamental compromise:

- freight trains,
- passenger trains,
- high-speed vehicles,
- and maintenance vehicles

must all operate on the same geometric infrastructure.

58.5. Classical Railway Limitation

Remark 58.17. Classical railway engineering therefore evolved toward strongly coupled design rules:

$$\kappa(s) \leftrightarrow \phi(s).$$

Remark 58.18. In many traditional approaches, curvature and cant evolution are treated almost as inseparable quantities.

Remark 58.19. This coupling simplifies:

- construction,
- maintenance,
- and standards.

Remark 58.20. However, it also limits:

- operational flexibility,
- vehicle-specific optimization,
- and adaptive reinterpretation of existing infrastructure.

58.6. The AIM Perspective

58.6.1. Pose3 Interpretation

Remark 58.21. Within AIM, cant is not interpreted merely as static geometric annotation.

Remark 58.22. Instead, cant contributes to a runtime-derived operational state:

pose3.

Remark 58.23. This enables explicit interpretation of:

- vehicle-space behaviour,
- center-of-gravity motion,
- dynamic comfort,
- and operational envelopes.

58.7. From Fixed Geometry to Operational Envelopes

Remark 58.24. Classical railway optimization often attempts to derive:

$$\kappa(s)$$

from:

$$\phi(s).$$

Remark 58.25. Within AIM, the inverse interpretation becomes possible:

- existing curvature geometry remains largely fixed,
- cant becomes partially adjustable,

58. Railway Dynamics versus Aircraft Dynamics

- and admissible velocity envelopes are evaluated dynamically.

Remark 58.26. Thus, the operational question changes from:

“Which geometry should be built?”

to:

“Which operational behaviour can be supported by existing geometry?”

58.8. Center-of-Gravity Interpretation

Remark 58.27. Different railway vehicles possess different:

- center-of-gravity heights,
- suspension systems,
- roll behaviour,
- and passenger comfort limits.

Remark 58.28. Thus:

- freight trains,
- passenger trains,
- tilting trains,
- and metro systems

may require different operational interpretations of the same physical track geometry.

Remark 58.29. This naturally motivates:

- vehicle-specific runtime evaluation,
- operational velocity envelopes,
- and realization-aware cant interpretation.

58.9. Operational Consequence

Remark 58.30. The operational opportunity is not necessarily:

“new geometry”

but often:

“better interpretation of existing geometry.”

58. Railway Dynamics versus Aircraft Dynamics

Remark 58.31. Small cant adjustments, sometimes only within millimeter or centimeter ranges, may significantly affect:

- admissible speed,
- passenger comfort,
- and operational stability.

Remark 58.32. This creates the possibility of:

- operational upgrades,
- speed improvements,
- and realization-aware optimization

without requiring major reconstruction of alignment geometry.

58.10. AIM Interpretation

Proposition 58.33. *Within AIM, railway alignment is not merely static infrastructure geometry.*

Proposition 58.34. *Railway alignment forms an operationally evaluated dynamic guidance system for constrained vehicle motion.*

Proposition 58.35. *Cant, curvature, vehicle-space behaviour, and runtime evaluation must therefore be interpreted jointly.*

58.11. Connection to AIM

Summary 58.36. Aircraft and railway systems share common dynamic principles related to:

- inertia,
- curvature,
- center-of-gravity behaviour,
- and passenger comfort.

The essential difference lies in the degree of spatial freedom.

Within AIM, this insight motivates:

- runtime-based vehicle-space interpretation,
- realization-aware dynamics,

58. *Railway Dynamics versus Aircraft Dynamics*

- and operational evaluation of existing infrastructure.

Thus, railway alignment becomes interpretable as a dynamically evaluated guidance system rather than merely static track geometry.

59. Examples

Motivation

Remark 59.1. Theoretical concepts become meaningful only when illustrated by concrete examples.

Remark 59.2. The following examples demonstrate how different transition concepts can be interpreted within the curvature-based framework.

59.1. Example 1: Clothoid Transition

59.1.1. Definition

Definition 59.3. The clothoid is defined by linear curvature:

$$\kappa(s) = as.$$

59.1.2. Properties

- $\kappa'(s) = \text{const}$
- $\kappa''(s) = 0$

Remark 59.4. The clothoid introduces curvature at a constant rate.

59.1.3. Interpretation

Remark 59.5. In the curvature-based framework, the clothoid represents a linear control function.

Remark 59.6. Its simplicity explains its historical dominance in railway design.

59.2. Example 2: Polynomial Transition

59.2.1. Definition

Definition 59.7. A polynomial transition may be defined as

$$\kappa(s) = a_1s + a_2s^2 + a_3s^3.$$

59.2.2. Properties

- $\kappa'(s)$ is variable
- $\kappa''(s)$ is controllable

59.2.3. Interpretation

Remark 59.8. Polynomial transitions allow shaping of higher-order behaviour.

Remark 59.9. They can be used to reduce dynamic excitation by smoothing curvature variation.

—

59.3. Example 3: Comparison of Curvature Profiles

Remark 59.10. Consider two transitions:

- a clothoid,
- a cubic polynomial.

Remark 59.11. Both achieve the same boundary conditions:

$$\kappa(0) = 0, \quad \kappa(L) = \kappa_0.$$

59.3.1. Observation

Remark 59.12. Although geometrically equivalent at the endpoints, the internal behaviour differs.

- clothoid: constant curvature growth,
- polynomial: variable curvature growth.

Remark 59.13. This leads to different dynamic responses.

—

59.4. Example 4: Influence of Speed

Remark 59.14. Lateral acceleration is given by

$$a_{\text{lat}} = v^2 \kappa(s).$$

Remark 59.15. For identical geometry, increasing speed amplifies dynamic effects.

Remark 59.16. This highlights that transition design cannot be separated from operational conditions.

—

59.5. Example 5: Cant Interaction

Remark 59.17. Cant modifies the effective lateral acceleration experienced by the vehicle.

Remark 59.18. The combination of curvature and cant determines the system behaviour.

Remark 59.19. Transition design must therefore consider both geometric and cant evolution.

—

59.6. Example 6: Interpretation as Control Problem

Remark 59.20. Consider $\kappa(s)$ as an input function.

Remark 59.21. The system produces:

- forces,
- vehicle response,
- wear.

Remark 59.22. Different transition curves correspond to different control strategies.

—

59.7. Example 7: Numerical Comparison of Two Transition Laws

59.7.1. Setting

Remark 59.23. Consider two transition curves of equal length

$$L = 100 \text{ m}$$

connecting a straight segment to a circular arc with final curvature

$$\kappa_1 = \frac{1}{1000} \text{ m}^{-1}.$$

Remark 59.24. Thus, both transitions satisfy

$$\kappa(0) = 0, \quad \kappa(L) = 0.001.$$

59.7.2. Case A: Clothoid

Definition 59.25. The clothoid is given by

$$\kappa_A(s) = \frac{\kappa_1}{L} s = \frac{0.001}{100} s = 10^{-5} s.$$

Remark 59.26. Its curvature derivative is constant:

$$\kappa'_A(s) = 10^{-5} \text{ m}^{-2}.$$

59.7.3. Case B: Cubic Polynomial Transition

Definition 59.27. Consider the normalized cubic law

$$\widehat{\kappa}(u) = 3u^2 - 2u^3, \quad u = \frac{s}{L}.$$

The corresponding physical curvature law is

$$\kappa_B(s) = \kappa_1 \left(3 \left(\frac{s}{L} \right)^2 - 2 \left(\frac{s}{L} \right)^3 \right).$$

Remark 59.28. This transition satisfies

$$\kappa_B(0) = 0, \quad \kappa_B(L) = 0.001,$$

but its curvature derivative is not constant.

59.7.4. Comparison of Curvature Growth

Remark 59.29. At the midpoint $s = 50$ m, the clothoid gives

$$\kappa_A(50) = 10^{-5} \cdot 50 = 0.0005.$$

Remark 59.30. For the cubic transition, with $u = 0.5$,

$$\kappa_B(50) = 0.001 \left(3 \cdot 0.5^2 - 2 \cdot 0.5^3 \right) = 0.001(0.75 - 0.25) = 0.0005.$$

Remark 59.31. Hence both transitions have the same midpoint curvature, but they differ in the way curvature is introduced.

59.7.5. Comparison of Curvature Derivatives

Remark 59.32. For the clothoid,

$$\kappa'_A(s) = 10^{-5} \text{ m}^{-2}$$

is constant over the whole interval.

Remark 59.33. For the cubic transition,

$$\kappa'_B(s) = \kappa_1 \left(\frac{6s}{L^2} - \frac{6s^2}{L^3} \right).$$

With $L = 100$ and $\kappa_1 = 0.001$, this becomes

$$\kappa'_B(s) = 0.001 \left(\frac{6s}{10000} - \frac{6s^2}{1000000} \right).$$

59. Examples

Remark 59.34. At the start and end points:

$$\kappa'_B(0) = 0, \quad \kappa'_B(100) = 0.$$

At the midpoint $s = 50$:

$$\kappa'_B(50) = 0.001 \left(\frac{300}{10000} - \frac{15000}{1000000} \right) = 0.001(0.03 - 0.015) = 1.5 \times 10^{-5} \text{ m}^{-2}.$$

59.7.6. Interpretation

Remark 59.35. The clothoid introduces curvature with constant rate from the very beginning. The cubic transition starts more gently, reaches a higher curvature growth rate in the middle, and then tapers off smoothly toward the end.

Remark 59.36. This shows that identical boundary conditions do not imply identical dynamic behaviour. Even when two transitions connect the same geometric states, their internal curvature evolution may differ substantially.

59.7.7. Dynamic Implication

Remark 59.37. For a vehicle speed v , the lateral acceleration is

$$a_{\text{lat}}(s) = v^2 \kappa(s).$$

Hence the rate of change of lateral acceleration is proportional to

$$v^2 \kappa'(s).$$

Remark 59.38. The numerical example therefore illustrates a central point of this manuscript: transition curves should be compared not only by endpoint conditions, but by their full curvature evolution and its derivatives.

—

59.8. Example 8: Speed and Dynamic Effects

59.8.1. Setting

Remark 59.39. We consider the same two transition curves as before:

- clothoid (linear curvature),
- cubic transition.

Remark 59.40. Let the vehicle speed be

$$v = 200 \text{ km/h} = 55.56 \text{ m/s}.$$

59.8.2. Lateral Acceleration

Proposition 59.41. *The lateral acceleration is given by*

$$a_{\text{lat}}(s) = v^2 \kappa(s).$$

Remark 59.42. With

$$v^2 \approx 3086 \text{ m}^2/\text{s}^2,$$

we obtain for the final curvature

$$a_{\text{lat},\text{max}} = 3086 \cdot 0.001 = 3.086 \text{ m/s}^2.$$

59.8.3. Rate of Change of Acceleration

Definition 59.43. The rate of change of lateral acceleration is

$$\frac{da_{\text{lat}}}{ds} = v^2 \kappa'(s).$$

59.8.4. Clothoid

Remark 59.44. For the clothoid:

$$\kappa'(s) = 10^{-5}.$$

Remark 59.45. Thus:

$$\frac{da_{\text{lat}}}{ds} = 3086 \cdot 10^{-5} = 0.0309 \text{ m/s}^2/\text{m}.$$

Remark 59.46. The acceleration increases at a constant rate along the transition.

59.8.5. Cubic Transition

Remark 59.47. At the midpoint $s = 50 \text{ m}$:

$$\kappa'_B(50) = 1.5 \times 10^{-5}.$$

Remark 59.48. Thus:

$$\frac{da_{\text{lat}}}{ds} = 3086 \cdot 1.5 \cdot 10^{-5} = 0.0463 \text{ m/s}^2/\text{m}.$$

59. Examples

Remark 59.49. At the start and end:

$$\kappa'_B(0) = \kappa'_B(L) = 0$$

and therefore

$$\frac{da_{\text{lat}}}{ds} = 0.$$

—

59.8.6. Interpretation

Remark 59.50. The clothoid introduces acceleration immediately with a constant rate.

Remark 59.51. The cubic transition starts smoothly (zero rate), increases the rate in the middle, and returns smoothly to zero.

—

59.8.7. Engineering Implication

Remark 59.52. From a passenger comfort perspective:

- clothoid: predictable but abrupt start,
- cubic: smoother entry and exit, but stronger variation internally.

Remark 59.53. From a system perspective:

- clothoid: simple control,
- cubic: distributed control with reduced boundary effects.

—

59.8.8. Key Observation

Proposition 59.54. *Even for identical geometry and speed, different curvature laws produce different acceleration profiles.*

—

Summary 59.55. Dynamic behaviour depends not only on curvature magnitude, but on its rate of change.

The numerical example shows that transition curves act as control functions shaping the evolution of forces in the system.

This supports the central thesis that alignment design is fundamentally a problem of curvature control.

—

59.9. Example 9: Optimization Sketch

59.9.1. Motivation

Remark 59.56. Traditional transition design selects predefined curve families.

Remark 59.57. In contrast, the curvature-based framework allows direct optimization of the transition function.

—

59.9.2. Problem Formulation

Definition 59.58. We seek a curvature function

$$\kappa(s), \quad s \in [0, L],$$

such that:

$$\kappa(0) = 0, \quad \kappa(L) = \kappa_1.$$

Remark 59.59. Additionally, we may impose smoothness conditions such as

$$\kappa'(0) = 0, \quad \kappa'(L) = 0.$$

—

59.9.3. Objective

Definition 59.60. A simple objective is to minimize the maximum rate of curvature change:

$$\min_{\kappa} \max_{s \in [0, L]} |\kappa'(s)|.$$

Remark 59.61. This corresponds to minimizing peak dynamic excitation.

—

59.9.4. Candidate Solutions

Remark 59.62. Two candidate solutions are:

- clothoid: constant κ' ,
- cubic transition: zero boundary derivatives.

—

59.9.5. Qualitative Comparison

Remark 59.63. The clothoid distributes curvature growth evenly over the interval.

Remark 59.64. The cubic transition reduces boundary effects but increases curvature variation in the interior.

59.9.6. Optimality Consideration

Proposition 59.65. *An optimal transition balances boundary smoothness and internal variation of curvature.*

Remark 59.66. The optimization problem therefore involves trade-offs between:

- peak curvature derivative,
 - smoothness at endpoints,
 - distribution of curvature change.
-

59.9.7. General Optimization Framework

Remark 59.67. More general formulations may include objectives such as:

- minimizing jerk,
- minimizing energy,
- minimizing wear-related forces.

Remark 59.68. These objectives can be expressed as functionals of $\kappa(s)$ and its derivatives.

59.9.8. Interpretation

Remark 59.69. The optimization problem is defined directly in curvature space.

Remark 59.70. This eliminates the need to select a predefined transition family.

Remark 59.71. Instead, transition curves emerge as solutions of an optimization problem.

59.9.9. Connection to AIM

Remark 59.72. Within the AIM framework, the curvature function becomes a parameterized object subject to constraints and optimization.

Remark 59.73. This enables integration with numerical methods and automated design processes.

59.9.10. Key Result

Theorem 59.74. *Transition curves can be derived as solutions of optimization problems in curvature space rather than selected from predefined families.*

59.10. Summary

Summary 59.75. The present example is formalized in the optimization chapter, where classical transition families are derived as minimizers of curvature-based functionals.

The optimization sketch demonstrates that transition design can be formulated as a mathematical optimization problem.

This shifts alignment design from rule-based construction to model-based optimization and provides a natural link to numerical methods and automated engineering workflows.

59.11. Summary

Summary 59.76. The examples demonstrate that:

- transition curves can be compared via $\kappa(s)$,
- different functional forms lead to different system behaviour,
- geometry alone does not determine performance,
- transition design is inherently a control problem.

Two transitions may connect the same straight and circular arc and still exhibit different internal behaviour.

The clothoid is characterized by constant curvature growth, whereas the cubic transition provides a smoother start and end, but stronger variation in the interior.

This confirms that transition design is fundamentally a problem of curvature control rather than mere endpoint matching.

60. Use Cases

Motivation

Remark 60.1. A theoretical modelling framework becomes substantially more convincing when it is connected to concrete engineering scenarios.

Remark 60.2. The following use cases illustrate how the proposed AIM framework can be applied to realistic railway alignment problems.

Remark 60.3. The examples are not meant as software demonstrations only. They are intended to clarify the interaction between:

- data,
- structured alignment models,
- dynamic interpretation,
- and optimization.

60.1. Use Case 1: Reconstruction of an Existing Alignment

60.1.1. Scenario

Remark 60.4. An existing railway alignment is available only through measured points, legacy file formats, or partially structured engineering data.

Remark 60.5. Typical input may consist of:

- a measured polyline,
- imported TRA/GRA data,
- LandXML fragments,
- or mixed data of uncertain quality.

60.1.2. Task

Remark 60.6. The task is to reconstruct a mathematically coherent alignment model from heterogeneous data.

60.1.3. Pipeline

raw data
→ parsing
→ grabbeltisch sorting
→ sparse / pose3 state reconstruction
→ initial guess
→ numerical refinement

60.1.4. Interpretation

Remark 60.7. This use case emphasizes that railway geometry is not simply “read from a file”. Instead, it is reconstructed as a structured state object.

Remark 60.8. The AIM framework interprets imported data as evidence for an alignment, not as unquestionable truth.

60.2. Use Case 2: Smoothing of an Existing Track

60.2.1. Scenario

Remark 60.9. A measured or existing alignment is geometrically feasible, but exhibits undesirable local irregularities.

Remark 60.10. Examples include:

- noisy curvature estimates,
- abrupt cant changes,
- locally degraded transition zones.

60.2.2. Task

Remark 60.11. The task is to derive an improved alignment preserving the overall route while reducing local dynamic excitation.

60.2.3. Objective Structure

Remark 60.12. Typical objective terms include:

- fitting to the original track,
- smoothness of curvature,
- smoothness of cant,
- reduction of jerk.

60.2.4. Interpretation

Remark 60.13. This is one of the clearest examples showing that alignment should be treated as an optimization problem rather than as a purely geometric description.

60.3. Use Case 3: Design of a New Transition Zone

60.3.1. Scenario

Remark 60.14. A new transition between straight track and circular arc is required.

Remark 60.15. Classically, one would choose a predefined transition family such as a clothoid. In the present framework, the transition may instead be derived through optimization.

60.3.2. Task

Remark 60.16. The task is to construct a transition satisfying:

- boundary curvature conditions,
- cant conditions,
- dynamic smoothness criteria.

60.3.3. AIM Perspective

Remark 60.17. In the AIM framework, the transition is not primarily selected as a named curve family. Rather, it is derived as a controlled curvature evolution.

Remark 60.18. This use case directly illustrates the central thesis that transition curves are control functions rather than merely geometric connectors.

60.4. Use Case 4: Coupled Optimization of Curvature and Cant

60.4.1. Scenario

Remark 60.19. A railway section is to be improved with respect to comfort and dynamic performance.

60.4.2. Task

Remark 60.20. The task is to optimize both:

$$\kappa(s) \quad \text{and} \quad \phi(s),$$

instead of treating cant as a secondary afterthought.

60.4.3. Objective Terms

Remark 60.21. Relevant objective terms include:

- effective lateral acceleration,
- jerk,
- curvature smoothness,
- cant smoothness.

60.4.4. Interpretation

Remark 60.22. This use case demonstrates the transition from pose2-based geometry to pose3-based railway state modelling.

Remark 60.23. It also shows that railway alignment design is inherently a coupled problem.

60.5. Use Case 5: Schwerpunkt-Trassierung

60.5.1. Scenario

Remark 60.24. Instead of designing only the track centerline, one seeks to design the motion of a representative point of the vehicle system.

60.5.2. Task

Remark 60.25. The task is to optimize the trajectory of the center of mass, or of another dynamically relevant reference point, by adjusting curvature and cant.

60.5.3. Model

Remark 60.26. The design variables are:

$$\kappa(s), \quad \phi(s),$$

while the induced trajectory

$$\gamma_c(s)$$

becomes the primary object of evaluation.

60.5.4. Interpretation

Remark 60.27. This use case represents the conceptual extension from classical alignment design toward the new idea of Schwerpunkt-Trassierung.

Remark 60.28. It is one of the clearest expressions of the originality of the present framework.

60.6. Use Case 6: Multi-Track and Network Optimization

60.6.1. Scenario

Remark 60.29. A railway system rarely consists of a single isolated alignment. Instead, one must often deal with:

- parallel tracks,
- switches,
- branching structures,
- coupled corridor geometries.

60.6.2. Task

Remark 60.30. The task is to optimize several alignments simultaneously under topological and geometric coupling constraints.

60.6.3. Typical Constraints

- coincident switch nodes,
- tangent continuity,
- parallel track spacing,
- compatible cant and geometry transitions.

60.6.4. Interpretation

Remark 60.31. This use case shows that the AIM framework extends naturally from single curves to entire railway systems.

60.7. Use Case 7: Interactive Optimization from the Grabbeltisch

60.7.1. Scenario

Remark 60.32. A user imports multiple heterogeneous data sources and arranges them in the grabbeltisch interface according to their semantic role.

60.7.2. Task

Remark 60.33. The system must transform this user-guided sorting into a mathematically meaningful optimization input.

60.7.3. Pipeline

```
drop files
→ detect / parse
→ sort on grabbeltisch
→ build alignment input
→ construct initial guess
→ optimize
→ preview
→ commit to model
```

60.7.4. Interpretation

Remark 60.34. This use case is particularly important because it connects the abstract framework to actual engineering workflow.

Remark 60.35. It also demonstrates that user interaction and mathematical modelling are not opposites, but complementary parts of the same system.

60.8. Use Case 8: Standards and BIM Integration

60.8.1. Scenario

Remark 60.36. An optimized alignment must be exchanged with other systems through standardized or semi-standardized formats.

60.8.2. Task

Remark 60.37. The task is to translate the internal alignment state into layered representations suitable for engineering and BIM workflows.

60.8.3. Interpretation

Remark 60.38. This use case highlights the value of a structured internal model. Only such a model can be mapped consistently to multi-layer data models containing horizontal alignment, profile, and cant.

60.9. Comparative Summary of Use Cases

Use Case	Primary Input	Primary Output
Reconstruction	raw / imported geometry	structured alignment state
Smoothing	measured / degraded track	improved alignment
Transition design	boundary conditions	optimized transition law
Curvature–cant coupling	route + speed + cant	pose3-consistent state
Schwerpunkt	vehicle-based reference	optimized reference trajectory
Network optimization	multi-track graph	globally consistent corridor
Grabbeltisch workflow	user-sorted imports	optimization-ready input
BIM integration	structured state model	exportable layered model

60.10. Conclusion

Summary 60.39. The presented use cases demonstrate that the AIM framework is not limited to theoretical alignment descriptions.

It supports:

- reconstruction,
- optimization,
- dynamic interpretation,
- user-guided engineering workflows,
- and system-level railway design.

Together, these use cases show that alignment can be treated as a structured, dynamic, and optimization-ready object.

61. BIM Alignment Modelling

Motivation

Remark 61.1. While BIM provides a powerful framework for infrastructure data integration, the modelling of railway alignment within BIM remains structurally fragmented and computationally limited.

Remark 61.2. The AIM framework suggests that alignment should not be treated as a secondary geometric component, but as a primary organizing structure.

Remark 61.3. This chapter introduces the concept of BIMalignment as a synthesis of BIM data modelling and alignment-centred computational representation.

61.1. Concept of BIMalignment

Definition 61.4 (BIMalignment). BIMalignment is an alignment-centred modelling approach in which infrastructure geometry, semantics, and dynamics are organized around a canonical alignment state representation.

Remark 61.5. It combines:

- BIM data structures (IFC, LandXML, etc.),
 - AIM state representation,
 - computational and optimization capabilities.
-

61.2. Core Representation

Remark 61.6. The central object in BIMalignment is the alignment state:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 61.7. This representation:

- unifies horizontal and vertical geometry,

- integrates cant,
 - enables dynamic interpretation,
 - supports optimization.
-

61.3. Alignment as Reference Structure

Remark 61.8. In BIMalignment, infrastructure elements are defined relative to the alignment.

- track geometry follows alignment,
- platforms reference stationing,
- signals are positioned along chainage,
- structures are aligned to geometry.

Remark 61.9. This establishes alignment as the primary coordinate and reference system.

61.4. Transformation Pipeline

IFC / LandXML / legacy formats
→ parsing
→ normalization
→ AIM state model
→ BIMalignment model
→ analysis / optimization
→ export to BIM

Remark 61.10. This pipeline separates:

- data exchange (BIM),
 - computation (AIM),
 - application (BIMalignment).
-

61.5. Bidirectional Integration

61.5.1. Import

Remark 61.11. BIM data is interpreted and transformed into the AIM state model.

Remark 61.12. This includes:

- reconstruction of alignment geometry,
 - interpretation of stationing,
 - extraction of cant and profile information.
-

61.5.2. Export

Remark 61.13. Optimized or modified alignments are exported back into BIM formats.

Remark 61.14. This ensures compatibility with:

- existing design tools,
 - project workflows,
 - lifecycle data systems.
-

61.6. Advantages of BIMalignment

61.6.1. Unified model

Remark 61.15. BIMalignment replaces fragmented representations with a unified alignment-centred model.

61.6.2. Computational capability

Remark 61.16. Unlike traditional BIM, the model is directly usable for:

- dynamic evaluation,
 - constraint checking,
 - optimization.
-

61.6.3. Consistency

Remark 61.17. A single state representation ensures consistency between:

- geometry,
 - dynamics,
 - constraints,
 - and derived quantities.
-

61.6.4. Extensibility

Remark 61.18. The framework allows integration of:

- new transition models,
 - advanced vehicle dynamics,
 - additional constraint libraries,
 - network-level interactions.
-

61.7. Relation to Existing BIM Workflows

Remark 61.19. BIMalignment does not replace existing BIM processes, but augments them.

Remark 61.20. It introduces a computational layer that enables:

- deeper analysis,
 - design optimization,
 - improved data interpretation.
-

61.8. Role of the Grabbeltisch

Remark 61.21. The interpretation of BIM data often requires human input.

Remark 61.22. The grabbeltisch serves as an intermediate layer where:

- imported data is classified,
- semantic roles are assigned,
- alignment components are structured.

Remark 61.23. This enables robust handling of heterogeneous and imperfect data.

61.9. Challenges

61.9.1. Standard complexity

Remark 61.24. BIM standards such as IFC are complex and require careful mapping to alignment-centred models.

61.9.2. Data quality

Remark 61.25. Real-world data is often incomplete, inconsistent, or noisy.

61.9.3. Adoption

Remark 61.26. Integration into existing workflows requires:

- tool support,
 - user acceptance,
 - compatibility with established practices.
-

61.10. Outlook

Remark 61.27. BIMalignment provides a foundation for future systems in which:

- alignment is the central modelling object,
 - geometry and dynamics are unified,
 - optimization becomes an integral part of design,
 - BIM evolves from static representation to computational modelling.
-

61.11. Summary

Summary 61.28. BIMalignment extends BIM by introducing an alignment-centred, computationally enabled modelling approach.

It bridges the gap between data representation and engineering computation, enabling consistent, dynamic, and optimization-ready handling of railway alignment within BIM environments.

62. Contribution

Motivation

Remark 62.1. Railway alignment has traditionally been approached as a geometric design problem, supplemented by engineering rules and dynamic considerations.

Remark 62.2. However, these aspects are often treated separately, leading to fragmented models and limited theoretical coherence.

62.1. Central Hypothesis

Theorem 62.3 (Curvature as System Variable). *A railway alignment can be represented as a curvature-driven system in which*

$$\kappa(s)$$

acts as the primary state variable.

Remark 62.4. All geometric, kinematic, and dynamic properties of the alignment follow from $\kappa(s)$ and its derivatives.

62.2. System Interpretation of Alignment

Proposition 62.5. *The behaviour of a railway alignment can be described by the mapping*

$$\kappa(s) \longrightarrow \text{forces} \longrightarrow \text{system response} \longrightarrow \text{performance}.$$

Remark 62.6. This establishes alignment as a system rather than a static geometric object.

62.3. Unification of Transition Curves

Theorem 62.7 (Unified Transition Representation). *All transition curves can be expressed as curvature functions*

$$\kappa(s)$$

within a normalized domain.

Remark 62.8. This allows comparison of historically distinct transition concepts within a single mathematical framework.

62.4. Transition Curves as Control Functions

Theorem 62.9 (Control Interpretation). *Transition curves act as control functions governing the evolution of curvature and thus the dynamic behaviour of the system.*

Remark 62.10. This interpretation replaces the traditional view of transition curves as purely geometric connectors.

62.5. Structured Alignment Model

Proposition 62.11. *An alignment can be represented as a structured sequence of elements:*

- *fixed curvature elements,*
- *transition elements,*
- *auxiliary alignment layers such as cant.*

Remark 62.12. This structure separates representation, geometry, and system behaviour.

62.6. Coupling with Engineering Variables

Remark 62.13. Alignment design is governed by coupled variables including:

- curvature,
- cant,
- speed,
- and their rates of change.

Proposition 62.14. *Dynamic constraints can be expressed as conditions on*

$$\kappa(s), \quad \kappa'(s), \quad \kappa''(s).$$

62.7. Numerical and Optimization Perspective

Remark 62.15. The curvature-based representation allows direct application of numerical integration and optimization methods.

Proposition 62.16. *Alignment design can be formulated as a parameterized optimization problem in curvature space.*

62.8. Integration with Data Models

Remark 62.17. Modern railway data models represent alignment as a multi-layer structure consisting of horizontal, vertical, and cant components.

Proposition 62.18. *The structured alignment model proposed here is compatible with such multi-layer representations.*

62.9. Main Contributions

Summary 62.19. The contributions of this work can be summarized as follows:

- A curvature-based system interpretation of alignment.
- A unified representation of transition curves.
- The interpretation of transition curves as control functions.
- A structured alignment model separating geometry and behaviour.
- Integration of geometry, dynamics, and engineering constraints.
- A foundation for alignment-based information modelling (AIM).

The numerical comparison of transition laws shows that identical boundary conditions may still lead to substantially different internal curvature evolution and therefore different dynamic behaviour.

62.10. Scientific Positioning

Remark 62.20. The presented approach does not introduce a new transition curve, but a new conceptual framework for understanding alignment.

Remark 62.21. It connects previously separated domains:

- geometry,
 - dynamics,
 - engineering design,
 - and data modelling.
-

62.11. Outlook

Remark 62.22. The framework provides a basis for future work in:

- dynamic simulation,
- alignment optimization,
- network-level modelling,
- and BIM integration.

63. Discussion

63.1. Purpose of the Discussion

Remark 63.1. The previous chapters introduced the mathematical foundations, the state-based model, the optimization framework, the software-oriented architecture, and the intended engineering use cases of the AIM approach.

Remark 63.2. The present chapter discusses the significance, scope, and limitations of this framework.

Remark 63.3. Its purpose is not only to summarize previous results, but to position the AIM approach with respect to classical railway alignment design, modern computational methods, and infrastructure data modelling.

63.2. From Geometry to System Modelling

Remark 63.4. A central shift introduced by the AIM framework is the transition from pure geometry to system modelling.

Remark 63.5. Classical alignment theory typically treats the railway path as a curve to be constructed, checked, and documented.

Remark 63.6. In contrast, the AIM framework interprets alignment as a structured state object whose behaviour is governed by curvature, cant, and their derivatives.

Remark 63.7. This leads to a more general view: alignment is not merely a shape, but a functional object with geometric, dynamic, and operational meaning.

63.3. Transition Curves Reinterpreted

Remark 63.8. One of the most important conceptual consequences of the AIM framework is the reinterpretation of transition curves.

Remark 63.9. In many traditional presentations, transition curves appear mainly as named geometric objects, such as clothoids or engineering variants.

Remark 63.10. Within AIM, transition curves are instead understood as control functions for the evolution of curvature and therefore for the dynamic behaviour of the railway system.

Remark 63.11. This reinterpretation is significant because it unifies geometric, dynamic, and optimization-based viewpoints.

63.4. Advantages of the AIM Perspective

63.4.1. Unified internal representation

Remark 63.12. A major strength of AIM lies in the use of a canonical internal model. External data sources may differ substantially in structure and semantics, but can still be mapped into a common state-based representation.

Remark 63.13. This reduces dependence on file formats and enables consistent processing across heterogeneous sources.

63.4.2. Optimization readiness

Remark 63.14. The AIM representation is designed not only for description, but also for computation.

Remark 63.15. Since curvature, cant, and derived quantities are represented explicitly, the model is naturally suited for least-squares methods, SQP, and related optimization approaches.

63.4.3. Dynamic interpretation

Remark 63.16. The framework incorporates dynamics directly through effective lateral acceleration, jerk, cant deficiency, and related quantities.

Remark 63.17. This strengthens the engineering relevance of the model, especially for high-speed and comfort-sensitive applications.

63.4.4. Extensibility

Remark 63.18. The architecture is modular:

- new transition families may be added,
- new constraints may be introduced,
- objective functionals may be reweighted,
- network-level coupling may be extended.

Remark 63.19. This makes AIM suitable not only for one fixed tool, but as a long-term framework.

63.5. Comparison with Classical Approaches

Remark 63.20. Classical railway alignment design often follows a decomposition into:

- horizontal alignment,

- vertical profile,
- cant,
- subsequent rule checking.

Remark 63.21. This decomposition is practical and historically well established. However, it tends to separate modelling from evaluation and often treats dynamics only in a downstream manner.

Remark 63.22. The AIM approach differs by integrating:

- state representation,
- dynamic evaluation,
- and optimization

into a single conceptual structure.

Remark 63.23. This is not intended to invalidate classical methods, but to provide a more general framework into which they can be embedded as special cases.

63.6. Relation to BIM and Data Standards

Remark 63.24. AIM is not designed as a competitor to standards such as LandXML, IFC, or railML.

Remark 63.25. Instead, it serves as a normalization and computational layer between heterogeneous data models and actual engineering reasoning.

Remark 63.26. This is especially important because most standards are primarily designed for exchange and documentation, whereas AIM is designed for interpretation, analysis, and optimization.

Remark 63.27. In that sense, AIM may be viewed as a computational core that complements BIM-style information structures.

63.7. Role of the Grabbeltisch

Remark 63.28. An important practical insight of the present work is that fully automatic interpretation of imported data is neither realistic nor always desirable.

Remark 63.29. The grabbeltisch concept acknowledges that engineering understanding is often needed to assign semantic roles to imported objects.

Remark 63.30. This user-guided intermediate layer is not a weakness of the framework, but a deliberate design choice.

Remark 63.31. It allows the system to combine:

- machine-based parsing,
- human interpretation,
- mathematical reconstruction.

63.8. Limits of the Present Formulation

63.8.1. Physical simplifications

Remark 63.32. The dynamic model introduced in the present manuscript is intentionally simplified.

Remark 63.33. Although effective acceleration and jerk provide meaningful engineering proxies, they do not yet replace full multibody simulation, detailed wheel–rail contact models, or full vehicle-specific dynamics.

63.8.2. Standards formalization

Remark 63.34. While many railway rules can be written as constraint residuals, the full formalization of national and operator-specific regulations remains an open and demanding task.

63.8.3. Topological complexity

Remark 63.35. The extension from single alignments to complete railway networks is one of the strongest features of the framework, but also one of its most challenging aspects.

Remark 63.36. Switches, crossings, branching structures, and corridor-wide coupling introduce substantial mathematical and computational complexity.

63.8.4. Industrial robustness

Remark 63.37. The framework is conceptually strong, but practical deployment requires robust implementations, extensive validation, and careful interface design.

Remark 63.38. In particular, optimization quality depends strongly on:

- good initial guesses,
- stable numerical methods,
- suitable weighting of objectives and constraints.

63.9. Scientific Positioning

Remark 63.39. The contribution of the AIM framework is not the invention of yet another transition curve.

Remark 63.40. Its real contribution lies in the proposal of a broader conceptual framework in which:

- alignment is a state-based object,
- transition curves are control functions,
- geometry and dynamics are treated jointly,
- imported data is interpreted rather than merely accepted,
- and optimization becomes a natural component of design.

Remark 63.41. This positioning places AIM between several established fields:

- railway engineering,
- differential geometry,
- optimization,
- BIM and infrastructure data modelling,
- and interactive engineering software.

63.10. Philosophical Perspective

Remark 63.42. At a deeper level, AIM reflects a shift in engineering modelling: from static representation toward functional interpretation.

Remark 63.43. In this sense, the track is not only a line in space, but a rule-bound, state-dependent object whose meaning arises from its behaviour.

Remark 63.44. This is particularly visible in the move from pose2 to pose3 and from simple curve design to Schwerpunkt-Trassierung.

63.11. Outlook of the Discussion

Remark 63.45. The present discussion suggests that AIM should be understood not as a finished final system, but as a foundational framework.

Remark 63.46. Its long-term value lies in its ability to host:

- richer vehicle models,
- deeper optimization strategies,
- stronger network formulations,
- tighter BIM integration,
- and increasingly automated engineering workflows.

63.12. Summary

Summary 63.47. The AIM framework reinterprets railway alignment as a structured, dynamic, and optimization-ready object.

Its main strengths are the unification of geometry and dynamics, the reinterpretation of transition curves as control functions, and the ability to connect data interpretation, engineering constraints, and numerical optimization.

Its current limitations lie mainly in the simplifications of the dynamic model, the partial formalization of standards, and the remaining challenge of full industrial robustness.

Nevertheless, the framework provides a strong foundation for a new alignment-centred perspective on railway infrastructure modelling.

Teil XIII.

Discussion

Discussion

What do we already know?

Remark 63.48. The previous parts developed AIM as a complete alignment-based information framework.

Remark 63.49. The framework now encompasses geometry, states, runtime evaluation, reality, modelling, optimization, engineering constraints, and practical applications.

Remark 63.50. Together, these components provide a coherent theoretical and practical foundation for railway alignment modelling.

Remark 63.51. The framework itself is therefore largely complete.

What is still missing?

Remark 63.52. A complete framework still requires interpretation.

Remark 63.53. New concepts must be related to existing engineering traditions.

Historical approaches must be re-evaluated.

Underlying assumptions must be examined.

Consequences beyond the immediate scope of the framework must be considered.

Remark 63.54. The missing element is therefore reflection.

Remark 63.55. The question is no longer how AIM works, but what AIM means within the broader context of railway engineering and information modelling.

What comes next?

Remark 63.56. This part provides an interpretive perspective on the AIM framework.

Remark 63.57. The focus shifts from framework construction toward framework understanding.

Remark 63.58. The chapters that follow relate AIM to:

- historical railway engineering approaches,
- established modelling traditions,
- previous optimization systems,

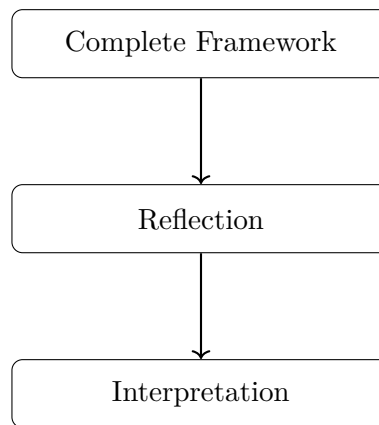
- and broader implications for alignment-based information modelling.

Remark 63.59. Particular attention is given to reinterpreting earlier concepts through the perspective developed throughout this thesis.

Remark 63.60. The objective is not merely comparison, but understanding.

Relation to the AIM Roadmap

Remark 63.61. Within the AIM roadmap, the present part corresponds to the transition from framework construction toward framework interpretation.



Remark 63.62. The progression

Complete Framework $\rightarrow$ Reflection $\rightarrow$ Interpretation

marks the transition from developing AIM toward understanding its position within the broader landscape of railway engineering and information modelling.

Remark 63.63. Discussion therefore serves as the conceptual mirror of the framework, revealing its assumptions, implications, and relationships to established practice.

Principle 63.64 (Interpretation Principle). Within AIM, engineering frameworks should not only be constructed and applied, but also interpreted in relation to historical practice, alternative approaches, and their broader conceptual consequences.

Summary 63.65. Applications demonstrate usefulness.

Discussion provides understanding.

The present part therefore reflects upon the AIM framework, its origins, its consequences, and its place within the broader evolution of railway engineering and information modelling.

64. AXTRAN Reinterpretation

Motivation

Remark 64.1. Classical alignment optimization systems already moved beyond purely manual curve construction.

Remark 64.2. Within AIM, this tradition is reinterpreted through runtime-state trajectory optimization.

64.1. From Geometry Optimization to Runtime-State Optimization

Remark 64.3. AIM generalizes classical alignment optimization toward realization-aware runtime-state trajectory design.

Remark 64.4. This is not a rejection of classical systems, but an extension of their geometric core toward operational state-space optimization.

64.2. Connection to AIM

Summary 64.5. AXTRAN-like optimization can be interpreted as an important predecessor of AIM-style trajectory shaping in curvature and runtime-state space.

Teil XIV.

Outlook

Outlook

What do we already know?

Remark 64.6. The previous parts developed and analyzed the AIM framework in its present form.

Remark 64.7. The foundations, geometry, states, runtime services, realization concepts, modelling structures, optimization methods, engineering constraints, applications, and broader interpretations have now been established.

Remark 64.8. Together, these elements form a coherent framework for alignment-based information modelling.

Remark 64.9. The present work therefore provides a complete description of AIM as it currently exists.

What is still missing?

Remark 64.10. No framework is ever truly complete.

Remark 64.11. New technologies emerge.

Engineering workflows evolve.

Information systems become increasingly integrated.

New requirements arise from automation, digital twins, infrastructure management, and future transportation systems.

Remark 64.12. The missing element is therefore the future perspective.

Remark 64.13. The question is no longer what AIM is, but what AIM may become.

What comes next?

Remark 64.14. This part places AIM into a broader context and explores possible directions for future development.

Remark 64.15. The chapters that follow investigate:

- BIM-related perspectives,
- alignment-centric information models,
- future engineering workflows,

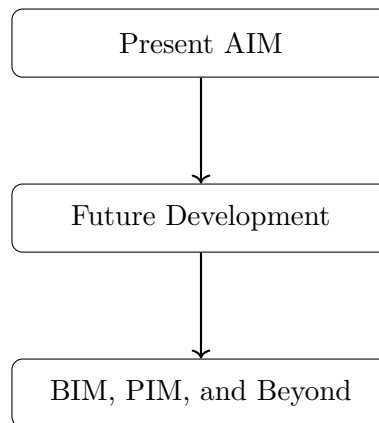
- extensions toward PIM concepts,
- and open research questions.

Remark 64.16. The objective is not to finalize the framework, but to identify opportunities for growth and integration.

Remark 64.17. Particular attention is given to the possibility that alignment may evolve into a first-class information entity comparable to the role currently occupied by building-centric BIM concepts.

Relation to the AIM Roadmap

Remark 64.18. Within the AIM roadmap, the present part corresponds to the transition from the present framework toward future developments.



Remark 64.19. The progression

Present AIM $\rightarrow$ Future Development $\rightarrow$ BIM, PIM, and Beyond

marks the transition from describing the current framework toward exploring its future evolution.

Remark 64.20. The outlook therefore serves not as a conclusion, but as an invitation to continue the development of alignment-based information modelling.

Principle 64.21 (Open Framework Principle). Within AIM, alignment-based information modelling should be understood as an evolving framework whose future development is expected to extend beyond the concepts presented in this thesis.

Summary 64.22. Discussion explains what AIM means.

Outlook explores what AIM may become.

The present part therefore identifies future directions, broader applications, and open questions that extend beyond the scope of the current work while building upon its foundations.

65. BIM and Alignment Modelling

Motivation

Remark 65.1. Building Information Modelling (BIM) has become a central paradigm in modern infrastructure projects.

Remark 65.2. Its goal is the integration of geometry, semantics, and lifecycle information into a unified digital representation.

Remark 65.3. However, despite its broad scope, the representation of railway alignments within BIM remains conceptually and technically limited.

65.1. BIM as a Data Integration Paradigm

Remark 65.4. BIM is fundamentally concerned with:

- object-based modelling,
- semantic enrichment of geometry,
- interoperability across disciplines,
- lifecycle management of infrastructure assets.

Remark 65.5. In this context, geometry is typically associated with objects such as:

- buildings,
- bridges,
- track elements,
- terrain models.

Remark 65.6. Alignment geometry is often embedded implicitly within these objects, rather than treated as a primary modelling entity.

65.2. Alignment in BIM Standards

65.2.1. IFC Alignment

Remark 65.7. Recent extensions of IFC introduce explicit alignment entities.

Remark 65.8. These aim to represent:

- horizontal alignment,
- vertical alignment,
- cant (superelevation),
- stationing.

Remark 65.9. While this is a significant step forward, several issues remain:

- separation of components,
 - lack of unified state representation,
 - limited support for dynamic interpretation,
 - complexity of schema usage.
-

65.2.2. Layered Representation

Remark 65.10. In most BIM-related formats, alignment is decomposed into layers:

- horizontal geometry,
- vertical profile,
- cant definition.

Remark 65.11. This decomposition is practical for data exchange, but does not directly support computation or optimization.

65.3. Limitations of Current BIM Alignment Modelling

65.3.1. Static representation

Remark 65.12. BIM primarily represents geometry as static data.

Remark 65.13. Dynamic properties such as acceleration, jerk, or vehicle response are not part of the core model.

65.3.2. Fragmentation

Remark 65.14. The separation of horizontal, vertical, and cant components leads to a fragmented representation.

Remark 65.15. This fragmentation complicates:

- consistency checks,
 - dynamic evaluation,
 - optimization-based design.
-

65.3.3. Lack of computational core

Remark 65.16. BIM standards focus on data exchange and documentation.

Remark 65.17. They do not provide a unified computational model for alignment analysis and optimization.

65.4. AIM as a Computational Layer for BIM

65.4.1. Conceptual Role

Remark 65.18. The AIM framework can be understood as a computational core that complements BIM data models.

Remark 65.19. Instead of replacing BIM, AIM operates between:

- external data formats,
 - and internal engineering reasoning.
-

65.4.2. Canonical Representation

Remark 65.20. AIM introduces a unified representation:

$$(\gamma(s), \theta(s), \kappa(s), \phi(s)).$$

Remark 65.21. This representation:

- integrates horizontal geometry,
- integrates vertical and cant behaviour,
- supports dynamic evaluation,

- is directly usable in optimization.
-

65.4.3. Transformation Pipeline

BIM / IFC / LandXML

- parsing
- fat representation
- normalization
- AIM state model
- analysis / optimization
- export

Remark 65.22. This pipeline separates:

- data exchange concerns,
 - computational modelling,
 - engineering reasoning.
-

65.5. Alignment-Centred BIM

65.5.1. Shift of perspective

Remark 65.23. The AIM framework suggests a shift toward alignment-centred modelling.

Remark 65.24. Instead of treating alignment as one component among many, it becomes a primary organizing structure.

65.5.2. Implications

- infrastructure elements are referenced to alignment,
 - geometry is derived from alignment state,
 - dynamic behaviour becomes part of the model,
 - optimization becomes a native operation.
-

65.6. Integration with BIM Workflows

65.6.1. Import

Remark 65.25. Existing BIM data is imported via:

- IFC parsing,
 - LandXML parsing,
 - legacy format interpretation.
-

65.6.2. Processing

Remark 65.26. Within AIM, the imported data is:

- normalized,
 - reconstructed,
 - enriched with dynamic interpretation,
 - potentially optimized.
-

65.6.3. Export

Remark 65.27. Results may be exported back into:

- IFC alignment structures,
- LandXML files,
- other engineering formats.

Remark 65.28. This ensures compatibility with existing BIM ecosystems.

65.7. Relation to BIMinfra

Remark 65.29. Infrastructure BIM (often referred to as BIMinfra) extends BIM concepts to linear infrastructure such as railways and roads.

Remark 65.30. AIM can be interpreted as a specialization within BIMinfra focusing on:

- alignment-centric modelling,
 - dynamic interpretation,
 - optimization-driven design.
-

65.8. Discussion

Remark 65.31. BIM provides a powerful framework for data integration, but lacks a strong computational alignment model.

Remark 65.32. AIM addresses this gap by:

- introducing a unified state representation,
- enabling dynamic evaluation,
- supporting optimization,
- and integrating heterogeneous data sources.

Remark 65.33. Together, BIM and AIM can be seen as complementary:

- BIM for data integration and lifecycle,
 - AIM for computation and alignment reasoning.
-

65.9. Summary

Summary 65.34. Current BIM standards provide important structures for data exchange and semantic modelling, but remain limited in their treatment of railway alignment.

The AIM framework complements BIM by introducing a unified, state-based, and optimization-ready representation of alignment.

This enables a transition from static data handling toward dynamic, computational, and alignment-centred infrastructure modelling.

66. Future Work

66.1. Purpose of this Chapter

Remark 66.1. The AIM framework presented in this manuscript establishes a mathematical and conceptual foundation for alignment-centred modelling.

Remark 66.2. However, many aspects remain open and provide opportunities for further research, development, and practical implementation.

Remark 66.3. This chapter outlines potential directions for future work, ranging from mathematical extensions to software architecture and engineering applications.

66.2. Extension of the Dynamic Model

66.2.1. From simplified dynamics to vehicle models

Remark 66.4. The current formulation relies on simplified dynamic quantities such as effective lateral acceleration and jerk.

Remark 66.5. Future work should extend this toward more realistic vehicle models, including:

- multi-body vehicle dynamics,
- wheel–rail interaction,
- suspension behaviour,
- track elasticity.

66.2.2. Speed-dependent modelling

Remark 66.6. The integration of variable speed profiles

$$v(s)$$

opens the possibility for more realistic evaluation and optimization of alignments under operational conditions.

66.3. Schwerpunkt-Trassierung

66.3.1. Conceptual development

Remark 66.7. The idea of designing the trajectory of a dynamically relevant point, such as the center of mass, represents a fundamental extension of classical alignment design.

Remark 66.8. Further work is required to:

- formalize the mapping from track geometry to vehicle trajectory,
- incorporate full dynamic models,
- define appropriate objective functions.

66.3.2. Integration into optimization

Remark 66.9. A key research direction is the coupling of Schwerpunkt-Trassierung with optimization methods, enabling:

- comfort-driven design,
 - dynamic optimization,
 - direct evaluation of vehicle response.
-

66.4. Advanced Optimization Methods

66.4.1. SQP extensions

Remark 66.10. While Sequential Quadratic Programming provides a powerful baseline, future work may include:

- tailored SQP formulations for sparse alignment structures,
- improved constraint handling,
- adaptive weighting strategies.

66.4.2. Global optimization

Remark 66.11. Alignment optimization problems are often non-convex.

Remark 66.12. Future research may explore:

- global optimization techniques,
 - multi-start strategies,
 - hybrid deterministic–stochastic methods.
-

66.5. Network-Level Modelling

66.5.1. From single alignment to network

Remark 66.13. Real railway systems consist of interconnected alignments forming complex networks.

Remark 66.14. Future work should extend the AIM framework to:

- branching structures,
- switches and crossings,
- corridor-wide optimization.

66.5.2. Topological constraints

Remark 66.15. A systematic treatment of topological constraints remains a major challenge.

Remark 66.16. This includes:

- node consistency,
 - multi-track coupling,
 - network-wide feasibility conditions.
-

66.6. Constraint and Objective Libraries

66.6.1. Constraint formalization

Remark 66.17. Although many railway rules can be expressed as constraints, a complete formalization of regulatory frameworks remains an open task.

Remark 66.18. Future work should aim at:

- systematic encoding of standards,
- validation against real projects,
- parameterization for different railway systems.

66.6.2. Objective design

Remark 66.19. The design of objective functions is crucial for optimization-based alignment design.

Remark 66.20. Potential extensions include:

- cost models,

- maintenance optimization,
 - energy efficiency,
 - lifecycle considerations.
-

66.7. Integration with BIM and Data Standards

66.7.1. Deeper BIM integration

Remark 66.21. The integration of AIM with BIM workflows can be further developed by:

- tighter coupling with IFC alignment models,
- bidirectional synchronization,
- embedding AIM states within BIM objects.

66.7.2. Standard extensions

Remark 66.22. The AIM framework may also inspire extensions to existing standards, for example:

- inclusion of dynamic properties,
 - support for optimization metadata,
 - representation of control-based alignment models.
-

66.8. Interactive and Real-Time Systems

66.8.1. Live optimization

Remark 66.23. One of the most promising directions is the development of systems that allow real-time feedback during alignment design.

Remark 66.24. This includes:

- live preview during optimization,
- interactive constraint adjustment,
- immediate visualization of dynamic effects.

66.8.2. User interaction

Remark 66.25. The grabbeltisch concept may be extended to:

- interactive constraint definition,
 - visual debugging of alignment states,
 - intuitive editing of transition parameters.
-

66.9. AI and Assisted Design

66.9.1. Data-driven approaches

Remark 66.26. Machine learning techniques may complement the AIM framework by:

- learning from existing alignments,
- suggesting initial solutions,
- identifying patterns in design choices.

66.9.2. Hybrid systems

Remark 66.27. A promising direction is the combination of:

- physics-based modelling (AIM),
 - data-driven methods (AI),
 - interactive user control.
-

66.10. AXTRAN Revival and Beyond

Remark 66.28. Classical systems such as AXTRAN demonstrated the feasibility of optimization-based alignment design.

Remark 66.29. Future work may aim to:

- reconstruct and generalize such systems,
 - extend them to network-level problems,
 - integrate modern numerical and computational techniques.
-

66.11. Software Architecture and Deployment

66.11.1. Scalability

Remark 66.30. Practical applications require scalable implementations capable of handling large datasets and complex networks.

66.11.2. Multi-view systems

Remark 66.31. Future systems may include:

- synchronized multi-window views,
 - combined 2D/3D representations,
 - integration of GIS and CAD functionality.
-

66.12. Long-Term Vision

Remark 66.32. In the long term, the AIM framework may contribute to a new paradigm of infrastructure modelling in which:

- alignment becomes the central organizing structure,
 - geometry, dynamics, and optimization are unified,
 - engineering design becomes an interactive computational process.
-

66.13. Summary

Summary 66.33. The AIM framework opens a wide range of research and development directions.

These include extensions of the dynamic model, integration with BIM, network-level optimization, real-time systems, and AI-assisted design.

Together, these directions point toward a future in which railway alignment is no longer treated as static geometry, but as a dynamic, computational, and optimizable system at the core of infrastructure engineering.

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